

Murray Stahl: Selected Writings (reader draft)

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- Subject index (stub)
- Chronological bibliography (vault book_candidate=yes)
- Editorial consistency pass (checklist)
- Note on textual variants

Murray Stahl: Selected Writings

A **reader's edition** compiled from Horizon Kinetics research publications, compendia, letters, and attributed interviews.

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Edition

See `book/EDITION.md` for versioning and how to roll forward new quarterlies.

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You also maintain a **single compiled PDF** in the same vault:

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Editorial expectation

- The compilation will **overlap** many pieces that already appear as separate chapters. It is a **convenience / provenance / completeness** anchor, not a second pass of the full book body. When a chapter's headnote cites a specific compendium or commentary PDF, that file remains the **primary** line of attribution; this volume is the **shelf copy** that bundles much of the same material under one filename.

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Book VI spine in this edition

- `python book/scripts/build_compilation_spine.py` →
`book/sources/compilation_inventory.csv`, `compilation_coverage.csv`, and `part-book-vi-studies-in-absurdity-spine.md`.
- `python book/scripts/emit_appendix_book_vi_compilation_full_text.py` → `appendix-b-book-vi-compilation-full-text.md` (full extract body for the export).

Preface — How This Edition Was Built

This manuscript is a **reader's edition**, not a substitute for the original publications. The aim is to gather Murray Stahl's public and firm-attributed analytical writing in one continuous reading experience: essays, letters, commentaries, and (in later parts) interview answers and transcribed remarks, each with a clear **source line** so you can return to the primary PDF or web file.

The underlying archive lives alongside this book folder: hundreds of Horizon Kinetics PDFs, plus selected downloads mirrored from the firm's website and, where necessary, the Internet Archive when a live link was broken but a historical file could be recovered. A **single compiled PDF**, 830350067-Compilation-of-Murray-Stahl-s-Writings.pdf, also sits in the vault; it overlaps many chapters but is kept **in the audited corpus and extracted** as a reference volume (see manuscript/00-compiled-murray-stahl-writings-source.md). Text was extracted mechanically and then **lightly edited for readability**—running headers, page numbers, and obvious extraction artifacts—without changing the substance of the arguments.

Co-authored and firm-authored material. You instructed that co-authored pieces may be included. Where a chapter is a "Founders' Letter" or similar, the voice is often first-person plural and reflects Horizon Kinetics as a firm. Those chapters remain in the edition because they are part of the published record Stahl co-authors; headnotes state the genre plainly.

Interviews and video. Interview chapters will reproduce **Stahl's answers only**, with the interviewer and publication named. Video appearances are treated the same way once transcribed; embed URLs are collected separately so transcripts can be verified against the recording.

What this edition does not claim. It does not claim completeness relative to every paywalled magazine, every conference panel, or every private communication. It claims fidelity to the texts that could be located in the archive at the time of assembly, and a transparent method for adding more as they surface.

The **step-by-step assembly procedure**—extract, curate, outline, draft, index—is written out in book/00_STEPS.md so the project can continue methodically even if work pauses for months between sessions.

Exported PDF. The reader PDF (book/build/export/Murray-Stahl-Reader-Draft.pdf) includes an auto-generated **Contents** page at the front (major # headings only). **Appendix B** repeats the full compilation extract on purpose, alongside chapters that already excerpt some of the same material—see manuscript/98-editorial-consistency-pass.md for redundancy notes and formatting spot-checks.

The first chapter proper begins immediately after this preface: a short Murray's Musings piece on bonds in an era of suppressed yields—a useful on-ramp because it is self-contained and illustrates how later, longer quarterlies will be presented.

Chapter 1 — The Transformation of the Fixed Income Asset Class

Source. Murray's Musings, published in the *Fixed Income & Cryptocurrency Compendium*, Horizon Kinetics LLC, © 2020.

PDF in vault. TheTransformationOfTheFixedIncomeAssetClass.pdf (parent hk_pdfs/ folder).

Extract used for assembly.

book/build/text/TheTransformationOfTheFixedIncomeAssetClass.txt

Minor presentation edits only: removed repeated compendium running heads, page banners, and page-break markers.

The Transformation of the Fixed Income Asset Class

Fixed income is an investment that is a type of loan. A borrower is obligated to make payments of a predetermined amount, and the principal must be paid on the maturity date. The legal predetermination of the payments is the origin of the term fixed. In other words, fixed means predetermined. This legal definition will not change.

An exceedingly low level of interest rates changes the character of this type of investment in extraordinary and not so obvious ways. It should be evident that a fixed income investment that pays a coupon of less than 2% is, both before and after taxes, a negative real (after inflation) return instrumentality. For a New York City resident of means, even a fixed income instrument that yields 4% or less, is a negative real return instrumentality. Although there is surprisingly little discussion of this reality, it should be self-evident to investors.

It is less obvious—although nonetheless true—that bonds, in this historically low interest rate period, if sold short, are actually incredibly inexpensive put options. An example will serve to illustrate this point. Consider that the Bank of America 4.44% bonds due January 20, 2048 are

now priced, on a yield-to-maturity basis, at 3.31%. They trade well above par value and at maturity must be worth par value. Prior to maturity, which is almost 30 years from now, a variety of circumstances could occur in which these bonds could be worth considerably less than par.

If, for instance, U.S. government bonds for some reason yield 5% instead of the present level of 2.2%, the Bank of America bonds would probably yield 7.45%—even assuming that credit quality does not deteriorate.¹ Quite obviously, the bonds would trade well below par value.

Hence, two possibilities exist for these Bank of America bonds. The first is that interest rates will never increase. As a result, the Bank of America bonds will not be priced at par value until the maturity date of January 20, 2048. The short seller will simply lose the 3.31% annual yield to maturity. A short seller in the 50% tax bracket (Federal, State, and local) will effectively lose half that sum after the benefit of tax deductions. Consequently, one would pay 1.66%—half of 3.31%—for a long-term put option on interest rates. Which is incredibly cheap—just try to price a multi-year put option on long-term interest rates with a broker-dealer.

The other possibility is that interest rates do increase. In that event, the bonds might trade substantially below par value—depending upon how high interest rates rise. These bonds are effectively a put option for which the premium is the yield to maturity. The potential loss for a bond (absent default), or in this case for a near-30-year put option on the bond, is limited to the yield to maturity. In other words, the loss is limited to the amount of the current price premium for the bond, just as for the option premium for a conventional put option.

This similarity to a put option is problematic on a broader systematic basis, because it changes the incentives for investing in bonds. Consider the matter from the perspective of an investor who might purchase a U.S. Treasury bond due in 2048 that has a yield to maturity of 2.23%. What if, alternatively, that bond were used as collateral to establish a small short position in a Bank of America bond, say a short equal to 10% of the long position. The investor puts \$100 into the U.S. Treasury bond yielding 2.23%, then takes \$10, or 10%, shorting that amount of Bank of America debt for which they must pay a yield to maturity rate of 3.31%. In terms of that investor's new risk/reward profile, the first aspect of this hedged position is that the net yield on it is actually a positive 1.9% carry.

An investor considering such a trade must contemplate whether or not it is better to sacrifice 33 basis points of annual income in exchange for the possibility that the creditworthiness of Bank of America might deteriorate. If Bank of America creditworthiness deteriorates, as occurred in 2008, the bank's debt might trade at 50% of par value, as it did at that time. A 10% short position with a loss of possibly 60% in price would produce a portfolio-level trading gain of 6%.

One might argue that such a trade is outside of the ambit of a conventional bond investor and therefore unlikely to be made. But it is not outside of the ambit of an investor who would be inclined to sell short Bank of America common stock. The shares now yield 2.17%, so that is a cost of the short sale. Moreover, the shares can appreciate. Thus far, in 2019, the shares have increased in value by 32.67%. That would have been the loss to a short seller.

Why might a short-seller have been attracted to the Bank of America shares? The net income for the first nine months of 2019 was \$19.25 billion—a decline of 2.07% versus the comparable period in 2018, when it was \$19.657 billion.

What if a hypothetical short seller were sufficiently astute to predict that in 2019, the net income of Bank of America would decline. This hypothetical short seller would have predicted, as well, the bank's difficulty with overnight borrowing in September of that year. Although this investor might reasonably have anticipated that Bank of America shares would decline, what could not have been reasonably anticipated is that its stock would in fact appreciate by 32.67%.

Establishing a hedge against a banking-related issue with systemic dimensions can be very expensive, carry costs aside, in terms trading risk, until the problem finally manifests itself. Selling short a bank bond substantially eliminates the trading or directional risk, since the bond will ultimately mature at par value. That is to say, if an investor is worried about possible banking risks, short sale of bank stock is too blunt an instrument to use as a hedge, there being many circumstances in which bank shares could rise considerably in price, creating an extremely expensive hedge.

Surveying Both Idiosyncratic and Systemic Risks in Low-Yield Bonds

There exist many long-term bonds that, in principle, have high price convexity paired with extraordinarily low yields to maturity. One example might be San Francisco City & County Water 6.95% due November 1, 2050, which trades at a price of 159.21 with a 3.22% yield to maturity. Apart from higher interest rates, variables that could cause the bonds to decline are fluctuations in the creditworthiness of the city or an earthquake that could severely damage the water system, requiring extensive repairs.

All U.S. bonds trade in relation to Treasury bond yields. Yet, with less than a 100-basis point spread over Treasuries, it is clear that municipal bonds, project finance bonds, or corporate

bonds have unique risks that are not being priced into them. For example, consider the Comcast 4.00% Notes due November 1, 2049, trading at 110.82 with a 3.41% yield to maturity. Major changes are clearly evident in Comcast's delivery of content to subscribers and in the company's ability to sell advertising. Although no investor is in a position to predict the financial conditions of Comcast over the 30-year life of this bond, the valuation is predicated upon the belief that there is no realistic possibility of a significant deterioration in its financial position.

The same statement might be made about AT&T, which is actually a fairly leveraged enterprise. Its 5.7% bonds due March 1, 2057—a 38-year life—are priced at 125.38 with a yield to maturity of 4.32%.

Universities are now issuing debt, despite their huge endowments, because the cost of debt capital is so low, while the likelihood is high that the university will ultimately repay the debt in inflated dollars. As an example, Princeton University 5.70% bonds due March 1, 2039 trade at 139.32 with a yield to maturity of 3%. Similarly, Harvard has issued a 4.875% bond due October 15, 2040 now trading at 130.80 with a yield to maturity of 2.9%.

If one sells short the Princeton or Harvard bonds, all one needs for a substantial profit is a modest rise in interest rates. A rise in long-term Treasury yields of perhaps 150 basis points for any reason would cause the Harvard bonds to trade at par value. A 30-year U.S. Treasury now yields 2.2%; 150 basis points higher, and it would yield 3.72%. If the Harvard bond were to trade a yield spread of 115 basis points above the Treasury, it would be priced at par value almost 31 points lower. In fact, because any meaningful rise in interest rates would probably severely reduce the market value of the Harvard endowment, the assessment of Harvard's creditworthiness would probably be reduced as well, such that the spread to Treasuries would probably rise as well.

The State of Texas 5.52% bonds due April 1, 2039 are priced at 138 with a yield to maturity of 2.82%. Surely, one could envisage a diminution of creditworthiness if, for some reason, extraction of fossil fuels were to be prohibited in the United States. In a less extreme scenario, the same outcome might be achieved by a severe reduction in oil and gas prices. The bond market does not appear very concerned about this possibility, at least judging by the present yield of the State's debt.

If the production of coal were to be forbidden in the U.S., it would negatively impact the fortunes of Union Pacific Corporation (UNP). Union Pacific managed to issue a 4.375% bond due November 15, 2065 which now trades at 109.85 with a yield to maturity of 3.91%. In fact, even a severe recession could lower an investor's perception of the credit worthiness of Union Pacific. The company also issued 3.84% bonds due March 20, 2060 presently trading at 102.32 with a yield to maturity of 3.73%.

Altria Group, Inc. (MO) managed to sell 6.20% bonds due February 14, 2059, presently trading at 117.33 with a yield to maturity of 5.16%. The mere existence of such a bond is more than a statement about the long-term real rate of return on bonds. It is effectively a statement that large-scale smoking in the U.S. will continue past the year 2059.

Here is where it becomes yet more interesting.

The University of California issued 4.77% bonds due May 15, 2115, now priced at 126.36 with a yield to maturity of 3.75%. It is virtually a certainty that the person buying this bond will not live to the maturity date. Ohio State University managed to sell 4.80% bonds due June 1, 2111, now trading at 129.45 with a yield to maturity of 3.68%.

Guardian Life Insurance Company of America has issued 4.85% bonds due January 24, 2077. they trade at 124.87, a yield to maturity of 3.78%. A life insurance company is an interesting case relative to long-term bonds. If interest rates increase, it would surely have a negative impact on the market values of the debt. On the other hand, if rates decrease sufficiently from this point, the company might experience difficulty earning enough from its bond portfolio to pay its annuity liabilities. Viewed from the perspective of the bond buyer, the best-case scenario is that interest rates remain at present levels and neither materially increase nor decrease for many decades.

If an investor buys a 57-year bond from a life insurance company whose entire investment portfolio is comprised almost entirely of long-term bonds, and if one of the many long-term bonds in its investment portfolio loses value, then the collateral for your bond will lose value, as well.

These bonds are only a small sample of many that have a similar character. The common denominator is that the rate of compensation above that of Treasury bonds is less than 200 basis points per annum and sometimes even less than 100 basis points, despite the fact that many of the bonds have maturity dates far longer than the 30-year Treasury. The University of California bond will mature in 91-years or 3x the maturity of the longest-term U.S. Treasury bond. It does not appear that much, if any, marginal compensation is paid for the extra 60-year life of the bond. In the light of such pricing, there is no room for any rise in interest rates or any deterioration of credit quality. Yet, history demonstrates that such eventualities frequently occur.

The expense of selling such bonds short is small relative to the reward possible if history repeats

itself. Ergo, such bonds and many others are now effectively low-cost, low-risk put options with heretofore unimaginably lengthy times to expiration.

If shorting, I prefer to sell short bonds—not stocks—because they are a much more reasonable hedge against uncertainty. There is a limit to how much money one can lose, and one knows what that limit is—as opposed conventional hedge instruments, in which the potential loss is infinite.

Q. I have a hypothetical question relative to the circumstance of a 75-year-old investor who expects to live ten additional years and who invests in long-term bonds that yield 4.8% with good credit, as with the University of California issue. Playing devil’s advocate, what would be wrong with that, if rates stay roughly the same over that period?

A. What would be wrong with that? If that investor has to live off the principal—not just the income—and the bond declines in market value, there would be plenty wrong with it. That person would be forced to sell the bond for money to live on, and the sale might not produce adequate funds. That person might die without enough money. An alternative risk is that this investor is fortunate enough to be in very good health and live longer than the expected additional 10 years, but with each passing year, the cost of living rises above the purchasing power of that bond’s coupon income. That investor would not be very happy with that outcome. That is what’s wrong with it.

End of chapter. Next: Chapter 2 — “The Achievement of Diversification” (chapter-02-the-achievement-of-diversification.md).

Chapter 2 — The Achievement of Diversification

Source. Murray’s Musings, *European Contrarian Report Compendium*, November 2020.
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PDF in vault. TheAchievementofDiversificationNovember2020.pdf
Extract used for assembly.
book/build/text/TheAchievementofDiversificationNovember2020.txt

Contributors (as in original). Murray Stahl, Chairman; Steven Bregman, President; Peter Doyle, Managing Director.
Presentation edits: removed compendium running heads, page banners, and the standard lengthy research disclaimer at the end of the PDF; merged words broken across line ends in the source extract.

The Achievement of Diversification

It would seem to be arithmetically self-evident that if various constituent members of an index have such low weightings as to be statistically irrelevant, then they cannot, arithmetically, play any role in diversification. Eighteen members of the S&P 500 index now have only a 1-basis point (0.01%) weight. There are 38 index constituents that have a 2-basis point weight, 41 that have a 3-basis point weight, 41 with a 4-basis point weight, and 46 at 5 basis points.

These 184 constituents of the 500 S&P Index are arguably statistically irrelevant. To go further, one could designate members with 6-basis point weights as also statistically irrelevant. If one draws the boundary of irrelevance in this manner, there are another 36 such positions, bringing the total number of statistical irrelevancies to 220, or 44% of the names in the index.

If one wishes to assert, as an issue of statistical irrelevance, that at least 50% of the S&P 500 names fall into this category, one has only to include the 7-basis point positions. There are 29 constituent members with that weight. Thus, the grand total of statistically irrelevant positions is 249, or 49.8% of the names in the S&P 500 index.

A summary of the index weightings of approximately half of the constituent members of the S&P 500 is shown in the following table.

Table 1: Selected S&P 500 Index Weightings

Number of companies	Weighting (basis points)	Total weight (basis points)
18	1 bp	18 bp
38	2 bp	76 bp
41	3 bp	123 bp
41	4 bp	164 bp

Number of companies	Weighting (basis points)	Total weight (basis points)
46	5 bp	230 bp
36	6 bp	216 bp
29	7 bp	203 bp
Subtotal	249 companies	1030 bp (~10.3% of index cap.)

Source: iShares

Roughly one-half of the number of corporations in the S&P 500 index have, in aggregate, 10.3% of the index's market capitalization.

Interestingly, there are only 17 positions with an 8-basis point weight, and another 28 9-basis point positions. If these positions, in addition to those listed above, were considered as statistically irrelevant, the total number of irrelevant positions would equal 295, or 59% of the index and would have a collective weight of 14.27%. One could continue in this manner, because there are 20 10-basis point positions.

As noted, there are 295 positions of 9-basis points weight or less that comprise 59% of the number of members of the S&P 500 index and 14.27% of its market value. To assess the importance of these firms to the index, assume that—for only the 295 companies in question—a recession occurs that is equivalent in severity to the recession of 2008. In this scenario, assume that these 295 companies as a group decline in share price by (37)%, which was the S&P 500 return for 2008. The result is a 37% loss on 14.27% of the index or (5.28)%—which is not a significant decline during a major recession.

This is hardly the same thing, though, as saying that these companies are unimportant businesses or are statistical irrelevancies within the U.S. economy. One way of understanding index weight versus economic impact is to select ten well-known firms with statistically immaterial S&P 500 weights and calculate their economic impact by a metric other than stock market capitalization. I chose these ten companies that are familiar to investors: Discovery, Inc., News Corp., Sealed Air, Newell Brands, Franklin Resources, Mohawk Industries; American Airlines, Carnival Corporation, Textron, and Lumen Technologies, otherwise known as CenturyLink, Inc.

Table 2: Ten Firms with Immaterial Weights in the S&P 500

Ticker	Name	Weighting
DISCA	Discovery, Inc. – Series A	3 bp
NWSA	News Corporation Class A	3 bp
SEE	Sealed Air Corporation	2 bp
NWL	Newell Brands	2 bp
BEN	Franklin Resources, Inc.	2 bp
MHK	Mohawk Industries, Inc.	2 bp
AAL	American Airlines Group, Inc.	2 bp
CCL	Carnival Corporation	3 bp
TXT	Textron Inc.	3 bp
LUMN	Lumen Technologies, Inc. (CenturyLink, Inc.)	3 bp

Source: iShares

Those ten firms collectively have a 25-basis point weight in the S&P 500. One might compare them to Apple Inc. (AAPL), which has a 6.45% index weight.² In other words, the weight of Apple in the index is 25.8x the weight of these ten well-known companies.

The metric I have chosen to assess these firms' economic impact is the number of their employees. This statistic is significant, because in a horrendous recession some of the employees presumably would lose their employment. Together, these firms have 485,800 employees. In contrast, Apple has 137,000 employees.

Table 3: Number of Employees in Ten Firms with Immaterial Weights in the S&P 500

Ticker	Name	Number of employees
DISCA	Discovery, Inc. – Series A	9,000
NWSA	News Corporation Class A	28,000
SEE	Sealed Air Corporation	16,500
NWL	Newell Brands	49,000

Ticker	Name	Number of employees
BEN	Franklin Resources, Inc.	9,700
MHK	Mohawk Industries, Inc.	42,100
AAL	American Airlines Group	133,700
CCL	Carnival Corporation	120,000
TXT	Textron Inc.	35,000
LUMN	Lumen Technologies, Inc. (CenturyLink, Inc.)	42,800
	Total	485,800
AAPL	Apple, Inc.	137,000

Source: Company reports

Some argue that if the ten aforementioned firms were to have difficulties—and obviously some of these firms now face serious challenges—the economic impact would be heaviest from those firms with the largest number of employees. In industry terms, a 20% layoff rate from the diverse group with a 25-basis point collective weight would be far more damaging to the U.S. economy than if Apple were compelled to undertake a 20% layoff. If the object of diversification is to reflect economic impact, then surely one can argue that the weightings of an index should reflect the impact of a company on the economy. Market capitalization weights reflect valuation rather than economic impact.

Tesla, Inc. (TSLA) has a \$392 billion market capitalization and, viewed from that perspective, is by far the largest automobile company on the planet. General Motors (GM) has a \$52.7 billion market capitalization, and Ford Motor (F), \$32 billion. When Tesla enters the S&P 500 index, it will obviously overwhelm the weightings of GM and Ford. GM, in a reasonably good year, would build 7.7 million vehicles, Ford might produce 2.4 million vehicles, and Tesla, somewhat more than 400,000.

Thus, the question of the meaning of the term *diversification* arises. Diversification of the S&P 500 index is meant to reflect the U.S. economy, but the fact that the S&P 500 does not do so is plain to see. If diversification is based upon market capitalization, then diversification is effectively based upon valuation. The conclusion would logically follow that the index would have a bias in favor of higher valuation.

Valuation is not the only bias within a national stock market. Most large firms have become global multinational companies, regardless of their nation of origin. If the country of incorporation is a small one, then that country is probably financially irrelevant to a global multinational firm. The obvious example of this phenomenon is Switzerland. The large index members of the iShares MSCI Switzerland ETF (EWL) are surely not Swiss with respect to the geographic diversification of their businesses. The following table lists the ten largest positions in that index.

Table 4: iShares MSCI Switzerland ETF (EWL) — Ten Largest Positions

Ticker	Name	Weighting
NESN	Nestlé SA	21.07%
ROG	Roche Holding PAR AG	12.90%
NOVN	Novartis AG	9.73%
ZURN	Zurich Insurance Group AG	3.87%
ABBN	ABB Ltd.	3.70%
LONN	Lonza Group AG	3.56%
UBSG	UBS Group AG	3.47%
GIVN	Givaudan SA	3.04%
SIKA	SIKA AG	2.88%
CFR	Compagnie Financiere Richemont SA	2.82%
	Total	67.04%

Source: iShares

In 2019, Switzerland accounted for 1.26% of Nestlé's sales. For Roche, 1.94% of sales were attributed to Switzerland. The fact that these firms cannot possibly achieve an acceptable level of growth for their investors by confining their business activities to Switzerland should be apparent. For many years, Nestlé, Roche, and Novartis expanded outside of Switzerland. These three firms collectively represent 43.7% of the MSCI Switzerland index, which is still considered to be a diversified index. MSCI Switzerland is an index that cannot possibly be a successful investment without the success of Nestlé, Roche and Novartis.

The historical return of MSCI Switzerland, since inception on March 12, 1996, is 7.18% annually. Is this an acceptable return, with that level of concentration? The difficulty with weightings

based on market capitalization is that eventually some company within the index will either become the best performing investment or, alternatively, achieve enormous size by raising capital via share issuance. There exists no objective limiting principle to reestablish diversification other than placing an arbitrary limit on the size of position. Most index providers would not allow a single position to become 25% or more of an index. Another reasonably popular limiting principle is to restrict the number of 5% positions to about 50% of the portfolio. All indexes, nevertheless, seem to have accumulated statistically irrelevant smaller holdings.

MSCI Switzerland has only 39 positions, therefore, concentration might be inevitable. The smallest position in the index is Banque Cantonale Vaudoise, with a 44-basis point weighting.

An alternative approach, if one wants to invest in Switzerland itself on a diversified basis, might be to purchase Swiss real estate companies, on the theory that the real estate is, after all, located in Switzerland. Valuations of those companies must, therefore, reflect conditions in that country.

There are four primary publicly traded real estate companies in Switzerland: Swiss Prime Site, PSP Swiss Property, Allreal Holding and Mobimo Holding.

Table 5: Four Primary Traded Real Estate Companies in Switzerland

Ticker	Names	Market capitalization (CHF in billions)
SPSN	Swiss Prime Site AG	6.05
PSPN	PSP Swiss Property AG	5.15
ALLN	Allreal Holding AG	3.12
MOBN	Mobimo Holding AG	1.75

Source: Bloomberg

Of these four, only Swiss Prime Site, with a 74-basis point weighting, qualifies for inclusion in the MSCI Switzerland index. One could argue that these four real estate firms have collectively become a more representative and diversified Swiss market index than MSCI Switzerland, by virtue of the total number of properties they hold and of their diversification by geography and property type.

The following table lists the number of properties held by the four primary Swiss real estate companies.

Table 6: Holdings of Four Swiss Public Real Estate Companies

Ticker	Names	Number of buildings
SPSN	Swiss Prime Site AG	187
PSPN	PSP Swiss Property AG	160
ALLN	Allreal Holding AG	64
MOBN	Mobimo Holding AG	138
	Total	549

Source: Company filings

This table raises a profound question about the nature of indexation. If one wishes to invest in the nation of Switzerland, is this best achieved by building a concentrated portfolio of giant multinational firms that have proportionately little business activity in Switzerland? Or, alternatively, can one devise a weighting system among the four publicly traded Swiss real estate firms that have their entire investment located within Switzerland? The fact should be obvious that a portfolio of Swiss real estate firms is a superior means of investing in Switzerland.

The limiting factor is that if specific Swiss diversification is to be achieved in this manner, this diversification is not scalable. The aggregate market capitalization of these four Swiss real estate companies is only 16.07 billion CHF. Nestle, alone, has a market cap of 300 billion CHF. Mass-scale indexation must necessarily be global multinational investing, because only global-scale firms have sufficient equity liquidity to support indexation at the trillion-dollar level.

Concentrated—not diversified—investing can be rewarding during a period of declining interest rates and increasing globalization. If these economic variables are not dominant in the future, an entirely different form of indexation will probably take their place.

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Next: Chapter 3 — Inflation and Economic Thought.

Chapter 3 — Inflation and Economic

Thought

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Contributors (as in original). Murray Stahl, Chairman; Steven Bregman, President; Peter Doyle, Managing Director.

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Inflation and Economic Thought

Historians write about inflation in a very different way than economists.

Historians view inflation as a destabilizing force, whatever the cause. For instance, historians agree that a salient cause of demise of the Song Dynasty of China (960–1279) was the fact that it issued paper money, the first government to do so. The successor Yuan Dynasty lasted until 1368, and the inflation caused by fiat currency issuance is generally cited by historians as a factor in its demise. Similarly, the inflation caused by the influx of gold and silver into Spain from Spanish colonies in the New World in the 16th and 17th centuries is known to historians as “the price revolution.” The repercussions of the inflation caused by this dramatic rise in the supply of gold and silver undermined Spain’s international economic standing. There were other historical debasements, such as King Henry VIII of England continually reducing the gold and silver content of the coinage to pay for his lavish lifestyle. In British history, this is known as “the Great Debasement.”

None of these events or any of the many other historical debasements have seemed to attract the interest of even the greatest economists. Many financial experts regard John Maynard Keynes as the greatest economist of all time, yet in his magnum opus, *The General Theory of Employment, Interest and Money* published in 1935, the index shows only seven mentions of inflation in the entire book. Of these mentions, one simply states the fact of inflation in the Central Europe of the postwar era. The other references are largely definitional and used as foundations upon which the theory of employment is placed. Inflation basically is defined as the point beyond which monetary expansion does not produce increases in output. The word *debasement* is never used in the book. Keynes might be forgiven for such a view, since *The General Theory of Employment, Interest, and Money* was written in a period of history during which the challenge was to increase output and thereby increase employment.

In the magisterial three-volume biography of John Maynard Keynes by Robert Skidelsky, there are no references to inflation in the index to Volume I, only six references to inflation in the index to the 635-page Volume II, and a single reference in the index to Volume III, which is about as long as Volume II.³ Ironically, Keynes became very wealthy by speculating on the decline of the German currency in the aftermath of the First World War. In other words, Keynes became wealthy because of German inflation and his understanding of it. He was very interested in inflation as an investor. He appears not to have been vitally interested in the subject as a scholar.

A similar phenomenon is observed in *A Monetary History of the United States 1867–1960* by Milton Friedman and Anna Jacobson Schwartz.⁴ Since the book is about monetary issuance, one might expect some discussion of the subject of debasement. The word is not mentioned once in the index to this book that is 700-pages long excluding the appendixes. There are five references to inflation in the index. This should not be surprising. The object of the book is to study the optimal rate of currency issuance with a view to economic stimulus. Some degree of currency issuance makes government and individual debt more manageable, so used properly and in moderation, inflation is just a policy tool to increase economic growth.

In neither the Keynes book nor the Friedman book is there any advice as to what an investor might do to protect wealth from inflation’s destruction of purchasing power. Of course, neither book is designed for an investor. These books were written for policymakers and other economists. It would never have occurred to either economist that the policymakers be deprived of the power or policy tool to create money and, thereby, inflation.

Friedrich Hayek is unique among the major economists of the 20th century in that he advocated that national governments be deprived of the money creation function, which he describes in *The Denationalization of Money* (1976).⁵ The basic idea of the Hayek book is that governments cannot be trusted not to abuse the money-creation function. Therefore, Hayek advocates the creation of “private money.” The problem is that one cannot necessarily trust those in control of private money either, and as a consequence of this obvious objection, the Hayek book remained in obscurity until rediscovered by the proponents of cryptocurrency.

In the world of cryptocurrency and blockchain, it is possible that users of the currency can

continually verify its issuance status. As an example, if bitcoin will issue slightly less than 2.5 million coins between the time of this writing and the year 2140, that is objectively verifiable as a fact by means of the blockchain, which forms the permanent public record of every bitcoin transaction since its creation. Bitcoin ease of transfer has yet to rise to the level of being accessible to the average person, but that function is constantly improving. Private money might be the solution to the inflation problem. Unfortunately, it is not yet ready for mass usage. In any case, the community of academic scholars tends to avoid the subject of cryptocurrency.

In 1975, the economist John Kenneth Galbraith wrote a popular account of the history of money entitled *Money: Whence It Came, Where It Went*.⁶ Given the era in which the book was written, the author could hardly avoid discussing the subject of inflation. Galbraith strongly advocates for government wage and price controls or, alternatively, wage and price guidelines that are informal requests with which government pressure might force compliance even if they are not formal controls. Even a book that is directed towards the general public is really a call for action by policymakers.

The common factor among economists of a variety of philosophical orientations is that inflation is something that must be controlled by government action of some sort. The underlying assumption is that the inflation problem has nothing to do with existing government policy. Moreover, inflation is not considered to be a problem that can be confronted by the individual on behalf of the individual.

The problem with a government policy is that its effects on individual circumstances differ from person to person. As an example, consider a pension adjusted annually to a national consumer price index. Even if it is calculated in a manner to properly measure and account for inflation, it might not properly measure the increase in expenses of a given pensioner if that individual's expenditure is weighted heavily towards items experiencing above average inflation, such as health insurance or education.

The economic consequences of inflation, the historical documentation of which spans well over 2,000 years, are such that it merits its own asset class in investment planning. It should be self-evident that one cannot simply assume that common stocks *sui generis* are efficient or are a sufficient hedge against inflation. Labor is the most significant operating expense for most members of the popular stock indexes. Many people assume that a business can maintain profit margins by increasing wages at the same rate as general inflation. However, a wage increase of X% in a progressive tax system does not necessarily become an after-tax wage increase of that same X%, because in a progressive system, higher wages are taxed at a higher rate. The U.S., of course, follows a progressive tax structure. As a generalization, one requires a pre-tax wage increase above inflation to obtain an after-tax increase equivalent to inflation. Most businesses, under a condition of persistent inflation, are not able to maintain their margins once wage rates begin to inflate.

A good argument can be made for gold as an inflation hedge in a portfolio, but it must be remembered that gold can be highly idiosyncratic for very long periods of time. For example, from 1980 to 2000, gold actually declined, although inflation clearly existed during this period; the price of gold was unchanged from 1934 to 1967, as a consequence of government action, although price levels increased by 2.5x.

The Federal Reserve Bank of Minneapolis calculates inflation in the United States back to 1800 and the data illustrates the same point.⁷ The First World War years in the U.S. were highly inflationary, as is evident in the following figures.

Table 1: U.S. Inflation Rate, 1916–1920

Year	Rate
1916	7.7%
1917	17.8%
1918	17.3%
1919	15.2%
1920	15.6%

Source: Federal Reserve Bank of Minneapolis

The gold prices for those same years are as follows; clearly, the gold price did not keep pace with inflation at that time:

Table 2: Gold Prices, 1916–1920

Year	Price
1916	\$18.99
1917	18.99
1918	18.99

Year	Price
1919	19.95
1920	20.68

Source: piketty.pse.ens.fr

As a matter of historical curiosity, in 1717, during Isaac Newton's tenure as Warden and Master of the Royal Mint of the United Kingdom, he set the gold price at 3 pounds 17 shillings. It was still at that price in 1914, on the eve of the First World War.

On the other hand, one could have purchased gold in the year 2000 for \$279.11 an ounce and it now trades at \$1,908.07 an ounce. This is about a 10% annualized increase, considerably above most estimates of inflation for this period. If one can tolerate the idiosyncratic behavior of gold, it can be a good protection against inflation.

Far better than gold are the royalty companies that finance gold production by the purchase of revenue interests, or some effective equivalent, in precious metals mines. The two most successful are Franco Nevada Corp. (FNV) and Wheaton Precious Metals Corp. (WPM). Once the investment is made, the return is greatly enhanced if the price of the commodity rises, since the royalty companies experience no increase in expenses; theirs is simply a contractual relationship, and they require none of the capital equipment or physical field operations of a mining company. Other examples of gold royalty companies, are:

Table 3: Additional Gold Royalty Companies

Ticker	Company
RGLD	Royal Gold, Inc.
SAND	Sandstorm Gold Ltd.
OR	Osisko Gold Royalties Ltd.
MMX	Maverix Metals Inc.
MTA	Metalla Royalty & Streaming Ltd.

This is an example of a “hard asset” business model. The revenue is a contractual commitment by the producer of the commodity to the company providing the financing. The inflation beneficiary aspect is that the royalty company's operating expenses do not increase in the event of inflation; in the event of inflation, a revenue increase is essentially a pre-tax profit increase.

The challenge to asset allocators and other investors is that an insufficient number of such hard asset companies exist to form an index. Also, the aggregate market capitalization is very small in relation to the great amount of assets in the world that require some inflation protection.

Many investors in the practical world, therefore, are compelled to use leverage as an inflation hedge, on the premise that the cost of debt capital is low and, in any case, even modest inflation will eventually erode the purchasing power of the obligations that must one day be repaid. That is, for investors employing leverage, paying back “in cheaper money”: the amount of the bond or loan due on the maturity date in five years or 10 remains the same as on the first day of the loan, but the nominal income of the borrower has presumably inflated over time.

John Maynard Keynes did not write extensively about inflation, but his personal portfolio was structured in such a manner to benefit from consistent inflation inasmuch as he made use of copious leverage. The following table, taken from *Keynes' Way to Wealth* by John F. Wasik, will illustrate the point.⁸

Table 4: John Maynard Keynes — Leverage and Net Worth, 1919–1945

Year	Loans	Net assets	Year	Loans	Net assets
1919	£15,498	£16,315	1933	£78,859	£55,222
1920	20,837	(1,837)	1934	165,343	146,007
1921	N/A	N/A	1935	188,271	220,619
1922	2,720	21,558	1936	299,347	506,522
1923	1,200	34,364	1937	190,035	215,244
1924	1,200	63,797	1938	106,470	181,547
1925	1,200	43,610	1939	109,136	199,495
1926	2,200	40,800	1940	60,655	171,090
1927	46,900	44,000	1941	28,753	205,281
1928	25,790	13,060	1942	19,720	254,073
1929	14,000	7,815	1943	31,643	312,271
1930	65,000	12,525	1944	46,167	355,310

Year	Loans	Net assets	Year	Loans	Net assets
1931	11,965	15,100	1945	38,886	411,238
1932	19,774	21,722			

Source: John F. Wasik, *Keynes' Way to Wealth* (New York: McGraw Hill Education, 2014), 119.

There were years where Keynes was enormously leveraged. For example, in 1920, Keynes had £20,837 of borrowings, and his net assets were negative £1,837. One could say technically he was insolvent. Similarly, in 1928 on the eve of the stock market crash, he had £25,790 of leverage and £13,060 of net assets. In the year 1929, it was £14,000 of leverage and £7,815 of net assets. One can see this involves a fairly large multiple. In 1930, he had £65,000 of loans, and £12,525 of net assets, he was over 5x leveraged. It was only near the end of his life, during the war years, that Keynes had fairly modest leverage in relation to his net assets. The last full year of his life, 1945, he had only £38,886 of borrowings against net assets of £411,238.

It should be noted that Keynes drew living expenses from his personal account, so that the years of decline in net assets are not entirely from investment depreciation. The net asset values are not meant to illustrate investment performance. In any case, it is obvious that Keynes frequently made use of enormous leverage relative to his net assets.

This process has continued among investors and business people to this day. The contemporary world might be the most leveraged society in history. That means that the constituency supporting debasement of debt and money is very large. Inflation is the potential savior of the debt leveraged. Therefore, it is prudent to develop inflation hedges.

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Next: Chapter 4 — Diamonds as an Investment (chapter-04-diamonds-as-an-investment.md).

Chapter 4 — Diamonds as an Investment

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Conflict / offering note (from original). An affiliate of Horizon Kinetics Asset Management LLC acts as co-sponsor to a private fund that invests in diamonds; the fund is available only to certain qualified investors.

Companion essay in this edition. *Wealth and Crisis: The Role of Diamonds in Asset Allocation* (European Contrarian, January 2023) → Chapter 8; vault PDF [European-Contrarian-Compendium-January-2023_Wealth-and-Crisis_The-Role-of-Diamonds-in-Asset-Allocation.pdf](#).

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Diamonds as an Investment

Although diamonds were known and gathered 4,500 years ago, diamond mining on a systematic basis could only proceed with the development of the modern sciences of geology and crystallography. This was possible only at the end of the 19th century, when systematic diamond extraction in South Africa commenced. This ultimately gave rise to the De Beers monopoly. De Beers controlled the diamond market until the end of the 20th century.

The De Beers monopoly was an obvious barrier to price discovery and transparency. Thus, the evolution of diamonds as an asset class only became possible in the 21st century. Even still, diamonds present valuation problems not ordinarily present in commodities. Gold and silver of given levels of purity, for example, are homogeneous commodities, while all diamonds are heterogeneous; no two are identical. Accordingly, there developed a detailed grading system. Diamonds are commonly divided by the four criteria of color, cut, clarity, and carats, each with its own measures.

That grading system has existed for over 50 years, but it is only lately that modern computing technology has enabled the ability to assemble a diverse set of diamonds into packets that each have equivalent value in much the same way that ETFs are designed to combine sets of

individual companies to be representative of an industry exposure. Such packets, then, are homogeneous, like other commodities. That is a significant change in the investment characteristics of diamonds as an asset class.

This section of the Compendium is intended to serve as a reference guide for and answer certain questions about what might be regarded as basic diamond investing. One such question might be: Where is historical data on diamond prices kept? Another might be: Is historical diamond price data accessible to the public?

Historical price data is available via the Rapaport Group, which is a large Belgian broker of diamonds. This is the only database that I have seen that makes diamond prices readily available. Rapaport publishes the price lists, and one can purchase them from the company. For instance, one can buy the price list for April 27, 2007 for \$150.

Rapaport also publishes and sells a Diamond Price Statistics Annual Report for each year, starting with 1994, which is approximately the time that the De Beers monopoly was effectively dismantled. It might be recalled that the U.S. government charged both General Electric Company (GE) and De Beers, in 1994, with industrial diamond price fixing.

It is fair to assert that diamond prices rarely decline, at least according to the data. A notable exception was during the 2008 financial crisis, but even then the decline was modest in comparison with other assets, such as stocks and bonds. A brief drop in diamond prices also occurred when the World Trade Center was destroyed in 2001.

Constructing a diamond price index would require making certain assumptions regarding index weightings, and such assumptions can be subjective. It will not be attempted in these pages. In order to comprehend the complexity of building an index of diamond prices, one might parse the problem by considering the impact on price just of the differences in diamond cut.

Petra Gems publishes a diamond buying guide that illustrates the complex nature of this challenge. Oval or round cut diamonds are the most desirable and therefore the most expensive diamonds if all other factors are held constant, which of course, is rarely the case. In order of desirability:

- Oval cut diamonds are priced 5–15% lower than round cut diamonds.
- Princess cut diamonds are priced 17–26% lower than round cut diamonds.
- Cushion cut diamonds are priced 22–30% lower than round cut diamonds.
- Radiant cut diamonds are priced 20–27% lower than round cut diamonds.
- Emerald cut diamonds are priced 19–27% lower than round cut diamonds.
- Pear cut diamonds are priced 15–22% lower than round cut diamonds.
- Marquise cut diamonds are priced 23–32% lower than round cut diamonds.
- Asscher cut diamonds are priced 21–31% lower than round cut diamonds.

Another complexity is that the Rapaport Price List is based on ask prices rather than actual realized prices. A dealer frequently will demand and receive a 15% or even 20% discount to the Rapaport ask price. The market still lacks a certain level of transparency, since it is difficult to know actual realized prices.

A further problem is that even realized prices are semilogarithmic with respect to certain attributes, such as carat size. To illustrate this relationship, let us assume the existence of a round cut diamond of 1 carat, with color D (highest color grade) and clarity FL (flawless). If the selling price is X , a 1.5 carat round cut diamond with color D and clarity FL would sell for about 2.14x the price of the comparable 1 carat diamond. To be clear: if one compares the 1.5 carat diamond to a 1 carat diamond with these characteristics in common, the price would not be 50% higher just because it has 50% more weight; it actually would be 2.14x as expensive as the 1 carat diamond.

The only difference between the two diamonds in this example is that the latter has 1.5x the weight of the former. The 2.14x greater price is because a 1.5 carat diamond is rarer than a 1 carat diamond. In other words, the value density of a 1.5 carat diamond is more than 1.5x as great as a 1 carat diamond's value density.

Value density will ultimately become extremely important as the concept of physical commodities increasingly dominates the world of commodity ETFs. A 3 carat round cut diamond with D color and FL clarity could sell for \$170,000.

To be clear about the value density, a 3 carat diamond weight is only 6 grams, and there are 31.1035 grams in a troy ounce. Thus, a diamond with the aforementioned characteristics and price is selling for $\$170,000 \div 6 \text{ grams}$, or \$28,333 per gram. One troy ounce of diamonds of this caliber would theoretically sell for $\$28,333 \times 31.1035$, or \$881,255. In contrast, one troy ounce of gold sells for about \$1,757. Therefore, one ounce of diamonds with the aforementioned quality features is 501.6x more value dense than gold ($\$881,255 \div \$1,757$).

The largest physically backed gold ETF is SPDR Gold Shares (GLD), with \$50.736 billion of assets under management. Using the current \$1,757/ounce gold price for the calculations, this fund holds roughly 28,876,494 ounces of gold. If confidence in the financial system were to be lost, the investors ostensibly could demand to collect their 28,876,494 ounces of gold. What would that entail?

A troy ounce of gold has the following dimensions: 0.95 inches long, 1.65 inches wide, and 0.08 inches thick, which equals 1.254 cubic inches. An ounce of gold is easily transported and easily hidden in a crisis. However, transporting and hiding 29 million ounces is a far more complex logistical challenge—that would be 1.999 million pounds. One Honda Accord weighs about 3,300 pounds.

Contrast that value density with diamonds. If an ounce of diamonds is worth \$881,255, achieving a valuation of \$50,736,000,000, the market capitalization of SPDR Gold Shares, would require only $\$50.736 \text{ billion} \div \$881,255$ or 57,572.43 ounces, or 3,985 pounds. A human being could easily transport or hide over \$8 million by holding a mere 10-ounce diamond packet.

One of the reasons investors wish to hold physical gold is that it cannot be easily debased like the U.S. dollar or other fiat currencies. However, all of the large central banks have extensive gold reserves, and it might be necessary for central banks to sell those reserves if a time comes when the fiat currency is not trusted and might not even be an acceptable medium of exchange. In that case, large gold sales might depress the price of gold. On the other hand, central banks do not have reserves of diamonds to sell.

Gold has been readily available in sufficient supply at the requisite level of purity to use as a medium of transaction for thousands of years. On the other hand, a market for diamonds could not exist that was fair in any meaningful sense of the word until the modern science of crystallography developed.

A factor in the performance of diamonds is the evolution of India's diamond industry, which is where much of the cutting and polishing of rough diamonds is now done. The Indian diamond industry employs 850,000 people. In fact, the Indian city of Surat in the western state of Gujarat has become known as Diamond City. Diamond cutting is a highly skilled art, and people who are expert in this field are well compensated in Western nations. Indian skilled labor is far less expensive. The Indian diamond export trade is centered almost entirely in the Bharat Diamond Bourse in Mumbai, which handles 98% of the export trade. It has a constructed area of 2 million square feet, with two basement complexes containing an extra 1 million square feet.

The transition of a rough diamond to a polished cut diamond is a time-consuming process. Indian labor has greatly reduced friction in that area. That inexpensive labor has served to restrain diamond prices in recent years.

The global market for diamonds is expanding, as a greater proportion of the world population enters the middle class and becomes part of the diamond consumers' universe. Existing diamond exchanges include the London Diamond Bourse, the Diamond Federation of Hong Kong, the Israel Diamond Exchange, the Dubai Diamond Exchange, the Amsterdam Diamond Bourse, the Bharat Diamond Bourse, the Tokyo Diamond Exchange, the Shanghai Diamond Exchange, the Antwerp Diamond Exchange, the Diamond Dealers Club of Australia, Borsa Diamanti d'Italia, and the Guangzhou Diamond Exchange.

A number of new diamond bourses are in the process of being established. These include the New York Diamond Dealers Club, Borsa Istanbul, Diamant Club Wien, Bangkok Diamonds and Precious Stones Exchange, the Diamond Exchange of Singapore, and the Korea Diamond Exchange. This does not include the World Federation of Diamond Bourses itself.

It is important to recall that De Beers had a global monopoly, for all practical purposes, and was involved in price fixing that was only effectively challenged in 1994. This was over 300 years after the Buttonwood Agreement of 1792 that became the foundation of the New York Stock Exchange. The first important revision of that agreement did not occur until 1817. The U.S. Securities and Exchange Commission did not commence operations until 1934. The Depository Trust Company (DTC), which does clearing, was only founded in 1973, as the central securities depository of the U.S. market system, 181 years after the New York Stock Exchange. Bitcoin began its existence as a white paper published on October 31, 2008, and was first used in 2009. In 2022, it still has no regulatory structure.

In other words, the modern diamond market is evolving more rapidly than the securities market.

The first recorded use of a diamond engagement ring was in 1477, when Archduke Maximilian of Austria proposed marriage to Mary of Burgundy. Diamonds were a rarity even for engagement rings until 1947, when De Beers engaged the now-defunct advertising firm N.W. Ayer & Son to create the campaign entitled "A diamond is forever." This was one of the most successful advertising campaigns of all time and created the modern diamond market. The universal diamond grading system was only created by the Gemological Institute of America (GIA) in the 1950s. The Gemological Institute of America was only founded in 1931.

The diamond asset class is developing very rapidly. However, it is very unlike ordinary asset classes like securities, in which creation can be infinite—which is to say, if a certain kind of "paper" is appealing to the public, issuers can and will produce, with virtual immediacy, as great a quantity of it as the market will bear. The global production of rough diamonds is in decline, and probably in inexorable decline. The Facts & Figures section in this Compendium contains more data on that.

An investable asset class is in the process of being created and, moreover, with an asset manifesting declining production supply. According to Statista, rough diamond production was 177 million carats in 2005 and 116 million carats in 2021. The most alluring returns from an asset class occur before it actually becomes an investable asset class.

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Next: Chapter 5 — Trends in Healthcare.

Chapter 5 — Trends in Healthcare

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Trends in Healthcare

According to Statista.com, in the calendar year 2022, there were 4.76 billion retail prescriptions filled in the U.S. The current U.S. population is 334.299 million. This amounts to about 14.24 prescriptions for every human being in the nation. This does not include pharmaceuticals dispensed in hospitals. It does not include pharmaceuticals dispensed in ambulatory care clinics. It does not include so-called over-the-counter drugs sold without a prescription.

This is one of the reasons that healthcare is a growth industry. The proportion of GDP devoted to healthcare is generally increasing. A recent calculation by the World Bank tabulates that in 2019, healthcare spending in the U.S. amounted to 16.8% of GDP. The Statista calculation for 2020 indicates that healthcare spending accounted for 18.8% of U.S. GDP. Obviously, the 2020 figure was impacted by extraordinary measures necessary to deal with the pandemic.

COVID-19 is a worldwide phenomenon, yet the United States is unique even among industrialized nations as for its healthcare spending as a proportion of GDP. Apart from the U.S., the highest such expenditures as a percent of GDP are by the following ten nations, the highest of which is Canada at 12.9%.

Table 1: Healthcare expenditure as a percentage of gross domestic product (2020)

Country	Healthcare expenditure / GDP
Canada	12.9%
Germany	12.8%
France	12.2%
United Kingdom	12.0%
Switzerland	11.8%
Austria	11.5%
Sweden	11.5%
Norway	11.3%
Netherlands	11.2%
Japan	11.1%

Source: Statista

According to Statista, Costa Rica spends 7.9% of its GDP on healthcare. Hungary spends 7.3% of its GDP on healthcare. Luxembourg spends 5.7% of its GDP on healthcare. Mexico spends 6.2% of its GDP on healthcare.

One measure, although certainly not the only measure, of the effectiveness of these levels of spending is life expectancy. The World Health Organization (WHO) calculates life expectancy figures for nations annually.

Table 2: Life expectancy by nation (2022)

Country	Years
United States	78.203
Canada	82.847
Germany	80.989
France	83.229
United Kingdom	82.156
Switzerland	84.255
Austria	82.412
Sweden	83.505
Norway	83.393
Netherlands	82.451
Japan	84.820
Costa Rica	77.320
Hungary	74.958
Luxembourg	82.591
Mexico	74.832

Source: World Health Organization

When comparing national expenditures as a proportion of GDP, it is important, for meaningful comparison, to take account of any wide differences in per-capita GDP. One can say that the U.S. spends 2½ times more on health care than Costa Rica, based on spending 18.8% of GDP on healthcare versus Costa Rica spending 7.3%. However, the per-capita GDP of the U.S. is much higher than that of Costa Rica. The estimated per capita GDP of the U.S. in 2022 is \$76,027, while for Costa Rica it's \$13,089.⁹ Those figures allow one to calculate the per capita dollar amount of spending on healthcare:

- U.S.: $18.8\% \times \$76,027 = \$14,293.08$
- Costa Rica: $7.3\% \times \$13,089 = \955.50

That reveals that the per capita expenditure of the U.S. on healthcare is 15.0 times that of Costa Rica. Nevertheless, the U.S. life expectancy is 78.203 years and the life expectancy of Costa Rica is 77.320 years. Viewed on a per capita basis, the vast excess of U.S. expenditure does not appear to be productive.

A problem with this type of calculation is that the various data sources do not agree on the life expectancy figures. The aforementioned life expectancy figure for Costa Rica is compiled by the World Health Organization. The Population Division of the Department of Economic and Social Affairs at the United Nations calculates the life expectancy of Costa Rica at 79 years. Macrotrends.net lists it at 80.56 years. The peer reviewed journal known as *Proceedings of the National Academy of Sciences* is the source of the macrotrends.net life expectancy figure.

In any case, in the U.S., national healthcare expenditure routinely expands more rapidly than GDP. According to CMS.gov (Centers for Medicaid and Medicare Services), the annual spending figures for the past 10 years are as follows.

Table 3: U.S. national healthcare expenditure

Year	Annual expenditure (in trillions)
2011	\$2.676
2012	2.783
2013	2.856
2014	3.002
2015	3.165
2016	3.207
2017	3.448
2018	3.604
2019	3.757
2020	4.114
2021	4.256

Source: CMS.gov

The compound annual expansion rate of spending is 4.75%. It is, of course, needless to say that the U.S. population does not grow at 4.75% per annum, and the GDP does not grow at that rate, either. The logical consequence of the trend is that the healthcare expense burden on the

economy will become increasingly heavy.

In common investment parlance, the healthcare industry is frequently used as a synonym for pharmaceuticals. However, the U.S. healthcare sector is far more diverse. This is evident in the sectoral diversification of the largest healthcare ETF, known as The Health Care Select Sector SPDR Fund (XLV). This ETF is divided into five sectors, as detailed in the accompanying table.

Of these, biotechnology is an interesting category. It is obvious that the sector commands significant market capitalization. However, it does not produce large quantities of revenue. In any analysis of healthcare expenditure, it is more meaningful to determine the respective shares of that expenditure exclusive of biotechnology due to its limited revenue relevance. If this were done, the proportionate share of each of the four sectors would be as follows. While the pharmaceuticals sector accounts for somewhat over a third of the total, the balance is made up of health care providers and services, and two separate categories of companies providing health care equipment and supplies, and analytical tools, instruments, clinical trial services and research services.

Table 4: XLV industry allocation

Sector	Weight
Health Care Equipment & Supplies	18.77%
Health Care Providers & Services	22.36%
Life Sciences Tools & Services	12.95%
Pharmaceuticals	30.28%
Biotechnology	15.64%

Source: ssga.com

Table 5: XLV industry allocation excluding biotechnology

Sector	Weight
Health Care Equipment & Supplies	22.25%
Health Care Providers & Services	26.51%
Life Sciences Tools & Services	15.36%
Pharmaceuticals	35.89%

Source: ssga.com

In order for this system to continue to expand at its historical rate, the typical human being would have to devote more time to medical examination and analysis in order to determine if medical intervention is required and, should such intervention be required, the degree of intervention and type of treatment. Ultimately, the growth proposition for health care spending can be reduced to the internet/technology growth proposition that has previously been propounded in Compendium discussions: that in order for the historical rate of industry growth to be sustained, the average person must spend more time on the internet. In the medical sphere, the average person would have to spend increasing amounts of time interacting with the medical system.

The OECD (Organisation for Economic Co-operation and Development) maintains data on the length of the average hospital stay for most nations. It will be recalled from a prior table that the U.S. spends more on healthcare, as a proportion of GDP, than other industrialized nations. The following table compares the length of the average hospital stay in the U.S. versus the other industrialized nations.

Using length of hospital stay as a metric indicates that the U.S. uses its healthcare system less intensively than all nations on the list, excepting Sweden and the Netherlands. Even in these two instances, the intensity of hospital use is only marginally less than that of the U.S. Nevertheless, the U.S. spends a far greater proportion of its GDP on healthcare than does either Sweden or the Netherlands. The relevant figures are that Sweden spends 11.5% of its GDP on healthcare, the Netherlands spends 11.2% of its GDP on healthcare, and the U.S. spends 18.8%. If one were to use life expectancy as a measure of healthcare outcomes, the U.S. obtains a less satisfactory result than either Sweden or the Netherlands, or even any other industrial nation.

Table 6: Length of average hospital stay (2021)

Country	Days
Canada	7.7
Germany	7.4
France	5.5
United Kingdom	6.2

Country	Days
Switzerland	6.9
Austria	6.4
Sweden	5.3
Norway	5.9
Netherlands	5.2
Japan	16.4
Costa Rica	6.8
Hungary	5.4
Luxembourg	7.5
Mexico	N/A
United States	5.4

Source: OECD

If U.S. healthcare expenditures were to remain at 18.8% of GDP and not rise beyond this point, it must logically follow that the healthcare industry would be growing at the rate of GDP. This would be the nominal figure and not the real figure. Thus, if real GDP were to average 2% growth per annum and inflation were held at the level of 2%, and the healthcare industry maintains its current share of GDP, it would be growing at a 4% rate.

At the time of this writing, The Health Care Select Sector SPDR Fund (XLV) trades at 16.32x earnings and 4.67x book value, and the 3–5 year estimated annual growth rate, as stated on the most recent fund factsheet, is 7%.

Those valuations make it fairly obvious that the stock market participants in this sector presume that the current circumstance, which is to say the historical rate of expansion, will continue. Implicitly, that embodies the presumption that the healthcare industry will represent an ever-increasing share of U.S. GDP. It presumes that the general populace will not be troubled by the fact that other nations have a healthcare industry that consumes a much smaller proportion of GDP; that foreign healthcare systems appear to be used more intensively by the population and with a far lower financial burden than is the case in the U.S. Moreover, the general populace will not be troubled that the outcome, at least as measured by life expectancy, however imperfectly, is lower in the U.S. than in other industrialized nations.

The alternative view to that of the investment community as expressed in the valuation of the health care sector, is that the current circumstance is unsustainable and that the unsustainability will have negative consequences for equity indexes, the U.S. GDP, and the national government solvency. In any case, the consensus does appear to be untroubled, an additional measure of that sentiment being the \$40.987 billion in assets under management of XLV, as of the most recent reckoning.

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Next: Chapter 6 — Bitcoin v. Bonds.

Chapter 6 — Bitcoin v. Bonds

Source. Murray’s Musings, *Devil’s Advocate Report Compendium*, November 2022. Horizon Kinetics LLC © 2022.

PDF in vault. Devils-Advocate-Compendium-November-2022_Bitcoin-vs.-Bonds.pdf

Extract used for assembly. book/build/text/Devils-Advocate-Compendium-November-2022_Bitcoin-vs.-Bonds.txt

Contributors (as in original). Murray Stahl, Chairman; Steven Bregman, President; Peter Doyle, Managing Director.

*Presentation edits: removed compendium running heads and page banners; merged words split across line breaks; omitted lengthy standard research disclaimer; tables in Markdown. One obvious stray “\$” before a percentage in the source (“\$0.87%”) is rendered as **0.87%**.*

Bitcoin v. Bonds

The bond market thus far in 2022 has produced results that generate questions about the nature of the asset class itself. These questions arise from the year-to-date returns for salient aspects of the asset class. The returns can be visualized using popular exchange traded bond index funds (ETFs), as follows:

Table 1: Returns for selected bond ETFs, Dec. 31, 2021–Oct. 21, 2022

Ticker	Fund	Return
TLT	iShares 20+ Year Treasury Bond ETF	(36.11)%
IEF	iShares 7–10 Year Treasury Bond ETF	(18.08)%
AGG	iShares Core U.S. Aggregate Bond ETF	(16.62)%
HYG	iShares iBoxx \$ High Yield Corporate Bond ETF	(14.09)%
TIP	iShares TIPS Bond ETF	(13.42)%

Source: iShares.com

The obvious question to be posed when noting the negative returns experienced this year is whether the long-term return prospects of the bond asset class are sufficiently large to compensate the investor during the inevitable declines in those periods when the Federal Reserve must increase interest rates.

One might begin to answer this question by studying the returns since inception for the aforementioned funds. This data appears in the table below.

Table 2: Annualized returns, selected popular bond ETFs, inception through 9/30/22

Ticker	Fund	Inception date	Return
TLT	iShares 20+ Year Treasury Bond ETF	July 22, 2002	4.50%
IEF	iShares 7–10 Year Treasury Bond ETF	July 22, 2002	3.52%
AGG	iShares Core U.S. Aggregate Bond ETF	Sept. 22, 2003	2.90%
HYG	iShares iBoxx \$ High Yield Corporate Bond ETF	April 4, 2007	4.08%
TIP	iShares TIPS Bond ETF	Dec. 4, 2003	3.42%

Source: iShares.com

It should be pointed out that the inception dates of all these funds to the most recent date coincide with the greatest bond bull market of all time by far. It is an observation to keep in mind as one reflects on these results.

An obvious problem with this type of analysis is the question of subjectivity. One might choose to compare these funds to the prevailing rate of inflation during the relevant time period, in order to understand what the real, inflation-adjusted returns have been. Unfortunately, the calculation of the rate of inflation has itself become controversial and no agreement on a universal inflation measure appears to exist. Nevertheless, the returns can certainly be compared to one another. Even this approach, however, is not very promising because the returns since inception are for different time periods and the periods may not be sufficiently long to arrive at meaningful conclusions. Also, the returns for the different periods and the different classes of bonds are not radically different.

One can say that the highest return is for long term U.S. Treasury bonds, at least for the time period in question. If duration is a primary measure of risk, then it is reasonable that the highest return category is long term U.S. government bonds. But, we cannot know if the greatest overall risk is to be found in the high-yield bond sector, which in prior eras would have been the natural presumption due to the inherent credit risk. That can no longer be presumed, because the Federal Reserve has pursued an extremely aggressive policy of monetary accommodation for most of the time period in question. The alteration of this stance only commenced in mid-June 2022. Some time must pass until the withdrawal of monetary stimulus results in a more historically normal incidence of credit defaults or near defaults.

Credit spreads will likely widen and higher yields will be available in the sector if defaults once again become a primary risk factor for high-yield bonds. In that case, after a period of low returns or quite possibly negative returns, the best performing bond segment might ultimately

become high-yield bonds.

On a year-to-date 2022 basis, high-yield bonds slightly underperformed inflation-protected government bonds. Credit problems have not yet made an appearance.

Another means of viewing these funds is through the prism of after-tax returns, since individual taxpaying investors frequently make use of ETFs.

Table 3: After-tax annualized returns, selected bond ETFs, inception through 9/20/22

Ticker	Fund	Inception	Return
TLT	iShares 20+ Year Treasury Bond ETF	July 22, 2002	3.06%
IEF	iShares 7–10 Year Treasury Bond ETF	July 22, 2002	2.36%
AGG	iShares Core U.S. Aggregate Bond ETF	Sept. 22, 2003	1.79%
HYG	iShares iBoxx \$ High Yield Corporate Bond ETF	April 4, 2007	2.07%
TIP	iShares TIPS Bond ETF	Dec. 4, 2003	2.20%

Source: iShares.com

The calculations in Table 3 were made by iShares, not Horizon Kinetics. To make it universally applicable to all U.S. residents, only federal taxes are assessed. Residents of various states with state income taxes or even municipalities with local income taxes will have different rates of return. However, the returns are at least standard in that none include state and local taxes. State and local income taxes would not apply to any of the three Treasury ETFs. For the iShares Core U.S. Aggregate Bond ETF and iShares iBoxx High Yield Corporate Bond, the yields in the table are shown without the impact of state and local taxes.

If one presumes merely a 2% annual inflation rate, the Core U.S. Aggregate Bond ETF did not provide a positive real rate of return after taxes. This is significant because this particular ETF is designed to represent the entirety of the bond market. The fund itself has \$75.64 billion of assets under management. It is the second largest bond ETF and the tenth largest of all ETFs in the United States. The largest bond ETF is the Vanguard Total Bond Market ETF (BND), with \$79 billion of AUM and an inception date in 2007. In composition and returns, it is very similar in character to the iShares Aggregate U.S. Bond ETF.

If the after-tax rate of return of the entire bond market is lower than inflation over a long period of time during which inflation was mostly at a low rate and relatively benign—in this case covering periods of 15 to 20 years—one must logically rethink one's approach to bond investing.

The composition of the Aggregate Bond ETF is as follows:

Table 4: Composition, iShares Core U.S. Aggregate Bond ETF

Bond type	Percentage of AGG
Treasury	41.72%
MBS	27.05%
Industrial	14.41%
Finance	8.16%
Utilities	2.06%
CMBS	1.85%
Agency	1.22%
Supranational org.	0.92%
Local Authorities	0.84%
Cash	0.48%
ABS	0.35%

Source: iShares.com

Segments of the bond market clearly did provide a positive real after-tax rate of return when

measured against the conventional CPI inflation statistic. If such is the case, yet the broad index underperformed inflation after taxes, this would constitute the first relatively long-run empirical argument against broad indexation in favor of an actively constructed portfolio.

The market value of all publicly traded fixed income securities outstanding as of year-end 2021, is \$53.017 trillion.¹⁰ Bear in mind this is not all debt; it is the value of all publicly traded debt securities, which is lower than the value of all debt. This is the largest asset class in the U.S. and it is considerably larger than the market capitalization of the S&P 500. In fact, fixed income issuance has continued to be robust in 2022 even though, at the moment, the market for initial public offerings of equity is almost nonexistent.

In principle, the best decision that could have been made over the course of the past two decades within the broad bond asset class to maximize returns would have been a 100% allocation to long-term treasury bonds. The after-tax annualized return is 3.06%. Assuming roughly 2% annual inflation, which, again is subject to some degree of dispute, the real rate of return was 1.06% per annum. In order to earn this rate of return, one had to accept a standard deviation of 14.03, an effective duration of 17.17 years, and a weighted average maturity of 25.58 years.

By way of comparison, the first reliable price quote for bitcoin was \$0.0495 on July 17, 2010, a bit less than 5 cents.¹¹ Bitcoin now sells for \$19,189. The current interim bitcoin inflation rate is 1.73% per annum and at the time the notes for this Compendium were prepared, 19,186,625 bitcoin were in existence. By the time this is published, there will be even more. A total of 1,813,375 bitcoin will be issued from the time the notes were prepared for this Compendium until the year 2140. The number yet to be issued will continuously drop. In 556 days, the bitcoin interim inflation rate will decline to 87 basis points per year. With each passing day, of course, there will be one less day until the bitcoin interim inflation rate drops to **0.87%**. Moreover, the bitcoin interim inflation rate will decline by 50% every four years in accordance with the bitcoin issuance protocol. The projected 0.87% annual inflation rate will decline to 0.44% after four years.

Profits on bitcoin are taxed at the capital gains rate, not the statutory federal income tax rate that applies to interest on debt. This is because the IRS considers bitcoin to be property. Also, unlike interest on debt, capital gains tax is paid after sale at a profit.

What if a holder of a bond ETF, such as iShares Core U.S. Aggregate Bond ETF (AGG), were to have made the most marginal change possible to the investment allocation plan by investing 10 basis points annually into bitcoin? Structured that way, in the worst instance, the after-tax rate of return would have declined from 1.79% annually to 1.69%. That basically assumes a complete failure of the Bitcoin project, that bitcoin would have gone to zero.

An investment in bitcoin made in July 2010 would have increased 387,797x in value by today. In other words, a \$100 investment would now be worth \$38,779,797. If the investment had been made with only \$100 of a \$1 million bond portfolio even once in 2010—that would be a 1 basis point investment, one-hundredth of 1%—the remaining \$999,900 that was invested in bonds compounding at an after-tax 1.79% per annum rate for 12 years would now be worth \$1,237,137. If the entire \$1 million had been invested in bonds for the 12 years under discussion, the total capital sum held at this point would be \$1,237,261, or \$124 greater. That quantifies the risk of investing \$100 in bitcoin in a bond portfolio as a loss of about \$124. That \$124 difference would be the amount of loss in principal value after 12 years if bitcoin had failed completely.

In actuality, those bitcoin returns are somewhat understated since the calculation gives no credit for the distribution value of the various bitcoin forks such as Bitcoin Cash and Bitcoin Gold.

The market capitalization of Bitcoin is now \$368.3 billion. Bitcoin is worth less than 7/10ths of 1% of the U.S. publicly traded bond market value. The value of the bond asset class is constantly expanding due to new issuance. If the bond asset class did actually protect investors against debasement caused by inflation, there would be no need for this instrumentality.

The bitcoin return should be genuinely extraordinary if, at some point in the future, the investment community comes to the conclusion that bitcoin will better protect an investor against inflation than the bond asset class which appears to provide no such protection.

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Next: Chapter 7 — The Importance of the Oil Service Industry.

Chapter 7 — The Importance of the Oil Service Industry

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Contributors (as in original). Murray Stahl, Chairman; Steven Bregman, President; Peter Doyle, Managing Director.

Presentation edits: removed compendium running heads and page banners; merged words split across line ends; omitted lengthy standard research disclaimer.

The Importance of the Oil Service Industry

The energy sector weighting of the S&P 500 at the time of this writing is 5.04%. Some two years ago, it was in the vicinity of 2.5%.

Almost one-third of that 5.04% is accounted for by Exxon Mobil Corp. (XOM), with a 1.35% weight. Chevron Corporation (CVX) accounts for 92 basis points. The ConocoPhillips (COP) weight is 47 basis points.

The largest energy service company weighting in the S&P 500 index is Schlumberger Ltd. (SLB) at 19 basis points. The next oil service company to make an appearance is Halliburton Co. (HAL), which comprises 9 basis points. It is the 255th largest company in the S&P 500. Baker Hughes Co. (BKR) is an 8-basis-point position. This group is the extent of the S&P 500 exposure to oil service. In total it amounts to 36 basis points.

The oil service industry obviously can be ignored as a purely statistical factor in the S&P 500 Index. However, without the seismic fracking and well construction services offered by these companies, oil could not be extracted, and the oil extraction companies are important in the index. The oil service industry, as a generalization, has earned very poor returns on capital since 1981. From 2013 until very recently, it was in crisis, with numerous bankruptcies and substantial losses by virtually all industry participants.

Schlumberger, by example, had \$39 billion plus of shareholders' equity in 2003, whereas by the end of 2020, it had dropped to modestly in excess of \$12 billion. By year-end 2021, book value was slightly in excess of \$15 billion. As to fixed assets less accumulated depreciation, generally understood to be plant, property and equipment, this amounted to \$15.096 billion at the close of 2013. By the end of 2021, this figure had declined to \$6.429 billion. It was even lower at the end of June 2022: \$6.386 billion.

In the first six months of 2022, Schlumberger produced a 17.9% annualized return on shareholders' equity. Nevertheless, the company's \$664 million of capital expenditures in the first six months of 2022 is only 62.4% of the \$1.065 billion of depreciation and amortization recorded in this period, which means the balance sheet value of PP&E is still declining at a rapid rate. Schlumberger is not investing in itself even though it is the leading oil service company, with services that are critical to the global extraction of oil and natural gas.

The figures for Halliburton are similarly profound. It had \$15.581 billion of shareholders' equity at the end of 2013, and PP&E net of accumulated depreciation was \$11.322 billion. At the end of 2021, shareholders' equity had declined to \$6.713 billion, and net PP&E had declined to \$4.326 billion. By June, this last measure had declined to \$4.165 billion. The annualized return on shareholders' equity thus far in 2022 is 10.69%. Nevertheless, despite its profitability, Halliburton is making capital expenditures of only about 85% of depreciation expense. This is exclusive of the fact that the company actually sold \$116 million of equipment, and it also excludes yet another \$366 million of impairments in the first half of 2022.

In other words, Halliburton is also disinclined to invest in its business. All of this is quite understandable in the context of Halliburton history. At the end of 2013, the company had debt of \$7.816 billion against shareholders' equity of \$13.581 billion. At the end of 2021, long term debt was a higher \$9.12 billion against half as much shareholders' equity of \$6.713 billion. At the end of June 2022, debt had declined to \$8.525 billion against shareholders' equity of \$7.12 billion. Halliburton is still a leveraged company, although it should be noted that net current assets amount to \$5.496 billion.

Schlumberger stock trades at more or less the price reached in the summer of 2005. The same can be said for Halliburton, even though its shares had a slightly better outcome for the time period in question.

It might be argued that modern oil rigs are more productive than the oil rigs of the past, so that fewer rigs are needed to produce a given quantity of hydrocarbons. The U.S. Energy Information Administration (EIA) calculates drilling rig productivity over time using "new well oil production per rig" as the key metric. The data is contained in a drilling productivity report per region. The data is not so clear on this question.

The EIA data shows that the Anadarko region of Western Oklahoma's new well oil production per rig has more or less quintupled since 2012. On the other hand, the productivity increase since 2012 in a region such as Appalachia has been minimal. This contrast might be compared to the roughly 8x productivity increase in the Bakken region of North Dakota, but with minimal increases in the Haynesville area of East Texas and Western Louisiana.

The Permian Basin is perhaps the most prolific energy producing region in North America, and new well oil production per rig has advanced approximately tenfold there since 2012. Productivity reached a peak in 2020, with significant shutdowns of producing wells in all regions. Only the most productive wells remained in operation. This phenomenon is known as high grading. Since 2020, productivity has been in decline. This is not necessarily indicative of a loss of productivity per rig; it is likely that less productive land is now being brought into production.

Both Schlumberger and Halliburton arguably have now achieved an acceptable level of profitability. This is not true for many oil service firms. Baker Hughes, for example, is profitable on an operating basis, with an operating margin of 6.68%. Unfortunately, the company still reports impairments in non-operating losses. The balance sheet has \$14.88 billion of shareholders' equity against \$6.625 billion of debt, and balance sheet cash amounts to \$2.928 billion. However, goodwill and other intangible assets account to \$5.741 billion and \$4.049 billion. This total of \$9.79 billion can be subtracted from shareholders' equity to arrive at a hard book value of \$5.09 billion. Viewed in this light, Baker Hughes might still be regarded as a leveraged company, so it still displays signs of the damage caused by the long energy depression from June 2014 to June 2021.

The contraction in the number of operating offshore oil rigs has been far more extreme, and the duration of the contraction now exceeds four decades. There were 250 U.S. offshore crude oil and natural gas rotary rigs in operation at the end of 1980.¹² By the end of 2007, just before the financial crisis of the following year, the number of offshore rigs in operation had dropped to 62. In the next 24 months, this figure shrank to 37. The number of rigs in operation offshore then began a gradual increase to 58 in June 2014, on the eve of what was to become an energy depression that lasted for eight years. Data for August 2022 is the most recent, and at that time 18 oil and natural gas rotary rigs were in operation in the offshore U.S. waters.

A shortage of oil may be developing right now. The U.S. Strategic Petroleum Reserve has been used to balance supply in relative to the embargo on Russian oil. The SPR has declined to 416.4 million barrels as of September 30th, 2022, after a drawdown of 201.4 million barrels in the past 12 months. This rate of diminution is not sustainable. One possible solution is a resolution of the Ukraine conflict. That does not seem likely in the foreseeable future.

The other obvious solution is a meaningful increase in U.S. oil production, but in order to accomplish that, the remaining oil service companies must increase investment in their facilities to facilitate that production increase. The government plans, though, are still to replace hydrocarbons with other forms of renewable energy once the current crisis is resolved. Effectively, the proposition to the oil service companies is that they would need to make long term investments to solve what the government hopes is a short-term problem. If the Ukraine conflict is resolved, energy commerce might return to normal, in which case any oil service company that made investments to handle temporary U.S. oil production increases would be left with enormous amounts of redundant equipment.

The oil service companies have experienced about four decades of excess oil service supply and endured enormous losses in order to establish low break-even operating levels. Significant capital expenditure would inevitably raise break-even points for oil production levels that world governments would rather not be sustained. In fact, it is the stated intention of those governments to reduce hydrocarbon production as soon as it can be practically achieved.

The surviving oil service companies, therefore, have absolutely no incentive to increase investment in their output capacity. In fact, the entire industry continues to disinvest in their plant and equipment. It will be difficult, if not impossible, for hydrocarbon production to increase while the oil service companies are reducing their capital investments.

This makes oil service a very interesting industry from an investment perspective. It is the only industry that is currently aligned operationally to endure a depression, simply because it has just experienced a depression. It is therefore a potential non-correlated asset in an equity portfolio. The reluctance of the industry to invest will eventually serve to greatly increase typical oil service company return on equity to very alluring levels.

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Next: Chapter 8 — Wealth and Crisis: The Role of Diamonds in Asset Allocation (European Contrarian, January 2023).

Chapter 8 — Wealth and Crisis: The Role of Diamonds in Asset Allocation

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Contributors (as in original). Murray Stahl, Chairman; Steven Bregman, President; Peter Doyle, Managing Director.

Conflict / offering note (from original). An affiliate of Horizon Kinetics Asset Management LLC acts as co-sponsor to a private fund that invests in diamonds; the fund is available only to certain qualified investors.

Companion essay in this edition. *Diamonds as an Investment* (December 2022) appears as Chapter 4.

Presentation edits: removed compendium running heads and page banners; merged words split across line breaks; shortened the standard research disclaimer to the conflict line above; light punctuation cleanup only.

Wealth and Crisis: The Role of Diamonds in Asset Allocation

The modern understanding of asset allocation is based on the premise of social stability. If the assumption of social stability is removed, the conceptual foundation of modern asset allocation vanishes along with it.

For example, a basic building block of asset allocation is the notion that short-term government securities represent a risk-free asset. This is obviously untrue in an inflationary environment, since the asset will lose purchasing power. It is also untrue in an environment of war or civil war.

Inflation has not usually been a problem in the developed world since World War II, with the great exception of the 1970s decade, although inflationary episodes did occur in the aftermath of World War II, as one might expect.

Globally, however, inflation is not uncommon. Trading Economics recently produced a list of 10 nations with very serious inflation:

Table 1: Nations with Highly Problematic Inflation

Country	Annual Inflation Rate
Venezuela	1198.0%
Sudan	340.0%
Lebanon	201.0%
Syria	139.0%
Suriname	62.3%
Zimbabwe	60.7%
Argentina	51.2%
Turkey	36.1%
Iran	35.2%
Ethiopia	33.0%

Source: *TradingEconomics.com*

This list is only comprised of the nations with the highest recorded rates of inflation. If problematic inflation is defined as a recorded rate that exceeds 10% per annum, this list would double. Twenty countries, or about 10% of the world's nations, are experiencing serious inflation.

Inflation, though serious, is not the most grave threat to wealth preservation. The most serious threats are war, civil war, and social unrest.

Civil war can last surprisingly long. In the Western world, investors generally ignore such conflicts. It is astonishing to learn that the Myanmar civil war has been raging since 1948.

The country, once known as the British colony of Burma, achieved independence that year. Since then, seven states of that nation—representing a variety of ethnicities—have engaged in a continuous armed struggle for either autonomy or independence.

This is not the only example. A left-wing armed revolt against the government of the Philippines has been continuing since 1969. An armed struggle between the government of Colombia and the drug cartels has been ongoing since 1964. The civil war in Syria began in 2011. In Sudan, the war in Darfur has been continuing since 2003. The civil war in Yemen has been happening since 2014. Conflict still roils Afghanistan, a nation that's been in a continuous state of armed conflict since 1978.

One might be tempted to dismiss these conflicts as irrelevant to industrialized nations, but one must note that the low-intensity Ukrainian civil war that commenced in 2014 has evolved into an actual Russo-Ukrainian war.

None of the long-term studies of wealth and investment returns really take into account the influence of war and social unrest. Few studies of the impact of war upon wealth appear to exist. Many books have been written on the French Revolution and the Napoleonic wars, but the focus is on military strategy and social change. Similar remarks might be made about the First and Second World Wars. One reason for this is that the period from the Battle of Waterloo in 1815 to the beginning of the First World War in August 1914 is usually considered to be a peaceful era, but it is only peaceful from a Eurocentric or American perspective. In fact, even the American perspective ignores the wealth impact of its own Civil War.

At least fifty 19th-century civil wars have been almost entirely ignored by historians. One example is the Seinan Senso or Southwestern War, so called because it was largely fought on the island of Kyushu in Southwestern Japan. Essentially, the samurai class was not initially opposed to the Meiji Reform that implemented imperial rule in Japan in 1868, but it became increasingly concerned with the direction the government was taking and established an independent military academy. This was viewed as a serious challenge to governmental authority, and eventually a series of incidents quickly deteriorated into civil war. The government eventually prevailed, but the cost of the conflict nearly bankrupted the country. The war forced Japan off the gold standard and effectively tripled the country's national debt.

This very brief survey of conflict should be sufficient to establish that the probability of destabilizing conflict is not zero. Therefore, an investor's asset allocation profile should contain one or more assets that could maintain value and possibly even increase in value in the event of such a conflict.

Cash, bonds, stocks, private equity, and real estate—the current standards in asset allocation strategies—are vulnerable in the event of war. Cash and bonds are vulnerable since they are obligations of the government, and the creditworthiness of the government is in question during such a crisis. Equities represent business enterprises. Amid conflict, an enterprise might be subject to much higher taxation. Its business prospects could be greatly diminished during war, and might even be seriously damaged by the ensuing violence. The same points would apply to private equity. Real estate can obviously be damaged during an armed crisis, as well. It is also a highly visible asset that is not movable and could be expropriated.

Given the problematic features of conventional assets during a true crisis, the obvious and conventional alternatives are gold and silver.

One could readily dismiss silver as a crisis asset on the question of value density—that is, the physical quantity of an asset needed to afford protection. To demonstrate, let us assume that the basic unit of account for wealth is \$1 million. Using the current silver price of \$23.56/ounce, 42,444.82 ounces of silver would be required to store \$1 million in value. Each troy ounce is 0.18 cubic inches, so this equals 7,674.8 cubic inches, or 4.44 cubic feet, of volume. The challenge of transporting 4.44 cubic feet of silver is not trivial; it would weigh 2,910 pounds, or about that of a Nissan Sentra 4-door sedan.

This is not to assert that silver wouldn't appreciate in a crisis, but the primary wealth management problem in a crisis is not the maximization of return, but the prevention of seizure.

In the case of gold, seizure can be prevented if it is held outside of the regulated custodial system. In comparison with silver, a gold price of \$1,788 per ounce requires only 559.2841 ounces of gold for \$1 million of value, thus mitigating the transportation problem to some extent. Although this is far simpler than transporting 42,444.82 ounces of silver, weighing 38 pounds it still requires a transportation conveyance and is not easily concealed.

Art, in the form of a painting, is potentially value dense. The transportation problem is not necessarily complex and it does not necessarily require specialized conveyance. However, a painting is very fragile. It could be concealed, but maintaining the proper storage conditions to preserve the painting's value in a crisis is not a trivial problem.

This leaves diamonds as an important crisis asset that has not yet become an accepted asset class. Diamonds are light. One carat weighs one-fifth of a gram. Stated in other terms, one carat is equivalent to 0.007 ounces, and there are approximately 2,265 carats to a pound. The average person should have no problem locating a one-carat diamond worth \$2,000 for purchase, although much more expensive stones can be found. Even with a \$2,000 average per-carat price, one pound of diamonds—or 2,265 carats—is worth \$4,530,000.

In contradistinction, gold's value density is much lower. A pound contains 14.5833 troy ounces. At \$1,788 per troy ounce, the value of one pound of gold is $14.5833 \text{ (troy ounces)} \times \$1,788 \text{ (price/troy ounce)}$, or \$26,075. Assuming the purchase of relatively inexpensive \$2,000 diamonds as the basis for the calculations, one pound of diamonds is worth \$4,530,000, whereas one pound of gold is worth \$26,074. The ratio of the two assets, \$4,530,000 versus \$26,074, means that diamonds are 172.58 times as value dense as gold. It should be noted that this exercise greatly understates the potential value density of diamonds.

Investors might not understand the importance of the concept of the value density of diamonds,

but the idea has been explored thoroughly in popular literature and film. For instance, in the short story titled “The Adventure of the Blue Carbuncle” by Arthur Conan Doyle, Sherlock Holmes investigates a theft of a priceless gemstone from the hotel suite of the Countess of Morcar. The story would have been impossible to write, had the premise been that the equivalent value of gold were stolen: there would be no reason to have such a quantity of gold in a hotel suite. Moreover, Doyle would’ve had to explain the logistics of removing a large quantity of gold from a hotel suite without being observed. Similarly, in the Alfred Hitchcock movie *To Catch a Thief*, a cat burglar steals jewels, not gold. The jewelry theft is the MacGuffin, or plot device, upon which Hitchcock weaves the story based on David Dodge’s novel of the same name. A burglar’s escape by rooftop while carrying a large quantity of gold simply couldn’t happen.

Accepting the value density aspect of diamond investing leads to the logical conclusion that diamonds should be the asset of choice for the circumstance when social cohesion is in question. Social cohesion has been in question many times in history, and it is now unraveling in at least 10% of the world’s nations. Nevertheless, diamonds have yet to attract institutional interest.

This is probably because most investing in the contemporary era is institutional investing, and institutional investing can only exist with sufficient social cohesion for the institution to continue to function. If a financial institution’s investment committee were ever to turn its attention to the potential loss of social cohesion, it almost necessarily follows that the committee also must reflect on the possible inability of the committee and the institution to function. For such a committee considering that possibility, it would essentially be preparing for its own demise, in which case it might have an enhanced interest in a diamond portfolio to access in a genuine state of emergency. That would be much like planning for beneficiaries while simultaneously imagining the demise of the management.

As difficult as it might be to plan for the loss of social cohesion, the great number of breakdowns of the institutional order in many nations just in the first part of the 21st century makes an allocation necessary to an asset that can endure, is universally accepted, is transportable, can be hidden and protected from seizure, and appreciates over the course of time. Such an asset is diamonds.

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Next: Chapter 9 — The Rise of the Artificial Asset Class (Part II).

Chapter 9 — The Rise of the Artificial Asset Class

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Placement. Opens **Part II — Indexation, concentration, and “the artificial asset class”** in this reader’s edition.

Presentation edits: removed compendium running heads and page footers; merged words split across line breaks; replaced one garbled formula block with a short editorial note (see below); light punctuation cleanup only.

The Rise of the Artificial Asset Class

The Limits of Idiosyncratic Risk Avoidance in Indexation

Modern asset allocation is practiced as if it were a science. The objective is to balance risk and reward in accordance with individual risk tolerance. Careful attention is paid to individual goals and time horizon. This is accomplished by investing in different genres of equity and fixed income, and even cash equivalents. Thus, the risk of U.S. large capitalization versus emerging market equities is considered. The system considers the risk/reward ratio of high yield versus U.S. Treasuries. One now commonly finds foreign currencies invested alongside U.S. dollar cash equivalents.

One result of this systematized approach to portfolio construction is the Robo-advisor, which makes these allocations via computer algorithms. Hence, in practice, Wall Street has achieved a self-driving portfolio before Google has achieved a self-driving car. In fact, Wall Street has achieved a self-driving portfolio before scientists at the MIT Artificial Intelligence Laboratory have invented a robot that will properly clean your home.

Is the “rule set” of portfolio construction far simpler than the “rule set” for driving an automobile? This is a good question, actually. In each case, the obvious problem is to map all

possible idiosyncratic circumstances and program a proper response. In the case of driving an automobile, this is quite difficult, but there are definable conditions that precede accidents so that a sufficiently fast computer could undertake avoidance strategies. In contrast, in accordance with Modern Portfolio Theory, news of idiosyncratic circumstances such as earnings misstatements, corruption, terrorist attacks, oil spills, and many other calamities are assumed to be instantaneously reflected in stock prices. How can asset allocation programs react, since individual securities change the character of the asset class?

For instance, if Exxon had spilled oil into the Gulf of Mexico rather than British Petroleum, one could argue that the risk of holding the S&P 500 would be higher, given that Exxon is the third-largest position in the S&P 500. However, the S&P 500 portfolio beta would remain 1, by definition. Upon what basis, if any, could an asset allocation program react?

Indexation itself is designed to mitigate the idiosyncratic security risk problem by owning so many securities that no individual security, however idiosyncratic, can alter the volatility characteristics of the index. Indeed, at the moment, Exxon is only 2% of the S&P 500 Index.

Yet, as asset allocation becomes more specialized, there is a corresponding increase in specialized indexes used to craft just the right risk/reward balance. An example is the iShares U.S. Energy ETF (IYE), which holds in excess of \$1.2 billion in assets under management. Exxon is a 25.84% position in the index. Surely no one would care to defend the proposition that this index would be immune to the idiosyncratic aspects of Exxon if it were to become involved in a disaster similar to that which befell British Petroleum.

It is extraordinarily difficult to create an index that is sufficiently liquid to accommodate trillions of dollars of investment without being dominated by the weight of a relatively small number of huge global multinational companies. The paradox is that the scope of operations of such firms is so vast across the globe that most idiosyncratic developments on the international stage have some bearing upon the potential profits of these giant enterprises.

The Rise of Artificial Asset Classes

Consequently, parallel to the development of indexation, there has been a movement to devise artificial cash-settled investment vehicles in which the outcome, positive or negative, rests upon the calculation of a completely transparent number. This is made possible in real time by enormous advances in computers, telecommunications, and semiconductor design.

An example of such an artificial index is the VIX Index. The VIX is merely a weighted average of near-term options on the S&P 500. VIX futures were first listed on March 26, 2004. The open interest is now 479,510 contracts with a contract multiplier of 1,000. This is no doubt a considerable sum of money. In any case, considered within the context of contemporary indexation, which accommodates trillions of dollars, VIX is a mere rounding error. Moreover, it is ultimately a derivative of the S&P 500 and, therefore, intrinsically contains individual security risk.

Given the direction of advances in computational power and the need for new asset classes, it was inevitable that a wholly artificial asset class would be created. In the case of the Satoshi white paper in 2008, that artificial asset class was Bitcoin. It is perhaps coincidental that all equity asset classes, all commodity classes, and many classes of fixed income securities simultaneously collapsed in value in 2008.

The basic idea of Bitcoin as an asset class, leaving aside the questions of Blockchain technology, is that it would be accepted as a store of value like money, with the singular difference that the units of money would be fixed so as not to exceed 21 million units. The rate of issuance until the maximum is reached is expressed by a geometric series in the original publication.

[Editorial note: the PDF extract garbled the displayed series and summation notation; the narrative below restates the schedule in prose.]

The 210,000 in the formula is the number of starting coins; the 50 quantity in the formula is the starting Bitcoin proof of work reward. This figure is halved every four years. It will be 12.5 Bitcoin on or about July 11, 2016 insofar as can be calculated. Ultimately, an issuance of 21 million will be attained.

This is structurally different from all other currencies, which are so-called fiat currencies. In the latter instance, the central banks, not infrequently under great pressure from the central government, expand the amount of currency to the degree considered necessary.

The consequence is that money as a store of value — or even in many instances bank deposits, government bonds, and corporate bonds — generate negative real rates of return. This has not been true in the United States and most industrial developed nations since 1981 because interest rates declined. However, it may well become true in the foreseeable future due to the low level of bond rates as well as taxation of whatever little interest income is generated. In most nations of the world, fixed income securities or cash deposits have experienced negative rates of return in the past. As of April 30th, Tradeweb has reported that there are \$7.85 trillion of negative yielding sovereign debt in the world, including the majority of the debt of Switzerland, Japan, Germany, the Netherlands, Austria, France and Denmark.

In essence, Bitcoin is a permissionless instrument. In other words, it can function as money or a store of value if a sufficient number of people accept it as such. In principle, anything can be accepted as money. For example, in the Roman Republic, 20 sesterii, or 5 denarii, was legally equal to 18.22916 and two-third grains of gold. In the time of Nero, the denarius contained 99% silver. Under Trajan and the Antonines, the denarius first contained 5% copper, and eventually it contained 25% copper. Under the Emperor Severus, the denarius was 50% copper.¹³

In history, accepted instruments of exchange and store of value have included barley in Mesopotamia, cowry shells in Aboriginal Australia, tobacco leaves, and other commodities. In post-World War II Germany, cigarettes assumed this function for several years. Of course, gold and silver have been used as money and stores of value.

The common feature of every medium of exchange or store of value used thus far throughout history has been that the supply would generally increase. Sometimes, this was within the ability of the government to control; sometimes, it was left to circumstance. For example, historians believe that the impact of the California Gold Rush of 1849 was such that, in effect, the supply of money was suddenly increased and this stimulated economies all over the world, not just in the United States. Some would argue that the Transcontinental Railroad was built, in part, by demand for transportation services stimulated by the Gold Rush.

On the other hand, the inflationary pressures of the 1970s were certainly responsible for the increase in the price of gold during that decade. This, in turn, stimulated exploration, which then enhanced supply. Eventually, the price of gold declined precipitously.

Bitcoin, in contradistinction, is fixed in supply. If it were accepted as an asset class and, therefore, as a store of value, why should not this store of value be equivalent in capitalization to other stores of value? In other words, if supply is fixed, the only rationing mechanism remaining is price.

It should be recognized that almost all serious students of central banking and national monetary policy consider Bitcoin to be something akin to a Ponzi scheme. However, as an instrument, Bitcoin offers no promise of profit, which is the animating element of a Ponzi scheme.

Bitcoin has a market capitalization of roughly \$9 billion as of this writing. On its World Debt Clock, *The Economist* calculates that the total global government debt that trades has a U.S. dollar value of \$58.891 trillion.¹⁴ This is just government debt. What if Bitcoin were worth an amount equal to this ever-increasing sum? This would represent a coefficient expansion of 6,654x, or 665,400%.

It is perhaps worthy of note that in March of 2006, the Federal Reserve ceased publishing M3, the broadest money supply measure. It still publishes M2, which was about \$12.652 trillion in April 2016. In the first four months of 2016, U.S. M2 expanded by 2.61%, or an annualized rate of 8.04%. Should one add the value of M2 to world government debt as a somewhat better measure of store of value? If one were to do this, it would still be hopelessly inadequate as a calculation, since it does not account for all the other national M2 calculations.

In any event, if one merely added U.S. M2 and global government debt, the total is \$71.54 trillion, or 7,949x the current market value of Bitcoin. Euro M2 is another €10 trillion. Yen M2 is ¥934.765 trillion, or about \$8.5 trillion. No one has ever needed to calculate the worth of every instrumentality with a store of value function. It would surely be a huge number.

If the various stores of value in the world are engineered in such a manner as to generate negative real rates of return, surely the holders of such instruments would find this objectionable. Historically, if all central banks simultaneously enacted similar real purchasing power destruction policies, the investors had no recourse other than the political process.

Now there is an alternative that is accepted by very few and rejected by most investors. If the predilection of the majority of investors changes to be favorable to Bitcoin, a market capitalization of \$9 billion would hardly be adequate to express those preferences. Indeed, a market capitalization of \$1.227 trillion might not be adequate. After all, this figure is merely the aggregate market capitalization of the four “FANG” stocks: Facebook, Amazon, Netflix, and Google. If an artificial asset class such as a crypto-currency, of which Bitcoin is only one example, is accepted generally, this is a once-in-a-century event in terms of rate of return.

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Next: Chapter 10 — Russell 2000® index construction white paper; then Chapters 11–12 (S&P float / bond substitute commentary).

Chapter 10 — Russell 2000® Index Construction: When Small Caps Became a

Big Problem

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Contributors (as in original). Research team listing on cover: Murray Stahl, Steven Bregman, Therese Byars, Ryan Casey, James Davolos, Derek Devens, Peter Doyle, Michael Gallant, Matthew Houk, Utako Kojima, Eric Sites, Fredrik Tjernstrom, Steven Tuen.

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Russell 2000® Index Construction: When Small Caps Became a Big Problem

The Russell 2000® Index, a subset of the Russell 3000® Index, measures the performance of the small-cap segment of the U.S. equity universe and includes approximately 2000 of the smallest securities based on a combination of their market cap and current index membership. The index is unmanaged and cannot be invested in directly. Russell Investments is the owner of the trademarks, service marks and copyrights related to its indexes. Past performance is not a guarantee of future results.

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Introduction

Passive investing via exchange-traded funds (“ETFs”) has risen at an enormous rate over the past several years, as evidenced by the total assets under management held by ETFs. From 2001 to 2011, ETF assets have grown from \$82.9 billion to \$1.0 trillion, representing a compound annual growth rate of 28.8%. During 2012, assets rose an additional 25% to \$1.3 trillion¹, which, as we will see, has deep implications not only for equity ETF investors, but for all equity investors.

This change occurred even as total assets invested in U.S. equity mutual funds have declined since the 2008 Financial Crisis, in favor of bonds. It might be surprising that the net outflows from U.S. equity mutual funds during the first 10 months of 2012 were \$99.3 billion, exceeding the \$80.1 billion outflows during the same period in 2011 (for the full year of 2011, total net outflows reached \$129.6 billion). At this rate, the amounts withdrawn from equity mutual funds over the 2011-2012 time frame (during which the S&P 500 Index (“S&P 500”) produced positive returns) will easily eclipse the \$228.5 billion of assets withdrawn during the 2008 crisis year. Yet, within a net outflow market for equity mutual funds, equity ETFs have received large scale, ongoing net inflows. One might be especially surprised that even while equity mutual funds experienced large outflows in 2008, equity ETFs enjoyed inflows of over \$143 billion.

This may be one of the greatest flows of funds on record. This paper will examine whether this phenomenon is of sufficient scale to alter security and sector clearing prices and distort the character of the critical benchmarking tools upon which investors rely: equity indexes. If so, then the very tools that investors employ to measure risk and return have become unreliable and perhaps, in the extreme, dangerous. We will commence with two of the largest, most prominent equity indexes: the Russell 1000 and Russell 2000 Indexes (“Russell 1000 and Russell 2000,” respectively), although distortions within smaller and more sector-specific indexes will be shown to be far more dramatic.

This phenomenon is not strictly a result of retail investor demand. In fact, many institutional investors, such as pension funds and endowments, have increasingly utilized ETFs as a means to gain exposure to multiple asset classes, including fixed income, commodities, and more esoteric ones such as volatility. These highly sophisticated investors utilize quantitative tools to establish asset allocation policies in accordance with Modern Portfolio Theory, which is based on a belief in the applicability of the Efficient Frontier.

Those familiar with the process of establishing institutional investment policy will understand that a portfolio's allocation into various strategies, such as “large cap equity,” “small capitalization equity (“small cap”),” and “foreign equity,” among others, is established only after careful consideration, often utilizing complex calculations that consider data on risk, returns,

and volatility. Thus, it fundamentally follows that the investment vehicles chosen to represent a desired asset class must actually own securities from that asset class; otherwise, the investment characteristics expected of the investment would not actually be represented. For example, if an investment vehicle were chosen because it purported to invest in “value” stocks, but instead held “growth” stocks, the entire investment policy would be invalidated.

¹ Source: Investment Company Institute With regard to active managers, the discrepancy described above is referred to as “style drift.” To institutional investors, style drift is a serious offense and is grounds for dismissing the investment manager responsible for perpetrating it. However, what if the style drift is committed by indexes and, by extension, index funds? What would the recourse be if the index is a nationally-recognized standard for scores of billions of dollars of investment capital? Indeed, given that investors typically do not scrutinize index funds in the same way they do active managers, one must assume that the issue of style drift is never mentioned when index funds are utilized.

Consider the Russell 1000 and Russell 2000. These are a subset of the Russell 3000 Index (“Russell 3000”), which contains the largest 3,000 companies in the United States and captures approximately 98% of the U.S. equity market. By definition, the Russell 1000 consists of the 1,000 largest companies as measured by market capitalization (“market cap”), and is universally accepted as an appropriate representation of large cap (“large cap”) companies. The following 2,000 companies, that is, the 1,001st largest capitalization company to the 3,000th largest capitalization company, make up the Russell 2000, which is considered a proxy for small cap stocks. The construction of these indexes is fairly transparent and easily understood.

So far, so good. Except for one problem: the Russell 1000 doesn’t actually represent the largest 1,000 companies and the Russell 2000 doesn’t hold the 2,000 smallest companies. For example, the last company in the Russell 1000 by index weight is Groupon Inc. (Nasdaq: GRPN), which has a market cap of \$3.14 billion. ² The largest security in the Russell 2000 by index weight is Ocwen Financial (NYSE: OCN). Its market cap is \$4.72 billion, or 1.5x as large as Groupon. On a float-adjusted basis (which is the market cap excluding shares held by company insiders), the discrepancy is even more glaring. Groupon’s float-adjusted market cap (i.e. the total market cap of its equity not held by insiders) is \$1.25 billion, while Ocwen’s is \$2.83 billion, or about 2.25x as large. Can one persuasively argue that Groupon deserves to be in the large cap index and Ocwen Financial in the small cap index?

Lest one suspect that this discrepancy is an isolated event, expanding the observation to the bottom 50 companies in the Russell 1000 and the top 50 companies in the Russell 2000 yields a similar conclusion. The combined market cap of the bottom 50 companies in the Russell 1000 totals \$110 billion. In contrast, the top 50 companies in the Russell 2000 have an aggregate market cap of \$136 billion.

How, one might reasonably ask, can the aggregate market cap of the top fifty companies in the small cap index exceed the aggregate market cap of the bottom fifty components of the large cap index?

Clearly, a distortion exists in the construction of the Russell indexes (and, we suspect, in many others) that is not recognized by the investment community. Although the construction of these indexes is simple in theory, it is more complex in practice. The result is that investment products are not quite what they purport to be. Given the vast sum of investment dollars affected, which measures in the trillions ³, this is a serious issue that requires study.

² References to market prices are as of December 14, 2012. ³ Russell reports assets under management of \$159.1 billion, and \$3.9 trillion of assets invested in products that are benchmarked to the Russell U.S. indexes.

Therefore, this paper attempts to provide greater clarity with regard to the risks lurking in index-based passive investment products—risks that are not immediately obvious even to professional investors. However, because this paper is intended to be introductory, and the subject matter is complex, we will only touch on the most glaring problems, with less breadth or depth than is truly required. We will explore the central theme—distortions in index construction—as well as highlight a number of other, no less serious, issues arising from the ETF phenomenon.

Index Construction and Methodology According to Russell Investments In order to highlight the distortions in index construction, one must first be familiar with their stated methodology. In the case of Russell Investments (“Russell”), the creator of the Russell indexes, the Russell 3000 (and by extension, the Russell 1000 and Russell 2000) methodology is disclosed in the paper “Russell U.S. Equity Indexes Construction and Methodology.” ⁴

According to Russell, the indexes “are constructed to be unbiased and comprehensive, with each index incorporating the entire market segment it is designed to represent.” Additionally, the process is intended to be transparent and rule-based, in that all rules regarding the construction and maintenance of the indexes are published and publicly available. The indexes are designed to be a comprehensive representation of the investable U.S. equity market and to serve three main purposes: to act as performance standards for active managers, to serve as proxies for asset allocation purposes, and to become purchasable and replicable vehicles for passive investment strategies.

To ensure that the market segments are represented accurately, the indexes are reconstituted—

that is, rebuilt—annually. This allows for the inclusion of securities in the appropriate market cap (and style) index as they change in size and style over time. We will see that the annual reconstitution is the starting point of the distortion. One will also observe that most of the steps outlined below seem entirely straightforward and logical.

Basic Methodology

Index construction begins with the Russell 3000 Extended (“Russell 3000E”), which is intended to be the broadest U.S. index and is comprised of the largest 4,000 U.S. companies. It is from this comprehensive universe that the various domestic Russell indexes are created, including the similarly-named Russell 3000, which holds the largest 3,000 securities. All companies that are determined to be part of the U.S. equity market are included in the Russell U.S. indexes. Those determined to be non-U.S. are included in the Russell Global ex-U.S. Index.

Inclusion in the U.S. index can be determined based on a number of criteria. These are:

☐ Home-country indicator—if a company incorporates in, has a stated headquarters location in, and also trades in the U.S., it is assigned to the U.S. index; however, ADRs and ADSs are not eligible. ☐ Trading requirement—all securities for inclusion must trade on a major U.S. exchange; bulletin board, pink sheet or over-the-counter traded securities are not eligible for inclusion. ☐ Minimum closing price—a stock must have a closing price at or above \$1.00 on the last trading day in May.

4 http://www.russell.com/indexes/data/US_Equity/russell_US_indexes_methodology.asp

☐ Minimum total market cap—companies with a total market cap of less than \$30 million are not eligible. ☐ Minimum float requirement—companies with less than 5% available float (those shares available to the public; i.e. excluding shares owned by insiders/management) will not be eligible for inclusion. ☐ Corporate structure—companies with the following structures are excluded from inclusion: royalty trusts, U.S. limited liability companies, closed-end investment companies (although business development companies are eligible), blank-check companies, special-purpose acquisition companies, and limited partnerships. ☐ UBTI—companies that produce unrelated business taxable income (“UBTI”) are restricted from inclusion in Russell U.S. indexes. ☐ Share exclusion—the following share types are not eligible for inclusion: preferred and convertible preferred stock, redeemable shares, participating preferred stock, warrants, rights, and trust receipts.

Once eligible members are identified, Russell calculates the market cap of each security and sorts them in descending order. Companies number 1 through 1,000 are then included in the Russell 1000, which is the company’s large cap index offering. Companies number 1,001 through 3,000 are allocated to the Russell 2000, the company’s small cap index. Numbers 2,001 through 4,000 become members of the Russell Microcap Index (i.e. the smallest 1,000 securities in the Russell 2000 plus the next 1,000 securities).

The market cap level that separates the 1,000th and 1,001st company is deemed the “breakpoint” between the Russell 1000 and Russell 2000. Given that the breakpoint is based upon the demarcation line between companies 1,000 and 1,001, and not a predetermined market cap level, the breakpoints vary year to year, as market caps invariably fluctuate. This is evident from the following table.

Exhibit 1: Historical Breakpoints for Russell 1000/Russell 2000 Year 1,001st Breakpoint 2012 1,943,992,920\$ 2011 2,066,827,500\$ 2010 1,714,702,220\$ 2009 1,296,536,150\$ Source: ITG

Accordingly, Russell’s definition of large cap stocks, as expressed in the Russell 1000 has ranged from as low as \$1.29 billion to as high as \$1.94 billion within the past four years. Investors typically define large cap stocks as companies with market caps over \$10 billion. If this definition is used, then the Russell 1000 represents a deviation from the commonly accepted notion.

The Russell indexes follow market cap weighting rules, which means that the index weighting of each member is based upon the size of its market cap relative to the entire universe. In such a scheme, the largest companies (i.e. Exxon Mobil Corp. and Apple Inc.) have a commensurate weighting in the indexes, with an additional proviso that the weightings be adjusted to reflect free float.⁵ The method by which this is achieved is as follows:

☐ Sort the Russell 3000 members in descending order by total market cap. ☐ After membership is determined, adjust a security’s shares to include only those shares available to the public. ☐ Calculate the total market cap of the Russell 3000 by summing all members’ float-adjusted market caps. ☐ Calculate the percentiles for each company in the Russell 3000 by dividing the float-adjusted market cap associated with each member by the total market cap of the Russell 3000.

Within the index world, the adoption of float-adjusted ranking is relatively new and represents a material methodological change. Although correlation is not necessarily indicative of causation, it is interesting to note that several major index providers moved to float-adjusted weightings within a few years of one another, just as the flow of capital into ETFs began to surge. For example, FTSE, the British-based provider of global indexes, moved to a free float methodology in 2000. In 2002, the investment analytics and index company MSCI made the change. This

was followed by Standard & Poors, which made the decision to move to a free float system for its U.S. indexes, including the S&P 500, S&P MidCap 400, and the S&P Small Cap 600 Indexes, in 2005. Between 2000 and 2005, ETF assets under management swelled from \$65.5 billion to \$300.8 billion, a 4.5-fold increase. There are important implications of such methodological changes that are not immediately obvious.

One will therefore find that the companies at the top of the indexes virtually dominate the rest of the index members. For example, the top 10 companies in the Russell 1000, combined, represent 17% of the index, while the bottom 100 have an aggregate weight of only 0.8%, which equates to less than 1 basis point, or 1/100 th of 1%, each.

Reconstitution The process by which the indexes are constructed occurs once every year, which is called the Russell Reconstitution, a rebalancing in layman's terms.

Reconstitution begins in May. On the last trading day in May, all of the eligible securities are ranked by their total market cap, in accordance with the process delineated above. In the beginning of June, preliminary lists for the additions and deletions to the indexes are communicated to the marketplace. In late June (June 22 nd in 2012), the changes go into effect.

The exception to the reconstitution process involves corporate actions that occur between the rebalancings. For example, securities that leave an index for any reason (i.e. mergers, acquisitions, bankruptcy, among others) are not replaced. Specifically in mergers, the company that survives remains in the index, with the acquired company's market cap moving to the acquiring company's stock. Additionally, newly-created companies following spin-offs and initial public offerings are added as they occur, provided they meet the requirements of the index.

5 The purpose of this adjustment is to exclude the capitalizations that are not available for purchase and therefore not part of the investable opportunity set.

Overall, the methodology that Russell follows in constructing its indexes is fairly straightforward. However, it did introduce a slight change—yet one with great implications—in 2007. Prior to that year, Russell followed a “hard cutoff,” in that the ranking was strictly arbitrary. In other words, the 1,001 st stock in the ranking became the first stock in the Russell 2000.

In 2007, Russell began to use a “capitalization band,” in which the company retained the flexibility to override the process. It works as follows: Russell calculates a range of five percent around the newly determined market cap breakpoints, this being a range of 2.5% above and 2.5% below the breakpoints. Those companies that would automatically be deleted from an index due to a change in their ranking (e.g., from #995 [Russell 1000] to #1,050 [Russell 2000]), would stay in the current index. Such banding, according to Russell, is intended to reduce “unnecessary turnover.”

Theory versus Practice

“In theory there is no difference between theory and practice. In practice there is.” -Yogi Berra

In theory, if one desired to replicate the Russell 1000/2000/3000, it would be a relatively straightforward exercise, given the rules that must be followed. Each company would be listed in order of its market cap (float-adjusted), and one could easily discern where the breakpoint should be.

However, if one were to examine Russell's own data, one would find that the theoretical list and the actual list diverge markedly, as shown in the following table:

Exhibit 2: Russell Index Historical Market Capitalization Ranges (1995-2012) Hi Lo Hi Lo Hi Lo
 2012 540,213\$ 101 \$ 540,213\$ 1,354 \$ 2,608 \$ 101 \$ 2011 411,180 130 411,180 1,624 2,971 130
 2010 283,061 112 283,061 1,256 2,274 112 2009 338,408 78 338,408 829 1,688 78 2008 468,981
 167 468,981 1,363 2,751 167 2007 468,500 262 468,500 2,500 2,500 262 2006 368,500 218
 368,500 1,970 1,960 218 2005 386,900 183 386,900 1,800 1,800 183 2004 317,800 176 317,800
 1,600 1,600 176 2003 286,600 117 286,600 1,200 1,200 117 2002 309,500 128 309,500 1,300
 1,300 128 2001 486,700 147 486,700 1,400 1,400 147 2000 520,200 178 520,200 1,600 1,500
 178 1999 407,200 178 407,200 1,400 1,300 178 1998 271,600 222 271,600 1,400 1,400 222 1997
 198,000 172 198,000 1,100 1,100 172 1996 138,000 162 138,000 1,000 1,000 162 1995 98,400
 104 98,400 748 750 104 (\$ in millions) Source: Russell Investments Russell 3000 Russell 1000
 Russell 2000 The above exhibit shows that from 1995 until 2007, the separation between the
 Russell 1000 and Russell 2000 was discernible: the low range of the Russell 1000 matched very
 closely to the high range of the Russell 2000 for each respective year. For example, in 2006, the
 Russell 1000 contained companies with a market cap of \$36 8.5 billion at the upper end and
 \$1.97 billion at the lower end. The Russell 2000 then contained companies with market caps of
 \$1.96 billion at the upper end and \$218 million at the lower end. These are the types of ranges
 one would expect.

However, following the introduction of banding, the market cap ranges between the Russell 1000 and Russell 2000 began to overlap. During 2008, the Russell 1000 contained companies with a market cap of as little as \$1.36 billion. Meanwhile, the Russell 2000 held companies with

market caps of as much as \$2.75 billion. In other words, the small cap index included companies that were almost twice as large as those in the large cap index. As a result, the indexes are no longer true to their original mandates.

This naturally leads to the question: “Why have they changed?”

Answer: Because they are no longer theoretical constructs; they are now substantial asset management businesses.

The Transformation of the Index into a Business Recall that Russell cites three specific goals for its indexes: 1. to act as performance standards for active managers 2. to serve as proxies for asset allocation purposes, and 3. to become purchasable and replicable vehicles for passive investment strategies.

Historically, the index industry fulfilled the three purposes to varying degrees, with the greatest emphasis on being a performance standard for active managers. This purpose is self-explanatory: without a credible benchmark, it is difficult for investors to evaluate their investment managers. As well, the Russell indexes were utilized as proxies for asset allocation purposes. It is for this reason that Russell offers multiple indexes: by combining groups of equities with different characteristics (i.e. small cap and large cap; domestic and foreign), diversification can presumably be achieved, and in doing so, returns can be enhanced and risk reduced for the entire investment portfolio.

However, the index industry has increasingly evolved from one of measurement into one of providing investment products—the third leg of the index business “stool.” Although passive investing has existed for many years, such as through index mutual funds, over the last several years it has virtually exploded with the rise of ETFs. According to the Investment Company Institute, in only five years (2006-2011), assets under management in equity ETFs essentially doubled, from \$387.3 billion to \$754.4 billion, representing a compound annual growth rate of 14.2%. The benchmarks are now vigorously competing for investment dollars with active managers. As the following table will illustrate, the strategy is winning.

Exhibit 3: Net Cash Flows (2006-2011)

Year Equity Mutual Funds Equity ETFs 2011 (129,650) \$ 68,585 \$ 2010 (36,910) 80,031 2009 (138) 42,086 2008 (228,498) 143,644 2007 74,151 128,116 2006 148,494 59,792 (\$ in millions)
Source: Investment Company Institute Net New Cash Flows With this transformation, distortion of the index itself is unavoidable, because indexation as a business is a low-fee endeavor. The expense ratio of the iShares Russell 2000 Index (ticker IWM), for instance, with \$15 billion of AUM, is all of 0.2%. Accordingly, in order to remain a viable business, the indexes must be highly scalable. A given ETF's asset gathering capacity, though, will be limited by its least liquid member.

Let's illustrate with a simplified example: an ETF with, say, 100 constituents, all equally weighted at 1%, its smallest constituent having a \$150 million market cap, and limited to owning not more than 4.9% of a constituent's shares. The allowable limit for each holding would then be \$7.35 million. This ETF will be able to accept only \$735 million of assets.

If the next-largest constituent has a \$300 million stock market cap, and if the rule set can be altered to exclude the smallest company, then the asset gathering limit rises to \$1.5 billion. Alternatively, if that smallest company has inside ownership of 50%, and if the ETF adopts a float-adjusted weighting methodology, then it can be reduced from a 1% to 0.5% position, and the asset gathering limit again rises to \$1.5 billion. Thus, it is the liquidity issue that influences index construction, and the fixed mechanical rules are not really fixed anymore.

What is the result? Those companies that belong at the top of the Russell 2000, but which have limited float, are instead placed at the bottom of the Russell 1000. In this way, the weighting of the security (and thus the amount of stock required to be traded) is reduced greatly, since the indexes are capitalization weighted. Whereas the top 100 companies in the Russell 2000 typically have an average weighting of approximately 19 basis points, the bottom 100 companies in the Russell 1000 have weightings of less than one basis point.

In return, companies that would normally be long at the bottom of the Russell 1000 but which have ample liquidity are moved into the top of the Russell 2000, where the weighting is some 20 times larger than if they were in their proper place in the Russell 1000. But even this does not fully capture the potential rate of expansion, since there is far more money allocated to the Russell 2000 than the Russell 1000. Judging by the respective iShares ETFs, the differential is 6 \$150 million market cap x 4.9% ownership = \$7.35 million. 7 \$7.35 million per holding x 100 members = \$735 million.

on the order of 2.5-to-1. Effectively, then, the dollar volume of demand for companies placed at the top of the Russell 2000 is some 40x larger than if placed at the bottom of the Russell 1000.

The following table highlights a few of the companies whose placement appears to be biased in favor of the Russell 1000, even though their float-adjusted market caps would argue against their inclusion in that index.

Exhibit 4: Sample Companies in Russell 1000

Float-adjusted Company Market Cap Owned by % Ownership Market Cap U.S. Cellular (USM) 3,092.0 \$ Telephone & Data Systems 84.3% 507.2 \$ Echostar (SATS) 2,936.9 Charles Ergen 50.6% 1,330.0 Kronos Worldwide (KRO) 1,905.4 Valhi 50.5% 340.3 Titanium Metals (TIE) 2,883.9 Various Harold Simmons Entities 54.3% 1,316.0 Clear Channel Outdoor (CCO) 2,443.4 Clear Channel Communications 88.7% 273.6 (\$ in millions) Source: Bloomberg

It can be seen that these companies have substantial insider ownership, and therefore the float-adjusted market caps are substantially less than the stated market caps. Indeed, several of these (U.S. Cellular, Kronos Worldwide, Clear Channel Outdoor) have float-adjusted market caps of \$500 million or less.

If these securities were instead included in the Russell 2000, it would be very difficult to execute the index along the lines of the current methodologies, since the scalability of the index would be limited by the float of these companies and others like them. However, because these securities are overshadowed by the likes of Apple and ExxonMobil, which occupy the lion's share of the aggregate market cap of the Russell 1000, they are assigned to the lower reaches of the Russell 1000, where they have essentially zero weight. Even one hundred of such companies, and even if they were all rapidly expanding and very successful, would make no statistically meaningful impact upon the performance of that index (although, within the Russell 2000, such a cohort could have a dramatic impact).

Compare the above with the following constituents of the Russell 2000:

Exhibit 5: Sample Companies in the Russell 2000

Float-adjusted Company Market Cap Market Cap Pharmacyclics (PCYC) 4,380.0 \$ 3,476.4 \$ Two Harbors Investment Corp. (TWO) 3,376.4 3,329.6 athenahealth Inc. (ATHN) 2,747.2 2,648.8 Dril-Quip Inc. (DRQ) 2,838.5 2,304.5 Coeur d'Alene Mines (CDE) 2,084.5 2,072.8 (\$ in millions) Source: Bloomberg Comparing Exhibit 4 with Exhibit 5, one can easily observe the inherent flaw in the actual implementation of the indexes. There is no reason—other than the attempt to accommodate massive demand—why a company such as Two Harbor s, with a market cap of \$3.4 billion and a float-adjusted market cap of \$3.3 billion should be included in the small cap index while a company such as Kronos Worldwide, which has a market cap of \$1.9 billion and a float-adjusted market cap of only \$340 million, should be in the large cap index.

In the process of filling that demand, the fund flows will ultimately influence the price of the underlying stocks in significant ways. Take, for example, the following table, which contains data on the bottom 50 companies in the Russell 1000 and the top 50 companies in the Russell 2000.

Exhibit 6: Comparative Valuation Metrics Difference Total Market Cap.: 109,984 \$ 135,540 \$ Total Book Value: 71,182 \$ 56,118 \$ Total Sales: 94,849 \$ 68,936 \$ Total Net Income: 5,062 \$ 6,595 \$ Price/Book Value: 1.55 x 2.42 x 56% Price/Sales: 1.16 x 1.97 x 70% Forward P/E: 21.73 x 20.55 x -5% (\$ in millions) Source: Bloomberg Bottom 50 of the Russell 1000 Top 50 of the Russell 2000 Collective Data for the

When comparing the Russell 1000's bottom 50 companies with the top 50 of the Russell 2000, we see that the aggregate market cap of the top 50 Russell 2000 companies actually exceeds that of the bottom 50 in the Russell 1000. As we said before, this is quite extraordinary given the market cap-based weighting rules the indexes purportedly follow.

In terms of equity valuations, the top 50 Russell 2000 companies clearly command a valuation premium. For instance, the aggregate price to book value multiple for the Top 50 is 2.4x, while the Bottom 50 trades at a much lower multiple of 1.5x. The price to sales measure is equally lopsided: the Top 50 companies have a price to sales multiple of almost 2x, which is much higher than the Bottom 50's multiple of only 1.1x. Only on the forward P/E measure are the two groups comparable, owing to a number of companies with negative earnings in the Bottom 50. If we were to exclude these unprofitable companies (since negative P/Es distort the aggregate valuation), the valuation discrepancy becomes even clearer, as shown below:

Exhibit 7: Comparative (Adjusted) Valuation Metrics

Difference Total Market Cap.: 84,904 \$ 132,690 \$ Total Book Value: 63,292 \$ 55,895 \$ Total Sales: 77,324 \$ 68,732 \$ Total Net Income: 5,635 \$ 6,650 \$ Price/Book Value: 1.34 x 2.37 x 77% Price/Sales: 1.10 x 1.93 x 76% Forward P/E: 15.07 x 19.95 x 32% *Adjusted to exclude companies with negative earnings (\$ in millions) Source: Bloomberg Collective (Adjusted) Data for the Bottom 50 of the Russell 1000 Top 50 of the Russell 2000*

The issue of index construction is not merely academic. Companies that should be in the top of the Russell 2000 but which have fallen to the bottom of the Russell 1000 because of liquidity concerns are heavily penalized. Were they to remain in their proper location, that is, the top of the Russell 2000, their share prices would be higher, all else being equal. Conversely, companies that are fortunate enough to be placed at the top of the Russell 2000 solely because of their available float have an unfair advantage. The reason for this occurrence is that the Russell must accommodate the needs of the ETF industry.

Unintended Consequences Aside from the problem of style drift, the way in which the Russell

indexes are currently constructed also introduces other unanticipated and unintended consequences that would adversely impact returns. We highlight five that are especially relevant.

Indexes vs. Insiders The purchase or sale of company stock by an insider, such as the chief executive officer, chairman, or founder, is a closely watched affair. Those companies in which insiders make significant purchases are considered to be attractive candidates for investment. Conversely, when insiders sell, outside observers may infer that the shares might be overvalued or that the company might be encountering problems.

This is a rational strategy that attempts to profit from asymmetric information: insiders are much more knowledgeable about the prospects of their companies than the average outside minority investor, and this information is manifest in their stock transactions. As well, one can assume that wealthy entrepreneurs and owner-operators who have personal capital at risk in their companies, tend to make a distinguished series of risk/reward decisions (i.e. capital deployment) that ultimately should result in excess share price returns over time relative to various broad market benchmarks. There is ample statistical support of this phenomenon.

According to this theory, large insider purchases should be a positive signal that argues for purchase of the shares. Yet, by the rules of the float-adjusted weighting system, the opposite would occur. As insiders buy shares, the available float declines, prompting a reduction in the company's weighting in the index, all else equal. This is counter-rational behavior and, we suspect, is not even noticed by the index investor.

At the other extreme, when companies issue more shares, it is typically considered to be negative for existing shareholders, as dilution reduces the intrinsic value of their equity. However, an increase in a company's available shares that leads to a higher float-adjusted market cap will induce the indexes to purchase more shares, even as informed insiders sell.

Ultimately, as a result of the index methodology, the emphasis is not on the qualitative features of the index members, but simply on their liquidity.

Small Cap vs. "Small Cap"

In light of the evidence presented thus far, one might legitimately ask, "What is a small cap stock?" This question is a serious one because, if a small cap stock is merely an enterprise that has the same characteristics as a large cap stock except that it has fewer shares outstanding and is, therefore, less liquid, then the method of indexation as practiced by Russell (and others) is entirely appropriate. However, if one views small capitalization stocks as smaller businesses and, consequently, as a source of incremental optionality (as we discuss below), then it means indexation as currently practiced is far from appropriate.

In this discussion, optionality refers to the phenomenon of investing in small cap companies which become extraordinarily successful. Indeed, the allure of holding small companies lies in the possibility of producing a return on the order of 10, 20, 30, or even 40 times one's invested capital. The whole concept of small capitalization investing, therefore, is to develop a structure in which it is possible to take the risk of allowing one or more companies to flourish.

However, if the structure of the index emphasizes market cap as dictated by available float, then the same dynamic would not apply. The index would include "small caps," which would only be small caps by the narrow definition of indexation, but which are really large (or larger) companies without the same potential for expansion as true small cap stocks, as dictated by the law of large numbers. Wal-Mart has \$445 billion of annual revenue, Microsoft \$74 billion; it has been a long time since they were capable of a sustained double-digit rate of revenue expansion. The index-altered definition is not small cap investing in the traditional sense; it's something else, and it's not likely to produce a very robust rate of return.

Gresham's Law 8 and IPOs The investment banking community is beginning to realize that an initial public offering ("IPO") issued with sufficient liquidity and market cap will be included in a variety of indexes. Therefore, one effectively has a mathematically certain aftermarket bid, which is a phenomenon that had not previously existed, since the construction rules, as one may recall, allow for the addition of IPOs outside of the reconstitution time frame.

⁸Gresham's law is an economic principle that states that when a government compulsorily overvalues one type of money and undervalues another, the undervalued money will leave the country or disappear from circulation into hoards, while the overvalued money will flood into circulation. It is commonly stated as: "Bad money drives out good."

If this pattern continues, it is mathematically possible that the Russell 2000 will gradually, but inexorably be populated by IPO companies. An IPO-type of Gresham's Law will crowd out the smaller cap, more reasonably valued small positions within the index.

Since index funds must follow their respective index construction rules (i.e. avoid any tracking error), they will be required to purchase such IPOs. Yet, not only would they have no information about the company in question, they would have no desire for any information other than its market cap and float. In other words, these funds would be the ultimate uninformed buyer.

Turnover as a Source of Risk

What does it mean for investors when the liquidity—the dollar volume of buying and selling— of a derivative instrument such as an ETF is greater than that of its constituent parts?

The average daily trading turnover for IWM is equal to roughly 20% of its market cap; phrased differently, annual turnover is on the order of 5,000%, which is, frankly, absolutely astounding. For 2011, ETFs alone were responsible for over \$20 trillion of trading, which means that turnover for the entire group was 20x their \$1 trillion of assets under management, or about 2,000%. As a result, there is greater trading volume in IWM than in many of the companies comprising it. It has been calculated that if IWM had to create a sufficient number of new units (provide liquidity) for the aggregate volume of short positions in that fund, it would take the ETF more than 180 days to buy all the component securities if it limits its buying to 10% of the average daily volume of each holding.⁹ This extreme does not apply to the more liquid Russell 2000 holdings, and there are other mechanisms for providing liquidity in a continuous market, such as derivatives, but clearly the daily liquidity needs of the ETF in creating new or destroying old units cannot always be met in the open market.

Because the liquidity and daily trading volume of some of the largest ETFs now exceed the liquidity of the stocks that comprise their underlying indexes, this phenomenon has reached a critical point: indexes, which are meant to measure the performance of, and provide exposure to groups of stocks, have come to distort the prices of the stocks they are meant to measure. Paradoxically, the trauma of the 2008 financial crisis, which instigated a surge by investors into these instruments as a tool to diversify their holdings in order to reduce overall portfolio volatility has, by the magnitude of these efforts, actually contributed to the increasingly lockstep movements among indexes.

Hard Indexes vs. Soft Indexes, the Nasdaq 100 Index, and the Top-Heaviness Problem

Hard indexes and soft indexes are terms invented for this paper. A “hard index” is one with fixed mechanical rules for inclusion and weighting, such as the Russell indexes or the Nasdaq 100 Index (“Nasdaq 100”). The Nasdaq 100, it should be remarked, is deceptively simple: the 100 largest domestic and international non-financial securities (based on market cap) listed on the Nasdaq.

⁹ Primary source: J.P. Morgan Equity Derivatives Strategy; cited by Harold Bradley and Robert E Litan, for the Ewing Marion Kauffman Foundation in their October 19, 2011 prepared testimony before the U.S. Senate Committee on Banking, Housing and Urban Affairs, Subcommittee on Securities, Insurance and Investments. However, the ETF Industry Organization has estimated ETFs’ proportion of total notional U.S. equity volume at 33% (<http://www.etf-ia.com/sites/default/files/Notional-Volume.pdf>).

A “soft index” is one that allows for human judgment. The Dow Jones Industrial Average and the S&P 500 are examples of soft indexes because apart from any mechanical rules, there are committees that can decide to include or exclude given securities for those two indexes.

There is an inherent problem with hard indexes: they eventually become top-heavy. One of the constituents within the hard index is going to be the best investment, even if the index itself does not rise in value. In the case of the Nasdaq 100, that constituent would be Apple Inc. So successful was Apple that it was necessary to limit that company’s weight in the Nasdaq 100. At present, its weighting in the index is approximately 16%. Were an active manager to report a 16% concentration in a single stock, it would almost certainly attract due diligence attention.

To have a single company representing 16% of an index that purports to be a diversified portfolio representing the largest companies that trade on Nasdaq is reason enough for concern. However, that’s only the beginning.

It should be noted that Microsoft Corp. has a 7.5% weighting. Intel, the semiconductor company, accounts for 3.4% of the Nasdaq 100. Including Apple, these three companies are battling for dominance in the personal electronic device market. Other companies in the Nasdaq 100 that also compete in the personal computer industry include Adobe Systems (0.62% weight), Dell Inc. (0.61%), Symantec Corp. (0.44%), Seagate Technology (0.37%), NVIDIA Corp. (0.26%), Research in Motion (0.24%), and Micron Technology (0.23%).

What is the implication? It means that the Nasdaq 100, with the above companies accounting for one third of the index, is effectively a personal computer index, even though it’s supposed to have much broader representation. Thus, it negates the basic conception of indexation, which is that a panoply of companies will be owned such that no single element can unduly affect the investment result. However, it is obvious here that one single investment element not only affects the result, but determines the result in every conceivable scenario. No other industry in the Nasdaq 100 is as large as that of personal computers.

Conclusion The theory behind indexation is very simple: asset managers generally do not beat the indexes, and securities markets are generally efficient in any case. With fees and transactional friction, active managers might even subtract value.

ETFs enable investors to invest in passive indexes in an efficient manner. The advantages are well known: they have low expense ratios, are easily traded (with liquidity available throughout the day), provide broad diversification, and are tax efficient. However, much like any other

investment product, buyers should conduct due diligence before purchasing.

While the concept of indexes is straightforward, implementation is another matter. There is ample evidence to suggest that, for business reasons, indexes have been manipulated in ways that belie their stated goals. However, because they are taken at face value, investors do not invest the time needed to fully understand the compromises that must inevitably be taken in order to make the product viable. In other words, what one intends to buy may not be what one actually gets, and that may mean the difference between meeting or missing one's investment goals.

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Next: Chapter 11 — Not Your Grandfather's S&P.

Chapter 11 — Not Your Grandfather's S&P

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Extract used for assembly.

book/build/text/20160323071835_6bf4d9d1_Not_20Your_20Grandfather_s_20_20SP_FINAL.txt

Presentation edits: flattened multi-column layout into linear paragraphs; merged line-break splits; table of 30 largest S&P revenue changes preserved from extract.

Not Your Grandfather's S&P

While much of the recent market commentary offered by the financial media outlets is focused on the reasonableness (or not) of the market P/E of about 17x and about the reasonableness of that P/E in relation to "consensus" estimates of corporate profit growth, whether 3% or 6%, it is interesting to note that the largest corporations in the S&P 500 Index do not actually appear to be growing much, if at all. Of the 30 largest companies in the S&P 500, only one-third experienced any revenue growth in 2014, and only one-half had revenue growth of 2% or more. A traditional growth company such as Coca-Cola was negative, Merck was negative, as were IBM, Cisco Systems and General Electric. The aggregate sales growth of these 30 largest U.S. companies was only 4%; exclude just Facebook and Google, and growth was barely above 1%. Perhaps the top tier of the S&P is now too large to grow at a meaningful rate, which has implications for investors in the largest companies; or perhaps it was just a bad year.

There is another question, though, more subtle yet with deep implications for any investor who selects equities, makes asset allocation decisions, or conducts retirement planning or pension plan funding projections. What if the historical returns of the S&P 500, which form the basis for projecting future stock market returns, and therefore powerfully influence the proportion of a pension plan or 401k that will be allocated to stocks or bonds, are not actually representative of what investors are purchasing if they invest in the index today?

The nature of the S&P 500 today — the way the performance is measured — is different than it was before 2005. The difference came about this way: in order to accommodate greater quantities of assets for management in index-based investment products linked to the S&P 500 — that is, provide greater liquidity — Standard & Poor's shifted to a float-adjusted weighting system for the index constituents. The necessity grew out of the way indexes are constructed: all positions were weighted based on their total, or gross, stock market values. Once indexation began to be truly popular, though, with truly large inflows of funds, the problem for the index manufacturer was that if 40%, say, of the shares of a company were owned by insiders, there might be an insufficiency of publicly available shares, or float, to accommodate the volume of assets destined to buy those shares without inflating the share price and valuation. In practical terms, float-adjustment means that a company like Oracle or Microsoft would have its weighting in the index reduced by the proportion of inside ownership. The higher the inside ownership, the smaller the weighting would be set, so that there would be a proportionately lower index-based demand for the shares.

The entrepreneurial companies of today, those with substantial inside ownership (such as Tesla or Google) are accorded lower weights in the S&P 500 than would have been the case prior to 2005. Likewise, their impact upon the Index's returns will be lower than they would have been prior to 2005, when they would have been given full weight based upon their stock market capitalization. Therefore, the historical returns of the S&P 500 — upon which our notions of historical rates of return are based — are no longer continuous with or representative of what the S&P 500 is today.

Accordingly, how big would the impact have been on the S&P 500 had the entrepreneurial companies of the past been float-adjusted, as they are today? Here are two examples known to all: Wal-Mart and Microsoft, which span two generations of innovative service and technology business development in the U.S.

Wal-Mart was added to the S&P 500 in August 1982. Its initial weighting was a mere 0.25%. Sam Walton died in the first week of April 1992. During that decade, the Wal-Mart stock price appreciated by 28.0x, which translates to over 40% per year, and its Index weighting expanded to 2.0%. Inside ownership in 1982 was 38%, as it was after Sam Walton died. Had the S&P 500 applied its current float-adjusted rules back then, 38% of that truly spectacular return would not have been part of the historical S&P 500 return.

Just two years later, in 1994, Microsoft was added to the S&P 500, at a 0.85% weighting. Insiders owned about 40% of the shares. When Bill Gates announced his departure a dozen years later in 2006, the shares had risen in value 8.3-fold, or by over 19% per year (and the S&P 500 by 9.1% per year, inclusive of dividends). Microsoft's weighting in the Index had expanded to 2.0% (and was as high as 2.6% in 2004). A large part of this return would also have been lost from the S&P 500 had it been float-adjusted at the time. The liquidity-motivated float-adjustment mechanism, even if applied only to these two companies, would have had a measureable negative impact upon the Index's returns. What of the others: the Apples, the Intels, the Hewlett-Packards, the Nikes, the Starbucks, the Schwabs? We believe that the S&P 500 today cannot provide what it did in the past, to the degree that any great future entrepreneurial ventures will find their way into the Index.

As an aside, there is something interesting about the pattern of the Microsoft share ownership. When added to the S&P 500, the share price was about \$2.41. At January 1999, when the stock was \$44.75, insiders still owned 31% of the shares. By September 1999, near the peak of the Great Technology Bubble, when the shares were \$47.50 (when Microsoft's weight in the S&P 500 exceeded 4%), inside ownership had been reduced to 26%, and by early September 2000, the very threshold of the collapse of that bubble, when the stock was \$35, inside ownership had dropped to 19%. Today, 15 years later, the shares are only \$44.37 — a 1.6% annualized return — and were quite a bit lower than \$35 only two years ago. Whether by fortune or perception, insiders were dramatically reducing their holdings going into a decade-plus period of decline and stagnation. What did outsiders do? Had the float-adjusted index weighting method been in place at the time, then as insiders sold, such that the float increased, the Index rules would have increased the Microsoft weighting, and mutual funds and other index investors would have been buying more — more of what the insiders were selling.

Although most invest in indexes on the basis of the name on the wrapper, the name doesn't really tell you what's inside. It's not your grandfather's S&P 500.

12-Month change in revenue, 30 largest S&P 500 companies

Company	Change	Company	Change
Apple Inc.	6.72%	Coca-Cola	-2.85%
Microsoft	11.69%	Merck	-6.89%
Exxon Mobil	-8.09%	Citigroup	-3.30%
Johnson & Johnson	5.92%	Facebook, Class A	54.69%
Berkshire Hathaway	12.12%	Google, Class A	19.56%
General Electric	-1.69%	Google, Class C	19.56%

Company	Change	Company	Change
Wells Fargo	1.31%	Gilead Sciences	16.72%
Procter & Gamble	0.58%	PepsiCo, Inc.	1.41%
JPMorgan Chase	7.51%	IBM	-4.55%
Verizon	4.06%	Comcast	3.34%
Chevron	-4.88%	Walt Disney	8.37%
Pfizer Inc.	-12.55%	Cisco Systems	-3.01%
Intel Corporation	-1.19%	Wal-Mart	1.52%
AT&T Inc.	1.03%	Oracle	2.95%
Bank of America	1.96%	Philip Morris Int'l	-1.32%

Average change: 4.1%. Excluding Facebook and Google: 1.1%. Source: Company reports, Bloomberg (as in original PDF).

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Next: Chapter 12 — Equity as a Bond Substitute.

Chapter 12 — Equity as a Bond Substitute

Source. Horizon Kinetics market commentary, June 2015. Horizon Kinetics LLC © 2015.

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Presentation edits: removed running “MARKET COMMENTARY” headers; merged hyphenation across line breaks; Table 1 converted to prose-friendly layout from the extract’s monospaced block.

Equity as a Bond Substitute

It is quite understandable, in the world of low interest rates, that income-dependent investors would be open to bond alternatives as a means of generating income. One such popular alternative is income-producing common stocks as, for instance, those held in the iShares Core High Dividend ETF (HDV). This fund has \$4.7 billion in assets under management, a very reasonable 12-basis-point fee, and a trailing 12-month yield of 3.4%.¹⁵

In early June, the 10-year Treasury bond was yielding 2.19%; hence, it is not difficult to see the allure. HDV has produced more than a 13% compound annual rate of return in the three years ended 03/31/2015, and an 8.6% rate of return in the last year. In contrast, in the three years ended March 31, 2015, the iShares 7-10 Year Treasury Bond ETF (IEF) produced an annual rate of return of about 3.5% (though the 1-year return of 8.8% is comparable to that of HDV).

How much does this return data really inform you? Are these statistics sufficient, in the absence of any qualitative information, to suggest that the future returns will reprise those of the past? It is important to consider the HDV holdings. There are 74, many of which are important positions in the S&P 500, as is evident in Table 1. Consequently, the performance of this fund has relevance to the performance of the S&P 500 Index.

Since the index is not very different from the S&P 500, the concept of income stocks as a bond substitute is not very different from the concept that stocks can be used interchangeably with bonds. In this framework, ExxonMobil, often the largest position in HDV, is a bond substitute. By contrast, using this logic, British Petroleum (BP), which is a European company belonging in the asset class of European equities, is not a bond substitute. In fact, British Petroleum, despite a dividend yield twice that of ExxonMobil, is not even a member of the iShares International Select Dividend ETF (IDV).

HDV is considered a relatively low-risk investment. This is simply because its equity beta is 0.80, that is, its share price has varied about 80% as much as the stock market during some measurement period. The dividend yield is 3.4% and its trailing 12-month P/E ratio is 17.87x, according to iShares. Consider, however, that 24.5% of the fund is invested in the energy sector, and that the price of oil declined by approximately 25% in the six months ended May 31, 2015—it is no surprise that the energy companies reported lower earnings in the first quarter of 2015, which lower earnings may well persist for the remainder of 2015. Consequently, HDV is at least potentially composed of fairly high-P/E stocks, depending on what happens to the price of oil. In any case, the ultimate P/E may well be substantially more than the current measure of 17.87x trailing earnings, unless the energy companies are somehow able to revert to their 2014 profit levels despite far lower oil prices. The energy companies, in aggregate, reported earnings above

analyst estimates, but below earnings last year.

Of course, the most important question pertaining to index-trackers such as HDV relates to the risk relative to that of owning bonds. The index that HDV tracks, after all, was created as a bond substitute even though a stock, unlike a bond, has no maturity date and is, in fact, a long-duration asset.

An investor who thinks of the iShares Core High Dividend Index as a long-term bond might consider several factors. The iShares 20-Plus Year Treasury ETF (TLT) has produced a total return in the last 36 months of slightly more than 8% on a compound annual basis. By contrast, HDV has produced a return roughly 1.6 times that amount. That performance might tempt one to assert that the companies comprising HDV have been growing their businesses significantly during those 36 months, with this growth accounting for the substantial rate of return. However, the revenue growth for the year ended December 31, 2014 was a modest 2.7%. If oil prices don't increase substantially, we know what's going to happen in 2015. The three-year rate of revenue growth of the 50 companies is, cumulatively, 9.7%, on average, or 2.1% annualized.

There seems to be little doubt that HDV, essentially, is populated with mature firms that dominate their industries yet collectively have little prospect of expanding their businesses much beyond the rate of inflation. Of course, these businesses are subject to cyclical forces, such as changes in the price of oil. They are also potentially subject to permanently disruptive forces, such as the growing impact of solar energy on electric utilities.

Table 1 (as in original extract layout). Revenue growth for HDV constituents; source line per PDF.

Year-over-Year	3-year cumulative	Year-over-Year	3-year cumulative
1 XOM Exxon Mobil Corp.	-6.35%	41 AEE Ameren Corp.	3.68%
2 T AT&T Inc.	4.52%	42 WEC Wisconsin Energy Corp.	2.78%
3 GE General Electric	1.74%	43 COH Coach Inc.	-9.77%
4 VZ Verizon Communications Inc.	5.42%	44 MXIM Maxim Integrated Products Inc.	-0.96%
5 CVX Chevron Corp.	-7.38%	45 CA CA Inc.	-3.31%
6 PFE Pfizer Inc.	-3.84%	46 SCG Scana Corp.	10.14%
7 PG Procter & Gamble	-2.50%	47 RSG Republic Services Inc.	4.41%
8 PM Philip Morris International Inc.	0.10%	48 POM PEPCO Holdings Inc.	4.54%
9 MO Altria Group Inc.	0.23%	49 MCHP Microchip Technology Inc.	12.27%
10 COP ConocoPhillips	-4.69%	50 HAS HASBRO Inc.	4.78%
11 MCD McDonalds Corp.	-2.37%	51 CINP Cincinnati Financial Corp.	9.14%
12 KMI Kinder Morgan Inc.	15.32%	52 PNW Pinnacle West Corp.	1.07%
13 LLY Eli Lilly	-15.13%	53 LNT Alliant Energy Corp.	2.23%
14 OXY Occidental Petroleum Corp.	-4.25%	54 LEG Leggett & Platt Inc.	8.77%
15 DUK Duke Energy Corp.	5.14%	55 ORI Old Republic International Corp.	-0.67%
16 SO Southern	8.08%	56 OGE OGE Energy Corp.	-14.47%
17 KRFT Kraft Foods Group Inc.	-0.07%	57 WR Westar Energy Corp.	9.70%
18 LMT Lockheed Martin Corp.	0.53%	58 WDR Waddell & Reed Financial Corp.	16.64%
19 D Dominion Resources Inc.	-5.21%	59 STR Questar Corp.	-2.54%
20 EMR Emerson Electric	-0.83%	60 FMR FirstMerit Corp.	8.62%
21 NEE Nextera Energy Inc	12.45%	61 VR Validus Holdings Ltd.	-3.29%
22 KMB Kimberly Clark Corp.	0.83%	62 PNY Piedmont Natural Gas Co.	14.95%
23 CTL CenturyLink Inc.	-0.35%	63 FII Federated Investors Inc.	-2.16%
24 BAX Baxter International Inc.	11.39%	64 CBRL Cracker Barrel Old Country Store	4.26%
25 GIS General Mills Inc.	-2.49%	65 TRMK TrustMark Corp.	2.85%
26 AEP American Electric Power Inc.	10.83%	66 SJI South Jersey Industries Inc.	21.34%
27 SE Spectra Energy Corp.	6.98%	67 NWN Northwest Natural Gas Co.	-0.66%
28 PPL PPL Corp.	-1.89%	68 GHL Greenhill & Co.	-4.18%
29 RAI Reynolds American Inc.	2.85%	69 FI Frank S International NV	6.96%
30 NOV National Oilwell Varco Inc.	11.54%	70 EE El Paso Electric Co.	3.15%
31 IP International Paper	0.57%	71 IPHS Innophos Holdings Inc.	-0.59%
32 WM Waste Management Inc.	0.09%	72 CCOI Cogent Communications Holdings Inc.	9.20%
33 XEL Xcel Energy Inc.	7.06%	73 AVX AVX Corp.	-4.34%
34 MAT Mattel Inc.	-7.11%	74 OB OneBeacon Insurance Group Ltd.	1.85%
35 NYCB New York Community Bancorp Inc.	-2.31%	Average	2.69%
36 NUE Nucor Corp.	10.78%	Annualized	2.13%
37 K Kellogg	-1.45%		
38 PAYX Paychex Inc.	9.33%		
39 DTE DTE Energy	27.33%		
40 CLX Clorox	-0.66%		

Source: iShares, Company reports, Bloomberg. As of December 2014 (or closest month for companies with a different reporting cycle) — per original.

In summary, then, viewed as a bond substitute, the companies that comprise HDV are subject to various business risks. Furthermore, the ETF as a whole is subject to the valuation risk that inures to any perpetuity when interest rates rise, as eventually will be the case. As a bond

substitute, HDV is, in effect, a super-convexity index. Its historical volatility characteristics—upon which investors have in large measure based their investment decisions—do not yet include periods of rising interest rates. (Indeed, there was no such index in the 1970s.) HDV’s standard labeling statistics— the equity beta of 0.80 and its standard deviation of 9.1%— upon which buyers rely, do not reflect the risk characteristics of the index.

Moreover, since the firms held in HDV account for a substantial portion of the S&P 500, it follows that the potential volatility characteristics of even that index are poorly understood. The basic idea of asset allocation is that there are three basic asset classes — bonds, equities, and cash-equivalents— that have different levels of risk and return, meaning that they should behave differently over time. The creation of equity bond substitutes, and the manner in which these are valued of late, runs contrary to the historical practice.

Indexation is said to be based on modern portfolio theory. But the providers of these products don’t follow modern portfolio theory. They simply say that they do.

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Next: Chapter 13 — International diversification (Under the Hood, June 2015).

Chapter 13 — International Diversification: Bet You Don’t Know How Much You’ve Got

Source. *Under the Hood: What’s in Your Index?* (An Ongoing Series, June 2015). Horizon Kinetics LLC © 2015.

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Presentation edits: merged words split across line breaks from PDF extraction; recreated allocation tables from the opening pages; preserved numerical claims and punchlines; light punctuation only.

International Diversification – Bet You Don’t Know How Much You’ve Got

The central tenet of modern diversification is to eliminate the unsystematic risk events in a portfolio, so that the positive performance of some securities will offset the negative performance of others. The policy is to own many investments, as opposed to simply owning a few, and estimate future rates of return by extrapolating past rates of return. This seemingly scientific and statistically based reasoning is extended beyond the individual security level to categories of investments. It is therefore also the practice to establish some meaningful international exposure as a risk diversifier for a U.S. equity portfolio. Amongst institutions, this orthodoxy has already been accomplished. It is also becoming the norm for individual investors, with software-generated asset allocation models (“robo advisers”) suggesting optimal mixes of ETFs. The now standard mode for accomplishing this is via indexes. Even sophisticated institutions like a large state pension fund might manage three-quarters of their internally managed equities (that portion not assigned to outside advisers) through indexes.

Displayed below is a typical asset allocation model for a large-scale pension plan that could be responsible for tens or hundreds of billions of dollars of assets, as well as one suggested by the American Association of Individual Investors and another by a robo adviser.

Table 1: NY State & Local Retirement System, recent asset allocation

Segment	Share
U.S. stocks	38%
Foreign stocks	17%
Bonds, cash, mortgages	21%
Inflation-indexed bonds	6%
Alternatives (private equity, real estate, other)	18%
Total	100%

Source: New York State & Local Retirement System, Annual Financial Report (as cited in original).

Table 2: Recommended asset mix — American Association of

Individual Investors*

Segment	Share
U.S. stocks	50%
Foreign stocks	20%
Intermediate bonds	30%
Total	100%

*For moderate-risk investor: age 35–55, 20+ year investment horizon. *Source: American Association of Individual Investors (as cited in original).*

Table 3: Recommended asset mix — robo adviser*

Segment	Share
U.S. stocks	33%
Dividend stocks (also U.S.)	6%
Total U.S.	39%
Foreign stocks	27%
Natural resources	5%
Municipal bonds	29%
Total	100%

*For a risk tolerance of 5.0 on a scale of 10. *Source: A top-10 robo adviser (as cited in original).*

The institutional portfolio, shown in Table 1, is allocated about 55% to common stocks; 31% of the stocks are allocated to foreign equities. Bonds and cash are about 21% and, having the sophistication and scale, the institutional account also has 18% of the portfolio in a variety of alternative asset classes like private equity and real estate. Despite the differences in resources between, say, a \$175 billion pension fund with institutional trading desks and teams of analysts, and a \$100,000 IRA owner, the asset allocation output is essentially similar.

The individual investor, at least in the case of a self-defined profile for moderate risk and return objectives, will be advised to employ various ETFs to achieve about a 66% to 70% weighting in equities, of which about 30%–40% will be in foreign stocks, as in Tables 2 and 3. AT&T, as an example of a major corporate pension fund, currently has 38% of its equity investments in international stocks.

So, those are portfolio construction rules based on statistical attributes. Now for a qualitative inquiry. Let's start with the leading domestic equity index, the S&P 500, which would be the building block for any of these allocation models. Its largest 50 positions, being only 10% of all the companies in the index, nevertheless account for 45% of the market value of the entire index; we'll presume that this is substantially representative of the balance. This exercise is truncated to pay some respect to the 80/20 rule of efficiency.

As to the S&P's true geographic diversification, even a domestic index is not purely domestic, in that a large multinational company such as Coca-Cola, despite being a truly all-American and U.S.-domiciled business, certainly sells a lot of its product outside our borders. So what do we have?

On average, **44%** of the revenues of these 50 companies are derived from outside the U.S. Even if surprising, this is actually a conservative figure inasmuch as some companies only list "North American" revenues, which would include Mexico and Canada, as opposed to U.S. revenues. Other companies, like AT&T, do not state their foreign revenues, so AT&T is counted in the exercise as zero, despite having significant non-U.S. activities. One piece of information AT&T does provide is that it has \$26 billion of foreign currency swaps (notional value) on its balance sheet, which may be compared with \$129 billion of total revenues—presumably, its non-U.S. revenue is not trivial.

The largest 100 companies in the S&P 500 Index, 20% of the list by number, comprise 61% of the market value; their average proportion of foreign revenues is **39%**. There is a different way to calculate this: not on a simple average basis for each company, but weighted by each company's market value; in this instance the average value-weighted foreign revenue contribution is **26%**. If weighted, instead, by asset value, the figure might well be higher.

But those fine points don't matter, do they? We now know that investors, whether major institutions that, effectively, advise themselves, as well as individuals who are advised by institutions, are being told to allocate roughly 30% to 40% of their stock investments, which might otherwise be in the S&P 500 or essentially similar indexes, to non-U.S. companies. **Yet, by holding the S&P 500 those investors are already fully allocated internationally.** Say what???

One of the paradoxes here is that the major U.S. companies are quite globally diversified in terms of sources of business, if anyone would care to read their annual or quarterly reports.

Historians will one day be baffled as to how well-educated people were convinced to invest billions, and perhaps hundreds of billions of dollars, in companies of which they know nothing, often located in places they would be afraid to visit, all in search of diversification that was already achieved by the American firms that were being sold to fund the international diversification effort in the first place.

As a final note, without going into any further detail, would it be surprising to learn that the **10 largest holdings** in the basic international equity index that one is recommended to purchase in order to internationally diversify the already internationally diversified S&P 500 Index, namely the **MSCI EAFE** Index, get **30%** of their revenues from the U.S.? There must be a better way. Step 1, perhaps, is to understand that investing and economics are social sciences and cannot be reduced to simple or programmatic mathematical and statistical methodologies.

Upcoming *Under the Hood* reviews (as teased in the original): What's in your small cap index? What's in your emerging markets index? How safe is your utility index?

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Next: Chapter 14 — Further Basic Principles of Croupier Investing.

Chapter 14 — Further Basic Principles of Croupier Investing

Source. *Contrarian Research Report*, Horizon Research Group, April 27, 2006. © Horizon Research Group, 2006.

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Research staff (as listed on cover). Steven Bregman, Michael Gallant, Vivek Rattan, Murray Stahl, Peter Doyle, Naveen Kumar, Peter Schweitzer, Arul Thareja, Thomas Ewing, Tina Larsson, Eric Sites, Fredrik Tjernstrom, Steven Tuen.

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Studies in Absurdity — Subject: Further Basic Principles of Croupier Investing

I — The Fundamental Theorem of the Croupier

Given any economic growth rate for a society, it appears that the socio-economic inter-relationships among the various constituent members of that society multiply more rapidly than the economic growth rate. If true, this would be a function of specialization; of course, it is specialization that makes possible productivity improvement that, in turn, makes possible the expansion of the production possibility frontier that, in turn, makes possible a higher standard of living. In a specialized society, the failure of even a small portion of the constituent members to perform economic functions as expected or required can entail widespread and possibly unpredictable consequences for the economic systems. The consequence for securities markets is that the reasons for market value fluctuation of any individual security or set of securities is multi-causal and probably theoretically infinite. Thus, alternatively expressed, the risks to a set of securities cannot be completely enumerated. One can only more completely enumerate the risks than might have been possible hitherto. This yields that which will be called the Fundamental Theorem of the Croupier: Given any risk control strategy for a portfolio, it should always be possible to devise a more precise and refined risk control strategy.

II — The Fundamental Premise of the Fundamental Theorem

It will be observed that the fundamental theorem is based, in part, upon a notion that might be referred to as the fundamental premise. This is stated at the onset of this paper in Section I and can be more briefly formulated as follows: The complexity of a given society grows more rapidly than the given society itself. It is perhaps for this reason that the trading volumes of both the 2-year and 5-year U.S. Treasury future on the Chicago Board of Trade each exceeded \$10 trillion in 2005. The entire U.S. national debt is only at the level of \$8.9 trillion. Hence, if the U.S. national debt were only comprised of 2 and 5 year notes (which is obviously false) and there were no trading in the actual underlying instruments (which is also obviously false), the turnover rate on U.S. Treasury debt would be at the level of 224%. The turnover rate is actually much greater. In terms of risk, the 2- and 5- year Treasury Notes may be considered to be one of the lower risk security types that exist in the world. The turnover figure should be compared with NYSE 2005 turnover of 102%. (source: NYSE Factbook). It seems reasonable to suppose that there is a greater need to control risk for the companies listed on the New York Stock

Exchange than there would be for the 2- and 5-year U.S.

1 Source: CRB Commodity Yearbook 2005

Treasury securities. In fact, according to the U.S. Bureau of Public Debt, the total amount of marketable U.S. debt outstanding as of March 31, 2006 is \$4.34 trillion.² Approximately \$1.042 trillion of this amount is Treasury Bills that mature in less than one year. Treasury notes of all maturities are only outstanding at the level of \$2.4 trillion. Thus, viewing only the 2- and 5-year Treasury Futures traded on the Chicago Board of Trade and without reference to any trading in the underlying instrument, it is fair to say that turnover in these instruments easily exceeds the annual figure of 900%. One might wonder why relatively shorter term U.S. Treasury securities generate as much futures activity as has been the case. The answer can only be reasoned supposition. The analytical tools used for risk control in the field of bonds are far more precise than those used for equities. For instance, a bond portfolio with a given convexity and a given duration can be expected to generate a predictable return in accordance with the interest rate environment. One can know the effect of small changes in portfolio duration. The corresponding equity tools such as net long/short exposure or effective beta are far less precise. Ergo, it seems that portfolio activity varies in proportion to the precision of the available risk control tools rather than in proportion to the risk of the underlying instruments. Hence, one can state the following working supposition: Working Supposition – A croupier can stimulate trading activity by making available risk control tools. Clearly, the complexity of a society, or in this case the complexity of the banking system, is a significant factor. The modern bank is a large holder of U.S. Treasury Securities. It answers not only to shareholders but also to stakeholders such as depositors, borrowers, regulators, correspondent banks as well as legislators. These stakeholders will tolerate very little volatility of shareholders' equity. The failure of the bank to properly function can be disastrous for wide swathes of a given society. Given interest and principal payments as well as the continual passage of time, a bank portfolio is constantly changing. Therefore, much activity is required merely to maintain convexity and duration at a constant level.

III — The Logical Consequences of the Working Supposition

A. Activity Stimulation in Commodities Due to Competition The New York Mercantile Exchange, known as NYMEX, is the dominant North American crude oil contract trading venue. On April 21, 2006, the NYMEX set an all time daily volume record of 1,420,734 futures and options contracts. Of course,

² Over \$4 trillion of the U.S. national debt is in the form of non-marketable securities held by the Social Security Administration, in the form of U.S. Savings Bonds or held by various state treasuries as escrow for bond redemption.

NYMEX trades other commodities aside from oil. In addition, recent record energy prices have contributed to interest in commodity trading. However, in recent years the Intercontinental Exchange (NYSE - ICE) has created competition for NYMEX. The Intercontinental Exchange has become a significant energy futures exchange. This is partly the result of the Intercontinental Exchange purchase of the International Petroleum Exchange in 2003. On April 5, 2006, the Intercontinental Exchange reported its third consecutive month of record volume with volume for March 2006 at a level that was 115% greater than March 2005. Over 50% of ICE volume is in the Brent Crude Contract. This contract directly competes with the primary NYMEX contract which is based upon Light Sweet Crude 1,000 barrels per contract delivery point Cushing, Oklahoma. The Brent contract traded on ICE is based upon light, sweet North Sea crude that is often thought of as an alternative to the NYMEX primary light sweet crude oil contract. Commentary with regard to the NYMEX and ICE usually makes reference to the competition between these two similar contracts. However, even such competition can present opportunity. For example, NYMEX now trades contracts based upon the spread between the West Texas and Brent contracts. If trading moves towards the ICE Brent contract, the resultant demand might create a price differential between the West Texas and Brent contracts and this would be reflected in a change in value in the spread contract. NYMEX now trades spread contracts on the differential between the light sweet crude oil futures contract and the Canadian Bow River crude at Hardisty, Alberta. NYMEX also trades contracts based upon the differential between light sweet crude oil and the following four domestic U.S. grades of crude oil: Light Louisiana Sweet, West Texas Intermediate, West Texas Sour and the so-called MARS blend. Thus, the differences between the NYMEX and ICE futures as well as the differences between NYMEX and Canadian Futures Exchanges actually have become the basis for new futures contracts. The competition actually creates the possibility of new trading securities. It should be noted that there are significant differences between NYMEX and ICE. NYMEX is primarily a futures exchange. A futures exchange trades commodity contracts with standard uniform characteristics. In contrast, nearly one half of ICE is an OTC or over the counter exchange. OTC contracts have non-uniform or customized terms. Thus, there can be many different variations of an OTC Brent contract. Moreover, a futures contract may ultimately be settled by actual delivery of an underlying commodity. An OTC contract is always by cash payment. Each contract is a non-standardized arrangement between two counterparties. Furthermore, many OTC contracts are actually differential or spread contracts. For instance, an OTC contract may be based upon the spread differential of energy product between two delivery points such as Henry Hub and Cushing. Roughly 45% of the ICE business is OTC. The OTC

business in notional amount is growing at 235% per annum. Hence, ICE is

not merely an institution devoted to the capture of market share from NYMEX. It appears that the mere availability of new trading instruments is a stimulus to usage. The fact that each exchange can continually devise new trading instruments is an illustration of the Fundamental Theorem of the croupier as stated in Section I. B) Activity Managed Exchange Traded Funds According to the Mutual Fund Factbook 2005 edition published by the Investment Company Institute, there were 8,044 U.S. mutual funds in existence in 2004. This accounted for over \$8 trillion of net assets. There are over \$12 trillion of purchases of mutual funds as well as over \$12 trillion of redemptions each year. If the mutual funds industry were an exchange it would record 150% annual turnover and be larger than the NYSE. Naturally, none of this activity takes place upon an organized exchange. It is processed by a handful of custodian banks. These figures do not include the rapidly growing ETF's or exchange traded funds that actually do trade upon organized exchanges. According to the Investment Company Institute, Exchange Traded funds have (as of February 2006) net assets that total approximately \$315 billion. The comparable figure for U.S. mutual funds in February 2006 was \$9.2 trillion of net assets. Hence, the ETF's represent only 3.4% of total mutual fund assets. If conventional mutual funds could trade during the trading day as do the ETF's, it is obvious that an enormous quantity of activity would be created that would be of direct benefit to various croupier companies. In fact, entirely new types of transactions would take place. It would become possible to create basis trades among managers. One could conceivably establish a long position in a fund with good management and a short position in a fund with poor management. One could further refine the trade by limiting the selection universe to managers of similar styles. In fact, it is not absurd to imagine that manager basis trades might evolve into an entirely new asset class. It would certainly have less risk than pairs trading since there is no possibility of an acquisition at a premium of the fund held in short position. Consequently, this type of basis trade would be at the very least viewed as a pairs trade with no exogenous event risk. The funds would almost certainly trade at net asset value, as does any conventional exchange traded fund, since there would never be a shortage of supply given that demand would be satisfied by the open-end nature of the actively managed exchange traded fund. In fact, manager skill level basis trades are not the only hedging or risk control that could be envisaged. In principle, a fund could be long or short one or more of its competitors. It is already possible to do this with passive exchange traded funds. Of course, croupiers would enjoy an enormously expanded level of business activity. The reader perhaps experiences at this point the feeling of futility that often accompanies any exercise in pure fantasy. However, this is not an exercise in fantasy.

The creation of actively managed Exchange Traded Funds is under consideration by the Securities and Exchange Commission. Interested readers should consult SEC release No. IC-25258; File No S7-20-01 entitled Actively Managed Exchanged-Traded Funds. The release is a request for public comment. Passive Exchange Traded Funds were only authorized in 1993. The first such fund was based upon the S&P 500 and is known as SPDR Trust. It trades on the American Stock Exchange. The SPDR Trust has \$52 billion of net assets and recently has been trading in excess of 70,000,000 shares per day. Daily trading value therefore exceeds \$9 billion, or 17.6% turnover every day. Therefore, over the course of a year, given perhaps 245 trading days, there should be the astonishing turnover figure of 4,330%. This is clearly an important product for the American Stock Exchange. In order to comprehend this turnover figure it may be useful to examine the trading characteristics of ExxonMobil which is the largest company in the S&P 500. ExxonMobil has an average trading volume per day of 18.4 million shares. This results, at current prices, in an average recent daily trading value of \$1.17 billion. During the course of a year at this rate ExxonMobil will have experienced 72% turnover. Similar calculations can be performed for all of the leading S&P 500 companies and a similar result will be obtained. For instance, General Electric daily trading value is only \$993 million. This would result in an annual turnover rate for the GE share of 68.7%. If one were to ignore double counting of volume on NASDAQ and accept without adjustment the reported Google volume figure, then Google has daily trading value that is equivalent to approximately \$5.7 billion. If continued at the current rate for a trading year, this would generate a turnover rate of 971.5%. In reality, the true turnover rate is lower. Yet this is higher than GE and Exxon. It is very small in relation to the SPDR turnover rate. A useful comparison of Google without adjustment for NASDAQ double counting of volume is to compare these figures with the comparable unadjusted volume figures for the NASDAQ 100 Trust, otherwise known as the QQQQ. This trust has a share price of 42 and has recent daily volume, as reported by NASDAQ, of 130 million shares. Thus daily trading value equals \$5.46 billion. The net assets of the NASDAQ 100 are, according to NASDAQ, roughly \$21.4 billion. Thus the NASDAQ 100 would, in principle, generate an annual turnover rate of 6,150%. It must be emphasized again that due to NASDAQ double counting of volume, neither the Google nor the NASDAQ turnover rates are as high as enumerated for the purposes of this exercise. However, since double counting affects both instruments the ratio between the Google and the NASDAQ 100 turnover figures may be examined to obtain a sense of the differences in activity levels between the two instruments. NASDAQ 100 seems to have over 6x the activity of Google. This is similar to the U.S. Treasury futures activity level observed in relation to the activity level of the New York Stock Exchange (section II). In fact, the turnover rate of

the SPDR Trust traded on the American Stock Exchange is far greater than the turnover rate of the New York Stock Exchange even though most of the constituent members of the SPDR (S&P 500) actually trade on the NYSE. As a consequence another working supposition may be stated

as observed in Section II of this paper: Working Supposition # 2 – It seems as if trading activity is a function of the perceived precision of the instrument rather than of the risk of the underlying instrument. Readers may object to this supposition upon the basis that trading activity is really a function of liquidity. Yet, liquidity is merely a synonym for trading activity. Thus, one would say a given security trades at great volume because it is liquid and it is liquid because it trades at great volume. The more interesting point is that ExxonMobil, with a market value of \$396 billion, will perhaps generate trading activity of \$1.17 billion, whereas the SPDR with a market value of \$52 billion will generate trading activity with a value that exceeds \$9 billion. One might therefore expect that an actively managed exchange traded fund will generate much trading activity. The reader might wonder why anyone might propose the creation of an actively managed exchange traded fund for any reason apart from the generation of mere trading activity. In fact, many well reasoned arguments can be advanced. One such argument relates to proper expense allocation. As noted earlier in this section, the turnover of open end mutual fund assets, based upon the figures made available by the Investment Company Institute, suggest a turnover rate that easily exceeds 150%. In order to accomplish the purchase and redemption of shares in a mutual fund, the fund itself must engage the services of a custodian to keep shareholder records. In any fund, there are those who trade often, which means frequently purchase and redeem, and those who are simply content to be long term investors. Since the expenses associated with purchase and redemption are paid by the fund, these costs are shared by all fund participants. The most significant costs are never included in the expense ratio. These are the commission and trading friction costs that the fund experiences when the fund itself trades to accommodate admission and withdrawals. Although the operating costs of the fund must necessarily be shared by all fund participants, an actively managed exchange traded fund would provide a mechanism whereby investors could purchase and sell units outside of the fund on an exchange and thus reduce activity within a fund much to the benefit of those long term investors who merely wish to be continually invested in a given fund. Naturally, in order to accomplish frequent purchase and sale, the investors who wish to engage in this activity would need to make use of brokers, perhaps electronic brokers and exchanges. In other words, new business would be generated by a croupier. Of course, it could be argued that the increase in trading in the actual ETF would be exactly balanced by the reduction in investor induced turnover at the mutual fund level. Ergo, there should be no net change to systemic activity levels. However, this does not

appear to be the actual ETF experience given the data presented regarding the SPDR and the NASDAQ 100. Actively managed exchange traded funds might soon become a reality. If this does occur, it should prove to be a most entertaining laboratory for the inventive and imaginative faculties of the croupier employees. The proposal envisages that each actively managed mutual fund would create an ETF class. C) One Chicago and Single Stock Futures

One Chicago is the exchange upon which trade single stock futures as well as futures upon so-called narrow-based indexes. One Chicago is jointly owned by the Chicago Mercantile Exchange, the Chicago Board of Trade and the Chicago Board Options Exchange. Trading in single stock futures formally commenced on November 8, 2002. On that day 3,000 contracts were traded. A single stock future typically represents 100 shares of underlying stock. In November 2002 there were single stock futures on 21 stocks. The exchange currently lists 206 single stock futures. In principle, the need for single stock futures is not different from the need for commodity futures. For instance, a pension fund or a foundation may have knowledge of future cash flow or transactions and wish to guarantee the price of a security or group of securities. In the case of a foundation, it may have been awarded a significant amount of a single stock upon the demise of a benefactor. Yet, the foundation cannot actually receive the shares until the estate of the benefactor emerges from probate. The foundation wishes to guarantee the price of the stock some number of months in the future. It can accomplish this by the sale of the stock future. Of course, this could be accomplished as well by short sale of the actual stock. However, single stock futures have a margin requirement of only 20%. This is in comparison to the customary 50% margin requirement. Consequently, futures on single stocks are merely a device to own or control the underlying instruments with more leverage than is ordinarily available. It should be evident that this can become a very dangerous activity. Consequently, one should not be surprised to discover that this area has one of the most rapid growth rates of any of the newly created financial instruments. In March 2006, 395,988 such contracts traded on One Chicago. The compound annual growth rate of this activity since inception has been 88.1%. The compound annual growth rate for the past 36 months has been over 70.8%. Open interest has essentially doubled in the past 12 months. It is evident that single stock futures are more attractive than margin debt. Margin debt, as calculated by the NYSE was at the level of \$222 billion in February 2006, which is the most recent available figure. This is consistent with underlying equity value of \$444 billion, assuming a 50% margin requirement. This figure can be translated into an equivalent open interest figure for One Chicago if all margin debt

became single stock futures open interest. In such a circumstance, One Chicago open interest would be roughly 10 million contracts as opposed to the current open interest of roughly one million contracts. It would entail a 10-fold expansion at One Chicago. At the current rate of growth this could be accomplished within 48 months. At the current rate of activity, open interest turnover is at an annual rate of 540%. This is roughly 5 times the level of current New York Stock Exchange turnover. Most participants in single stock futures trading assert that the various instruments are being used for hedging purposes and not for speculative purposes. It is

merely interesting to observe that hedging or risk control activity seems to generate more turnover than ordinary investment activity. However, this evidence is entirely consistent with the experience of bond hedging activity conducted at the Chicago Board of Trade as noted in Section II of this paper.

IV — The Theory of Hedging Equivalence

It may seem that the continued use of the term “croupier” in preference to the other less pejorative and more obvious terms betrays the bias that the development of new alternative trading or hedging securities is not necessarily a long term positive development for financial markets. Indeed, it should be manifest that possibility for widespread abuse with more unfortunate economic consequences does exist. Moreover, the emergence of a “croupier” class of entrepreneurs whose only function is to encourage financial market participants to engage in ever more exotic and larger financial transactions may serve to concentrate capital upon zero sum trading actions rather than wealth creation activities. Nevertheless, it can be asserted with equal vigor and perhaps even more persuasiveness that new “hedging” instruments are merely mechanisms with which investors can properly manage portfolio risk. The increased trading volume associated with such instruments only contributes to the depth and liquidity of financial markets and as such is an element of enhanced market stability. Of course, it is also a source of enormous return on equity for the typical croupier. For instance, the Chicago Mercantile Exchange might well achieve a 34% return on equity in 2006. The expected return on equity for Google in 2006 should slightly exceed 25%. However, none of this is important for the purposes of this paper. If these new instruments enhance the risk/reward balance of portfolios and produce a positive good, then these will become universally used. If these new instruments are simply opportunities to engage in fruitless trading at best and serious abuse at worst, then ultimately these will fall into disuse. Thus, the world of portfolio management cannot remain only in part users of such devices. The field must either adopt universal usage or reject such usage. Therefore, another theorem can be stated as follows:

Risk control as currently practiced will either be universally accepted or universally rejected. If the instrumentation and impedimenta of modern risk control achieves universal acceptance, the already high returns on equity of many “croupier” companies will markedly expand from already elevated levels.

Afterthought

Whatever might occur in the future with regard to publicly traded exchanges will probably occur quickly and upon a very large scale. Consider these few recent developments:

1. The Chicago Board of Trade in cooperation with the Singapore Exchange plans to open a joint Asian Derivatives Exchange in the third quarter of 2006, to be hosted upon the CBOT electronic trading platform. If successful, the CBOT will effectively become a 22-hour Trans Pacific Derivatives Exchange.
2. The New York Mercantile Exchange (NYMEX) plans to combine with the Dubai government to create the Dubai Petroleum Exchange. NYMEX will also make its energy trading futures available for trading on the Chicago Mercantile Exchange Globex Platform.
3. The London Stock Exchange might merge with either NASDAQ, Euronext or possibly even the New York Stock exchange. The exchanges are establishing the ability to conduct business upon a far more expansive scale. Moreover, it appears that, at least among the primary global exchanges, there is a concerted effort to cooperate rather than to compete.

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Next: Chapter 15 — A Review & Update of the Croupier Business Model (2008).

Chapter 15 — A Review & Update of the Croupier Business Model

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A Review & Update of the Croupier Business Model

To review my definition of croupiers, they are facilitators of financial transactions. Some examples of croupiers are securities exchanges, custodians or processing companies. As a

generalization, the croupier does not put its own capital at risk. They are analogous to a dealer in a gambling casino who does not place his or her own capital at risk. Croupiers are to be distinguished from financial services firms, whose capital, generally speaking, is always at risk. An example of such a company would be Citigroup. The fact that Citigroup is not a croupier should in no way lead one to conclude that I am recommending or not recommending Citigroup. I merely use it as an example of a financial services company that does not qualify as a croupier in the sense that I've just defined the term.

At the outset, it needs to be recognized that the world exists today in a pecuniary society. Pecuniary refers not only to its ordinary, narrow definition relating to money, but more broadly in terms of negotiable securities. It is not possible to have a modern industrial economy without a pecuniary society, though it is possible to have a good economy without a pecuniary society. In Carl Sandburg's biography of Abraham Lincoln, he describes how, as a young man, Lincoln lived for an entire year without ever seeing money.

During Lincoln's youth, it was possible to sustain oneself in the U.S. by hunting, fishing and farming without ever participating in the pecuniary society. Today, that would be impossible, because the industrial enterprises that sustain the standard of living we have today are beyond the ability of even the wealthiest individuals to finance; for example, to build a utility power plant could conceivably cost many billions of dollars. Even the largest companies, in terms of market capitalization, would not allocate such a very large portion of their market value for investment in one utility project.

Such a move for a company, or for very wealthy individuals, would dramatically undiversify their portfolios, and they would be unlikely to make such a choice. As a result, it is necessary to have the ability to agglomerate the capital saved by many different individuals and businesses into one project. Representation of the value deposited by each can only be in the form of a negotiable instrument like a security.

To attract the savings of individuals and businesses, the invested savings must be available to be drawn upon as needed; in other words, those negotiable instruments must freely circulate, and be exchangeable; there is no alternative. In the current era, perhaps the most important reason for the apparent breakdown of the financial mechanisms is that the authorities permitted the evolution and construction of exchange mechanisms that were outside the domain of any reasonable rule system. This vast sphere of portfolio exotica, including interest rate swaps, credit default swaps and the like, exist outside of any organization.

The thrust of modern regulation is clearly and inexorably to bring transparency to those instrumentalities. That goal will likely be accomplished on some sort of organized exchange or clearinghouse. Beyond any question, announcements and implementation to that effect will take place within months, or possibly within weeks, though that shorter time frame is less likely. This situation represents the greatest opportunity that the croupiers could ever have had, because the scope of their activity is going to increase several fold. Beyond that, it will continue to increase exponentially, because there are vast regions of the world that are developing economically, and they will enter the pecuniary economies.

It's interesting to note that when the unregulated business of negotiable instruments begins to perform in a dysfunctional manner, it forces liquidation of securities in the regulated businesses as a way to fund the resultant collateral calls and collateral postings of the unregulated businesses. The result is a bizarre correlation amongst all sorts of asset classes that were never before correlated. It's a correlation that I think is almost at its end.

Let's use exchange traded funds (ETFs) as examples, which, incidentally, now embrace much more than the commonly understood stocks and bonds; they also embrace commodities. In the energy arena, for example, the ETF that contains the Dow Jones AIG Energy Index is down 66.8% since June 30, 2008. Looking at another commodity, in the same time period, the Dow Jones AIG Agricultural Index is down 50%. The Dow Jones AIG Industrial Metals ETF, yet another group of commodities, is down 58.3%.

Correlation can be found even when looking at subsets within the various ETFs; for example, the Dow Jones AIG Grains Index is down 54.3% since June 30, 2008. Grain is clearly a subset of the agricultural index. This situation is almost unheard of in the world of commodities, because one would imagine that the supply and demand imbalances, to the extent that there are imbalances, should logically be different at various points in time between energy commodities, agricultural commodities, and industrial metal commodities. There should also be differences among the agricultural commodities, but they are not.

The collapse of the unregulated, pecuniary economy acts like a suction pump that drains liquidity from all sources. One would have thought that, in the midst of such a liquidity crisis, the demand for the exchange of securities in the regulated pecuniary economy should be lessening, as it did in prior cases, but that is definitely not the case at the moment. For example, ProShares has Ultra Long and Ultra Short indexes that are weighted to perform 2x the underlying indexes on any given day (that's the definition of "ultra"). In the last month, a company known as Direxion created the innovation of a 3x, or super-ultra, ETF. At the moment, there are the Direxion Large Cap Bull 3x ETF (BGU), the Direxion Large Cap Bear 3x ETF (BGZ), the Direxion Small Cap Bull 3x ETF (TNA) and the Direxion Small Cap Bear 3x ETF (TZA).

In one month, the following volumes have been achieved: in the Large Cap Bull 3x ETF the volume is now at 13.9 million shares daily. The volume at the end of one month for the Large Cap Bear 3x ETF is 6.9 million shares daily. The Small Cap Bull 3x ETF, also one month old, has achieved the volume of 9.3 million shares daily, and the Small Cap Bear 3x ETF, the laggard of the group, has volume of 3.5 million shares daily.

In this connection, it is worthwhile noting that the original ProShares 2x Ultra has had undiminished volume throughout; if anything, it has increased. The appetite among market participants for risk may well have decreased, but the demand for liquidity has increased, and it's just at its beginning. Another example is the Lehman 20+ year Treasury ETF that now trades 4.5 million shares daily, an amount that is historically unprecedented for that particular instrumentality. The S&P SPDR (SPY) has 50% turnover daily. For the PowerShares QQQ (QQQQ) which is, of course, the NASDAQ 100, volume is recorded in a double counting mode; nevertheless, it trades over 200 million shares daily.

The market has spoken. It demands liquidity at every hour of the day on a 24 hour basis. The croupiers are merely processing companies; they are no more, and no less. They do not risk their capital other than in the plant and equipment necessary to facilitate liquidity transactions. They earn copious amounts of cash flow which, in the fullness of time, will increase. Over the last year, credit markets have essentially become frozen; they basically refuse to function. In the world of commodity exchanges, however, interest rate futures trading has diminished only somewhat vis-à-vis last year, which is an incredible phenomenon. While the entire securitization market has disappeared, including mortgages, student loans, credit card receivables, and any type of securitization, the interest rate futures volume is down only slightly.

At the moment, various governments of the world are creating more fixed income securities than have ever been created in the history of civilization. Though the United States is a leader in that effort, it is but one among many. This creation of fixed income securities is being done at record low levels of interest rates. Is it not reasonable to assert that, in some number of months, the market participants, the owners, or in some cases the former owners, of these securities will wish to hedge themselves by the purchase or sale of interest rate futures? Is it not likely that that market, which is already very large, will grow enormously? The likelihood of that scenario is the market opportunity that stands before the croupier.

In my humble opinion, the croupiers are on the verge of a golden era that is just now starting. Of course, given the low valuations of croupiers, most market participants would disagree. Time will tell which of us is right. As of the most recent reckonings, the available empirical data suggests that the croupiers' business is growing, not diminishing.

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Next: Chapter 16 — Q&A on trading volume, innovation, and Treasury / Fed tactics (January 2009).

Chapter 16 — Q&A: Trading Volume, Innovation, and the Treasury & Federal Reserve (January 2009)

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Trading volume trends

Q: With respect to your enthusiastic view of trading volume trends, and hence what the future

holds for exchanges, do you have an opinion on whether or not those volumes currently have a natural ceiling? If they do, what are the constraining factors? Do they include computing power, regulatory angst, contract/instrument types, etc.?

A: I don't believe the trading volumes have any natural constraint. Historically, processing power was one of those constraints because, up until the 1960's, everything was done by paper. At the moment, the new evolving standard, which will be with us in no more than several months, is the ability to transact at the sub-millisecond level. It's worthwhile reviewing that a millisecond is one-thousandth of a second. Whenever the ability to transact more frequently has combined with increased information reported by the markets, the result has been more frequent transactions; it breeds more transactional activity. The more venues available for trading activity in like instruments, the greater the probability that there will be an exploitable, if slight, differential in price between them. That price difference is essentially what drives the trading volume.

I think it's not an exaggeration to say that the London Stock Exchange has more business now than it had prior to the inception of the Chi-X platform, even though the Chi-X platform has obtained significant market share from the London Stock Exchange. As a matter of fact, when the first securities were introduced to Chi-X on a test basis, for example Lloyds TSB, even in those first few days, there was an increase in Lloyds TSB trading on the London Stock Exchange.

I believe it is true that on days when the London Stock Exchange is closed, Chi-X, which in a regulatory sense could be open, chooses to be closed. In my opinion, Chi-X exists as a potential arbitrage point of venue relative to the London Stock Exchange; the greater the volume in Chi-X, the greater the volume will be in the London Stock Exchange. If there were ten other exchanges like the Chi-X, I think that the volume would be multiplied accordingly.

The regulators are interested in liquidity, securitization and allowing investors to exit asset classes or individual securities that the owners consider problematic; for example, auction rate preferreds. Taking even the greatest security disasters of the last year that include Bear Stearns, Lehman Brothers, Fannie Mae, Freddie Mac, and AIG, in each and every case, the fixed income instruments in those securities were difficult to trade. The causes of the decline in securities prices, which were largely in the form of derivative instruments in the securities inventories, were found to be absolutely untradeable even though the underlying instruments were, in many cases, AAA rated mortgage backed securities. In terms of market scope those underlying securities are many, many times the size of the putative market values of Bear Stearns, et al.

Let's reflect on that statement. There is a \$14 trillion mortgage market in the United States of which at least \$7 trillion has been securitized. That \$7 trillion market became absolutely untradeable; yet, a stock like Bear Stearns, say what one will about it, never stopped trading. The owners were very displeased with the prices obtained, but there was always liquidity available. Therefore, it seems logical that the regulators would encourage the listing of many more items.

One of the great problems in the economy at the current moment is that the credit market is frozen, because there's no liquidity. There's no liquidity, because there's no price discovery; no one knows what the price is, and there is no contact between those who want to sell and those who might be interested in buying the securities. The entire housing market in the United States is not functioning, because liquidity doesn't exist. I think that a remedy for that liquidity problem is the listing of those securities. In any event, it appears fairly likely that the regulators are proceeding along those lines.

Golden age of financial innovation

Q: Is it possible that we are at the end of a Golden Age of financial innovation that began 30 years ago with the advent of personal computers and programs like Lotus 1-2-3? If so, would this situation alter your view on the desirability of exchanges as investments at the current prices?

A: The answer is in the question. If we were at the end of a Golden Age of financial innovation, obviously, croupiers would not be the wonderful investment opportunities that I believe them to be. In fact, I believe that we're at the beginning of an age of financial innovation. Let's take municipal bonds as just one of the many examples I could use.

Municipal bonds always existed, and they were always very difficult to trade, because the natural owners of them, insurance companies, for instance, would buy large portions of issues and hold them to maturity. If an individual wished to restructure a portfolio, add or subtract to an exposure, or change the nature of a portfolio ladder, it was extraordinarily difficult and created transactional friction, which is very expensive.

Recent months have seen the evolution of Exchange Traded Funds (ETFs) that own diversified portfolios of municipal bonds. In theory, these ETFs make it possible to buy one share of a diversified municipal bond portfolio. In an ideal world, that portfolio should function not very differently from a bank account, by avoiding the denominational problem, which is the minimum lot size at which it's necessary to buy or sell. Typically, in the world of municipal bonds, a \$100,000 face amount is considered to be an inadequate lot size; round lots, as a practical matter, are in the millions of dollars.

A person who wishes to withdraw \$5,000 or \$10,000 from a municipal bond account doesn't want to have to wait for an interest payment, and ETFs make it possible. If one wishes to own securities in foreign markets, it's extraordinarily difficult to transact except for in the most liquid securities, especially in the Asian markets. It's much easier to transact in an ETF. I believe that there are well over 500 ETFs in registration at the moment. Consider the number of publicly traded securities that exist in the world (which I think well exceeds 50,000 specimens), and imagine the many combinations and permutations possible with 50,000 specimens of securities incarnated as ETFs. That explanation is but one way to support the case that we're at the beginning of an age of financial innovation.

Another argument to support that case is that many markets in the world are still developing. A deep and rich publicly traded debt market does not exist anywhere in the world. The debt markets are all dealer markets in which the quotations, price discovery and debt are almost inscrutable. The debt market is many times the size of the equity market, and it's going to become listed and tradable.

In addition, there are the many companies in emerging markets that are not yet listed. In terms of financial innovation, the emerging markets stand the way the American financial market stood 100 years ago. The access to business activity is clearly shifting to the Pacific Basin and not a great quantity of the Chinese GDP is listed anywhere, or is even tradable. The same might be said of the Indian GDP, and the GDPs of those countries are likely to increase dramatically. Similar comments could be made about the Russian GDP, or about the Eastern European markets.

The changing tactics of the U.S. Treasury & the Federal Reserve

Q: Do the dark forces for evil, i.e. the bear raiders, exist? If they do, have the Treasury and the Fed, via their actions with the GSEs and AIG, made a terrible mistake by underwriting the bear raiders' playbook vis-à-vis finding a large leveraged company with opaque assets and assumed counterparty risk, short the stock, fire up the rumor mill, and ramp up the CDS price, spook the rating agencies and counterparties, and wait for the Treasury and the Fed to wipe out the equity?

A: The first way to respond to the question is that there are always going to be short sellers, and the short sale thesis, as outlined in the question, is not an unreasonable one for a short seller to engage in. Should they be called "the dark forces for evil?" I don't know if that's appropriate. A short seller is merely a market participant who tries to make a profit and, in particular, to profit from the short sale of securities.

It's very clear that sometime in mid-September, the government made an enormous change in its tactics. From the advent of the Bear Stearns crisis in early March 2008 to the AIG crisis in September 2008, the government was willing to lend assistance to various enterprises but, to establish moral hazard, the government exacted a terrible price from the equity shareholders, and that price was massive dilution.

In the case of Bear Stearns, it wasn't dilution on the part of government action, it was merely that J.P. Morgan was able to acquire the company at a large discount to book value. In the case of Fannie Mae, Freddie Mac and AIG, it was massive dilution. In the case of Lehman Brothers, although the government did render assistance initially, it did not act to preserve the enterprise. These are the documented actions of the government. It was not unreasonable for the short sellers to have presumed that the government would act in the same manner with all the other companies. However, the government very quickly changed its tactics and began to provide capital on fairly reasonable terms, given the environment.

The government can exact moral hazard against some market participants, but it's not reasonable to impose moral hazard against all participants in the marketplace. That approach would be tantamount to destroying the financial system, which the government must act to preserve. I don't think that the short sellers fully understand the implications of the government's actions. The government's now assisting the various firms by any means possible. Providing capital is not the only action the government will take; it is merely the first.

Through the Federal Reserve, the government is enormously increasing the money supply. From an arithmetical perspective, that's a debasement of the currency, but it's designed to assist those market participants who suffer from excess indebtedness. Such a policy has limited effectiveness in the short run but, in the long run, though it has many aftereffects that are not very pleasant, it will have the desired effect, which is to help the indebted companies.

In addition, the government plans a massive intervention in various types of markets, not the least of which is the government's plan to buy over \$600 billion of agency paper with a view to orchestrating the mortgage rate to be somewhere around the 4.5% level. The two GSEs, which probably jointly account for 21% of all the foreclosures in the United States, are rapidly restructuring defaulted mortgages, and the banks are doing likewise. There is massive assistance being rendered, and I think that there is more on the way. The government will act to preserve the system. The announcements literally say (and these are not merely approximations of the words; these are the words), "We will do whatever we must to defend the system." In my

opinion, that's not a good sign for the short sellers, who may not have appreciated that fact yet, but I think they will soon.

Investing alongside the Treasury & Fed

Q: At this stage, how wise is it to become a minority shareholder alongside the Treasury and the Fed in Fannie, Freddie or AIG? In theory, at the right odds, any bet is a good one, but do you think that the Treasury and the Fed will be predisposed to a future trampling on the minorities as a tonic for moral hazard?

A: The government always has the option of trampling on the minorities, but let's not forget about the majority. The government has made, and is making, a substantial investment in Fannie Mae and Freddie Mac. The government has a vested interest in seeing to it that the value of its investment is maximized. At the end of the day, all entities act in their own self-interest. It would be senseless to create moral hazard by destroying the value of the government's own investment in the GSEs. I think that the point about moral hazard has already been made in the case of Fannie Mae and Freddie Mac, so I don't think that anything would be accomplished by further trampling on the rights of the minorities.

At the moment, there are no viable substitutes for Fannie Mae and Freddie Mac. The original plan announced by the authorities in August was that, over time, Fannie Mae and Freddie Mac would liquidate. We've seen that, if those GSEs were allowed to liquidate, the consequences for the mortgage market would be disastrous. The private market has neither the means nor the willingness to provide an adequate substitute for Fannie Mae and Freddie Mac; therefore, according to the recent announcement, those entities will need to buy many more mortgages. Despite what might've been announced in August, Fannie Mae and Freddie Mac are going to, in my view, enormously increase the balance sheet assets, both in the actual mortgage book, and in the guaranteed book.

I believe that Fannie Mae has on its mortgage book and its guarantee book, probably about \$6 trillion of assets. At the moment, it has a market capitalization of about \$4 billion. Let's assume that on a \$6 trillion book, Fannie Mae could earn one basis point, or \$600 million. If Fannie Mae, as an entity, could merely earn one basis point on its book then it would be trading for, roughly, 6.7x earnings. If the company could earn two basis points on its book, which is not an extraordinary amount of money to earn, then Fannie Mae would be trading at 3.3x earnings. If Fannie Mae could earn 10 basis points on its book, which would be much less than its historical experience, it would be trading for less than 1x earnings.

If one were to assert that Fannie Mae could earn no more than 10 basis points on its book, which is probably a very poor assumption, and if Fannie Mae were not going to grow its book which, in light of recent announcements from the government, I think is also a poor assumption, then Fannie Mae is now trading at well below 1x earnings. Let's say that Fannie Mae were now trading at 0.6x earnings; if that were to increase to 6.6x earnings, an investor would make 10x his or her money. Even if it took five years to happen, the compounded annual rate of return would be 58.5%.

It shouldn't be difficult to imagine what might happen if the return on book, measured in basis points, were higher, and what would happen if the book itself became larger. One would see some truly extraordinary rates of return. Of course, since the government owns 80% of Fannie Mae, it would achieve 80% of those returns.

The government does not wish to own all of the mortgages in the United States. If it had wished to, it would've completely nationalized Fannie Mae, that is, owned 100% of the entity. However, the government did not wish to do so, because it would have become the effective guarantor of all the Fannie Mae debt. Even though there's no explicit guarantee, analysts viewing the U.S. Government's balance sheet would begin to include the agency debt in their calculations of the U.S. debt. That situation could affect the ability of the United States government to obtain funding.

One would think that the U.S. Government should logically wish to avoid having the agency funding added to the U.S. debt. That reason, and no other, is why 20% of Fannie Mae is still in the public domain.

If the minorities are trampled upon, then where is the moral hazard? If the government essentially obliterated the minority interest in Fannie Mae by issuing a sufficient number of warrants, since it would be disastrous for Fannie Mae to place its bondholders in moral hazard, where would be the moral hazard? It can only exist if there are publicly traded stockholders that are at risk; therefore, the minority interest serves a useful social function from the point of view of the government. I don't think that the minority interest is likely to be trampled upon; I think it's much more likely that it is going to make a great deal of money.

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account holdings and performance will vary depending on the size of an account, cash flows within an account, and restrictions on an account. Portfolio holdings are subject to change daily. Under no circumstances does the information contained within represent a recommendation to buy, hold or sell any security and it should not be assumed that the securities transactions or holdings discussed were or will prove to be profitable.

Next: *Chapter 17 — Exchanges: Less Talk, More Figures (August 2009).*

Chapter 17 — Exchanges: Less Talk, More Figures

Source. Horizon Asset Management, Inc. commentary, August 2009.

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09_20Croupiers_20Less_20Talk_20More_20Figures.pdf

Extract used for assembly. book/build/text/20160703154353_b8f56eae_8-21-

09_20Croupiers_20Less_20Talk_20More_20Figures.txt

*Presentation edits: removed letterhead and page footers; reconstructed select tables from the PDF extract; left the long historical options volume grid as a **verbatim extract block** where column alignment was unreliable in OCR/text extraction.*

The exchanges — less talk, more figures

Profitability

First, as to profitability, we wrote frequently last year, as the credit crisis wreaked havoc upon the financial system broadly and the equity markets specifically, that the daily transactional activity of the exchanges appeared to remain fairly robust. There were, of course, somewhat mixed results reported from the exchanges, and a certain degree of earnings diminution was inevitable, particularly in Europe and Asia, as exchanges in these regions adopt the ad valorem method of pricing, which bases fees on the market value of the transaction. Clearly, as equity markets fell by 50% in nearly every region of the world last year, such companies would be expected to report a larger degree of revenue diminution than would an exchange that bases its fee structure on volume alone. This latter pricing method is almost exclusively used by the U.S. exchanges.

With over one year of data in hand, the draconian margin compression scenario so widely expected to occur last year, and especially this year during recession, has not appeared. The most recent profit margins of the exchanges in the Core Value strategy are presented below.

Operating profit margin

Exchange	Fiscal 08	Fiscal 07
CME Group (1)	42.5%	39.4%
Hong Kong Exchange	78.5%	85.7%
London Stock Exchange (2)	50.4%	52.9%
Nasdaq OMX Group	17.5%	15.0%
NYSE Euronext	18.1%	16.3%

1. Before amortization of purchased intangibles & impairment of long-term investment.
2. Fiscal March 09 and 08. Before amortization of purchased intangibles and goodwill impairment charge.

Undoubtedly, these companies mostly reported lower earnings versus year ago periods. Yet, the current and reduced level of profitability still remains extraordinarily high relative to general corporate profit margins, perhaps the highest of any other industry. If it were remarked last year that a group of companies dependent on the health of the capital markets would be able to sustain 20%–75% operating margins throughout the credit crisis and any recessionary effects that would follow, this most likely would have been met with abundant laughter (or derision).

Given their resiliency throughout one of the worst operating periods for financial companies in history, as well as our expectation for their future prospects, we maintain a large weighting in this sector.

Competition

Among the most frequent questions, if not criticisms, is that of competition among the exchanges. Yet by a universal standard, there is much less competition today than there was only a few years ago. What would be the typical reaction if an investor could somehow be given foreknowledge that the constituents of a given industry would undergo a massive consolidation, to the degree that perhaps 20 or more competitors would be absorbed into five? It has happened

already in the North American securities exchange market. Three years ago, a long list of independent venues existed—including, among others, the NYSE, Amex, CBOT metals complex, Archipelago, ISE, NASDAQ, PHLX, Boston, IDCG, CME, CBOT, NYMEX, COMEX, Swapstream, ICE, NYBOT, Winnipeg, Toronto, Montreal, BOX, and Oxen (Alberta power)—and the commentary tracks how **76%** of those names were ultimately absorbed by **four** North American exchanges and **one** European exchange (with additional NYSE Euronext and Nasdaq OMX activity abroad noted in the original).

A more pointed criticism of exchanges as an investment is the loss of market share by major exchanges, such as by the London Stock Exchange, NYSE Euronext and others. Such figures are reported monthly, along with other statistics, by many of the exchanges. And it is true. There is unquestionably a loss of market share by many exchanges, sometimes of large magnitude, and it has been occurring almost without surcease for a long period of time. Witness the severe loss of market share by the New York Stock Exchange in stock trading volume during the past several years; it is virtually monotonic in its decline, and since 2005 amounts to about **38** percentage points, from over **80%** to under **45%**:

NYSE handled volume — market share (% of consolidated NYSE volume)
(Fifth column: 2009 where reported in original; sixth column on June row: point change 2005–2009 per extract.)

Month	2005	2006	2007	2008	2009
December	77.8%	69.1%	59.7%	47.9%	—
November	78.3%	69.8%	60.1%	47.4%	—
October	79.7%	70.9%	60.3%	47.7%	—
September	79.8%	71.2%	62.2%	46.7%	—
August	80.0%	71.8%	65.5%	46.5%	—
July	80.1%	74.2%	66.3%	48.3%	—
June	81.5%	76.5%	68.7%	51.0%	43.5%
May	81.0%	76.9%	68.1%	51.8%	42.9%
April	82.9%	75.5%	68.5%	53.6%	43.4%
March	81.7%	76.1%	69.9%	56.2%	44.7%
February	81.6%	77.5%	69.9%	58.0%	46.5%
January	82.3%	76.6%	69.0%	58.4%	46.9%

NYSE handled volume (millions of shares) — verbatim from the extract (same source file):

	2004	2005	2006	2007	2008	2009
December	33,222	34,546	34,669	39,241	52,300	
November	32,357	36,467	40,848	52,835	54,921	
October	33,313	40,956	41,763	47,107	77,383	
September	28,438	36,986	36,732	38,123	66,768	
August	27,924	34,670	38,115	64,319	42,443	
July	30,335	30,886	36,737	50,009	60,956	
June	29,370	34,698	44,981	48,271	50,215	50,612
May	30,574	32,894	44,579	46,458	42,329	55,731
April	32,780	36,907	34,464	41,273	46,508	59,622
March	34,764	38,265	40,828	50,004	53,429	72,400
February	28,815	30,882	32,102	38,890	45,877	55,589
January	34,204	33,360	39,760	41,869	60,780	50,390
Total	376,096	421,517	465,579	558,400	653,910	344,346
% Change	12.1%	10.5%	19.9%	17.1%	15.1%	

Witness also the climb in handled volume alongside the share decline: the narrative conclusion in the piece is that **market share loss and volume growth are not mutually exclusive**—competition is associated with increased volumes.

We described this phenomenon in a *Studies in Absurdity* review in September 2006. Lest this seem a product of an anomalous few years, the abbreviated table below describes the share volume growth at the New York Stock Exchange over the past nearly 50 years since 1960. On a decade-by-decade basis, the result is remarkably consistent at nearly **15%** per annum (the subdivision of the 2000 to 2008 period is due to the change in tabulation method between the NYSE and Euronext upon their merger).

NYSE annual share volume expansion

Period	Annualized
1960 – 1970	14.4%
1970 – 1980	14.5%
1980 – 1990	13.3%

Period	Annualized
2000 – 2003	10.3%
2004 – 2008	14.8%

Options and “Securities Exchange Moore’s Law”

This odd behavior among securities exchanges is even starker in an entirely different security type, options. The original piece presents long-run **total U.S. options industry volume** and **CBOE** share and growth statistics. The machine extract preserves rows but not perfect column alignment; the following block is **verbatim** from book/build/text/20160703154353_b8f56eae_8-21-09_20Croupiers_20Less_20Talk_20More_20Figures.txt (page 7 of the extract).

```

1980 - 1990 8.1% 1980 - 1990 9.4%
1990 - 2000 13.2% 1990 - 2000 9.7%
2000 - 2008 22.0% 2000 - 2008 17.6%
2005 - 2008 42.5% 2005 - 2008 36.6%
2007 - 2008 36.9% 2007 - 2008 26.4%
Year
Total Industry
Options Volume Year Total Volume
Yr/Yr
Change
% of
Industry
Vol
1973          1,119,245 1973          1,119,177 100%
1974          5,682,907 1974          5,682,907 408% 100%
1975         18,103,018 1975         14,431,023 154% 80%
1976         32,373,925 1976         21,498,027 49% 66%
1977         39,637,328 1977         24,838,632 16% 63%
1978         57,231,018 1978         34,277,350 38% 60%
1979         64,264,863 1979         35,379,600 3% 55%
1980         96,728,546 1980         52,916,921 50% 55%
1981        109,405,782 1981         57,584,175 9% 53%
1982        137,306,205 1982         75,735,739 32% 55%
1983        150,056,075 1983         82,468,750 9% 55%
1984        196,437,361 1984        123,273,736 49% 63%
1985        232,910,547 1985        148,889,091 21% 64%
1986        289,211,135 1986        180,357,774 21% 62%
1987        305,168,935 1987        182,112,636 1% 60%
1988        195,948,591 1988        111,784,045 -39% 57%
1989        227,016,660 1989        126,765,253 13% 56%
1990        209,922,748 1990        129,500,018 2% 62%
1991        198,801,600 1991        121,689,918 -6% 61%
1992        201,995,757 1992        121,467,604 0% 60%
1993        232,662,095 1993        140,348,955 16% 60%
1994        281,382,402 1994        183,934,483 31% 65%
1995        287,296,909 1995        178,533,465
-3% 62%
1996        294,797,702 1996        173,944,877 -3% 59%
1997        353,823,118 1997        187,243,741 8% 53%
1998        406,343,198 1998        206,865,991 10% 51%
1999        507,891,483 1999        254,331,851 23% 50%
2000        726,727,939 2000        326,359,531 28% 45%
2001        492,615,219 2001        306,667,851 -6% 62%
2002        457,717,950 2002        267,616,496 -13% 58%
2003        535,143,977 2003        283,946,495 6% 53%
2004        815,899,814 2004        361,086,774 27% 44%
2005        1,234,309,674 2005        468,249,301 30% 38%
2006        1,705,689,052 2006        674,735,348 44% 40%
2007        2,611,445,719 2007        944,471,924 40% 36%
2008        3,574,445,814 2008        1,193,355,070 26% 33%
Annualized GrowthAnnualized Growth
Total CBOE Options Contract Volume
1973 - 2008
Historical Options Industry Volume
1973 - 2008

```

The essay’s conclusion in prose: between **2000** and **2008**, CBOE **market share dropped dramatically** while **transaction volume rose dramatically**, on the order of **36%** per year; in **2008**, volumes rose by **26%**. The same sorts of results might be produced for trading in financial futures or commodities. Whatever one might say about securities exchanges, the empirical data, whether employing a multi-decade time frame or examining the tumultuous past two years, does not support the notion that market share and volume growth are mutually exclusive; if anything, the results are consistent with the opposite viewpoint, that competition is associated with increased volumes.

Perhaps there is a special form of Moore’s law that operates for securities exchanges, a sort of **Securities Exchange Moore’s Law**. (Intel co-founder Gordon Moore noted in a 1965 paper that since the invention of the integrated circuit in 1958, the number of transistors that could be economically placed on an integrated circuit had increased exponentially, doubling roughly

every two years. That phenomenon has continued for the almost one-half century since that paper’s publication.)

What could be the various enabling factors for exchange volumes, analogous to the process, physical chemistry and engineering advancements that facilitate the operation of Moore’s law? First, it should not be ignored that the mere expansion of the world economy, business activity, corporate profits and private savings will result in an expansion of the market capitalization of companies and the number of shares that can be traded, and of the volumes of commercial exposure that must be hedged via futures contracts, and so forth. Beyond that, it would appear that virtually any development in regulations, technology, portfolio or trading practice, or product innovation that lowers a barrier to trading, whether it is via increased transaction speed, lower transaction costs, better or more diverse tools for hedging, diversifying, reducing risk or acquiring risk, will invite more trading. Accordingly, one could hardly list all of the variables that have historically contributed to trading volume expansion in the absence of a comprehensive study.

As to future innovations, some of these can only be imagined — and they will be imagined. We believe that the pace of innovation has accelerated and that securities exchanges are now at the doorstep of a golden age. We might examine a few that are right now in the process of explosive development, in a future communication.

While the current economic environment is not without its challenges, we continue to believe the current investment environment offers a once in a generation opportunity for the long-term investor. The prior comments attempt to demonstrate there are companies that have only experienced share price compression in the past year, not fundamental diminution of value. Accordingly, we continue to monitor our businesses, as well as search for new ideas. We remain committed to our belief that the patient, long-term investor will ultimately be rewarded with superior returns.

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Next: *Part III — inflation, royalties, and real assets (01_TABLE_OF_CONTENTS.md).*

Part — Volume 3 Book VI: Studies in Absurdity (compilation spine)

This block inventories every essay listed on the **printed table of contents** of 830350067-Compilation-of-Murray-Stahl-s-Writings.pdf (Book VI — *Studies in Absurdity*), then shows where that material appears in this reader’s edition.

Machine-generated files (re-run python book/scripts/build_compilation_spine.py after vault or manuscript changes):

- book/sources/compilation_inventory.csv — **28** rows from the printed TOC (plus any PDF outline rows).
- book/sources/compilation_coverage.csv — coverage matrix (**3** rows matched to a chapter that cites a vault PDF).

Full anthology text (Book VI) — every page of the compilation extract is also concatenated in:

- book/manuscript/appendix-b-book-vi-compilation-full-text.md (emit via emit_appendix_book_vi_compilation_full_text.py).

Printed TOC — coverage

#	Date (TOC)	Title	TOC page	Matched vault PDF	Chapter
bv6-001	May. 2004	Reflections on Dell Computer and Other Matters	5	–	–

#	Date (TOC)	Title	TOC page	Matched vault PDF	Chapter
bv6-002	June 2004	The Logical Consequences of Yahoo!	9	–	–
bv6-003	July 2004	Erratum: The Logical Consequences of Yahoo!	17	–	–
bv6-004	July 2004	Joseph Stalin's Hedge Fund	19	–	–
bv6-005	Aug. 2004	Distressed Utilities from the Short Seller Perspective	31	–	–
bv6-006	Aug. 2004	Utilities from a Buyout Perspective	35	–	–
bv6-007	Aug. 2004	A Reader Responds to "Joseph Stalin's Hedge Fund"	41	–	–
bv6-008	Sept. 2004	The Other Housing Markets	45	–	–
bv6-009	Oct. 2004	eBay: The Problems Created by Consensus	55	–	–
bv6-010	Oct. 2004	Google and Yahoo as Toll Roads	59	–	–
bv6-011	Nov. 2004	Gresham's Law in Corporate Finance	67	–	–
bv6-012	Nov. 2004	Benford's Law as a Tool For Investors	79	–	–
bv6-013	Nov. 2004	Zipf's Law as a Tool For Investors with Brief Remarks Upon Bradford's Law	83	–	–
bv6-014	Nov. 2004	Gold Related Stocks Versus Gold Indexed Exchange Traded Funds: An Exercise in Comparative Valuation	93	–	–
bv6-015	Dec. 2004	High Yield Investing In the Absence of High Yield	101	–	–
bv6-016	Mar. 2005	The Misapplication of Investment Logic		–	–
bv6-017	Mar. 2005	The Misapplication of Investment Logic II		–	–

#	Date (TOC)	Title	TOC page	Matched vault PDF	Chapter
bv6-018	June 2005	The Economics of Sin or The Logic of Socially Irresponsible Investing		–	–
bv6-019	June 2005	Exchanges Versus Indices		20160703154353_b8f56eae_8-21-09_20Croupiers_20Less_20Talk_20More_20Figures.pdf	chapter-17-exchanges-less-talk-more-figures.md
bv6-020	Aug 2005	Exchanges Versus Indices Part II		20160703154353_b8f56eae_8-21-09_20Croupiers_20Less_20Talk_20More_20Figures.pdf	chapter-17-exchanges-less-talk-more-figures.md
bv6-021	Aug 2005	The Dilemma of the Drug Industry		–	–
bv6-022	Sep. 2005	Homebuilders and Google		–	–
bv6-023	Oct. 2005	Winston Churchill & Real Estate		–	–
bv6-024	Oct. 2005	Utilities Then & Now		–	–
bv6-025	Nov. 2005	Energy Trading for Fun and Profit		–	–
bv6-026	Mar. 2006	The Basic Principles of Croupier Investing		–	–
bv6-027	Apr. 2006	Croupiers Versus Hedge Funds		–	–
bv6-028	Apr. 2006	Further Basic Principles of Croupier Investing		20160512031900_7a5f9d04_Further_20Croupier_20Basics_20-_20_204-27-06.pdf	chapter-14-further-croupier-basics.md

Editorial rule

Where a **numbered chapter** already presents an essay from a standalone vault PDF, that chapter is the **primary** reading copy. **Appendix B** still carries the *entire* Book VI machine extract so nothing in the anthology is absent from the export bundle.

Appendix B — *Studies in Absurdity* (full machine extract)

Source PDF (vault). 830350067-Compilation-of-Murray-Stahl-s-Writings.pdf

Extract file (identical text to the body below). book/build/text/830350067-Compilation-of-Murray-Stahl-s-Writings.txt

This appendix prints the **entire** machine extract for *Volume 3 Book VI — Studies in Absurdity* so the reader export contains **all** anthology pages, even when a numbered chapter already presents an essay from a standalone vault PDF. Prefer the **chapter** copy for edited reading; use this appendix for completeness, search, and diff against the shelf PDF.

For a row-level map from the printed table of contents to vault PDFs and chapters, see `book/manuscript/part-book-vi-studies-in-absurdity-spine.md` (generated by `build_compilation_spine.py`).

BOOK VI—STUDIES IN ABSURDITY

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It is generally believed that Dell Computer pi oneered the concept of selling electronic goods directly to the customer without the intervention of a retailer. It may surprise some readers to learn that this is erroneous. The first electronic good firms of this type was known as the Electro Importing Company. It was founded in 1905 by Hugo Gernsback. Its address was 69 West Broadway in New York City. * The company sold radios by mail order, since the Internet had yet to be invented. Batteries c ould be obtained from the GeeDee Dry Battery Company that was also owned by Gernsback. In fact, Gernsback was far ahead of his time as he also manufactured a wireless signal device called Telimco Wireless. The device transmitted messages by Morse Code. Hugo Gernsback might have become a technology baron with a company as important as Dell Computer had he applied himself to the task. However, his interests changed to Science Fiction.

Hugo Gernsback is considered to be one of the originators of modern Science Fiction. Moreover, the annual prize awarded in the U.S. for the best science fiction of the year is known as the Hugo Award in honor of Hugo Gernsback. He is best known for his novel entitled "Ralph

124c 41+: A Romance of the Year 2660.”

Of course, it will never be known if Gernsback would have been successful as a technology entrepreneur. In order to achieve success, he would have needed to know much about modern accounting practice. This brings us to our subject of Dell Computers. Let us consider the following facts. A. During the past five fiscal years Dell recorded net income in the following amounts: 2004 \$2.645 Billion 2003 \$2.122 Billion 2002 \$1.246 Billion 2001 \$2.177 Billion 2000 \$1.666 Billion This is a total of \$9.856 Billion

* Martin Gardner – On Science, Literature & Religion (N.Y. Prometheus Books 2000 p. 105 4

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B. During the past five fiscal years Dell has repurchased its own common stock for the following dollar amounts: 2004 \$2.000 Billion 2003 \$2.290 Billion 2002 \$3.000 Billion 2001 \$2.700 Billion 2000 \$1.061 Billion This is a total of \$11.051 Billion C. The share repurchase program resulted in the following Dell share count at various fiscal year-ends on a fully diluted basis. 2004 \$2.619 Billion 2004 \$2.644 Billion 2003 \$2.726 Billion 2002 \$2.746 Billion 2001 \$2.728 Billion 1999 \$2.772 Billion The total reduction in fully diluted shares outstanding is 153 million shares. This was done at a cost of \$11.051 billion. Simple division would lead one to the erroneous conclusion that the company repurchased 153 million shares at \$72.23 per share. However, this is incorrect since Dell did not trade at that level during the share repurchase program. Of course, Dell did not repurchase, and could not have repurchased, shares at \$72.23 each. Far more than 153 million shares were repurchased. However, many shares were also issued under the corporate stock option program. Shares issued under the stock option program are not required to be expensed under current accounting regulation. Thus, the question naturally arises as to whether Dell Computer should be considered to have earned \$9.856 billion in the 2000- 2004 period of time. One manner of viewing the situation is to state that the company expended \$11.051 billion on the purchase of 153 million of its own shares at \$72.23 when those shares now trade at \$34 each, for a paper loss of \$5.849 billion. This paper loss is coincidentally not far removed from the current Dell Computer (as of January 30, 2004) shareholders' equity of \$6.28 billion. This would invite the seemingly absurd conclusion that Dell has lost \$5.849 billion of its \$6.28 billion of shareholders' equity so that it currently has very little shareholder equity. On the other hand, if the company had invested in anything other than its own shares and had sustained a paper loss of \$5.849 billion, it would unquestionably be required to record the loss.

Of course, the company would argue, perhaps not without reason, that the shareholders do derive sustained benefit from this activity since this is a means of compensation for employees who would otherwise need to be paid in cash. It is indeed difficult to avoid this conclusion. 5

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6 Nonetheless, one cannot necessarily conclude that the employees would have accepted \$5.849 billion of cash compensation in lieu of stock options. It is impossible to know what figure would have satisfied the employees, since those employees have some expectation of a value increase in Dell options. This expectation is incalculable. The company was not required to repurchase its own shares. If it had not done so it would now possess over \$11 billion of cash over and above its already substantial balances minus cash received from employees due to option exercise. However, it would also have many more shares outstanding. One could make all sorts of calculations about earnings on \$11.051 billion of marginal capital versus the potential dilution from stock option exercise. These would merely be theoretical. It can all be reduced to the observation that \$11.051 billion of the shareholders' funds was used to repurchase shares that currently have a market value of approximately \$5.2 billion. This should not be the road to prosperity. Yet, in an accounting sense, it is the road to prosperity. On January 28, 2000, Dell Computer recorded shareholders' equity of \$5.3 billion. In fiscal year 2001, it earned \$2.122 billion on that sum for a return on beginning shareholders' equity of 40%. On January 30, 2004, Dell Computer recorded \$6.28 billion of shareholders' equity. This was an advance from the \$4.873 billion of equity on January 31, 2003. The company earned \$2.645 billion for a return on beginning equity of roughly 54.3%. During the course of fiscal 2004, the company expended \$2.0 billion to retire 25 million shares. As noted previously, it actually repurchased many more of its own shares, but the 25 million share count reduction is net of share issuance. As a pure arithmetical proposition it repurchased 25 million shares at a cost of \$2.0 billion or \$80 per share. In footnote number one to its January 2004 financial statements, Dell does calculate that the options issued during the year cost \$829 million which is not recorded on its financial statements. This calculated amount is based on the Black Scholes option assumptions of a 3.8 year option term to expiration, a 0.99% risk free interest rate and a 43% share price volatility. The \$829 million cost figure is an abstraction based upon these assumptions. Moreover, this figure is calculated net of tax benefits which are calculated by the company to have been \$181 million in fiscal 2004. If the reader has followed all of these calculations, it is now possible to understand why Hugo Gernsback decided to write “Ralph 124C 41+: A Romance of the Year 2660.” This sort of work, as does modern accounting, requires a very well developed sense of imagination.

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June 29, 2004

Subject: The Logical Consequences of Yahoo!

I – Introduction

The human fears error to the extent that this fear is only surpassed by the fear of being perceived to be in error by others. This observation is quite relevant with regard to a security such as the Yahoo equity. The believers in the future of the shares must contend with the past collapse of the share price in the aftermath of the Internet bubble subsequent to March 10, 2000. Perhaps the shares will once again collapse in value due to excessive valuation. The risk is that one will be perceived to be in error. On the other hand, the unbeliever does not dare sell these shares short. The months preceding the Internet bubble peak on March 10, 2000 might be repeated. The valuation of the Yahoo shares might easily exceed anything recently recorded. Consequently, a short seller will risk being perceived to be in error. The most numerous group is, of course, the agnostics. However, unlike the agnostic in matters of faith, an agnostic in the world of equity investing is not permitted the luxury of indifference or doubt. In fact, the agnostic in the world of investing is not even permitted the luxury of privacy of conscience. This is because investment holdings and investment performance are exposed to observation and critique. The agnostic in Yahoo is revealed as such by the failure to own the shares. In the case of this share, failure to own may well be the difference between outperformance and underperformance of the benchmark S&P 500 index. Naturally, Yahoo is included in the index and now ranks 48th within the index in terms of market capitalization. An agnostic will therefore own Yahoo shares within a managed portfolio in proportion to its weight within the index. As is not the case in matters of religion, the investment agnostic attends services regularly and prays fervently.

II – The Self-Perpetuating Cycle

If one agrees that there are investment agnostics and since there are quite obviously equity indices, then if a given firm can issue a sufficient number of shares to create a large market capitalization, that firm is very likely to be included in one or more equity indices. If the shares are part of various indices, then there will certainly be a ready market for those shares among the vast number of agnostics who must purchase these shares to avoid the possibility of being thought to be in error by failure to outperform an index in a given time. 8

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9 If there is a ready market for the shares, the issuing firm now effectively has possession of a form of currency. The firm might then decide to issue more shares to its employees in the form of stock options as a means of compensation. Current accounting rules provide that the cost of the options need not be reflected as an expense of the corporate income statement. Since some portion (usually large) of employee compensation expense is not recorded as an expense, earnings are higher than would otherwise be the case and this is adequate grounds for a share price increase which, in turn, makes the issuance of further shares all the more irresistible. As more shares are issued and as the value of these shares increases, the agnostics must further increase purchases in order to maintain the proper portfolio weight. This is a cycle that seems to be rather self-reinforcing.

III – The Position of the Believer

The agnostic has now made the position of the believer very uncomfortable. In religion, the reward of belief is everlasting afterlife. In the world of investments, the only reward is more money to manage. Yet, in order to achieve this, the believer must be distinguished from the agnostic through the process of “adding value”. This is usually done by increasing the proportionate weight of Yahoo shares held in a portfolio to a level well in excess of the index weight. It therefore logically follows of the position and actions of the believer merely reinforces the already self-reinforcing cycle set in motion by the agnostic. In the case of Yahoo, this is an increase in share price.

IV – The Aftermath of the Bubble

Let us place ourselves in a rather uncomfortable position of the agnostic on the evening of March 9, 2000. The agnostic might well have owned shares of firms such as Yahoo, eBay and Amazon.com. At that moment, the valuations of these firms must have seemed to be excessive.

The temptation to sell must have been very strong. Of course, sale of these securities can expose the agnostic to the danger of being thought in error. Clients might withdraw allegiance to the agnostic equity manager. The more clever agnostics might have thought it wise to sell Yahoo, eBay and Amazon.com and to reinvest proceeds in "safer" or more mature firms such as Cisco Systems, Intel or Microsoft. The results of such an action are evident in the following table. Performance of Technology Bubble Stocks from March 10, 2000 to June 18, 2004

Group I Compound Annual Return Yahoo (21.23)% eBay 14.59% Amazon.com (6.74)%

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Group II Compound Annual Return Cisco Systems (22.11)% Microsoft (12.62)% Intel (16.33)%

It is obvious that the Group 2 or "safer" firms have underperformed the bubble stocks since the collapse after March 10, 2000. This tendency has continued in the one week from June 18, 2004 to June 25, 2004. Group I June 18, 2004 June 25, 2004 Yahoo \$32.07 \$34.91 eBay \$86.50 \$90.72 Amazon.com \$49.60 \$51.80

Group II

Cisco Systems \$23.42 \$23.43 Microsoft \$28.35 \$28.57 Intel \$27.64 \$27.78

The appreciation of the Group 1 shares from October 2002 has been so enormous as to efface the declines of the bubble aftermath period. Yahoo and eBay are now included in the S&P 500 index and rank of 48th and 37th positions in market capitalization, respectively. The agnostics have evidently purchased shares, and their aggregate purchasing power is quite considerable.

V – The Economics of a Bubble Share Short Sale

Let us assume that Yahoo were to decline by 90% to a share price of \$3.49. This would not be without historical precedent since Yahoo traded at less than \$5 per share at the end of September 2002. It would also not represent a valuation extreme. Yahoo, according to its own figures, is operating at breakeven levels if due allowance is made for the issuance of stock options. At a share price of \$3.49, Yahoo would trade at roughly 1.2x forecasted revenue while operating at breakeven. However, the actions of believer and agnostic have set in motion a self-reinforcing trend towards higher share prices. The 2% year-to-date return of the S&P 500 has witnessed a 27 basis point contribution from Yahoo as well as an approximate 27 basis point contribution from eBay.

These two equities account for 25% of the year-to-date S&P 500 return. Such shares are therefore indispensable. A short seller might witness a share price increase to a considerably higher level. 10

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11 Thus, let us assume that our imaginary short seller waits for a Yahoo share price of \$69.82 or double the current level. Let us further assume that the share collapses afterward to the \$3.49 level. The return to the short seller is 95% or only 500 basis points higher than would be the case if shares were sold short at a much lower price. Similarly, if the shares were to attain the level of \$139.64 or four times the current level and were then sold short, only to decline to \$3.49, the potential return to short seller only increases from 95% to 97.5%.

VI – Position within the S&P 500

Yahoo currently has the 48th largest market capitalization within the S&P 500. Since Yahoo trades at 109.1x the anticipated pro-forma earnings for 2004, investors or, more properly expressed, believers expect much growth from Yahoo. The consensus opinion on this subject is that Yahoo should be able to grow its earnings at over 50% per annum for the next several years.

The probability that any of the 47 firms in the S&P 500 with a higher market capitalization than Yahoo will grow profits at 50% per annum is essentially zero. Therefore, the belief that Yahoo will grow at 50% per annum is logically equivalent to the belief that it will increase its position in the S&P 500. At a 50% growth rate and no diminution of current valuation metric, Yahoo will have a \$384 billion market capitalization within five years. Of course, this calculation makes no allowances for the almost certain issuance of shares, which would result in an even higher market capitalization. General Electric, the largest company within S&P 500, currently has a market capitalization of \$328 billion. GE plans to spin off its insurance businesses, which might have a market capitalization of perhaps \$50 billion. Thus, post spin-off, the GE market value will be \$278 billion. Given a 12.5% growth rate, GE might attain a \$500 billion capitalization in five years. If Yahoo merely increases its share count by 5% per annum and grows at the expected rate without diminution of valuation metric, it should also attain a \$500 billion market capitalization. Therefore, the believers effectively but not actually argue that Yahoo will be the largest company in S&P in 60 months.

VII – The View of the Employee

The Yahoo employees are in a curious position. These receive copious amounts of Yahoo stock options as part of compensation. However, it is important to remember that these options have contingent value and not actual value. In order to have value from the perspective of an employee, the share must increase in price. Indeed, as noted previously, the Yahoo shares have substantially increased in price since October 2000. The increase is approximately sevenfold. The employee recipients of future options cannot reasonably expect a sevenfold increase in the shares in the next 20 months. Consequently, if the share does not increase in value at the recent rate, it logically follows that the compensation of the employees is being effectively reduced, albeit not by the actions of the company. It might be argued that the employees understand that options are a form of variable compensation and, as such, compensation will therefore vary. ¹¹

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¹² However, there is another aspect of this question. The economic value of Yahoo at \$34.91 per share is surely less than the economic value of Yahoo at a price below \$5 per share. The economic value of Yahoo shares at \$69.82 are even lower. Thus, viewed from the perspective of the company, X option grants at a \$69.82 strike price involve far less potential dilution than X option grants at \$34.91 at a \$34.91 strike price. The situation is not very different than paying in depreciating currency except that in the world of equity, as opposed to currencies, an appreciating equity is the mathematical equivalent to a currency where value is being eroded by inflation. The employees who accept this currency as a form of payment are practically engaged in a form of money illusion. If the employees ever become unwilling to accept this currency as a form of payment, the self-reinforcing cycle of share issuance and share appreciation will end.

VIII – The Perspective of the Short Seller

The economics of a bubble stock short sale have already been discussed in section V. The short seller does not improve returns by appreciable amounts if the short sales are undertaken at higher prices. However, if the short sales are not undertaken at extremely high prices, the damage to portfolio return can be very substantial. A short seller might at any moment have deployed 40% or 50% of total portfolio exposure toward short sales. Surely at 50% short exposure, one must reckon with at least a 10% adverse move in the short book. Arithmetically, this will cost the portfolio 500 basis points or 5%. It seems therefore reasonable to state that a short seller is willing to sacrifice at least 500 basis points per annum in order merely to maintain a short position of 50% magnitude. Realistically, most of the short sale equities will not decline to a price of zero. Also, realistically, these positions will not be covered at the low point. Thus, the expected return even in success mode is well below 100%. Further, this will occur in many instances over the fullness of time so that whatever return is achieved will be diminished on a compound annual basis if achieved in more than one year. Let us assume that one is willing to sacrifice potentially 500 basis points of return per annum in order to maintain a short exposure of 50%. If the average short were to decline by 80%, the expected contribution to portfolio return would be 40%, or 4,000 basis points. In that case, the following alternative might be of some interest. If one were to establish a 1.5% position in Yahoo October 2004 30 put options, the most recent price is \$0.95 per contract. If Yahoo were to decline to \$3.49, then the intrinsic value of the option would be \$26.51, for a return of 2,691%. This would add 40 percentage points or 4,000 basis points to portfolio return. Of course, if Yahoo continues its upward trajectory, as seems likely, or even fails to decline below 30 by October 15, 2004, the option would be worthless. If Yahoo reaches extremely high prices, the strategy becomes rather intriguing. If Yahoo were to rise to roughly \$93 per share, the trading price of a Yahoo put option roughly 15% out of the money can be accurately simulated by an eBay put, since eBay currently trades at a roughly \$93 as these words are being written. The eBay October 75 put options are \$1.15 bid, \$1.25 asked.

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¹³ The option is 19.3% out of the money. This is probably where a Yahoo option of similar characteristics would trade. Let us assume that Yahoo is \$93 per share, and that one purchases a 1.5% position in a Yahoo October 75 put option (note that this security has yet to exist, since Yahoo is not yet \$93 per share). Let us further assume that Yahoo declines to \$3.49 per share in one of the periodic bubble deflations that are caused by human dysfunctional behavior. In that case the intrinsic value of the put option would be \$71.51 per contract. The return is 5,620.8%. The contribution to portfolio return would be 84.30 percentage points or 8,430 basis points. If this happens, one has effectively hedged a 92% long exposure, even during a crash, with a 1.5% short or put position. The math becomes more interesting still if Yahoo ascends to \$140, \$150, \$160, or even \$170 per share. A disciplined investor who buys a Yahoo 15% out of the money put option with only four months to expiration has only to await a bubble burst once in order to achieve a very substantial portfolio impact. It must be noted that one is sacrificing 150 basis points of portfolio return every four months as Yahoo increased in value. However, this is less than would be sacrificed by a 50% short position in an environment of a 4% market increase. It

ought to further be observed that the risk of the short positions is at least theoretically infinite. The maximum loss of the position is finite and knowable. Another alternative is a shorter term put option with far less time value. The Yahoo August 2004 30 put options are \$0.40 bid, \$0.50 asked. In the case of a Yahoo decline to a price of \$3.49, the intrinsic value of the option is \$26.51. The return is 5,202%. Since August is only 2 months away from the current date, as opposed to four months in the prior example, one would establish a 0.75% position in the option as opposed to the prior 1.5% so as to limit the annual loss to 4.5%. There are six periods of two months duration within a year, so that $6 \times 0.75\%$ equals 4.5%. A return of 5,202% earned by a 0.75% position would contribute 39.02 percentage points to the portfolio return. In order to summarize the perspective of the short seller, one can state that a bubble share affords enormous hedging as well as profit potential if used as a small option position rather than as a short position. Moreover, unlike a short position, the return contribution to the portfolio from a position of constant size increases as Yahoo increases in price.

Summary and Recommendation

The human mind unjustifiably fears error. Sometimes insight actually emerges from error. For example, there is a very obvious error in the Jules Verne novel 20,000 Leagues Under The Sea. A League is equal to 1/20th degree of arc. Since the Earth is roughly but not quite spherical, it has 360° or 7,200 20th degrees of arc. If the circumference of the Earth is 24,901.56 miles, 1/20th degree of arc equals 24,901.56 divided by 7,200, or roughly 3.45 statute miles. In order to be 20,000 leagues under the sea, one would need to be $20,000 \times 3.45$ miles under the sea or 69,000 miles, which is 2.77x the circumference of the Earth. Therefore, it is quite impossible to actually a 20,000 Leagues Under the Sea. Nonetheless, none of this seems to detract from the delight of this novel. In addition, the novel does accurately predict the modern development of the submarine, among other insights.

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14 Similarly, it is almost certain that any forecast of the of the Yahoo share price will be in error. This is because there is no model that will successfully and accurately reckon with the human behavioral dimension of the “investment” decision process. Nevertheless, viewed from a mathematical perspective, the only thing that actually matters is that the bubble will one day burst. If much time and many option trade iterations are required, the mathematical consequence is that portfolio return contribution from this trade will increase, with much benefit to the hedge aspect of a portfolio. Therefore, a position in Yahoo put options of a near term character approximately 15% out of the money should prove to be a rewarding investment.

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16 July 1, 2004

Erratum

Subject: The Logical Consequences of Yahoo!

Only two days ago, a Devil's Advocate Report on the subject of Yahoo! was distributed under the heading above. Only one day ago, the following communication was received from one of our readers. His comment was prompt, apt and deserving of the wider attention of all of our readers and, so, is reprinted here along with what is hopefully a suitably apt response. Tuesday, June 29, 2004 7:45 AM Just so you know – the “20,000 leagues” in the Jules Verne novel refers to the distance traveled under the sea and not the depth below the sea.

July 1, 2004 You are, of course, quite correct. The voyage of the Nautilus is a distance of 20,000 leagues and not a depth a depth of 20,000 leagues beneath the sea. I should have said that since most people are unfamiliar with a formal definition of a league, it is believed by many that the title refers to a depth and not a distance. However, their pleasure in reading the novel is undiminished even though they realize that their initial impression was not correct. I certainly did not wish to imply a lack of scientific verisimilitude in the work of Jules Verne. It is my understanding that he insisted upon the highest possible standard of scientific accuracy. I only wished to make a point about Yahoo and, in my haste to do so, I inadvertently maligned Verne. My critical remarks, such as they are, are entirely confined to owners of Yahoo. I do not believe that Jules Verne would be a buyer of Yahoo at current prices. Thank you for bringing this to my attention. I am at least happy that some people find the essays sufficiently interesting to read to the end.

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18 July 20, 2004

Subject: Joseph Stalin's Hedge Fund

I – A Curious Circumstance

The title of this essay is not a mere literary device to attract the attention of the reader. Joseph Stalin actually had a hedge fund of sorts. More importantly, the story of Joseph Stalin's hedge fund should be of interest to investors.

The story begins in 1930 as Stalin ordered the forced collectivization of Agriculture. Peasant farmers now became collective farms and therefore employees of the state. As such, peasants were given production quotas of farm products that were required to be transferred to state control. Quite frequently, the state would establish a production quota equivalent to everything produced by a peasant. Since this logically implies that a peasant would receive no compensation for very hard labor and incidentally would be left without the means to feed himself or his family, the peasant quite properly lost all interest in production. * Ultimately, a widespread famine was to follow in the wake of this policy, resulting in widespread misery and death. The state responded to the inevitable decline in production by various punishments, including execution or deportation to Siberia. The modern reader might be astonished that the Soviet bureaucrats can actually have believed that famine, deportation and execution could ever be an effective means of increasing agricultural output. However, viewed from the perspective of the state bureaucrat, it seems no more than reasonable that high production quotas would result in high rates of production. If Stalin were a humanist, he would have undertaken to buy grain on the world market to feed his now starving agricultural workers. Yet, not only was Stalin no humanist, he was also a dictator lacking hard currency with which to purchase agricultural produce. Therefore, Stalin could not and would not import grain. Obviously, Stalin was in no position to export grain. Yet, none of this prevented Stalin from trading grain. In the United States and other developed nations grain prices were declining precipitously. If the problem in the Soviet Union was declining productivity, the problem in the United States was increasing productivity. During the 1920s, improvements in agricultural science, increasing use of mechanization, fertilizers and insecticides increased yields per acre. Improvements in roads and railroads resulted in more rapid transportation to market and therefore less spoilage. This is the arithmetical equivalent of increased production, as more produce is available to consume.

* The source for this policy is Khrushchev's Memoirs entitled *Khrushchev Remembers: The Last Testament*, translated by Strobe Talbot (Boston: Little, Brown & Co., 1974) p. 108 18

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19 Increasing electrification made possible increased use of refrigeration which, in turn, reduced spoilage and thereby further increased supply.

Of course the demand for food does not necessarily increase simply because supply increases. Consequently, by the iron law of supply and demand prices declined. Farmers could not sell grain at cost. Agricultural costs are generally fixed and thus the only alternative open to a farmer who wished to save his farm from foreclosure and his family from even worse poverty than was customary was to increase production. Naturally, this further reduced prices.

The Soviet and American examples are circumstances of self reinforcing cycles. Most economic cycles are intrinsically self equilibrating. For instance, lower prices usually stimulate increased demand. Nonetheless, there are cases in which production costs are fixed so that nothing is achieved by decreasing production when confronted with lower prices except decreasing revenue. Agriculture is an obvious example. The primary cost is the cost of land, which can either be expressed in rent or expressed in interest expense if the farmer chooses to own land with debt financing. In the American instance, an indebted farmer can only save the farm by increasing production, which results in more revenue. Unfortunately, increased production will also lower prices and result in less revenue. In the Soviet instance, there is an enormous philosophical distance between a high production quota imposed by the state and a high ratio of debt to land value chosen by the farmer. However, arithmetically these are both high fixed costs.

Although Stalin was a very poor judge of the effects of communism on agricultural production, he appears to have been a much better judge of world commodity markets. He could neither

import nor export grain. Yet, the market mechanism enabled him to trade grain. It seemed self evident that grain prices were caught in a downward self reinforcing spiral. Thus, the Soviets undertook to sell grain short on the Chicago Board of Trade.

In 1930, the American President was Herbert Hoover. It is his memoirs that serve as a primary source document of Soviet actions. The following extract from those memoirs relate events that then occurred:

Stopping Soviet Short Sales of Wheat

An interesting but momentarily alarming side issue arose in September 1930. We were informed that the Soviet government had been selling large amounts of wheat short on the Chicago market.

Investigation confirms this and I demanded that the Chicago Board of Trade and other markets prohibit transactions by foreign governments. In order to secure this action by them for the protection of our farmers, I was compelled to threaten federal control. Soon thereafter the Communist government began dumping large amounts of wheat upon the European markets and broke the price several cents a bushel. The Soviets, out of the short selling in Chicago, however, took a considerable sum which would otherwise have gone to our American farmer.

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** Herbert Hoover: The Memoirs of Herbert Hoover: The Great Depression 1929-1941 (New York, Macmillan, 1952) p.52-53 19

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20 The irony of this situation is something not even achieved in the best of Greek tragedy or in Shakespeare. This is because the irony functions at multiple levels. In the first instance, Herbert Hoover, one of the more well-intentioned and doctrinaire believers in the market mechanism, threatens government intervention to suspend the market mechanism. In the second instance, Joseph Stalin, the most ruthless and doctrinaire opponent of the market mechanism that ever lived, makes use of the market mechanism to earn, in the words of President Hoover, a considerable sum.

There is a second level of irony. The Soviet agricultural policy could not have been more different than the American agricultural policy. The natural consequence of the Soviet policy was widespread poverty, misery and famine. In the American case there was also widespread poverty, misery and some more than occasional examples of actual starvation.

The third level of the irony is that the reasoned application of the cold science of self interest prompted all market participants, including the communists, to engage in similar sell programs. In substance, the communists accepted capitalism although they offered various philosophical objections to the practice. Since all participants undertook similar actions, those actions became self reinforcing instead of self equilibrating. Self reinforcing cycles are inherently destructive since these ultimately must result in very extreme behavior.

II – A Possible Curious Circumstance

The basic tenet of all modern economic as well as investment theory is that cycles are inherently self equilibrating. In the sphere of investments, the equilibrating mechanism is the value oriented investor. If prices are too high, the value investor will sell. If prices are too low, the value investor will buy. This is the investment science expression of its former belief in 19th-century Marshallian economics. The word former is appropriate, since the late 20th century has propagated a refinement known as efficient market theory. Since the market is efficient, investors will not sell at prices that are too low and will not buy prices that are too high. Consequently, errors will be corrected before these have been, properly speaking, created.

A believer in this theory might very reasonably arrive at the conclusion that all investments should be indexed to the market. Indeed, many have arrived at that conclusion.

Unfortunately, such a view would logically entail the obsolescence of the professional money manager. If there were no professional active money managers, there would be very little need for consultants and asset allocators to evaluate such managers. Self interest now requires the application of efficient market logic in one of its most creative forms.

Although the market is efficient and active managers cannot surpass the averages, different stylistic attributes entail different volatility characteristics. Thus, the value oriented manager might be less volatile than a growth manager. A value oriented event driven hedge fund that is always market neutral would be still less volatile. There is now a virtually inexhaustible list of diverse strategies. These strategies can be properly blended in such a manner as to dramatically

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21 reduce volatility with little and, in some unusual cases, no sacrifice of return. This science in its ultimate refinement will create a portfolio of many managers that will never experience, or

perhaps seldom experience, negative returns.

As a science, this is much akin to deconstructing the human body and reassembling it in such a manner as to eliminate aspects of the central nervous system that activate the pain reflex. In economics, loss is the equivalent of pain. A human without a pain reflex might well engage in very self destructive behavior.

One therefore cannot help but wonder whether such an entity that now exists in the marketplace would not engage in the type of self destructive behavior of the Hoover-Stalin era. In theory, different investment styles should produce diverse portfolios. Is it possible that diverse strategies should produce similar portfolios? Fortunately, it is relatively easy to envisage such a circumstance.

A – The Curious Position of the Growth Investor

The Value Line Investment Survey each week publishes an index of those companies in its survey that usually encompasses about 1700 stocks. In the rear of that index, the reader will find a number of screens based upon growth or value criteria. The growth screen is located on page 39 of the July 16, 2004 edition. As noted in the heading of the screen, to be included, a company's annual growth of sales, cash flow, earnings, dividends and book value must have averaged 11% or more over the past 10 years and be expected to average at least 11% in the coming three to five years. The reader will readily observe that 23 of the 100 firms listed are part of the financial services/real estate sector. Another 14 are retail related firms. This is not without some significance, since there is an evident relationship between the willingness of financial service firms to lend money and the ability of consumers to buy real estate and basic consumer products. The names of these firms can be found in the appendix as Tables 1 and 2.

As such, there is nothing surprising in the fact that a portfolio of 100 issues should include 23 from the financial services/real estate sector and another 14 from the retail sector.

B - The Curious Position of the Income Oriented Investor

The income oriented equity investment style should, in principle, exhibit a form of exalted inconsistency with the growth investment style. Mature firms pay high dividends. Growing firms reinvest capital. The Value Line screen of high yield non-utility stocks can be found on page 38 of the index for the July 16, 2004 addition. There are 95 equities listed with a dividend yield ranging from 11.6% to 3.7%. The real estate and financial services related firms are 48 in number. The names of these firms can be found in the appendix as Table 3.

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22 C - The Curious Position of the Low P/E Investor

The low P/E investor should, in theory, display a certain inconsistency with the growth and dividend yield investors. A firm's share will generally trade at a low P/E because its earnings are not growing. The Company may exhibit a problem that prevents it from paying a high dividend. There are cases in which a low P/E share can also be a high dividend yield share. However, these are instances in which the investment community suspects, generally with good reason, that the dividend will be substantially reduced. Therefore, a high yield share does not usually remain on the low P/E list for a very long time. It will either reduce its dividend and therefore cease to be a high dividend yield share or, alternately, it will maintain its dividend and its price would appreciate since the danger of dividend reduction has been eliminated. In this instance, it will cease to be a low P/E high dividend yield share.

In any event, the low P/E screen in the July 16, 2004 edition can be found on page 35 of the index. It lists 100 equities with a P/E ratio range of 4.9x earnings to 11.0x earnings. A total of 41 firms from this list are engaged in real estate or financial services related activity. Another four firms are retail related. It is worthy of observation that very little imagination or ingenuity is required to expand this list beyond the 41 real estate and financial services firms already noted. For instance, Louisiana Pacific might well be included, since its basic product is lumber that is used in home construction. Similarly, an excellent argument can be made for the inclusion of Ford and General Motors as financial services shares, since profits from these activities are critical to the survival of these firms. Nevertheless, the list of Real Estate and Financial services related firms included in the Value Line P/E screen may be found in the Appendix as Table 4. The retail related firms are included in Table 5. D - The Curious Problem of Bargain Basement Stocks

Value Line includes a "Bargain Basement" screen in its weekly index. In the July 16, 2004 edition, this may be found on page 37. The formal definition of this screen is stocks with current price/earnings multiples and price-to-"net" working capital ratios that are in the bottom quartile of the Value Line universe. Net working capital equals current assets less all liabilities including long-term debt and preferred. This is very much a classical Graham and Dodd screen and is rather interesting. Ben Graham frequently earned profits by purchasing shares at a discount to net working capital.

The first observation that can be made about this screen is that it does not include any shares that trade at a discount to net working capital. The price to net working capital ratios range

from 125% of net working capital to 450% of net working capital. Nonetheless, the list, such as it is, includes twelve firms that are engaged in real estate or financial services. The list also includes four companies engaged in retailing. The entire screen includes only 29 firms. The firm in the 29th position, which happens to be Borders, trades at 450% of net working capital. The real estate and financial firms to be found in the "Bargain Basement" screen are listed in the Appendix as Table 6. The retail firms are listed as Table 7.

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23 E - Digression on the Curious Problem of the "Bargain Basement" Stocks

The most interesting aspect of the so-called Bargain Basement stocks is that, according to its own definition, there are not currently any bargain-basement stocks. This is very different than the ordinary market circumstance in which such shares do indeed exist. Readers of Benjamin Graham's book called "The Intelligent Investor" will recall the frequent reference made to such shares. For instance, on page 85 of the fourth edition (1973), Graham identifies 85 such firms as of December 31, 1957. In the next two years the subsequent gain for the portfolio was 75% as opposed to the 50% gain of the then referenced S&P 425 industrials.*** One might imagine that the absence of such investment opportunities might trouble value oriented investors. The natural rejoinder to the Bargain Basement share famine is that the market itself is not overvalued as it trades at perhaps 18 or 20x operating earnings. It would be conceded that it is somewhat high in light of historical norms but not high when considered in light of currently low rates of interest. Value Line does precisely the same thing when, in its July 16, 2004 index, page one calculates the median price earnings ratio of its universe of 1698 firms to be 18.2x profits. The median P/E calculation excludes 170 firms that are not profitable and therefore have P/E ratios that are not meaningful (or NMF in Value Line terminology).

The problem is that statements of this type are devoid of content insofar as these measure that which does not exist. The calculation is only comparable to a portfolio that will never include a company that will generate a loss. This is undoubtedly a goal that many would wish to achieve but all regard as an attainable. However, such calculations do provide reckless investors with a powerful rhetorical advantage.

Markets can never be excessively valued if it is permissible to exclude from valuation calculations the most excessive elements of valuation. It would seem more reasonable to believe that the most fertile portion of the investment opportunity set is always at the valuation fringe. Thus, if there are no "Bargain Basement" stocks, an important aspect of a normal investment opportunity set is lacking. Therefore, by definition, the situation cannot be normal.

F - A Further Digression on the Curious Problem of the Bargain Basement Stocks

The natural impulse of any observer of equities in the contemporary era is to attribute the absence of "Bargain Basement" stocks to the currently low interest rates that exist. This is undoubtedly a natural impulse. Nonetheless, it does not appear to be supported by the available data. The following table, to be known as Interest Rates: 1957, is taken from the Federal Reserve Historical Interest Rate Database. It is readily available to all those who wish to peruse such data on www.federalreserve.gov/releases.

*** Benjamin Graham: The Intelligent Investor (New York: Harper and Row, 4th Edition, 1973) p.85 23

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U.S. Interest Rates: 1957 December Federal Reserve Discount Rate 3.00% Prime Rate 4.00% 5 yr. Treasury 3.08% 10 yr. Treasury 3.21% 20 yr. Treasury 3.38%

It is worthy of note that by March 1958, the Discount rate had declined to 2.35%. In contrast, U.S. Interest Rates on July 16, 2004 were as follows: Federal Reserve Discount Rate 2.25% Prime Rate 4.25% 5 yr. Treasury 3.65% 10 yr. Treasury 4.49% 20 yr. Treasury 5.24%

The most substantial difference between the current interest-rate environment and the 1957 environment is that the yield curve is far more steep at present. In general, interest rates are now higher than was the case in 1957. The salient exception, of course, is the discount rate, which is now lower than was the case in December 1957. Any explanation for the absence of bargain basement stocks at the current time is necessarily conjectural. However, conjectural is not a synonym for unpersuasive or implausible. The most recent investment innovation is the advent of the hedge fund. This is arguably the most dangerous investment weapon ever wielded, since by the use of leverage and short selling it is actually possible to lose more than 100% of the given hedge fund investment. Interestingly, the proponents of hedge funds, which are many, argue that it is a device for the intelligent control of risk. Whether or not such a statement is true remains to be seen. Yet, presented in the manner of a risk mitigation device as opposed to a device with the inherent possibility of limitless loss, one must confess that the hedge fund has a considerable rhetorical advantage over any detractor.

In any case, the ordinary professional investor inadvertently creates a bargain basement stock when a serious problem is made known to the public with regard to a given company. It is usually not obvious that the problem will be rectified in a timely fashion. It is frequently not obvious that it can ever be rectified. Thus, the ordinary investor of the professional variety feels compelled to sell since the continued inclusion of such a share in a managed portfolio can only serve to reduce future return. Yet, no investor is necessarily compelled to buy. Since every share must be owned by someone, the seller must offer the buyer an inducement to purchase. This is the origin of the "Bargain Basement" discount. A company that sells at a discount to net working capital can presumably always be liquidated for at least net working capital and hence the buyer has the prospect of some measure of return. 24

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The hedge fund manager does not necessarily see a problem with ownership of a company that is problematic. This can be an opportunity. If one owns a company with poor prospects, one simply sells short a company with even worse prospects. The difference in performance between bad and worse is actually a source of positive return. It is the ultimate form of investment recycling. Good products can be made from bad, perhaps thoroughly rotten, raw materials. Thus, the problematic company is not sold.

This is not the only cause of the absence of the "Bargain Basement" stocks. Another explanation is the widespread use of index funds, exchange traded funds (ETF's), and baskets. Problematic firms simply remain in indices and, as such, must be held to mirror the index. If an asset allocation decision is made to own more of an index, shares of even problematic firms are purchased. Purchase price is irrelevant since any size is equally reflected in both the index and the portfolio that mirrors the index.

Consequently, there exists a dynamic equilibrium. There is no pressure to sell problematic investments and therefore the value oriented investor has no mechanism with which to bargain for a low purchase price from a potential seller. The focus is not even upon the given investment as such. It is upon portfolio standard deviation, hedge ratios, gross and net exposure, beta adjusted net exposure and average drawdown. This might one day prompt a very lively, but ultimately sterile debate about whether this is the classical serenity of the ancient Greek philosophers or merely an arithmetically induced form of complacency. The debate is ultimately sterile because the rhetorical advantage must be conceded to the group that asserts that it practices scientific risk control.

G - The Thread of the Argument Resumed or Joseph Stalin's Hedgy Fund Revisited

The most curious circumstance of the modern era is that all strategies seem to place a heavy reliance upon investment in financial services and related industries. This does not trouble anyone since financial services represent 20.4 % of the S&P 500 as of January 16, 2004. Of course, little reflection is required to see that financial services represent much more of the index. General Electric, General Motors and Ford are not included as financial services firms and yet financial services represent the centerpiece of the profitability of these firms. Other such nomenclature abuses exist in the S&P 500 sectoral definitions.

However, there is nothing objectionable in mere ownership of financial services and related firms. Indeed, there is nothing objectionable in a circumstance in which all managers chose to have large holdings of such firms. The only problem is when all such managers own large holdings of these firms and claim to be different. The managers might one day actually be asked to be different. The most obvious means of becoming different would be to sell the shares of the investments that cause the managers to be similar. The irony is that the managers would all be selling the same financial services related shares. Hence, managers with entirely different philosophies and motivations would undertake an identical action with the objective of being diverse. In economic terms, this is a self reinforcing as opposed to a self equilibrating cycle. In investment terms, it is the return of Joseph Stalin's hedge fund. 25

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Table I: Real Estate and Financial Services Included in the Value Line Growth Screen

ACE Limited Lowe's Cos. AFLAC Inc. MBNA Corp. American International Group Monaco Coach Brown & Brown Progressive (Ohio) Centex Corp Pulte Homes Countrywide Financial Ryland Group ElkCorp SEI Investments Fastenal Co. SLM Corp. Home Depot Standard Pacific Corp. Hovnanian Enterprise A Thor Inds. KB Home Total System Services Lennar Corp.

Table II: Retail Firms Included in the Value Line Growth Screen

AutoZone Inc. Best Buy Co. Christopher & Banks Claire's Stores Dollar General Corp. Family Dollar Stores Gap (The), Inc. Kohl's Corp. NBTY, Inc. Quiksilver, Inc. Ross Stores TJX Companies Wal-Mart Stores Williams-Sonoma 26

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Table III: Real Estate and Financial Services Related Firms
Included in the Value Line High Dividend Non-Utility Yield Screen

Annaly Mortgage Mgmt. Weingarten Realty Allied Capital Corp. Kimco Realty Thornburg Mtg. Avalon Bay Communities Apartment Investment Federal Realty Inv. Trust Equity Office Properties Trizec Properties New Plan Excel Realty Crawford & Co. B Capitol Fed. Fin'l Kimball Int'l B Hospitality Properties Washington Mutual Alliance Capital Management Prologis Health Care Property Bank of America Healthcare Realty Trust Catellus Develop. R.E.I.T. Penn. R.E.I.T. KeyCorp Crescent Real Est. AmSouth Bancorp. Mack-Cali Realty Bassett Furniture Archstone-Smith Tr. First Merit Corp. Liberty Property National City Corp. United Dominion Realty Rouse Co. Equity Residential Unitrin, Inc. BRE Properties BBOT Corp. Duke Realty Corp. Comerica, Inc. New York Community First Horizon National Regions Financial PNC Financial Serv. Washington R.E.I.T. Can Imperial Bank Simon Property Group Hudson United Bancorp

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Table IV: Real Estate and Financial Services Related Firms In the
Value Line Low P/E Screen

Beazer Homes Usa Aigon Ins. Group Sea Containers Ltd. A Allstate Corp. Kb Home Berkley (W.R.) Hovnanian Enterpr. A Lehman Bros. Holdings Standard Pacific Corp. Bear Stearns Centex Corp. Old Republic Ryland Group Loews Corp. Everest Re Group Ltd. UNUM Provident Corp. M.D.C. Holdings Thornbury Mtg. Horton D.R. Fidelity National PartnerRe Ltd. Amer. Financial Group Pulte Homes Greenpoint Fin'l ACE Limited Regions Fin'l Lennar Corp. MCIAInc. XL Capital Ltd. Hartford Fin'l Svcs. NVR, Inc. Nat'l Bank of Canada Annaly Mortgage Mgmt. National City Corp. Building Materials Chubb Corp. Countrywide Financial Morgan (J.P.) Chase Toll Brothers PNC Financial Serv. Fannie Mae

Table V: Retail Related Firms Included In the Value Line Low P/E Screen

Group 1 Automotive
Sears, Roebuck
Shopko Stores
Rent-Way Inc.

Table VI: Real Estate and Financial Services Related Shares

Included In the Value Line "Bargain Basement" Screen

Beazer Homes Usa KB Home Toll Brothers Centex Corp Standard Pacific Corp. Ryland Group M.D.C. Holdings Bear Sterns Knight Trading Group Edwards (A.G.) Investment Techn. Hertton D.R. 28

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Table VII: Retail Related Companies Included In the
Value Line "Bargain Basement" Screen

Enesco Group
Hudson's Bay Co.
Hancock Fabrics
Borders Group

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30 August 10, 2004

Distressed Utilities from the Short Seller Perspective Hedging, in its modern usage with regard to

equity investing, is known as the scientific control of risk. There seem to exist a virtually limitless array of strategies to extract return from equities and place risk upon someone else. The common denominator of such strategies is to simultaneously be long some equities and short some other equities. It is this seemingly simple idea that has created an investment management industry known as the hedge fund industry. This industry has in turn created other industries that service the hedge funds or, perhaps more properly expressed, act as suppliers. One of these suppliers is that industry known as the credit default swap industry.

The central idea is that one can purchase a bond of a seemingly lower credit quality and swap some or perhaps all of the default risk with a bank. The bank may or may not hold the risk for its own capital account. It may "hedge" its own risk by engaging in an offsetting swap transaction with yet another entity. However, at the end of this chain one will not necessarily find a willing holder or bearer of risk. It is claimed that the default risk can be readily hedged by an appropriate short position in the publicly traded share of the credit.

Readers of this essay will undoubtedly be aware that the Contrarian Research Report publication of Horizon Research has frequently recommended the share of various distressed utilities. All of these utilities are currently encumbered by excessive debt. Many of these debt instruments are high yield by the contemporary standard of low interest rates. Consequently many seekers of yield of the hedge fund variety endeavor to "capture" this yield by ownership of bonds and either purchase of credit default swaps or perhaps short sale of the individual equities. The distressed utilities group has the distinct honor to have accumulated remarkably large short interest in relation to trading volumes. The data for July 2004 is tabulated below.

Current Short Position Average Daily Volume 1. Calpine 125,006,771 8,462,043 2. El Paso 44,362,355 3,607,510 3. TXU Corp. 27,716,783 3,056,429 4. Sierra Pacific Rs. 26,715,956 1,299,257 5. Allegheny Energy 20,072,811 1,272,876 6. CenterPoint Energy 19,929,058 1,266,029 7. CMS Energy 17,628,843 832,029 8. Reliant Energy 17,408,923 1,861,024 9. Korea Electric Power 17,023,818 766,829 10. PG & E Corp. 15,821,238 1,313,686 11. Williams Cos. 15,779,217 2,517,462 12. Dynegy 24,999,924 2,370,262 13. Duke Energy 30,064,731 2,407,348 14. Teco Energy 20,427,322 1,259,286 30

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All of these short positions rank among the top 100 short position on the New York Stock Exchange for July 2004. However, this mere fact cannot convey the comparative size of the short positions. This can be properly understood by short position in relation to daily trading volume or days required to cover, this data is tabulated below:

Utility Short Position on the NYSE July 2004 Days Required to Cover (dtc)

1. Calpine 14.88 2. El Paso 12.30 3. TXU Corp. 9.07 4. Sierra Pacific Rs. 20.57 5. Allegheny Energy 15.78 6. CenterPoint Energy 15.74 7. CMS Energy 21.19 8. Reliant Energy 9.35 9. Korea Electric Power 22.19 10. PG & e Corp. 12.05 11. Williams Cos. 6.26 12. Dyregy 10.54 13. Duke Energy 12.48 14. Teco Energy 16.22

It is possible that reasonable minds may forecast an eventual bankruptcy filing for Calpine. However, it is certainly not likely that Allegheny Energy or Duke Energy will be bankruptcy candidates. However, since all of these companies are encumbered by debt and since bond investors wish to collect yield without default risk, the bonds are held with an associated short equity position. Incidentally, none of these issuers, with the possible exception of Calpine, are trading at yields that would suggest serious default risk.

For example, CMS Energy 9.875% due October 15, 2007, trade at 111.50 for a yield to maturity of 5.848%. The company is rated B. Similarly, CMS Energy 7.50% due January 15, 2009 trade at 104.75 with a yield to maturity of 6.253%. Allegheny Generating Co. 6.875% bonds due September 1, 2023 trade at 91.75 for a yield to maturity of 7.707%.

If one were to reject the notion that the short interest in the distressed utilities exists as a hedge for bond investors then one must accept an alternative reason for the short position, such as that these firms merely have poor fundamental qualities. In order to compare the magnitude of this short interest to other low quality firms, one should find other low quality firms with similar short interest ratios. Unfortunately, it is difficult to find firms with more than 10 days short coverage on the NYSE.

Among the top 100 short positions one could cite Delta Airlines (16.04 dtc), AMR (12.2 dtc), Xerox (13.99 dtc), General Motors (10.03 dtc), Silicon Graphics (16.05 dtc), Chunghwa Telecom (141 dtc), HCA Healthcare (18.25 dtc), Winn-Dixie Stores (22.82 dtc). Tenet Healthcare (19.03 31

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32 dtc), Albertson's (11.48 dtc), Orthodontic Centers of America (63.85 dtc), Krispy Kreme Doughnuts (15.64 dtc), Univision (14.00 dtc), Chico's FAS (17.92 dtc), Lyondell Petrochemical

(12.50 dtc), Allied Capital (21.44 dtc), Unumprovident (15.20 dtc), News Corp. (14.30 dtc) and Northfork Bancorp (10.73 dtc).

The most cursory view of the comparable short interest firms would suggest that the common denominator of this group is danger or near danger of insolvency. Thus, one can only conclude that a similar view is held by the community of investors that choose to be short the distressed utility shares. In any case, it is surely evident that an investment in distressed utility equity is a most Contrarian investment.

If the consensus view proves to be correct, as indeed might be the case, then it must follow that many bond investors in these enterprises as well as rating agencies are very misinformed. Of course, if the contrarian view is correct and these companies in the fullness of time repair their debt encumbered balance sheets, this should be the occasion for an interesting short squeeze. If the basis of the short interest is fundamental, then short interest might be gradually covered as the fundamental position improves.

On the other hand, if the short interest is caused by a hedged credit position, then the short interest will be repurchased in a more mechanical manner as debt ratings improve. In any case, many investors remark that the current investing environment lacks volatility. One cannot know the future with any degree of certainty. Yet, it does seem likely that the distressed utilities will be the source of much future volatility irrespective of whatever scenario will unfold.

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August 23, 2004

Utilities From A Buyout Perspective

At times various industries are subject to consolidation activities. The synonym most freely employed for this type of activity is buyout. This type of activity is commonly accompanied by a sort of speculative madness in which investors pay increasingly higher prices for publicly traded firms that are suspected to be candidates for buyout. This is so common a phenomenon that its occurrence is considered to be trivial. If this is so, then surely the absence of such madness should merit observation and notice. In the past eight months there have been eight utility buyout transactions. These are as follows: 1. In November 2003, Enron agreed to sell Portland General Electric to the Texas Pacific Group. 2. In January 2004, a private equity group organized by AIG Global Investment agreed to buy 25 power plants from El Paso. 3. In March 2004, Unisource (Tucson Electric Power) agreed to be acquired by Kohlberg, Kravis, Roberts. 4. In May 2004, Brascan agreed to acquire 769 megawatts of generating capacity from Reliant Energy for \$100 million. The assets are primarily hydro-electric and are located in upstate New York. 5. In July 2004, TNP Enterprises (Texas New Mexico Power) agreed to be acquired by PNM Resources (Public Service of New Mexico).

6. In July 2004, Centerpoint Energy announced the sale of its interest in Texas Genco to a group composed of The Blackstone Group, Hellman & Friedman LLC, Kohlberg Kravis Roberts, and Texas Pacific Group. 7. In August 2004, Arclight Capital Partners agreed to purchase the West Virginia Natural Gas operations of Allegheny Power. 8. Approximately four weeks prior to this transaction, Arclight Capital Partners raised \$1.6 billion for a power industry investment fund. The original Arclight fund raised \$950 million for a power industry investment vehicle in September 2002. This was at the height of the utility crisis. 34

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35 Until very recently, private equity funds did not generally invest in utilities. The inherent complexity of utility deals was sufficient to dissuade such investment. In any case, the presence of other investment opportunities made utility investing unnecessary. In the current investment environment, it is difficult to find a non-financial services firm that can be leveraged and that trades for less than 20x reliable earnings. Given current interest rates and the need to pay at least the customary 20-25 % control premium to undertake an acquisition, many forms of leveraged buyout are not currently practical. Nonetheless, the leveraged buyout firms have access to enormous amounts of capital. It has been calculated by Carlyle/Riverstone Group (yet another group formed to conduct energy buyouts) that only 2% of private equity funds have as yet been committed to this sort of utility buyout transaction. It is interesting that at the height of the utility crisis, there were few investors in the utility field.

In late 2002, First Reserve Corp., the oldest energy buyout firm, bought 33% of Quanta Services. This latter company is engaged in utility construction and maintenance. In addition, Mid

American Energy, a subsidiary of Berkshire Hathaway, was active at this time with purchases of the Kern River Pipeline, the Northern Natural Gas Pipeline and loans to such utilities as TXU as well as a convertible preferred investment in Williams Cos. However, Mid American is somewhat constrained as regards further utility investment by various provisions of the Public Utility Holding Company Act of 1935. Subsequent to late 2002, the most noteworthy buyout transaction was the purchase of the Williams Cos. general partner interest in Williams Energy Partners by Madison Dearborn Capital Partners and Carlyle/Riverstone. The transaction is not worthy of note because of its size; it is only worthy of note because it represents something of a strategic shift of focus for Madison Dearborn. This latter firm has four funds under management, with nearly \$7.7 billion in assets.

The first fund was launched in 1993. This is the first investment of this firm in a regulated business.

Consequently, over a very brief period of time, the focus of a very significant portion of the leveraged buyout community has been tuned toward the field of utilities. If the attention of this group of private equity firms had been directed towards any other industry group, it is reasonable to expect that the customary frenzy would have overtaken the investment community. Yet, the frenzy has not occurred. It is still possible to purchase a utility at a discount to book value.

Examples are Aquila Energy, El Paso, Sierra Pacific, Dynergy, CMS Energy, SEMCO Energy, Calpine and Reliant Energy. One might definitely wonder whether the absence of frenzy is not, in itself, a form of frenzy.

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36 Afterword I Kohlberg Kravis and Roberts is apparently content to pay 17.4x current earnings for Unisource.

This is also equivalent to a purchase price that is 1.5x book value. The Cascade LLC (Bill Gates's investment vehicle) is apparently content to invest in PNM Resources at 15.5x current earnings or 1.12x book value. PNM Resources claims that it will generate \$10 million of pre-tax synergies as well as \$40 million of pre-tax savings as a consequence of refinancing TNP Enterprises debt. The company should have \$68 million shares outstanding on a fully diluted basis subsequent to the merger. This is likely to be an accretive transaction. The Kohlberg Kravis and Roberts, Blackstone et al. buyout group for Texas Genco is apparently paying 13.4x current earnings and 1.16x book value. Moreover, the buyout group appears to be content to proceed with the exercise of the Texas Genco right of first refusal to acquire the 25.2% interest in the South Texas Project Electric Generating Station. This is a 2,500 MW nuclear power plant. The buyers appear content with their investment. Nonetheless, utility investing has not gained very many enthusiasts in the publicly traded equity community.

Afterword II On June 23, 2004, Nevada Power, a wholly owned subsidiary of Sierra Pacific Resources, agreed to buy a partially constructed 1,200 MW power plant from Duke Energy. The purchase price is \$182 million and it is estimated that an additional \$376 million will be required to complete the plant. The plant itself is located in the Moapa Valley, which is 20 miles northeast of Las Vegas.

The plant will not be operational until 2006. Sierra Pacific is considered to be in danger of insolvency because of \$336 million in power trading arrangements that Enron claims it is owed. The state commission (PUCN – Public Utilities Commission of Nevada) is expected to approve the transaction within 60 days. It is apparently satisfied that the utility can carry the construction burden of the yet to be completed plant. The investment community considers Sierra Pacific to be properly valued at 65% of book value. It should be noted in this connection, although it is not necessarily pertinent to the subject of this report, that the Federal Energy Regulatory Commission may well exert jurisdiction over the Enron-Sierra Pacific Power contract and may decide (unlike the Enron bankruptcy judge) that Enron is not entitled to \$336 million in termination payments. In this case, Sierra Pacific may well trade at a premium to book value.

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37 Afterword III The eight deals listed in the text of the main body of this essay perhaps leave the impression that this is the entirety of all electric power transactions. This is not the case. The main body transaction list was oriented toward the inclusion of leveraged buyout activity. A partial list of other transactions recently announced follows: 1. Ancora Management LLC will purchase the Cleveland Thermal District heating and cooling system that provides steam and chilled water to more than 200 buildings. The transaction was announced in July 2004. 2. Ameren Corp. will purchase Illinois Power from Dynergy for \$500 million. Ameren will also assume \$1.8 billion of debt. The transaction was announced in July 2004. 3. Edison International announced that it will sell 13 power generation plants to the British publicly traded firm International Energy for \$2.2 billion plus the assumption of \$3.2 billion of debt. The plants have an aggregate generation capacity of 5,400 MW. The transaction price is therefore

equivalent to \$1,000,000 per MW. The transaction was announced in July 2004. 4. American Electric Power agreed to sell 4 minority interests in Colorado cogeneration plants to a subsidiary of Bear Stearns. The sale price for the four minority interests was of \$156 million. This was equivalent to \$567,000 per megawatt for equity interest only, without consideration of de facto assumption of debt. The transaction was announced in March 2004 and closed in July 2004. 5. American Electric Power completed the sale of 3,813 MW located in Texas to a consortium of Sempra Energy Partners (San Diego Gas & Electric) and Carlyle/Riverstone. This occurred in July 2004. The transfer of the equity interest occurred at 1.62x book value. 6. Fortis, a Canadian utility, agreed to buy the Alberta and British Columbia assets of the U.S. based Aquila Energy for \$1.4 billion. Aquila originally paid \$600 million for the Alberta assets in 2000. The British Columbia assets are carried on the Aquila books for \$461 million. Fortis therefore purchased at a premium to book value. The transaction closed in May 2004. 7. Entergy announced that it would sell its Enetegy-Noch trading unit (a buyer has not yet been selected) and that it would spend \$1.5 over the next 24 months to repurchase shares. If Entergy-Koch is not sold as planned, the stock repurchase plan would be reduced to \$1 billion. At current prices, the more aggressive stock repurchase plan would reduce Entergy shares outstanding by over 10%. Entergy would be repurchasing shares of itself at 1.58x book value. Utility share repurchases are very unusual. This was announced on August 2, 2004. 37

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38 This is not an exhaustive list. There are many transactions that have not been cited including some by FPL Group, CMS Energy and NRG Energy. This list is sufficiently exhaustive to leave the impression in the mind of the reader that an extraordinary amount of acquisition activity is being undertaken in a fairly short period of time. Most interestingly, it is largely ignored by the participants in the world of publicly traded equities.

Afterword IV There is one aspect of utility investing that does not escape the keen scrutiny of the Wall Street community. This is the subject known as BPL. This acronym stands for Broadband Over Power Lines. It seems that until recently the inherent impedance and surge variations present in power lines have been an impediment to the development of high speed communications over power lines. However, modern advances in such things as digital signal processing have made BPL possible. Consequently, some two dozen utilities are conducting trials of high speed internet access. If high speed internet access is possible, then voice over internet protocol is also possible. Quite obviously, Wall Street is highly intrigued, especially as companies conducting trials are sizeable and respectable. These firms include Cinergy, PPL Progress Energy, PEPCO Holdings, and Consolidated Edison. In fact, it has been learned that these firms are cooperating in this matter with Earthlink. This latter company is threatened by the evolution of high speed access over cable and thus is developing a countervailing strategy in concert with electric utilities. It is believed by those with knowledge in this field that there are only two vehicles by which one might invest in this new technology. These are publicly traded firms known as Telkonet (TKO) and Ambient (ABTG). However, four companies in this field apparently are planning Initial Public offerings. These are eagerly awaited. The four private firms are known as: 1. Current Technologies 2. Amperion 3. Main.Net 4. Design of Systems on Silicon

Telkonet has a \$97 million market capitalization and \$2,000,000 of revenue. Thus, it trades at 485x revenue. Its P/E ratio is infinite since it has no earnings. Its price to book value ratio is 5.4x. This ratio is likely to increase since the primary component of book value is roughly \$18 million of cash that is being expended at the rate of almost \$5 million per quarter. The company has attracted some institutional ownership. 38

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39 Ambient has only a \$36 million market capitalization. Its market capitalization to sales ratio cannot be calculated since it has no sales. Therefore, it cannot have profits and consequently it cannot have a P/E ratio. Its price to book value ratio is 7.3x. This ratio is likely to rise at a more rapid rate than Telkonet since it is spending its limited cash resources at a much more rapid pace.

It is currently expending \$4.5 million per quarter and this is roughly equal to the most recently available cash balance. Its primary investor and apparent source of capital is Consolidated Edison (ED). This latter company trades at 15.4x current earnings and 1.4x book value. Evidently, the management of Con Edison has proven that at least some portion of the assets of a utility can trade at a substantial premium to book value. 39

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August 25, 2004

A Reader Responds to "Joseph Stalin's Hedge Fund"

Dear Horizon, While I think your point about growth, value, and income investors all buying financial services/real estate companies is very interesting, I don't think the historical analogy with the 1930's is a good one. First, modern historians have conclusively shown the Ukrainian famine was a deliberate act. Stalin's goal was not greater production, but genocide. He and his henchmen viewed the Ukrainian peasants as class enemies because they had a long tradition of owning private property and they generally favored an independent Ukraine (see Robert Conquest's *Harvest of Sorrow*.)

Second, it is a mistake to describe Herbert Hoover as "one of the more...doctrinaire believers in the market mechanism". Contrary to popular belief, Hoover was a proponent throughout his career of state intervention and many of the New Deal policies championed by FDR were actually just expansions of programs started by Hoover. (The best book on the subject is Murray Rothbard's *America's Great Depression*, but any recent biography of Hoover acknowledges this point.) Finally, the idea that declining prices forced American farmers in the 1930's to increase production, thereby further lowering prices, doesn't make much sense. It is a basic tenet of economics that most people will always maximize their profits and in the case of farmers that usually means maximizing profitable production. Your claim that at low commodity prices pushed farmers to ramp up production suggests that in prior years when prices were high farmers purposefully ease up and made less money than they otherwise would have. Sincerely, [A Contrarian Research Reader] Dear [Reader], Thank you for your letter. You raised a number of interesting points. I was not aware that any recent biography regards Hoover as a state interventionist. Certainly President Hoover does not portray himself as a state interventionist in his memoirs. Certainly, President Hoover was severely criticized by the Democratic Party as not sufficiently interventionist. Hoover himself 40

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41 devotes the second half of Volume 3 of his memoirs to severe critique of the interventionist policies of President Roosevelt. As an example, one might cite Chapter 35 of Volume 3, which is entitled "Usurpation of Power". In addition, Chapter 36 of the same volume is entitled "Collectivism Comes to the Currency – and its Consequences". Chapter 37 is entitled "Fascism Comes to Agriculture". Chapter 38 is entitled "Fascism Comes to Business – With Dire Consequences". Chapter 39 is entitled "Fascism Comes to Labor – With Consequences". Chapter 40 is entitled "Introduction to Socialism through Electrical Power". This chapter severely criticized the Tennessee Valley Authority. Chapter 42 is entitled "Collectivism by Thought Control and Smear". In Chapter 43 Hoover does applaud some Roosevelt initiatives such as the Export-Import Bank, Social Security, the Securities and Exchange Commission and the Fair Labor Standards Act. Nonetheless, the thrust of the Hoover memoirs is quite anti-state intervention. One might readily object that these memoirs, as perhaps all memoirs, are disingenuous. This may be correct. Nevertheless, I have found it difficult to discover any serious Republican contemporary of Hoover who would characterize him as a state interventionist. I am familiar with the work of Prof. Rothbard that you cite in your letter. It may be the best book on the Great Depression. I leave judgment of such matters to others. Personally, I find the views of that book rather extreme. However, since anything that I have to say on the matter is not likely to be convincing, I would like to defer to the opinion of William F. Buckley on the subject of Rothbard. Mr. Buckley is hardly a proponent of state intervention in business. There are those – I think of late Murray Rothbard – who cried out against politics of co-existence and liberation, but his perspective was so much the captive of an anti-statist obsession that his eyes squinted, and at the end he was incapable of distinguishing – he loudly professed – between the leaders of the Soviet Union and the leaders of the United States. On this matter, in those frenzied days, I counter-preached that the man who pushes an old lady into the path of an oncoming truck and the man who pushes an old lady out of the path of an oncoming truck, are not to be denounced even handedly as men who push old ladies around. The aforementioned quotation is from a speech that Mr. Buckley delivered at the Heritage Foundation on October 20, 1999. It is contained in the collected speeches of William F. Buckley in the volume entitled "Let Us Talk of Many Things" (Prime Publishing, © 2000, pg. 470). The Rothbard criticism of Hoover is thus not entirely surprising. A man who cannot distinguish between Stalin and Hoover is not to be expected to distinguish between Roosevelt and Hoover.

Of course, when reading history it is unfortunate that much polemicism interferes with scholarship. Hoover found it rather difficult to distinguish between Mussolini and Roosevelt. It is worthy of note that President Hoover established the Hoover Institution on the campus of Stanford University as much more than a mere presidential library. It is supposed to be dedicated to the study of social systems based upon private enterprise and personal freedom. Its mission statement supports the view that "...ours is a system where the Federal Government 41

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42 should undertake no governmental, social or economic action except where local government, or the people cannot undertake it for themselves..." In any event, surely the scholars of the Hoover Institution would not be equally pleased that their organization should be called the Roosevelt Institution. Your second point (it is actually your first point) relates to the motivation of Stalin in commencing upon the policy of collectivization. I did consult the work that you cite on the subject by Mr. Robert Conquest. Incidentally, it is one of the unusual ironies

that sometimes occur that Mr. Conquest was a scholar at the Hoover Institution at the time that the book was written. The last chapter does indeed attribute such motivations to Stalin. I myself find Stalin's motivations on anything to be somewhat inscrutable. It is known that he was one of the most ruthless despots in history. If someone dared to oppose him, or if Stalin merely suspected opposition where there was none, the individual would vanish. Mr. Conquest makes the argument that the Ukrainian peasants were prevented from leaving the Ukraine during famine as a means of eliminating class enemies. It must be rather difficult to starve someone on a working farm in a rural area unless one deploys sufficient well fed police to prevent peasants from a) growing food, b) hunting and fishing, c) stealing food from working farms on which they were employed. Surely it is far more efficient to execute the alleged class enemy in the customary manner. Indeed, Stalin must ultimately have come to this conclusion, since he did have literally countless millions of persons executed in the customary manner. If Stalin had wished to murder people by means of famine caused by the collectivization of agriculture, he must have been a sufficiently astute student of market economics to predict that the loss of incentive would cause production to fall. In any event, I do not think that a vast gulf separates myself from Mr. Conquest. I think that Stalin was a ruthless despot that murdered people and had no knowledge of market economics. Mr. Conquest seems to think that Stalin was a ruthless despot that murdered people and had profound knowledge of market economics. Your last point relates to the idea that declining prices forced American farmers to increase production. You state that my claim to this effect is counter to the basic tenets of economics. First, I am flattered that you use the word claim. It implies that I have seriously studied the subject. If I had studied the subject much more intensively than I have thus far I might be considered to be a dilettante. I am far from being a scholar on this subject as well as many others. I also agree that increasing production during a period of declining prices is counter to the basic tenets of economics. Nonetheless, the statistics produced by U.S. Department of Agriculture do manifest an interesting pattern. 42

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Acres Planted (000's acres)	Production (000 bushels)	Price Per Bushel
1926 60,712	832,213	\$ 1.21
1927 65,661	875,059	1.18
1928 71,152	914,373	0.998
1929 67,177	824,183	1.03
1930 67,559	886,522	0.663
1931 66,463	941,540	0.382
1932 66,281	756,307	0.375
1933 69,009	552,215	0.736
1934 64,064	526,052	0.839

As late as 1933, acres planted were actually increasing. Production itself achieved a record amount in 1931 when the price declined to 38¢ per bushel. Afterward, production did decline although acreage planted did not decline. Scholars are rather divided on the causes of this phenomenon. Some believe that unfavorable weather conditions explain production declines. Others believe that it may be attributable to seizure of many farms in mid planting season for mortgage foreclosure or non-payment of property taxes. It was the policies of the Roosevelt Administration, commenced in 1933, that deliberately sought to raise agricultural production. The statistics presented in this table are from the National Agricultural Statistics Services of the U.S. Department of Agriculture. Internet site reference is at www.nass.usda.gov. My own impression of the data is that given weather, taxes, debt and government policies, production and prices are determined by more than merely the decisions of farmers. Once again, I thank you for your letter. I do not expect that you will accept my view and I imagine that my propensity to error is rather high. Nonetheless, you may find that an explanation of my reasoning at least clarifies my point of view.

Sincerely yours,

Murray Stahl 43

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44 September 2, 2004

The Other Housing Markets

I. The Optimists Versus the Pessimists Among the various investment debates between optimists and pessimists, perhaps none is as polemical as the debate over the future course of the housing market. In the case of this essay, the term housing market refers to the shares of various housing related companies such as the homebuilders, home supply retailers such as Lowes, as well as suppliers of credit such as Fannie Mae. Although the ultimate result is not knowable, it does appear that the optimists have exhibited far more creativity with regard to the rhetorical presentation of their argument.

The pessimists are generally content with reference to the cyclical characteristics of the housing industry. The pessimistic argument is that a combination of low-interest rates and high home prices has created a sort of cyclical excess. As housing returns to more normal conditions a veritable collapse of industry profits will occur. The optimists have rather ingeniously bypassed this type of argumentation and have successfully rendered it rather sterile. The optimistic case may be found in two variants. The first variant states that the average price of a new home sold

in the U.S. has only increased by 5.0% per annum since 1980. This is hardly consistent with excess, especially since the 1980 reference point was a period of extraordinarily high interest rates and hence artificially low real estate prices. In fact, the average selling price for a new home only increased by 5.05% per annum since 1994. Moreover, it is asserted with some considerable justification and persuasiveness that the average selling price comparison is not an appropriate measurement since modern housing offers better features and more floor space than was the case historically. Therefore, one must make some adjustment for a quality differential. The National Association of Home Builders calculates a Quality Adjustment Index (QAI). This adjusts for changes in home size as well as amenities over time. Properly adjusted, the average new home price increased at a 2.55% compound annual rate since 1980. The QAI adjusted rate of price increase since 1994 is only 2.71% per annum. (see Appendix 1 for this data). This is a powerful and rather convincing argument. The second optimistic variant is yet more creative. This elegantly bypasses the macroeconomic or cyclical argument of the pessimistic camp. The view is that the publicly traded homebuilders have gained and will continue to gain market share from smaller, private homebuilding firms. Thus, even in the event that aggregate new home sales should decline, the public group can compensate for the decline to a significant degree by market share increase. The decline in 44

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45 profits, if this should occur, is likely to be quite modest. In any case, this decline is fully discounted in the low valuations of publicly traded home builders. The market share argument is difficult to refute since the available data clearly indicates that the publicly traded firms now have a larger share of the aggregate new home market than was previously the case. Estimates of current market share can be found in Appendix 2. The 19 publicly traded homebuilders listed in Appendix 2 have achieved an approximately 25% national market share. This share is sufficiently low so that it can be envisaged to further increase. The rhetorical force of this argument is that it makes any cyclical argument essentially irrelevant.

- II. A Different Housing Market Investors are accustomed to statements about the current strength of the housing market. It may therefore be rather surprising to learn that one aspect of the national housing market is in a decline of virtually unprecedented magnitudes. This is the manufactured housing market. In 1996, a generally strong year for housing in a historical context, sales of manufactured homes represented a very significant share of new home construction. In that year 363,411 manufactured homes were sold. This is to be compared with 758,000 new homes sold in the same year. In 2003, the last complete year for which there is available data, manufactured home shipments were 130,937. New single family homes sold totaled 1,087,000. The data from 1980-2003 is included in Table 1. 45

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TABLE 1 Manufactured Home Shipments vs. Single Family Homes Sold 1980-2003

Manufactured Homes	Single Family Homes	1980	221,091	545,000	1981	240,313	436,000	1982	238,808	412,000	1983	295,079	623,000	1984	294,933	639,000	1985	283,489	688,000	1986	244,660	750,000	1987	232,598	671,000	1988	218,429	676,000	1989	198,254	650,000	1990	188,172	534,000	1991	170,713	507,000	1992	210,787	610,000	1993	254,276	666,000	1994	303,932	670,000	1995	339,601	667,000	1996	363,411	758,000	1997	353,377	801,000	1998	372,843	885,000	1999	348,671	906,000	2000	250,550	877,000	2001	193,229	901,000	2002	168,411	972,000	2003	130,937	1,087,000
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National Assn. of Home Builders and Manufactured Housing Institute

The current cycle in manufactured housing is far more severe than the last decline experienced the real estate crisis of 1988-1991. The worst year for manufactured housing was 1991, when 170,713 manufactured homes were sold. In 2003, only 130,937 manufactured homes were sold. Interestingly, during the previous 1988-1991 real estate crisis, conventional new home sales declined. During the current manufactured housing crisis, new home sales are quite robust. Consequently, one cannot but wonder if the market share gains made by the publicly traded homebuilders are at the expense of the manufactured home builders. 46

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47 If the answer to this hypothetical question is no, then one must wonder why manufactured home sales have reached depression levels. If the answer to this hypothetical question is yes, then one must wonder if customers that formerly could only afford a manufactured home are actually able to afford a more conventional home. New single family homes are on average far more expensive than manufactured homes. The comparison of new single family home prices (average) and average sales prices for manufactured homes in a historical context is to be found in Table 2. TABLE 2 Comparison Between Single Family New Home Prices and Manufactured

Home Prices (averages)

Single Family Homes Manufactured Homes 1980 \$76,400 \$19,800 1981 83,000 19,900 1982 83,900 19,700 1983 89,800 21,000 1984 97,600 21,500 1985 100,800 21,800 1986 111,900 22,400 1987 127,200 23,700 1988 138,300 25,100 1989 148,800 27,200 1990 149,800 27,800 1991 147,200 27,700 1992 144,100 28,400 1993 147,700 30,500 1994 154,400 32,800 1995 158,700 35,300 1996 166,400 37,200 1997 176,200 39,800 1998 181,900 41,600 1999 195,600 43,300 2000 207,000 46,400 2001 213,200 48,400 2002 226,700 51,300 2003 240,000 est.
N/A Source:

National Assn. of Home Builders and Manufactured Housing Institute

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48 One of the many surprising variants of the law of supply and demand is that although manufactured home sales are significantly in decline, the average selling price is still being increased. In any event, if new home sales are capturing market share from manufactured homes, it is extraordinary that the former manufactured home buyer can afford a much more expensive dwelling. According to the Manufactured Housing Institute, the average buyer of a manufactured home earns \$28,900 per annum. It would not be easy for such an individual to carry the monthly payments (mortgage, insurance, property taxes, maintenance, and operating costs) of a \$240,000 home even with a 20% (\$48,000) down payment. It is, of course, possible that the former buyers of manufactured homes have become far more affluent since 1997. It is also possible that the customary excess liberality of credit extension makes purchase temporarily affordable by virtue of such devices as adjustable rate mortgages and other devices. If the latter is actually the case, then the argument could be made that in the next housing cycle the homebuilders lose share to manufactured homebuilders while aggregate industry single family home sales are in decline. Naturally, such an argument is only a theoretical abstraction since one can only make assertions. Definitive proof is not possible.

III. Another Housing Market

The consumer is not only offered the choice between a conventional single family home and a manufactured home, the consumer is also offered the choice between home ownership and rental in a multi-family residence. If one cannot accept the proposition that the single family home sales are increasing at the expense of manufactured homes, then one must search for another source of customers. This is because the national home ownership rate has been gradually increasing in recent years. TABLE 3 National Home Ownership Rate 1982-2003

1982 64.5% 1993 64.2% 1983 64.4% 1994 64.2% 1984 64.1% 1995 65.1% 1985 63.5% 1996 65.4% 1986 63.9% 1997 65.7% 1987 64.1% 1998 66.4% 1988 63.8% 1999 66.9% 1989 63.8% 2000 67.5% 1990 64.1% 2001 68.0% 1991 64.2% 2002 68.3% 1992 64.4% 2003 68.8% Source – U.S. Census Bureau

It does appear that at least some of the increased demand for new homes has been at the expense of multi-family residences. The evidence for this is the national rental vacancy rate for multi-family housing as tabulated by the U.S. Census Bureau. 48

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49 TABLE 4 National Multi-Family Rental Vacancy Rate 1982-2003

1983 5.7% 1993 7.3% 1984 5.9% 1994 7.4% 1985 6.5% 1995 7.6% 1986 7.3% 1996 7.8% 1987 7.7% 1997 7.7% 1988 7.7% 1998 7.9% 1989 7.4% 1999 7.1% 1990 7.2% 2000 7.0% 1991 7.4% 2001 8.4% 1992 7.4% 2002 9.0% Source – U.S. Census Bureau

The increase in rental vacancy rate has been rather severe in the past several years. Nonetheless, another one of the oddities of the law of supply and demand makes itself manifest. The median asking rent has been steadily increasing. The absorption rate for new multi-family units has been steadily decreasing. The formal definition of absorption is an apartment that is occupied within three months of completion. Table 5 – Median Apartment Asking Rent 1983 \$386 1993 \$573 1984 \$393 1994 \$576 1985 \$432 1995 \$655 1986 \$457 1996 \$699 1987 \$517 1997 \$729 1988 \$564 1998 \$738 1989 \$590 1999 \$790 1990 \$600 2000 \$815 1991 \$614 2001 \$881 1992 \$540 2002 \$905 Source – U.S. Census Bureau

Table 6 - Multi-Family Absorption Rate 1983 69% 1993 80% 1984 67% 1994 73% 1985 65% 1995 72% 1986 66% 1996 73% 1987 63% 1997 73% 1988 66% 1998 73% 1989 70% 1999 72% 1990 67% 2000 72% 1991 74% 2001 64% 1992 75% 2002 59% Source – U.S. Census Bureau

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50 One must wonder if the declining absorption rate will ultimately result in a decline in either asking rent or the rate of multi-family rental unit construction. Thus far, the declining absorption rate has not impacted the rate of multi-family rental unit construction. TABLE 7 Multi-Family Rental Unit Construction 5 Units or More 1969-2003 (000 units) 1969 486.0 1987 356.9 1970 490.8 1988 309.5 1971 684.6 1989 278.1 1972 846.8 1990 236.3 1973 853.6 1991 149.2 1974 511.3 1992 122.8 1975 299.4 1993 118.2 1976 304.9 1994 182.5 1977 399.3 1995 207.7 1978 462.2 1996 214.3 1979 423.4 1997 260.2 1980 313.1 1998 282.9 1981 255.3 1999 284.1 1982 278.9 2000 280.7 1983 420.8 2001 292.6 1984 431.0 2002 306.0 1985 468.4 2003 344.2 1986 442.7 Source -U.S. Census Bureau

The coincidence of increased construction as well as increased vacancy rate is surely not a manifestation of a normally functioning market. Increased rental rate as aforementioned must be considered to be yet another anomaly. IV. Another Investor The current housing cycle is most extraordinary, since different aspects such as manufactured housing versus single family housing are normally rather correlated and in the past several years have become negatively correlated. It is difficult to resist the conclusion that the market share gains of the traditional homebuilders have come at the expense of manufactured housing.

Moreover, it is difficult to accept that the typical buyer of a manufactured home has suddenly become far more affluent. In addition, the apartment market seems to prosper and yet sacrifice share to the single family housing market. It is difficult to believe that this state of affairs can continue. Nevertheless, many investors, including many very astute value investors, are attracted by the low p/e ratios of the homebuilder companies. Another notable value investor, the fellow who controls Berkshire Hathaway, appeared to be more attracted to the manufactured home builders.

On August 7, 2003 Berkshire Hathaway acquired Clayton Homes, the leading manufactured home builder, for approximately \$1.7 billion. The purchase price represented a p/e ratio of 15.6x trailing earnings. If recent sales of manufactured housing is any guide current conditions, the investment has yet to establish its merit. It will be interesting to see which investor has properly

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51 assessed the future. It does not appear likely that both sets of investors will prosper. If manufactured homes regain share because easy credit terms or conventional homes cannot continue indefinitely, the conventional home builders might experience a most unusual and serious negative cycle.

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Appendix 1

New Average Home Selling Price Quality Adjustment Index 1980 \$76,400 68.1 1981 83,000 73.5 1982 83,900 75.2 1983 89,800 76.8 1984 97,600 69.8 1985 100,800 70.7 1986 111,900 73.4 1987 127,200 77.4 1988 138,300 80.3 1989 148,800 83.5 1990 149,800 85.1 1991 147,200 86.2 1992 144,100 87.3 1993 147,700 91.1 1994 154,400 95.5 1995 158,700 98.2 1996 166,400 100.0 1997 176,200 102.9 1998 181,900 105.5 1999 195,600 110.7 2000 207,000 115.4 2001 213,200 119.5 2002 226,700 124.8 2003 240,000 N/A Source: US Census Bureau

It is worthy of note in connection with the Quality Adjustment Index that according to the National Association of Home Builders the medium square footage of a new home has increased as follows in the past 30+ years: 1970 1,385 sq. ft 1990 1,905 sq. ft 2002 2,113 sq. ft. 2003 2,123 sq. ft

Similarly, in 1990, 76% of all new homes had central air conditioning. In 2002, this figure had increased to 87%. Similar figures can be found for the presence of a two car garage, number of bedrooms or number of bathrooms. 52

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Appendix 2 Home Sales of 19 Publicly Traded Firms, Most Recent Annual Figures It should be noted that the sales figures of the home builders may not precisely correspond to the new home sales figures released by the U.S. government for two reasons. First, the U.S. government calculates home sales. The home builders count home delivered. A home sold might not necessarily be booked as revenue in a given accounting period since it might have a construction defect that is easily correctable and yet will prevent the home owner from occupying the residence until repairs or corrections are completed. Second, the home builders are not all on December fiscal years whereas the government statistics are for years ending December 31.

Nonetheless, the individual sales figures should, in aggregate, give a reasonable idea of market share. Beazer 15,800 MDC 11,225 Centex 42,823 KB Home ** 23,404 Horton 37,637 Standard Pacific 7,676 Hovnanian 11,531 Brookfield Homes 1,528 Pulte* 32,693 M/I Homes 4,148 Lennar 32,180 Meritage 5,642 Ryland 14,978 Technical Olympic 6,135 NVR 12,124 WCI*** 1,666 Toll Brothers 4,911 William Lyon 2,804 Orleans 1,753 270,658

* Domestic only

** Excluding France *** Homebuilding Revenue only; excludes Tower Apartments

This represents an approximate 25.2% national market share. In reality, this number might be larger since the publicly traded companies tend to be concentrated in those states with the most rapid population growth. These are Nevada, Arizona, Colorado, Utah, Idaho, Florida, Texas, North Carolina, Washington and Georgia. The home builders almost certainly have a much higher than 25.2% in many of these states.

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54 October 13, 2004

eBay: The Problems Created by Consensus History reveals that unhappiness and terrible strife frequently result when people differ. One of the ironies of economics and finance as a social science is that problems generally occur when people agree. This is because agreement in finance changes the rate at which future earnings are discounted. The earnings of a company upon which there is much disagreement are discounted at a very high rate. Conversely, the earnings of a company upon which there is no disagreement are discounted at a very low rate. Among those companies with a low future earnings discount rate there exists eBay. Of course the simple fact that earnings are discounted at a low future rate does not imply that shareholders will necessarily earn only this rate. A sufficiently high rate of profit growth will counterbalance a low discount rate. The consensus eBay forecast is that the future rate of earnings growth will be very high. The critical question is whether such an optimistic consensus is reasonable. Let us commence with the most wildly optimistic future possible for eBay. Let us presume that 20 years hence, eBay will have annual revenue that is roughly equivalent to the current annual revenue of Wal-Mart. In that case, in 2024, eBay should record \$272 billion of annual revenue. It may not be obvious to the reader that this is a wildly optimistic forecast. This is because the revenue of eBay is a function of the value of the merchandise that is sold on its network. The average fee might be in the region of 8 – 9 % of merchandise value. The eBay fee schedule is reproduced below: Insertion Fees Starting or Reserve Price Insertion Fee \$0.01 – 0.99 \$0.30 \$1.00 – 9.99 \$0.35 \$10.00 – 24.99 \$0.60 \$25 – 49.99 \$1.20 \$50 – 199.99 \$2.40 \$200 – 499.99 \$3.60 \$500 or more \$4.80

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Final Value Fees Closing Price Final Value Fee Items Not Sold No Fee \$0.01 – 25.00 5.25% of closing value \$25.01 – \$1,000.00 5.25% until \$25 plus 2.75% of the remaining balance from \$25.01 – \$1,000 Over \$1,000.01 5.25% of the initial value plus 2.75% of the initial \$25- \$1,000 plus 1.5% of the remaining value over \$1,000 Arithmetically inclined readers will perceive that an estimated revenue of 8-9% of merchandise value might be rather excessive. However, the purpose of the exercise is to be optimistic. Thus, if eBay were to record \$272 billion of revenue in the year 2024, it would need to sell through its system 11.76 x this amount (100/8.5%) or roughly \$3.2 trillion. This is equivalent to 29% of the current U.S. GDP.

Revenue figures are devoid of meaning without the application of a theoretical profit margin. The revenue comparison is with Wal-Mart. One might imagine that given this comparison, it might be appropriate to apply the Wal-Mart net profit margin of 3.5% and thereby derive an earnings figure. Although this is certainly reasonable, it is not wildly optimistic. A much more optimistic approach would be to apply to the 2024 revenue estimate the profit margin of Microsoft. This is currently 22.2% on a net basis. Such a figure is rarely, if ever, exceeded by a large enterprise.

Estimated revenue of \$272 billion in 2024, with a net profit margin of 22.2% would result in net income of \$60.384 billion in 2024. The next step would be to calculate an earnings per share figure. Towards this end, it is necessary to estimate the shares outstanding for eBay in 2024. This is rather difficult since eBay is generally increasing its shares outstanding each year. For instance, the eBay shares outstanding for each fiscal year on a fully diluted basis is provided below:

Estimated Shares Outstanding

1999- 2003 Fully Diluted 1999 546,066,000 2000 560,692,000 2001 561,190,000 2002 585,640,000 2003 656,657,000 55

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56 In order to remain optimistic, let us fix the number of shares outstanding in an amount equal to the fully diluted share count as it appears in the June 30, 2004 form 10-Q, which is the last available SEC filing. This is 682,421,000 shares outstanding. Thus, it is assumed that eBay will issue no shares for the next two decades and, therefore, shares outstanding in 2024 will be 682,421,000.

It will be recalled that net income in 2024 was estimated to be \$60.384 billion. Given this net income estimate and estimated 2024 shares outstanding, simple division reveals an estimated \$88.48 earnings per share figure in 2024. This is far in excess of the \$1.60 consensus earnings estimate for 2005. The consensus earnings estimate, it should be noted, does not count as an expense the costs, such as these might be, of options issuance. The optimistic case simply avoids this issue by the unrealistic assumption that options issuance will immediately cease. There remains the question of the P/E ratio to be applied to profits in 2004. Some might argue that this is best estimated by the Wal-Mart P/E ratio as this currently exists. This is a P/E ratio of merely 19.28x and, as such, too low to be utilized in an exercise in optimism. The Microsoft P/E ratio of 20.29x is not much superior for use as a basis for optimism. Let us therefore arbitrarily apply a P/E ratio of 35x.

The justification of this is that eBay in 2024 would certainly be the largest company in the S&P 500 Index. There does not appear to be much precedent for the largest company in the S&P 500 trading at such a multiple. However, General Electric and Cisco Systems appear to have briefly surpassed this level in 2000 at the bubble peak. Nonetheless, the largest member of the S&P 500 does not generally trade at a 35X P/E ratio. There can be no doubt that eBay will be the largest company in the S&P 500 Index by 2024, since it is already in the 33rd position. eBay currently has a market capitalization that is approximately equivalent to 3M (Minnesota Mining and Manufacturing.) If all this were to occur in 2024, then eBay would sell for \$3,096.80 per share in that year. The compound annual rate of return for those shareholders with the wisdom and foresight to buy those shares at the October 8, 2004 closing price of \$92.59 would be 19.19% per annum.

Naturally this is a very optimistic figure. If eBay were merely to trade at 19.28x its 2024 estimated profits, its 2024 trading price would be \$1,705.89. Its compound annual rate of return would be 15.68% per annum. This is rather interesting, since it is immediately apparent that the long term compound annual rate of return is far more sensitive to earnings growth than to P/E ratio.

Let us therefore alter the growth rate of the company. Let us assume that eBay in 2024 will sell through its system a quantity of merchandise equal to that sold by Wal-Mart at the current time. This is \$272 billion. However, since eBay is an intermediary, it will generate as revenue 8.5% of this amount. This equals \$23.12 billion and is 8.5 x the current eBay revenue. We shall assume a Microsoft-like profit margin on a net basis of 22.2%. The current eBay profit margin exclusive of any expense associated with options issuance is 24.57%. In accordance with the estimate here presented, eBay would earn 22.2% of \$23.12 billion of revenue – or \$5.43 billion. This would be equivalent to \$7.52 per share, assuming share issuance were to immediately cease. The application of a 35X P/E ratio in 2024 would generate a share price of \$263.24 in that year. This 56

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57 yields a future 20 year compound annual rate of return of 5.36%. Consequently one might say that this is the return available to shareholders if eBay were to become equivalent to the current Wal-Mart with a much higher P/E ratio than Wal-Mart. If eBay were to become equivalent in scale of operations to Wal-Mart with a 22.2% net profit margin (versus the current Wal-Mart 3.5%) and a Wal-Mart P/E ratio of 19.28X, the 2024 share price would be \$144.99. The 20 year compound annual rate of return would be only 2.27%. If eBay were merely to duplicate Wal-Mart, a Wal-Mart profit margin of 3.5% and a Wal-Mart P/E ratio of 19.28X, the 2024 share price of eBay would be calculated as follows:

Merchandise Value through EBAY systems: \$272 billion X 8.5% equals EBAY revenue, or: \$23.12 billion X Wal-Mart Profit Margin of 3.5% equals EBAY 2024 net income of \$809 million Divided by 682,421,000 shares equals: \$1.19 per share in 2024 X 19.28 P/E ratio equals: \$22.86 per share in 2024 as the eBay share price. The compound annual rate of return equals <6.76%> per annum or a loss of 6.76% every year for 20 years.

It would seem reasonable to conclude that each \$22.86 of the eBay share price equals the forecast of one Wal-Mart success. Therefore, in order to achieve the current \$92.59 eBay price in the future, eBay would need to create 92.59 divided by 22.86 successes of the magnitude of Wal-Mart, or slightly more than four Wal-Marts as these currently exist. A level of success that is less than four times the current Wal-Mart scale results in a negative equity yield curve. Since most reasonable observers would conclude that eBay, whatever its virtues, is not likely to exceed the scale of four Wal-Marts, one might well ask why it is as widely owned and traded. The daily trading volume of eBay is over 10 million shares, or 4X that of 3M, its current size equivalent in

the S&P 500. The answer can only be that investors anticipate large short term profits from eBay whatever might be its long term compound annual rate of return. Consequently, it would appear that the equity yield curve for eBay is inverted.

If the yield curve for eBay is actually inverted, then the critical variable becomes the sell decision, which can be reduced to the search for the point at which the yield curve becomes negative. Yet, unlike the bond yield curve, this point is not readily knowable. One can only be a sound agnostic and confess that one cannot know what one cannot know. Nonetheless, one can know that the point at which the equity yield curve becomes negative is the point, by definition, at which the consensus group will choose to sell. Consensus implies agreement. In finance and economics, when market participants act in agreement problems frequently occur. It seems that participants implicitly agree that the destiny of eBay is to be many times larger than the contemporary Wal-Mart. These forecasts recur with great regularity throughout history. Each reprise shows the same result. Most readers can easily imagine the result. 57

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58 October 19, 2004

Google and Yahoo as Toll Roads (To be viewed as Part III of the Yahoo, eBay and Google Trilogy) (Or Humbug and the Internet Made Simple)

Imagine a situation in which entrepreneurs could build tollgates in front of the retail establishments of the world. In this world the tolls would be assessed not against the shoppers but against the retailers based upon the number of individuals who entered given establishments. One might think, in the absence of other information, that this was a sort of organized crime racket. However, let us say that this was perfectly legal. In that case, one might envisage that the retailers would object. However, let us say that the entrepreneurs that operated the toll gates agreed, as a sort of quid pro quo, to guide potential shoppers to given stores if the stores agreed to pay a toll commensurate to those efforts. Thus, it could be argued that the store would actually benefit from this arrangement as it would receive more traffic than might have been the case without the toll. The owners of the store might nonetheless wonder how it would be possible to generate traffic without the exercise of coercion on the potential customers.

In such a world the toll gate operators would have the right to remove the street signs pointing to a competing retail establishment and, in its place, leave the signage of its new toll gate customer. This would be a fairly lucrative arrangement from the perspective of the toll gate operators. Yet, since this is only an exercise in imagination, let us suppose that the toll gate operators were still not content. The toll gate operators now undertook to produce road maps upon which the only roads labeled would be those where paying customers were located. Moreover, the toll gate operators were so clever that they paid monitors to stand at strategic crossroads and intersections to gently block both foot traffic and vehicular traffic in such a manner that potential shoppers are guided to shopping establishments rather than, let us say, parks where they might take a pleasant but nonetheless commercially useless stroll. Such an exercise should require very little imagination since these aforementioned toll gates, bizarre road maps and paid monitors are merely “pay for search” engines, and banner ads, particularly of the “pop up” variety. The highway is the Internet. The primary toll gate operators, although there are others, are Google and Yahoo.

One more imaginative effort is required in order to understand the Google/Yahoo circumstance. Let us suppose that in our bizarre world these toll gate operators could monetize their asset by a share listing. Quite obviously, the toll gate rights to the retailers of the world to the end of time would have a nearly infinite value. Indeed, at least some would assert that the actual toll gate operators do have such a value. The aggregate market capitalization of the trilogy suspects is roughly \$153 billion. This is far more than the market value of even the most successful global franchise. PepsiCo has a market capitalization of \$81 billion. Coca-Cola has a market 58

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59 capitalization of \$95 billion. WalMart is valued at \$222 billion. The aggregate market capitalization of the trilogy subjects is a mere 45% away from being equal to WalMart. In the hope that the imagination of the reader has been sufficiently stimulated, it is now possible to proceed to more analytical matters. The first point of interest is the toll gate fee. In modern parlance this is known as the cost per click (CPC). The toll gate CPC mechanism functions in the following manner. A potential seller of merchandise pays for a position in a Google or a Yahoo search. Each position in a search by a given keyword has a different cost per click. For example, if one quickly wished to know the CPC for the keywords “books”, one could visit the Overture.com website (Overture is owned by Yahoo). The specific web address is www.content.overture.com. On the right hand side of the page under the heading “tools” one then selects “View bid tools”. If one were to key in the term “books”, one will discover, perhaps surprisingly, that eBay is the high bid for books, at \$0.64 per click. Amazon.com and Barnes and Noble.com pay \$0.54 per click. It may interest the reader to know that Home Depot has purchased the 10th position under the books keyword at \$0.20 per click. WalMart has selected position number thirteen at \$0.19 per click. An interested reader could retrieve the same data

from Google Adwords. However, Google requires a \$5.00 account activation fee. The Overture.com bid rate tool is free.

It should be immediately apparent that the exercise of indefatigable industry on the part of Google or Yahoo could result in a very large advertising bill if Google and Yahoo actually obtain a large number of clicks.

For instance, let us make use of a keyword with which the reader is undoubtedly quite conversant. The keyword is "mutual funds". It is important to observe at this point that the cost per click can vary by date or even time of day. Consequently, on Saturday, October 16, 2004 the cost per click for given position rankings for the keyword mutual funds on Overture.com were as follows: Position Advertisers CPC

1 T. Rowe Price \$2.07 2 Fidelity \$2.06 3 Schwab \$2.05 4 Vanguard \$1.75 5 American Express \$1.01

Interestingly, on Sunday, October 17, 2004, the CPC and position rankings had changed as follows:

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60 Position Advertisers CPC

1 American Express \$2.36 2 Schwab \$2.34 3 T. Rowe Price \$2.34 4 Fidelity \$2.33 5 Vanguard \$1.75

According to the CIA World Fact book, the United States had 166 million internet users as of the end of 2003. Let us suppose that on each day 100 million such users wished to check either prices of mutual funds owned or some other data element and routinely were guided to this point by a search engine. If the average CPC was \$2.34 (rounded to the nearest penny), as was the case on Sunday October 17, 2004, the resulting daily bill for the industry would be 100 million x \$2.23, or \$223 million. Over the course of a year this would be \$81.395 billion. Of course, this assumes that each user only visits one site. If each user chose to visit five financial sites, the resulting bill would be \$406.98 billion.

In order to place this last number in its proper context it should be sufficient to observe that McCann Erickson, the subsidiary of the advertising agency Interpublic Group calculates that world advertising expenditures in 2003 were \$469.8 billion.

In actuality this circumstance could never happen. This is because it is common practice to place limitations on search engine clicks in the form of daily budgets. Indeed, the search engine tools are sufficiently advanced so that users can control cost per click. For example, Google operates an Adwords Discounter that automatically monitors the competition for a given advertiser and reduces the CPC to one cent above competitors in the case where a given advertiser is higher than competitors.

Another automatic way to control cost is by advertisement relevance. If the advertisement has a high position and yet it does not attract clicks, it will by definition have a lower click through rate. Rank is not solely determined by the bid CPC. It is actually an algorithmic combination of CPC and click through rate. Thus, a low click through rate will result in downward movement in rank. This prevents so-called "squatting" in the top positions by irrelevant users.

The most significant way of controlling the cost as well as the effectiveness of Internet advertising is by the use of either negative keywords or negative geography. For example, if a perfume dealer did not carry the Givenchy brand, the term "Givenchy" would be a means of excluding that dealer from search results if the user performed a search for Givenchy perfume. Similarly, a mutual fund group not registered in Oklahoma would not appear in the search results of Oklahoma residents.

The use of negative keywords and restrictive phrases is part of the effort to improve the conversion rate. This is the rate at which clicks become sales. This effort creates an interesting conflict of interest between the search engine firms and the advertisers. In traditional advertising, one pays for audience reach or that which is sometimes known as impressions. One will pay X 60

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61 dollars for a 30 second television advertisement that will be viewed by Y people. The concept of conversion rate still exists. In this case, it is known as the response rate. A certain percentage of the people who view the advertisement will buy the product. This can be measured by comparing sales with and without advertising. The cost of advertising or effectiveness can be controlled to a degree by the proper selection of an audience demographic and geography.

In the case of Internet advertising as opposed to television or print advertising, the advertiser can exclude certain Internet users or clicks based upon a variety of criteria. As the advertisers learn more about the audience, these advertisers should, in principle, be able to improve the conversion rate by excluding clicks that are unlikely to result in sales. In contrast with traditional advertising, Internet advertising can be made more effective (a higher conversion rate) by excluding clicks. The exclusion of clicks will reduce revenue to the search engines.

It could be argued that if the advertisers can improve their conversion rates, they should be

willing to pay more for each click or pay higher CPC's. This is arithmetically false as is evident from the following theoretical exercise.

Assume the existence of a manufacturer of a product with a unit cost of \$10 and a 10% net profit margin, or a profit of \$1 per each item sold. Further assume a CPC of \$0.10. The conversion rate will be, in this case, assumed to be 10%.

If there were 10,000 clicks, the profit for the manufacturer as well as the return on the Internet advertising investment could be calculated as follows:

Cost of Ad Program: 10,000 clicks at \$0.10 CPC = \$1,000

Profit on Sales 10,000 clicks x 10% conversion rate = 1,000 units sold 10,000 units sold at \$10/unit = \$10,000 revenue \$10,000 revenue x 10% net profit margin = \$1,000 profit

Profit on Ad Program

\$1,000 sales profit minus \$1,000 advertising cost = \$0 Return on \$1,000 investment = Breakeven, or 0%

Let us now assume that through the use of negative keywords, the manufacturer can increase the conversion rate to 20% by reducing the number of clicks to 5,000. There will still be 1,000 unit sales for a 20% click conversion rate. Therefore the cost of the Internet advertising campaign will decline to \$500 (5,000 clicks x \$0.10 CPC). Return on investment will now be \$1,000 profit in relation to advertising investment of \$500, or 200%. Note that in this case the search engine company simply endures a 50% erosion in revenue. The manufacturer has no incentive to pay more per click since this will not improve the profit position. Thus, if the manufacturer agrees to pay 20¢ per click the return on advertising investment would once again be zero.

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62 Fertile minds will respond that a competitor will ultimately observe this circumstance and be willing to pay \$0.15 per click once the negative keywords are known. Yet this only restores some lost revenue to the search engine. It does not result in growth. Thus, 5,000 clicks at \$0.15 per click is only \$850, not the \$1,000 previously received.

It is therefore the case that the search engine firms experience two dynamic variables. The first is growth in searches and pay-for-search advertising that increases revenue; the other is conversion rate improvements on the part of advertisers that serve to lower revenue. It must be true that the advertisers first need to engage in advertising by paid search and then gather data for use in ultimately increasing conversion rate and cost of search. Thus, revenue for the search engine companies must grow in order that data be available to be used in efforts to possibly reduce that revenue.

This is only one aspect of the search engine business. Another aspect is known as Traffic Acquisition Cost. This is the payments that Google and Yahoo make to the owners of web sites for directing their users to undertake searches with Google and Yahoo. For example, the Washington Post or Weather.com websites conduct searches powered by Google. Consequently, if one were using the Weather.com site and noticed that it will rain for much of the next 10 days, one might wish to purchase a new raincoat. If the reader were to search for raincoats, the search would be directed by Google software. A click on a given link would then result in revenue for Google which would be shared with Weather.com in this instance.

One of the elements that a buyer of Google or Yahoo shares should seriously consider is the ability of these generators of traffic to demand higher Traffic Acquisition Fees. If the providers of this traffic were to have such an ability, then the profit margin of Google and Yahoo would necessarily decline. In principle, if a given provider of traffic has sufficient volume, it could support a search engine on its website and charge advertisers for paid searches. In this regard, traffic acquisition cost as a percentage of Google revenue has exhibited the following historical trend:

Google Cost of Revenue as a Percentage of Revenue (Traffic Acquisition Costs are the Primary Cost of Revenue) 1999-2004 (See Google IPO Prospectus Page 57 Section: Cost of Revenues)

1999 412.7% 2000 31.8% 2001 16.5% 2002 29.9% 2003 42.7% 6 months 2004 47.5%

A high profit margin assumption is critical to a continued high Google share price valuation. Margin erosion caused by the continued demands for a greater share of revenue by traffic providers could ultimately destroy this valuation. The question is whether Google and Yahoo have a proprietary position in the Microsoft sense of the term or whether these firms are merely

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63 traffic intermediaries in the sense of the distribution terms that exist in all industries. In the latter case, the profit margins will ultimately prove to be very low.

In this connection, one might observe that paid search is actually a form of consumer education. If Fidelity, Schwab and Vanguard continually generate many millions of clicks with regard to Mutual Fund searches, then surely consumers should ultimately learn to "bookmark"

these web locations or otherwise learn to access these directly. This would eliminate the need for a search and therefore eliminate the fees that Google and Yahoo generate per click during paid searches. In any case, it is surely unreasonable to believe that consumers will continue to access the most popular websites in each product category by search engine as opposed to direct entry. Consequently, the most popular sites should eventually discover that more traffic is generated outside of the search engines and payment for clicks should not grow or even markedly decline. None of this is believed by the growing number of Google and Yahoo shareholders. These investors believe, as is evident from the Google and Yahoo valuations, that these two firms have virtually limitless growth possibilities. In order to test this presumption, let us suppose that Google and Yahoo were to grow at only a "conservative" 30% per annum for each of the next 20 years. It should be noted that the Google and Yahoo proponents anticipate higher rates of growth. The logical consequence of such a modest assumption might astonish investors. As a means of testing the reasonableness of such an assumption, one might select as a point of departure the global advertising expenditures as these currently exist and have existed in the past 14 years. The Universal McCann Insider's Report of McCann Erickson calculates that 2003 global advertising revenue was \$469.8 billion. The compound annual growth rate of advertising spending around the globe has been 4.1% as is evident from the accompanying table:

Worldwide Advertising Expenditures 1990-2003 (Source: McCann Erickson) 1990 \$275.9 1991 282.3 1992 299.2 1993 304.2 1994 332.0 1995 371.0 1996 390.2 1997 401.3 1998 411.9 1999 436.1 2000 474.3 2001 440.9 2002 450.5 2003 469.8

Advertising expenditure is expected to rise to \$489.4 billion in 2004. If compounding continues at the current rate, world advertising expenditure should be \$1.107 trillion in 2024. Google and Yahoo together are expected to record \$6.6 billion of revenue in 2004. If one were to assume that 63

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64 these two firms were to collectively grow by 30 % per annum for the next 20 years, then the revenue of those two firms in the aggregate would be \$1.254 trillion in 2024. This would be more than the global projected advertising budget. Of course, one could presume that Internet advertising is so effective that global advertising will simply grow at a rate much greater than the historical average. Yet, one might also say that if Internet advertising is indeed so much more productive than conventional advertising, companies will simply shift dollars from conventional advertising to do Internet Advertising.

One interesting way of viewing the estimated \$1.254 trillion Google/Yahoo aggregate revenue for 2024 is that this advertising spending is 4.6x the total volume of merchandise of WalMart as it currently exists. In fact, it is roughly twice as high as the aggregate current revenues of the following companies: Revenues of Large Companies That Might Advertise On the Internet Microsoft \$36.8 billion Coca-cola \$21.9 Pepsi \$28.5 MMM \$19.3 Target \$49.0 Home Depot \$69.2 Lowe's \$33.7 Dell \$45.7 Hewlett \$78.4 Apple \$8.3 Disney \$30.3 Starbucks \$4.9 Heinz \$8.5 Coach \$1.3 Tiffany \$2.1 Sony \$71.4 Palm \$1.1 Nokia \$37.2 NY Times \$3.3 MGM Mirage \$4.1 Royal Caribbean Cruises \$4.2 Procter & Gamble \$51.4 LM Ericsson \$17.5 Amazon.com \$6.0 eBay \$2.7 \$636.5 billion

Readers who do not question the reasonableness of a \$1.254 trillion 2024 revenue forecast for Google and Yahoo might try one of the following exercises. First, attempt to compose a list of companies that might currently generate \$1.254 trillion of revenue and might benefit from 64

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65 Internet advertising. It is rather difficult. In any case, it is pointless since obviously companies will not spend their entire revenue on advertising.

If the reader still does not question the reasonableness of the Google/Yahoo 2024 revenue estimate, the following exercise might promote some healthy agnosticism. Simply access the bid rate tool at www.content.overture.com. Select "view bids tools". Then have ready a copy of the Oxford English Dictionary. Begin at "A" and type each noun in alphabetical order into the view bid tool as these appear in the dictionary. It will be exceedingly difficult to find very many nouns, if any, that have not attracted a bid rate. If still not convinced, readers are advised to obtain copies of the Larousse French Dictionary, the Langenscheidt German Dictionary, the Mondadori Italian Dictionary, Cassell's Spanish Dictionary, the Kodansha Furigana Japanese Dictionary, the Oxford Russian Dictionary, the 1,726 page English Chinese Pinyin Dictionary, Ben Yehuda's Hebrew Dictionary and the Hippocrene Practical Arabic Dictionary. Be patient and type in nouns or key phrases if one is conversant in those languages. Do not doubt that agnosticism will eventually overwhelm the reader. 65

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November 3, 2004

I – A Curious Transaction

On September 30, 2004, there occurred one of the more bizarre investment banking transactions in recent memory. A company known as Calpine (NYSE-CPN) successfully priced a \$736 million unsecured convertible note issue. The notes were offered at 83.9% of par value. The notes are convertible into common shares at a price of \$3.85 and this represented a 23% conversion premium on the day of pricing. The notes are due in 2014. The interest rate is 6% except that the notes will not pay interest in years three, four and five. As a means of recompense, the accreted value of the notes will rise to par during that period. Upon conversion, or perhaps more properly stated, in the event of conversion, the company will deliver par or accreted value in cash. Share will only be issued to cover value, if any, that exceed par. The yield to maturity of the notes was 6% on pricing date. The reason that this transaction should be considered to be rather bizarre is that Calpine is a CCC credit. Many believe that Calpine is at serious risk of default. Yet, this is only 197 basis points more than the rate at which the United States Treasury is able to borrow for ten years. One might be tempted to say that it is evident that the conversion privileges of Calpine have considerable value. This view is undoubtedly correct. Yet the matter is far more complicated than is readily apparent. This is because at the time that the notes were being priced, Calpine entered into a ten year Share Lending Agreement with Deutsche Bank. This latter institution serves as the investment banker of the former entity. Calpine is lending Deutsche Bank 89 million of its own shares. The shares are to be used to facilitate short sales of Calpine stock so that investors in its bonds can hedge their transactions. Generally speaking, companies do not encourage the short sale of their own stock. In this particular case, Calpine sees fit to depart from tradition, and with good reason, as will be demonstrated shortly. Calpine does not anticipate that the borrowed shares will be considered to be issued and outstanding from an accounting perspective. It is presumed that this is because the borrowed shares will be ultimately returned to the lender. Thus, shares will be sold short, that don't actually exist, to hedge the purchase of convertible bonds. Yet, viewed from a certain perspective, this is a very reasonable sort of undertaking. One purchases a bond with a low coupon that has considerable risk of default. Simultaneously, one sells short sufficient shares to cover expected losses in the event of default. If the hedge

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ratio is properly established, one can profit in any scenario. For example, if the bonds do indeed default, these might have some value in bankruptcy. Yet the share will be worthless. Consequently, the profit in the stock short sale could exceed the loss in bonds and a profit will be earned. This profit will be enhanced since, until default, the hedge will enjoy interest income on the bonds and a small amount, such as it is, of rebate income on the short sale. Incidentally, those who are interested in an alternative assessment of the Calpine level of credit worthiness should study the Calpine 4 3/4% convertible unsecured notes due on November 15, 2023. These are puttable to the company at par on November 15, 2009. The yield to put is approximately 16% although the time to put (maturity so to speak) is roughly five years sooner than the 2014 notes. The conversion premium of the 4 3/4% notes is approximately 61%. In the weeks subsequent to the offering of the 6% 2014 notes, the Calpine stock has substantially declined in value. The conversion premium of the 2014 notes is now roughly 29% and these notes still trade at a price that is slightly above the original offering price of 83.9% of par value. Given the default risk of Calpine, the clearing price of the 2023 notes reflects a reasonable 16% yield to put. However, irrespective of the poor credit quality of Calpine, if one can find something even worse than Calpine notes (namely Calpine stock) that can be sold short, one would be willing in effect to provide financing to the company on very advantageous terms. It is interesting to observe in this connection that although the company might pay 6% per annum for capital, the provider of that capital might not actually earn 6%. This is because the provider of capital must commit an amount to Calpine upon which 6% will be earned as well as a given amount to be sold short in Calpine shares. This latter sum, pro forma, would only earn the short seller's rebate of, at most, 1%. Consequently, if a hedge transaction results in the purchase of \$1,000 face amount of bonds and short parity of the stock, the transaction would appear as follows: \$1,000 face amount of bonds \$839 cost, earning 6% = \$60 income Parity, or \$682.11, Calpine stock sold short Earning 1% = \$6.82 income

Total capital committed = \$1,521.11

Total income generated = \$66.82

Comprehensive yield = 4.39%

Naturally, this does not represent the profit or loss on the actual transaction. The profit or loss is obviously dependent upon the price movement in Calpine bonds and shares. However, Calpine does pay 6% for capital and the provider of capital does receive 4.39%. If Deutsche Bank pays Calpine a rate of interest for the loan of 89 million shares, the Calpine cost of capital will actually be lower than 6%. The constraint upon Calpine in its capital raising activity was the inability to borrow its shares.

Prior to the consummation of this particular transaction, the short interest in Calpine was roughly 130 million shares. As of the most recent reckoning dated October 26, 2004, the short interest on Calpine share has expanded to a total of 197.6 million shares. 67

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68 The essential principle in all of this is that one need not offer to the public a company with a good business plan or even favorable prospects. The instrument in question may even be very overvalued. All that is required is that there exist something or anything that can be sold short against this investment so that the risk can be passed to another investor. One might have thought that there should exist some functional limit to the number of shares that can be borrowed for the purpose of short sale. Apparently the only limitation is the creativity of the human intellect to devise ever newer means of creating capacity. Incidentally, this particular technique of creating capacity in a crowded environment can trace its origins to a technique practiced by the great showman, entrepreneur and deceiver of the public known as P.T. Barnum.

Non-American readers of this essay might not readily appreciate or understand this reference. American readers might find the Barnum reference amusing. Limitations of space prohibit the inclusion of more Barnum detail at this point in the essay. Interested readers are referred to the Appendix for more information.

II – Gresham's Law The term Gresham's Law is attributed to Sir Thomas Gresham (1519 – 1579). The law states that good money is driven from circulation by bad money. In the language of the times good money referred to money that retained its precious metal content. Bad money was money that had been "clipped". That is to say that some gold or silver had been shaved from the coin. The observation was that people would hoard those coins that had not been clipped and use in exchange those that had been clipped. The concept of Gresham's Law can also be applied to economies that permit the co-terminus existence of paper or fiat money and gold as currency. The premise is that the bad or paper money will drive the good or gold money from circulation. Some modern economists have attempted to refute the validity of Gresham's Law as it is supposedly based upon a false premise. The premise is that the buyer of goods and services has the choice of which currency, good or bad, to present in payment and that either will be accepted.

It is argued by the critics that the sellers of goods and services might just as easily refuse to accept bad money in payment so that good money will drive out bad money. This is merely an extension of the efficient market hypothesis that informed traders will insist upon fair value for their money or products. Indeed, there are examples of the reverse of Gresham's Law in operation in certain periods of history ¹.

The relevance of Gresham's Law in the current connection is whether or not there exists some sort of mechanism whereby so-called "bad" securities could drive "good" securities from circulation. Let us return to the example of Calpine. If it is possible for Calpine to issue securities at an advantageous cost of capital, certainly it will do so. Investors will make such an activity possible if there is the possibility of profit. That which makes this activity possibly profitable is the belief that the Calpine equity is worse than the Calpine debt. That is to say that the Calpine equity should decline more than the Calpine debt. In order to decline, surely it must have some flaw. The greater this flaw, the more the equity will decline. Naturally, a greater decline potential makes Calpine more desirable for the purposes of this type of transaction.

1 Of course, there are also examples of Gresham's Law in operation. 68

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69 Then, if such types of transactions are to be desired, this is logically equivalent to the statement that there is demand. If demand exists, then Calpine can readily provide supply². III - Paradoxes One of the interesting aspects of the Calpine transaction is that it is, at least in principle, a riskless transaction. One is long a convertible bond, one is short stock. If the stock should rise in value, it is almost certain that the bonds will rise in value. Conversely, if the bonds decline in value, it is quite likely that this is because the shares have declined in value. Since the bond is a legal claim, such as it is given the unsecured nature of the claim, and as the share has no such claim, the bonds might retain some value while the share retain no value. Thus, a profit would be earned on a bad company with a bad business plan and the investor would have no risk. If the reader finds this to be somewhat paradoxical, the following element is even more paradoxical. Calpine is a highly leveraged company. Much of its debt is high cost capital. Yet, all things being equal, if it can raise capital at a lower cost than its existing capital, it can eliminate those securities and replace these with more favorable issues. If it can thus lower its cost of capital, it will improve its interest coverage ratios, its cash flow and ultimately improve its earnings. If it can thus improve its earnings, its stock would then appreciate. Of course, if it was believed that the shares would appreciate investors might not undertake the type of transaction that is described in these pages. In any case, if Calpine could repeat its transaction for its 6% unsecured Convertible Debentures a sufficient number of times it would not be a bad credit but presumably become a good credit or at least a much better credit than is currently the case. If the transaction were to be replicated many times, then Calpine would need to lend to Deutsche Bank or some other investment banker many more shares to be shorted so that "riskless" convertible arbitrage transactions could be consummated. None of these shares would actually exist insofar as the Calpine shares outstanding is concerned. Yet, the short interest would dramatically expand beyond the current level of 197.6 million shares. These amount is equivalent to 44.8% of the company share float. It is equivalent to 19 days of trading volume at

the current volume rate. The intriguing aspect of this transaction is, of course, that the company can actually improve its position in a fundamental sense by being perceived as a problematic enterprise. This is surely a paradox. It is an example of that sort of paradox known as the Epimenides Paradox. An example of this sort of paradox is the following: The following sentence is false. The preceding sentence is true. 3

2 Charles Kindleberger – Historical Economics (Los Angeles: University of California Press, 1990) P. 307 3 More on paradoxes of this type can be found in Douglas Hofstadter – Godel, Escher and Bach: An Eternal Golden Braid (New York: Basic Books 1999 ed.) p. 21 69

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70 A bad company can make itself good by being bad. One of the problems with this type of analysis is that concepts such as good and bad or even profit and loss are somewhat subjective. If the company has losses and because of this the shares decline, the hedged purchasers of its convertible bonds would actually have a profit since the short sale component of the trade will have increased in value. Thus, the company should be provided low cost funds to be misused, since investors will profit. On the surface, this is an example of the so-called “Grellings Paradox”. In the Grellings paradox the problem occurs in categorizing between two terms of that are supposed to embrace all possible types. For example, a term can be either autological (self-descriptive) or heterological (non-self descriptive). Is “heterological” heterological?⁴ Can Calpine improve its capital structure and thereby reduce its risk and simultaneously provide its financiers with a riskless transaction? This is clearly impossible. Although, it is possible that Calpine can reduce its risk and improve its capital structure. It is also possible that the convertible arbitrageurs take little or even no risk. The risk is merely transferred. It is no more than obvious that if Calpine shares are sold short, these must be purchased by some other party. If Calpine improves its fundamental position, its shares may well increase in value. Conversely, if the short sellers and convertible arbitrageurs profit the long investors must realize losses. Calpine is merely transferring its balance sheet risk into the hands of long investors. Ultimately, as hedge transactions expand, there must be a greater fool. The mechanism is impossible without the existence of this fellow. That this fellow obviously does exist is evident from the fact that Calpine is only one of many bizarre hedge transactions. The basic principles of the free enterprise system entail that proper utilization of resources is rewarded. If it is possible to reward improper utilization of resources the system loses animating principle. The British writer G.M. Chesterton had a phrase for this state of affairs. He called it the “Utopia of Users.” EBay, Google, Travelzoo and Much Else There exist some misguided analysts, fortunately very few in number, that regard EBay as a preposterously overvalued equity. Nonetheless, it can be purchased with the assurance of profit if one were to simultaneously short the appropriate quantity of the shares of Yahoo. One might well ask why anyone would then undertake to purchase Yahoo, since it should be well understood that its shares are even more ludicrously valued than those of EBay. Such an undertaking can be justified upon rational economic grounds if one were to simultaneously short shares of Amazon.com. Indeed, almost any investment is justifiable, irrespective of the madness of its valuation if only there exist investors yet more mad who would accommodate an offsetting short sale. For instance, Google closed trading on Friday October 28, 2004 at a price of \$190.64. Its market capitalization was at least \$51.6 billion. One must use the qualification of “at least” since current accounting regulations prohibit Google from listing as outstanding shares that have

4 Ibid, p. 21 70

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71 been already granted to employees but not yet vested. It is reasonable to presume, though, that, in the fullness of time, the Google shares outstanding figure will increase. In any case, one could purchase Google with confidence if one were equally confident that Ask Jeeves is improperly priced relative to Google. None of this should, in principle, inhibit buyers of Ask Jeeves since it is quite possible that Overstock.com is inflated in valuation in relation to Ask Jeeves. The chain can be continued almost indefinitely with such issues as Stamps.com, 1-800-Flowers.com, Jupiter Media, Novatel Wireless, Viewpoint, SINA, SOHU, Netease, Findwhat.com, Monster.com, Internet Security Systems, CTRIP.com, Web Methods, Digital Insight, Alloy, Roxio, Infospace and Travelzoo. It can be asserted with justification that the long/short style of trading is merely an example of a free capital market function that will limit valuation extremes. The problem with the technique is that the mere existence of valuation extremes can change the risk profile of companies if these can be exploited to raise capital. A fascinating example of this is the strange case of Travelzoo.

This was an internet company with no traffic and therefore no revenue. The company therefore decided to give shares away to approximately 700,000 people who registered on the Travelzoo website. (See Travelzoo Form 10-K 2003). This created traffic which, in turn, created revenue which, in turn, created equity value. Naturally, the company has raised equity capital which, all other things equal, makes the shares less preposterously valued than was previously the case. In the past twelve months, the shares have increased in value by 765%. Moreover, companies can now improve their fortunes by the issuance of shares directly to employees in lieu of cash

payments. Many people believe that the value of this share or option payment is too difficult to calculate and hence no expense for this purpose should be tabulated on the company income statement. Whether or not this is a reasonable position is better left to accounting professionals. However, it cannot be denied that large companies that issue stock to employees are in a position to command substantial human resources. The following ten firms customarily pay employees in stock or options: Users of Equity Compensation Market Capitalization Yahoo \$ 50 billion Google 51 EBay 68 Qualcom 68 USA Interactive 15 Amazon.com 14 Cisco 128 Intel 143 Microsoft 304 Dell 88 \$ 929 billion

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72 If these companies were inclined to issue as much as 5% of their aggregate market capitalizations each year to employees, their combined purchasing power would be almost \$46 billion. At the rate of \$80,000 per individual, or twice the annual household income of the typical American family, these firms could collectively employ 575,000 persons or roughly 48% of the current employment of WalMart. View ed upon a more global scale, the average global per capita income is only \$7,500. Upon this scale, these companies could collectively employ 6,133,333 persons. In China, per capita GDP is only \$4,563 per person ac cording to the CIA World Fact book, so that the labor of 10,081,087 people can be purchased. In India theses shares would have far more purchas ing power since the average per capita GDP is only \$2,537 according to the CIA World fact book. A figure of 5% of aggregate ma rket capitalization of these ten firms would purchase the services of 18,131,651 persons.

Consequently, pairs trading should, in theory, moderate the valuation extremes created by market inefficiencies. However, if the companies can take advantage of these inefficiencies and either raise the capital or use shares to pay for services, theses may ultimately validate what were thought to be inefficiencies. The most overvalued would be those with the cheapest source of capital and therefore the most dangerous short sales. As a result, the companies could effectively transfer risk not to shareholders , but actually to short sellers. The potential liabilities of this latter group are theoretically infinite. Perhaps one day there will be a capital market crisis caused not by fools that are selling but fools that are buying. In any case, if hedge funds wish to exploit the actions of fools in a more or less riskless fashion, it is n ecessary that the fools ultimately cooperate by realizing that they are fools and sell ove rvalued shares. Yet, if the greater fool only becomes an even greater fool and continues to purchase overvalued shares, a crisis of a different sort might commence. V - Calpine and Barnum The animating principle of the aforementioned Calpine transaction is that profits can be created from poor investments that have perhaps greater return potential and lower risk than profits from good investments. In fact, even if the spread that could be earned from the Calpine transaction were rather low, since the trade theoretically does not involve substantive risk, the return can be appropriately enhanced with leverage, assuming that the Calpine convertible arbitrage spread is greater than the available cost of borrowing. In history, very few have been successful in changing deficiencies into applause. One of those was P.T. Barnum. His personal technique or an example thereof is revealed in Appendix 2.

Barnum himself had numerous techniques that could be used to attract the interest of the public.

One of those was to hang a double sided sign from a building w ith the following inscription facing the street:

PLEASE DO NOT READ THE OTHER SIDE OF THIS SIGN! Naturally this is exactly what virtually everyone proceeded to do. Therefore, dear reader, please do not read the Appendix. 72

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73 Appendix 1 Barnum and Crowds American readers will recall P.T. Barnum as the man famous for the quotation of “there’s a sucker born every minute”. According to Barnum himself the actual quotation was that “there is a customer born every minute”. Non-American readers might not be familiar with Am rican. P.T. Barnum was arguably the most famous Am erican entertainment entrepreneur of the 19 th century. He operated a museum of curiosities (today these would be regarded as monstrosities) known as the American Museum. It was located in New York City at Broadway and Vesey Street. Later in life it was he that founded th e famous Barnum Circus. Among other events, it was Barnum that arranged for the midget Tom Thumb to perform before Queen Victoria of Great Britain. In order to illustrate the personality of Barnum one might commence with the story of his first exhibit. This was a woman purported to

be George Washington's nurse. The year was 1835 and it was asserted that this nurse was then 161 years old. This did attract a large audience for a time.

Yet, the allure of this exhibit did eventually dwindle. However, Barnum was not one to accept defeat in the aftermath of success. Consequently he began to write anonymous letters to the press that denounced his own exhibit as a complete fraud, and indeed, it was actually a fraud. Nonetheless, the accusations did once again attract a crowd. Now people wished to determine whether or not the exhibit was genuine. Barnum made a fortune! The object of the Calpine transaction that was the subject of this essay was to attract a crowd.

The failure to attract the interest of the public is almost always a lost opportunity, as Barnum was to learn. The following excerpt is as it appears in the "complete" two volume 1871 edition of the Barnum memoirs entitled "Struggles and Triumphs". The autobiography, although quite obviously designed to be self-serving is one of the more delightful autobiographies ever to be written by a scoundrel. The author must have greatly enjoyed his life, as the following passage will illustrate. "On that fourth of July, at one o'clock, p.m., my museum was so densely crowded that we could admit no more visitors, and we were compelled to stop the sale of tickets. I pushed through the throng until I reached the roof of the building, hoping to find room for a few more, but it was in vain. Looking down the street, it was a sad sight to see the thousands of people who stood ready with their money to enter the museum, but who were actually turned away. It was exceedingly harrowing to my feelings. Rushing down stairs, I told my carpenter and his assistants to cut through the partition and floor in the rear and to put in a temporary flight of stairs so as to let out people by that egress into Ann Street.

By three o'clock the egress was opened and a few people were passed down the new stairs, while a corresponding number came in at the front. But I lost a large amount of money that day by not having sufficiently estimated the value of my own advertising, and consequently not having provided for the thousands who had read my announcements and seen my outside show, and had taken the first

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74 leisure day to visit the museum. I had learned one lesson, however, and that was to have the egress ready on future holidays. Early the following March, I received notice from some of the Irish population that they meant to visit me in great numbers on St. Patrick's Day in the morning.

"All right" said I to my carpenter, "get your egress ready for March 17", and I added, to my assistant manager: "If there is much of a crowd, don't let a single person pass out at the front, even if it were St. Patrick himself; put every man out through the egress in the rear". The day came, and before noon we were caught in the same dilemma as we were on the 4th of July; the museum was jammed and the sale of tickets was stopped. I went to the egress and asked the sentinel how many hundreds had passed out? "Hundred," he replied, "why only three persons have gone out by this way and they came back saying that it was a mistake and begging to be let in again". "What does this mean?" I inquired; "surely thousands of people have been all over the museum since they came in." "Certainly," was the reply, "but after they have gone from one saloon to another and have been on every floor, even to the roof, they come down and travel the same route over again". At this time I espied a tall Irish woman with two good sized children whom I had happened to notice when they came in early in the morning. "Step this way, Madame," said I politely, "you will never be able to get into the street by the front door without crushing these dear children. We have opened a large egress here and you can pass by these rear stairs into Ann Street and thus avoid all danger". "Sure," replied the woman, indignantly, "and I'm not going out at all, nor the children either, see, for we have brought our dinner and we are going to stay all day". Further investigation showed that pretty much all of my visitors had brought dinners with the evident intention of literally "making a day of it".

No one expected to go home till night; the building was overcrowded, and meanwhile hundreds were waiting at the front entrance to get in when they could. In despair I sauntered upon the stage behind the scenes, biting my lips with vexation, when I happened to see a scene painter at work and a happy thought struck me: "here" I exclaimed, "take a piece of canvas four feet square, and paint on it, as soon as you can, in large letters

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75 Seizing his brush he finished the sign in fifteen minutes, and I directed the carpenter to nail it over the door leading to the back stairs. He did so, and as the crowd, after making the entire tour of the establishment came poring down the main stairs from the third story, they stopped and looked at the new sign, while some of them read audibly: "To the Aigress (sic)." "The Aigress" said others, "sure that's an animal we haven't seen," and the throng began to pour down the back stairs only to find that the "aigress" was the elephant, and that the elephant was out of doors or so much of it as began with Ann Street. Meanwhile, I began to accommodate those who had long been waiting with their money at the Broadway entrance. 5 Barnum was a very serious student of human behavior. If the reader has reached this point, it may well be because of the success of the Barnum sign trick referenced at the end to the actual essay section. Readers interested in the sign trick and other Barnum devices are advised to consult another of the Barnum books. This is entitled "Dollars and Sense or How to Get On: The Whole Secret In a

Nutshell". The original work was published in 1890. Another of Barnum's works is entitled "Humbugs of the World" (1865). It is a veritable encyclopedia of schemes used to defraud the gullible and greedy. Barnum was evidently a student of financial manias. In this work the reader will find descriptions of the Dutch tulipmania of the 17th century, the Mississippi Bubble as well as the South Sea Bubble. The books should be required reading for serious investors.

5 P.T. Barnum – Struggles and Triumphs edited ed. (NY Penguin Publishers, 1981) p. 13 – P.T. Barnum – Struggles and Triumphs – Reprint of the 1871 edition Volume 1 (Kessinger Publishing Reprint of p. 138-140) 75

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76 Appendix 2 How to Deceive to the Delight of the Audience: Barnum Style Barnum relates another adventure in deception of the crowd in his autobiography entitled "Struggles and Triumphs." This passage relates to Barnum competing with someone who had a rival exhibition in Mesmerism. The passage is as it appears in the reprint of the 1871 edition. It devolved upon me to open a rival mesmeric performance, and accordingly I engaged a bright little girl who was exceedingly susceptible to such mesmeric influences as I could induce. That is, she learned the lesson thoroughly, and when I had apparently put her to sleep with a few passes and stood behind her, she seemed to be duly "impressed" as I desired; raised her hands as I willed; fell from her chair to the floor; and if I put candy or tobacco into my mouth, she was duly delighted or disgusted. She never failed in these routine performances.

Strange to say, believers in mesmerism used to witness the performances with the greatest pleasure and adduce then as positive proofs that there was something in mesmerism, and they applauded tremendously – up to a certain point.

That point was reached, when leaving the girl "asleep," I called up someone in the audience, promising to put him "in the same state" within five minutes, or forfeit fifty dollars. Of course, all my "passes" would not put any man in the mesmeric state; at the end of three minutes he was as whole awake as ever. "Never mind" I would say, looking at my watch; "I have two minutes more, and meantime, to show that a person in this state is utterly insensible to pain, I propose to cut off one of the fingers of the little girl who is still asleep." I would then take out my knife and feel of the edge, and when I turned around to the girl when I left on the chair she had fled behind the scenes to the intense amusement of the greater part of the audience and to the amazement of the mesmerists who were present. "Why! Where's my little girl?" I asked with feigned astonishment. "Oh! She ran away when you began to talk about cutting off fingers." "Then she was whole awake, was she?" "Of course she was, all the time." 76

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77 "I suppose so; and, my dear sir, I promised that you should be in the same state at the end of five minutes, and as I believe you are so, I do not forfeit my fifty dollars." 6

Calpine is creating an enormous short position in its own shares while using its ability to raise capital from the arbitrage community to improve its own balance sheet. If it can continue this practice for a sufficient number of iterations, it will improve its balance sheet and hence ultimately improve its share price. It will be recalled that the buyers of the 6% convertible will receive cash for par value and shares only to the degree the bonds exceed par in value. It is to be presumed that the arbitrageurs will need to repurchase the shares with the cash received. Of course this will be possible. The question is only one of price. Calpine may well have created the possibility of profit without risk. It is not entirely clear which party has the possibility of profit and which party carries the risk. In this manner, the hedge fund industry is now creating a multitude of risk transference transactions. The necessary precondition is only a greater fool willing to bear the risk. The companies are also using these to either raise capital or to acquire resources. The hedge fund investors as well as the companies wish to play the role of Barnum. As the reader of the autobiography will undoubtedly discover, there can only be one Barnum.

53 P.T. Barnum – Struggles and Triumphs-Reprint of the 1871 edition Volume 1 (Kessinger Publishing Reprint) p157-158

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78 Appendix 3 Transaction Alternatives There will come a day when the Barnum books will be required reading for investment professionals. Anyone who searches for these books at the current time will soon realize that these are rather rare. This is particularly true of 19th century editions. A different sort of hedge fund transaction might be to buy several 19th century editions and hold these in storage for the day, perhaps soon, when these will be in demand and therefore command a much higher price than is currently the case. Perhaps they can then be sold on eBay. The value realization catalyst might well be the day that Calpine repays its \$736 million worth of unsecured convertible debentures in cash and its holders will then need to purchase almost \$700 million of stock to cover a short position on a riskless transaction. The shares will

no doubt be covered at some price. Barnum would be proud. Once again a crowd would throng about the “egre 78

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80 Benford's Law As A Tool For Investors

The law of numbers known as Benford's Law is unfamiliar to most investors. This is rather odd since, as will be discussed shortly, it should be of particular interest to short sellers. Benford's law essentially states that in tables of statistics, the digit 1 tends to occur with an unusually high probability. Any number must begin with one of nine possible digits. These are the numbers one to nine. It is believed incorrectly that each number, such as the number one, occurs with equal and random frequency. That is to say that one occurs 11.1% of the time or one out of nine times. Actually, studies show that in tables of statistics, the number one appears in the first position more than 30% of the time. Although this law is called Benford's law, the phenomenon was actually discovered by the 19th Century astronomer Simon Newcomb. Mr. Newcomb was, among other things, a self-taught mathematician. He made the observation that tables of logarithms were much more worn in the first few pages than in the last few pages. This seminal observation remained nothing other than a mathematical curiosity until it was formulated as a law of numbers in 1938 by a physicist known as Frank Benford. The law states that the distribution of first digits will, on average, follow the following pattern: 1 30.6% 2 18.5% 3 12.4% 4 9.4% 5 8.0% 6 6.4% 7 5.1% 8 4.9% 9 4.7% Total 100%

Thus, the digit one will appear in the first position 30.6% of the time. Benford found that the law was generally valid for such diverse and unrelated statistics as length of rivers, physical constants, populations, specific heat, black body radiation, atomic weight, molecular weight, cost data, addresses, exponential expansions, American League baseball statistics and newspaper lineage. The actual article can be found by interested readers in the Proceedings of the American Philosophical Society Vol. 78 p. 551-572, 1938. The title of the article is “The Law of Anomalous Numbers”. The utility of this law for the investment analyst should be obvious. Sets of financial data should, over the fullness of time, display more ones, twos and threes than eights, nines and

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© Horizon Research Group, 2004 Page 81 sevens. If this is not the case, then there is likely to be some sort of contrivance or probably even a fraud involved.

The reason why the law works is rather simple to envisage. Imagine a newly created firm with one million dollars of shareholders' equity. Let us assume a constant growth rate of 10% and complete reinvestment of earnings (or no dividend payments). The next ten years will exhibit the following progression of shareholders' equity. year 1 \$1,100,000 year 2 \$1,210,000 year 3 \$1,331,000 year 4 \$1,464,100 year 5 \$1,610,510 year 6 \$1,771,561 year 7 \$1,948,717 year 8 \$2,143,589 year 9 \$2,357,948 year 10 \$2,593,742

Eight years are required until two becomes the first digit. Let us assume that the growth rate were still 10% and that the firm commenced operations with two million dollars of shareholders' equity. The progression of shareholders' equity would then be as follows: year 1 \$2,200,000 year 2 \$2,420,000 year 3 \$2,662,000 year 4 \$2,928,000 year 5 \$3,221,000

A three appears in the first digit in year five. It will be discovered that only four years are required to attain four in the first digit position if the starting value is three million dollars. If the starting value were nine million dollars, a period of time slightly in excess of one year is required to reach ten million. At this point a one is once again in the first position and the phenomenon will repeat itself.

It should be observed that the law applies only to numbers that have a dimension such as net profit. The law would not apply to numbers expressed as a rate, such as tax rate since, obviously, the government could theoretically legislate that the tax rate remain constant for many years. The law would apply to numbers that theoretically compound, such as shareholders' equity or current liabilities. 81

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© Horizon Research Group, 2004 Page 82 Users of Benford's Law in its fraud detection incarnation claim that cheaters seem to have a great affinity for digits five and six. These are used much more frequently than the 8% or 6.4% expected occurrence as suggested by Benford's

Law.

If someone were to examine the net income figures for the NASDAQ 100 group of companies subsequent to 1999, it does not appear that, on a quarterly basis, the number one appears in the first digit 30.6% of the time. Of course, the author of this essay may have simply made a computational error. Alternatively, perhaps Benford's Law is not universally valid. The top twelve companies in the S&P 500 as of November 3, 2004 recorded 5 or 6 in the first position on annual income figures 22.5% of the time during the past 10 years (1994- 2003). This would be 27 observations of 120 data points. The Benford expected frequency is 14.4%. The digit three appears in the first position 19 times in these 120 data points or 15.8% of the time. The expected frequency is 12.4%. The digit 9 appears seven times in 120 data points for an actual observed frequency of 5.8%. The expected frequency is 4.7%. The digit one appears in 25 observations of 120 data points or 20.8% of the time. The expected frequency is 30.6%. Readers will be left to judge whether or not Mr. Benford made an illuminating or merely a coincidental observation in his 1938 paper. In any case, the mathematics community accepts Benford's Law as valid. The investment community has yet to be convinced. 82

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Mathematically inclined readers might prefer the formulaic statement of Benford's Law which is: Digit D appears as the first digit with the frequency proportional to: $\log_{10} (1+1/D)$

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© Horizon Research Group, 2004 Page 84 Zipf's Law as a Tool for Investors, with Brief Remarks upon Bradford's Law

Large collections of quantitative data frequently display similar numerical attributes. These similarities have often been elevated to the level of so-called laws by mathematicians. It is perhaps more correct to use the deliberately less precise term "statistical tendency."

Zipf's law states that the frequency of occurrence of a given event (p) is a function of rank (i). The law specifies that P i is approximated by $1 / i^a$, where a is an exponent close to unity or one. In formulaic terms:

P

$i \sim 1 / i^a$

The law can be explained in common terminology by stating that in a statistical distribution a variable for a given rank multiplied by that rank will generally yield a similar number. For example, let us consider a ranked statistic such as armed forces personnel among countries. According to the International Institute for Strategic Studies, national armed forces personnel in the year 2000 were as follows:

Rank Nation Number of Military Personnel

1 China 2,810,000 2 Russia 1,520,000 3 United States 1,366,000 4 India 1,303,000 5 South Korea 683,000 6 Pakistan 612,000 7 Turkey 610,000 8 Iran 513,000 9 Vietnam 484,000 10 Egypt 448,000 11 Ethiopia 352,000 12 Burma 344,000 13 Syria 316,000 14 Ukraine 304,000 15 Thailand 301,000 16 Indonesia 297,000 17 France 294,000 18 Brazil 288,000 19 Italy 251,000 20 Japan 237,000 84

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If you were to multiply rank by personnel for each country, one would then derive the Zipf values as follows: (e.g. Russia = 2(rank) x 1,520,000 = Zipf value 3,040,000).

Rank Nation Zipf Value

1 China 2,810,000 2 Russia 3,040,000 3 United States 4,098,000 4 India 5,212,000 5 South Korea 3,415,000 6 Pakistan 3,672,000 7 Turkey 4,270,000 8 Iran 4,096,000 9 Vietnam 4,350,000 10 Egypt 4,480,000 11 Ethiopia 3,872,000 12 Burma 4,128,000 13 Syria 4,108,000 14 Ukraine 4,256,000 15 Thailand 4,515,000 16 Indonesia 4,752,000 17 France 4,998,000 18 Brazil 5,184,000 19 Italy 4,769,000 20 Japan 4,740,000

A Zipf rank is a way of noticing an irregularity that might not be noticeable in a standard

frequency distribution or rank order. It does not appear unusual that Brazil has 288,000 military personnel. Yet, it does seem unusual that Brazil would have the second highest Zipf value after India. One might reasonably wonder whether Brazil is expanding its military with a view to threatening its neighbors. Actually, the reason is a difference in the definition of military personnel. The Brazilian Federal Police are considered to be paramilitary. The U. S. crude equivalents, such as Border Patrol, FBI, Treasury ATF and Customs police are considered to be civilian. France also has a high Zipf value because the National Gendarmerie are counted as military. Thus, the Zipf value provokes an inquiry whereas the mere fact that France has 294,000 armed forces personnel provokes no inquiry.

If one noticed that India happens to have the highest Zipf value, one might now suspect that the Indian armed forces personnel data includes some paramilitary forces and is not strictly comparable with some of the other data. Indeed, this would be a correct assumption. The figures for the Indian armed forces include numerous paramilitary and police units such as the Border Security Force, National Security Guard, Indo-Tibetan Border Police, Special 85

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© Horizon Research Group, 2004 Page 86 Frontier Police, Ladakh Scouts, Central Reserve Police Force, Central Industrial Security Force and the Railway Protection Force.

It is quite natural to assume that, when viewing a compilation of statistical data, the data is homogeneous. Yet, this is evidently not always true. An analyst must have a tool that can be used to generate pertinent inquiries about the nature of the data under examination.

Zipf's Law is such a tool.

The inventor of Zipf's Law was not a mathematician, but a philologist. He worked at Harvard University as a linguistics professor. It was he who studied the frequency of occurrence of certain words in English language texts. The three most common words were "the," "of," and "to." The occurrences of these words plotted against rank appears to follow Zipf's Law. It was this rank order plotting that generates the Zipf observation.

These observations were then applied to a diverse range of statistical phenomena and appeared to have a universal applicability. Some of these observations are contained in a 1935 book by Zipf entitled "Psycho-Biology of Languages." However, his seminal work formulating Zipf's Law is entitled "Human Behavior and the Principle of Least Effort," published in 1949. Interested readers might also consult "Selected Studies of the Principle of Relative Frequency in Language," published by Zipf in 1932. Readers are cautioned that these books are rather rare and not very easy to obtain, albeit obtainable.

Such concepts can be applied to security analysis. In principle, the various constituent parts of the S&P 500 are efficiently priced. Yet, the Zipf values expose differences among companies. Table A displays the consensus estimated net profit for the 20 most profitable S&P 100 firms in terms of anticipated 2005 net profit. A complete table of estimated 2005 profits for all S&P 100 companies can be found in Appendix A.

Consensus Estimated 2005 Net Income Ranked By Income

(amounts in billions of dollars)

Rank Company 2005 Net Profit

1 Citigroup \$22.8 2 Exxon \$21.8 3 General Electric \$18.9 4 Pfizer \$17.5 5 Bank of America \$16.3
6 Microsoft \$15.1 7 AIG \$13.6 8 JP Morgan Chase \$11.9 9 Wal-Mart \$11.6 (cont'd, next page)
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© Horizon Research Group, 2004 Page 87 10 SBC Communications \$10.9 11 Altria \$10.5 12 Johnson & Johnson \$9.9 13 Chevron \$9.7 14 IBM \$9.2 15 Wells Fargo \$7.8 16 Fannie Mae \$7.3
17 Procter & Gamble \$7.3 18 Verizon \$7.3 19 Intel \$7.0 20 Cisco \$6.7

The next step is to calculate Zipf numbers based upon rank and net income. A list of the Zipf numbers for the estimated 2005 net income for each of the companies in the S&P 100 can be found in Appendix B. The following table displays the 20 lowest Zipf values. It will be observed from even casual scrutiny of Appendix B that the median Zipf number is slightly in excess of 140.

20 Lowest Zipf Numbers Based Upon 2005 Consensus Estimates Net Income

Rank Company Zipf Number

1 Citigroup 22.8 2 Schering Plough 35.2 3 Exxon 43.6 4 General Electric 56.7 5 Yahoo 67.3 6
Pfizer 70.0 7 Bank of America 81.5 8 Comcast 88.2 9 Microsoft 90.6 10 JP Morgan Chase 95.2
11 AIG 95.2 12 eBay 102.2 13 Wal-Mart 104.4 14 SBC Communications 109.0 15 Automatic
Data Processing 110.6 16 EMC 110.9 17 Altria 115.5 18 Fannie Mae 117.0 19 Wells Fargo 117.0
20 Johnson & Johnson 118.0 87

The Zipf number calculation creates an interesting sample devoid of the ordinary common denominators such as a P/E ratio screen or price to book value screen. Yahoo and eBay are selected as companies worthy of study. Note that the Zipf value does not suggest whether these are to be studied as long or short investments. The long buyer concludes that the growth rate of these two firms is so high that this warrants investment. Alternatively, the short seller might conclude that the high returns on equity of these companies are unsustainable and their valuations preposterously high. Of course, a conventional screen might have suggested these as worthy of study due to their growth rates. Yet, only a different type of screen would have suggested Fannie Mae as a subject worthy of study. It is possible that one might conclude that its P/E ratio is sufficiently low such that it discounts all possible negative earnings events. Alternatively, one might conclude that the accountancy issues are so severe that earnings cannot be accepted as presented in the accounts of Fannie Mae.

The other end of the Zipf number is, of course, the high values. These can be observed in Appendix B. Two possible topics for future study are Lowes (Zipf number 149.6) and Washington Mutual (Zipf number 148.8). Both companies are quite sensitive to the housing industry. Positively inclined investors might note the high growth rate of Lowes and the low valuation of Washington Mutual. Negatively inclined investors may regard these as exemplary instances of companies sensitive to the housing bubble. It must be emphasized that the Zipf number merely reveals subjects worthy of study.

General Electric has a Zipf number of 56.7. There does not exist a powerful statistical tendency that draws firms to the median Zipf number. As a pure matter of arithmetic, GE cannot improve its Zipf number by an increase, even if it were substantial, in net income. If GE were to record \$21.9 billion of net income in 2005 instead of the projected amount of \$18.9 billion, it would then be in the number 2 net income rank and its Zipf number would actually decline (i.e., $21.9 \times 2 = 43.8$). Nevertheless, this does not necessarily entail that the net profit of GE will decline, although this is an arithmetically obvious way to generate a Zipf number more comparable to the general tendency.

Another possible scenario is suggested by another mathematical theorem known as Gambler's Ruin. The theorem operates as follows: Assume the existence of two players that wager for pennies by guessing heads or tails. Each player obviously has a 50% probability of winning each wager. If the wagering process is continued for a sufficiently lengthy period of time, the probability that one of the players will become insolvent is 100%. The probabilities P_1 and P_2 , that players one and two, respectively, will become insolvent, is a function of the ratio of the amount of pennies of each player to the total quantity of pennies held by both players. The formula for this is:

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$$P_2 = n_1 / n_1 + n_2$$

Where n_1 = the number of pennies held by player 1 n_2 = the number of pennies held by player 2

General Electric is rapidly increasing its total assets by growth and acquisition. If GE were merely wagering for pennies, it would have an enormous advantage of size. For instance, if GE had 850 pennies and its adversary N2 had 150 pennies, then, by substitution, the probability that the GE adversary would eventually become insolvent would be:

$$P_2 = \frac{850}{850 + 150} = 85\%$$

Total assets of GE now exceed \$704 billion. Total assets of Citigroup equal the astonishing sum of \$1.4 trillion. The companies have enormous power over customers and suppliers. As these firms seek efficiencies with a view to increasing profit margins, it is conceivable that this can occur at the expense of suppliers, if not of customers. If this were the case, then the net profit of various companies would decline and their Zipf numbers would decline as a consequence. If the reported profit of the largest companies such as Citigroup and GE were to increase, their respective Zipf numbers would increase. Hence, the increasing dominance of several large S&P 100 firms is a potential scenario that would equilibrate the differences that currently exist in Zipf numbers.

Incidentally, calculation of Zipf numbers need not be confined to net profit figures. Quantities such as shareholders' equity, total assets, levels of property and equipment, debt capital or even inventories can be subject to Zipf number analysis. For example, although space limitations make a thorough analysis impossible, the Zipf number for J.P. Morgan Chase

appears unusually high in relation to the established pattern for the total assets variable. Consequently, if one accepts the validity of Zipf number analysis, one would select J. P. Morgan Chase as a subject worthy of study. Zipf number analysis is in no way any sort of substitute for conventional equity analysis. It is merely a means of selecting suitable subjects for such analysis. As a selection methodology, it does yield subject matter that is not likely to be suggested by conventional screening methods. For instance, the Zipf number for S&P 100 net income yields subject matter as diverse as low P/E stocks such as Washington Mutual, high P/E stocks such as Yahoo, a classical growth stock such as Lowe's or Wal-Mart, a conventional value stock such as J. P. Morgan Chase as well as a non-conventional value stock such as Comcast. 89

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A reader who accepts the validity of Zipf numbers may wonder why the Zipf pattern is a consistently observed feature of aggregations of number with regard to many sorts of phenomena. The truthful answer is that the cause is unknown. However, it is known that Zipf's law is an example of a power law. A power law is simply a relationship between two variables such as X and Y that can be written in the form:

$$y = ax$$

K

where a and K are constants. In physics, many such power laws exist such as the Stefan – Boltzman Law of energy radiated per surface unit of a black body in unit time. Another example would be the inverse square law of Newtonian gravity.

Variants of Zipf's law can be found in the so-called Pareto Principle. The establishment of the Pareto Principle actually occurred before Zipf wrote his books. The Pareto Principle is also known as the 80 – 20 Rule or the Law of the Vital Few or the principle of factor sparsity. It was observed by certain Italian economists that 80% of the property in the country was owned by 20% of the people. The S&P 500 tends to gravitate to a Pareto distribution structure. This is to say that 80% of the S&P market value is represented by 20% of the stocks. At the moment the 100 largest capitalizations account for about 66.2% of the index. (This is something that readers can consistently observe.) It will exhibit the tendency to restore itself to roughly the 80% level. This can theoretically be accomplished by the decline in value of many of the NASDAQ type firms with high valuations that have come to gradually populate the S&P 500. Of course, the principle of Pareto distribution might finally be invalidated by the continual ascent of those NASDAQ companies.

It is not known why certain quantitative patterns continually appear in natural phenomena. It is merely known that the basic Pareto distribution is a power law of the form:

$$F(x) = 1 - a^{-x}$$

a for $x > 1$ and $a > 0$

Readers even without mathematical expertise will note the similarity of this equation to the Zipf function on page one of this essay.

It might well be asked why students of markets should concern themselves with quantitative patterns in natural or statistical phenomena given that the causes of these patterns are unknown. The reason is that there exists yet another law known as Bradford's law that is used by librarians. It is a law that quantifies the exponentially diminishing returns from extending a given library search. For instance, the probability of finding something of interest in x books or journals decreased exponentially with the number of items searched. If one finds two items of interest in the first 10 books searched, perhaps a search of another 2090

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© Horizon Research Group, 2004 Page 91 books will be necessary in order to discover another 2 items of interest. In this case, the Bradford multiplier known as bm would be 2 (i.e. 20/10).

Bradford's law clearly has implications for investors who must search over increasing numbers of securities in order to find a worthwhile investment opportunity. Given that there is no substitute for the rigorous fundamental study of a certain company as investment possibility, one cannot achieve an efficiency by economizing on research. Consequently, in order that an analyst make most efficient use of available time, one must seek economies in the search process for possible subjects or companies to study. Hence, Bradford's law has relevance for the investment analyst.

Readers should excuse the length of this paper. It is rather difficult to express the remarkable patterns in quantitative data in concise, succinct form. Nevertheless, this has been done by such

an individual as Winston Churchill. A far superior substitute for this essay is merely the following Churchill quotation:

“Man will occasionally stumble over the truth, but most of the time he will pick himself up and continue on.”

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© Horizon Research Group, 2004 Page 92 Appendix A: S&P 100 Index Ranked by Consensus Estimated Net Income for 2005 1 Citigroup 22.8 2 Exxon 21.8 3 GE 18.9 4 Pfizer 17.5 5 Bank of America 16.3 6 Microsoft 15.1 7 AIG 13.6 8 J.P. Morgan 11.9 9 Wal-Mart 11.6 10 SBC Comm 10.9 11 Altria 10.5 12 Johnson & Johnson 9.9 13 Chevron 9.7 14 IBM 9.2 15 Wells Fargo 7.8 16 Fannie Mae 7.3 17 Proctor & Gamble 7.3 18 Verizon 7.3 19 Intel 7.0 20 Cisco 6.7 21 Wachovia 6.7 22 Conoco 6.4 23 Merck 5.7 24 Home Depot 5.6 25 Hewlett Packard 5.0 26 Coca Cola 4.9 27 Morgan Stanley 4.9 28 Freddie Mac 4.8 29 Tyco 4.6 30 U.S. Bancorp 4.5 31 Merrill Lynch 4.4 32 Pepsico 4.3 33 Goldman Sachs 4.1 34 Abbott Labs 3.9 35 Allstate 3.9 36 Dell 3.9 37 Wyeth 3.9 38 American Express 3.8 39 Eli Lilly 3.5 40 Ford 3.5 41 Amgen 3.4 42 Dow Chemical 3.4 43 Oracle 3.3 44 Time Warner 3.3 45 MMM 3.2 46 St Paul Travelers 3.2 47 United Health Group 3.1 48 Washington Mutual 3.1 49 United Technologies 3.1 50 MBNA 2.9 51 Bellsouth 2.9 52 Walt Disney 2.8 53 Viacom 2.8 54 Bristol Myers 2.7 55 Du Pont 2.7 56 Metlife 2.7 57 Medtronic 2.6 58 Lowe's 2.6 59 McDonald's 2.5 60 Caterpillar 2.4 61 Anheuser Busch 2.3 62 Qualcomm 2.3 63 Target 2.3 64 Motorola 2.3 65 Prudential Financial 2.2 66 Alcoa 2.2 67 Boeing 2.2 68 National City 2.1 69 Suntrusts Banks 2.0 70 First Data 2.0 71 UPS 2.0 72 Exelon 2.0 73 Nextel 2.0 74 Fifth Third Bancorp 2.0 75 Texas Instruments 1.9 76 Cendant 1.9 77 Sprint 1.9 78 Kimberly Clark 1.9 79 Gillette 1.9 80 Boston Scientific 1.9 81 Walgreen 1.7 82 BB&T 1.7 83 Honeywell 1.7 84 Carnival 1.7 85 Applied Materials 1.6 86 Bank of NY 1.6 87 Emerson Electric 1.6 88 FedEx 1.6 89 Southern 1.5 90 Illinois Tool Works 1.5 91 Schlumberger 1.5 92 Lockheed Martin 1.5 93 Duke Energy 1.4 94 Colgate Palmolive 1.4 95 EMC 1.2 96 Automatic Data Proc. 1.2 97 eBay 1.1 98 Comcast 0.9 99 Yahoo 0.7 100 Shering Plough

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© Horizon Research Group, 2004 Page 93 Appendix B: S&P 100 Ranked by Zipf Numbers for Consensus Estimated 2005 Net Income 1 United Technologies 151.9 2 Medtronic 150.4 3 Lowe's 149.6 4 Metlife 149.4 5 Washington Mutual 148.8 6 Kimberly Clark 148.4 7 Du Pont 148.4 8 Boston Scientific 148.0 9 Bellsouth 147.9 10 Target 147.7 11 McDonald's 147.7 12 Gillette 147.6 13 Viacom 147.3 14 Sprint 147.2 15 Motorola 147.1 16 MBNA 146.7 17 Cendant 146.2 18 Walt Disney 146.2 19 United Health Group 145.9 20 Bristol Myers 145.8 21 St. Paul Travelers 145.5 22 Qualcomm 145.4 23 Fifth Third Bancorp 145.3 24 Time Warner 145.2 25 Texas Instruments 145.1 26 American Express 144.4 27 Nextel 144.3 28 Boeing 144.1 29 MMM 144.0 30 National City 144.0 31 Honeywell 143.9 32 Prudential Financial 143.8 33 Caterpillar 134.5 34 Anheuser Busch 134.1 35 Carnival 142.8 36 BB&T 142.8 37 Wyeth 142.7 38 Exelon 142.7 39 Alcoa 142.7 40 Dow Chemical 142.5 41 UPS 142.0 42 Oracle 141.9 43 Bank of NY 141.7 44 First Data 141.5 45 Walgreen 141.2 46 Emerson Electric 140.9 47 Conoco 140.8 48 Amgen 140.7 49 Wachovia 140.7 50 Dell 140.4 51 Suntrust Banks 140.2 52 Applied Materials 140.1 53 Ford 139.4 54 Pepsico 137.6 55 Southern 136.9 56 Fedex 136.6 57 Allstate 136.5 58 Eli Lilly 136.5 59 Merrill Lynch 136.4 60 Illinois Tool Works 136.1 61 Goldman Sachs 135.3 62 U.S. Bancorp 135.0 63 Lockheed Martin 134.5 64 Home Depot 134.4 65 Freddie Mac 134.4 66 Cisco 134.0 67 Schlumberger 134.0 68 Tyco 133.4 69 Intel 133.0 70 Abbott Labs 132.6 71 Morgan Stanley 132.3 72 Verizon 131.4 73 Merck 131.1 74 IBM 128.8 75 Colgate Palmolive 127.6 76 Coca Cola 127.4 77 Duke Energy 126.8 78 Chevron 126.1 79 Hewlett Packard 125.0 80 Proctor & Gamble 124.1 81 Johnson & Johnson 118.8 82 Wells Fargo 117.0 83 Fannie Mae 117.0 84 Altria 115.5 85 EMC 110.9 86 Automatic Data Proc. 110.6 87 SBC Comm 109.0 88 Wal-Mart 104.4 89 eBay 102.2 90 AIG 95.2 91 J.P. Morgan 95.2 92 Microsoft 90.6 93 Comcast 88.2 94 Bank of America 81.5 95 Pfizer 70.0 96 Yahoo 67.3 97 GE 56.7 98 Exxon 43.6 99 Schering Plough 35.2 100 Citigroup 22.8 93

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Gold Related Stocks Versus Gold Indexed Exchange Traded Funds:

An Exercise in Comparative Valuation

It is customary that an essay begin with a statement of purpose. This essay must begin with a disclaimer. This is not intended to be either a purchase or sale recommendation of any individual gold related equity. It is not intended to recommend either purchase or sale of any gold related exchange traded fund. It is merely a theoretical exercise that compares the valuation at the current time of the recently issued Street Track Gold Trust as opposed to the valuation of a well

established gold company such as Newmont Mining. The exercise has as its premise the notion that investors are reasonable. As is well appreciated, this is the most dangerous action in all matters of finance. It is the source of much, and quite possibly of most, of the misfortune in the field of investments.

Street Tracks Gold Trust is an exchange traded fund (ETF) listed on the New York Stock Exchange. It is sponsored by the World Gold Council. Bank of New York is the fund trustee and State Street is the marketing agent. HSBC Bank is the fund custodian.

This fund is simply designed to hold gold in bullion form such that each share is equivalent to 1/10 of an ounce of gold. The fund commenced trading on November 18, 2004 with the issuance to the public of 2.3 million shares. Its price has thus far very closely approximated the daily gold spot trading price. For instance, London spot gold closed on November 26, 2004 at \$451.85 per ounce. The gold trust closed trading at 1:00 pm for the holiday shortened trading session of November 26, 2004 at 45.29. Any difference between fund price and London gold prices is easily attributable to the time difference between the two markets.

Since the Street Tracks Gold Trust is an ETF, it is to be anticipated that more shares will be issued so that the 2.3 million currently outstanding should not be taken to be representative of the shares of the gold related investment demand that is ultimately possible of this investment. At the moment, a salient point: that the fund seems to trade at its apparent net asset value. This had already had an effect upon a similar type of shares.

The similar share in question is known as the Central Fund of Canada (CEF). This fund trades on the American Stock Exchange. The fund holds gold and silver in bullion form.

Ordinarily, the fund traded at a 12-14 % premium to net asset value. This was not unreasonable viewed from the perspective of the individual investor. An investor in raw gold or silver bullion in the U.S. would need to pay a sales tax which would vary by place of residence as well as the customary bid/ask spread of the bullion dealer. The mere existence of the Street Tracks Gold Trust has made this entirely unnecessary.

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© Horizon Research Group, 2003 Page 95 Consequently, in the days between November 18, 2004 and November 26, 2004, the premium to net asset value of the Central Fund of Canada has essentially vanished. Its current NAV premium is slightly in excess of 2%.

The existence of the gold ETF has not yet had any impact upon the valuation of leading gold shares such as Newmont Mining. This latter company has five primary assets. These are:

- 1.) Approximately 100 million ounces of gold reserves, assuming that the gold price remains above \$375 per ounce. At a price of \$325 per ounce, reserves would be only 91 million ounces,
- 2.) Approximately 7.5 billion pounds of copper reserves,
- 3.) Approximately 480 million pounds of zinc reserves,
- 4.) Approximately 6% equity ownership of the publicly traded Canadian Oil Sands Trust (Toronto Stock Exchange COS-U),
- 5.) Royalty income from investments made by Franco-Nevada Mining before it was acquired by Newmont Mining. This income approximates at \$60 million per annum.

One might imagine that the valuation of these five assets is a rather straightforward proposition. However, it is quite subjective based upon the valuation premise utilized by a given analyst. For instance, it should be obvious, even at a glance, that gold is the primary Newmont asset. The amount of reserves is a function of the gold clearing price. Given gold at \$375 per ounce, the company asserts that reserves are about 100 million ounces.

However the gold price now exceeds \$450 per ounce so that it is reasonable to assume that reserves are higher than the level quoted by the company since previously uneconomic reserves at \$375 per ounce are presumably now economic.

In any case, let us utilize a reserves estimate of 100 million ounces since, it is the highest available estimate from a presumably reliable source. These reserves cannot be valued at markets since each ounce has a cash extraction cost estimated by Newmont to be about \$233 per ounce. Thus, the value of gold reserves could be said to be:

100 million ounces at \$452 each \$45.2 billion Minus cash cost of \$233 / oz \$23.3 billion

Gold Value of \$21.9 billion

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© Horizon Research Group, 2003 Page 96 This gold will not be produced in the current year. In light of current production rates, the reserve life is equal to 14.5 years. Hence, the 21.9 billion gold reserve value must be reduced by a present value calculation. This requires the choice of a

discount rate.

The choice is necessarily subjective. However, given the inherent uncertainties of gold production, there should be few objections to the selection of 10% as an appropriate discount rate.

A present value calculation will result in an \$11.336 billion valuation for the gold reserve. This calculation is based upon a cash production cost of \$233 per ounce. Yet, there are capital costs involved in gold production. It is arguable that a more appropriate valuation calculation would include total production costs. This figure is supplied by the company and is said to be \$296 per ounce. Therefore the value of gold reserves could more properly be said to be:

100 million ounces at \$452 each \$45.2 billion

Minus total production cost of \$296 / Oz \$29.6 billion

Gold Value of \$15.6 billion

This figure must also be subject to a present value calculation. The present value is \$7.576 billion, assuming a 10% discount rate.

Copper reserves can be valued in a similar manner with one complexification that will be addressed in due course. Copper reserves approximate 7.5 billion pounds. The current London Metals Exchange price for high grade copper per pound is \$1.48. The cash production cost is, according to Newmont Mining \$0.85 per pound. This exercise will ignore the total production cost since the purpose of the exercise is to create a realistic best case valuation for Newmont Mining. Consequently, the value of copper reserves could be assessed in the following manner:

100 billion pounds of copper at \$1.48 per pound \$11.1 billion Minus cash production cost of \$0.85 \$6.375 billion

Copper Value of \$4.725 billion

This figure must also be subject to a present value calculation. The reserve life of copper at current rates of extraction is about 17 years. The present value of the copper production at a 10% discount rate equals \$2.147 billion.

One of the complexifying factors is that the forward commodity price curve for copper is very different from that of gold. In the case of the latter commodity, gold for delivery in August of 2006 trades at \$475.8 per ounce. This is the price one might reasonably expect for gold given the time value of money. This implies a 2.6% interest rate. 96

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In contradiction, copper for delivery in August of 2006 trades at \$116.35 per pound. This implies a 21.4 % decline in copper prices. Given allowance for the time value of money. It would appear that the expected copper price is \$1.11 per pound. Thus, the mere act of valuing the copper at \$1.48 per pound is at variance with the commodity market expectation. Obviously, the use of \$1.11 per pound for the realized copper price would result in a much lower present value for copper.

The zinc reserves total 480 million pounds. Given the current production rate, the asset has a reserve life of perhaps four years. The current London Metals Exchange zinc trading price is \$0.51 per pound. The cash cost of production is \$0.38 per ounce. Hence, the zinc reserves can be valued in the following manner:

480 million pounds of zinc at \$0.51 per pound \$244.8 million Minus cash cost of \$0.38 per pound \$182.4 million

Zinc Value of \$62.4 million

The present value of this asset at 10 % is \$54.4 million. It clearly exerts negligible influence upon the trading price of Newmont Mining.

The shareholding in Canadian Oil Sands Trust is easy to value. Newmont owns 6 % of the company that, at current trading prices, is worth \$295 million.

The last asset is \$60 million of annual royalty income. One could simplistically assign a 20x p/e ratio to this quantity and thus value the asset at \$1.2 billion. However, this makes no allowance for present value. Given an estimated 20 year life (this could be rather generous) the asset is unquestionably worth less than \$600 million. The actual figure discounted at 10 % per annum is \$509 million.

In sum, the net asset value of Newmont Mining can be calculated in the following manner:

1.) Value of Gold (cash production cost approach) (present value) \$11.336 billion 2.) Value of

Copper (present value) \$2.147 billion 3.) Value of Zinc (present value) \$0.054 billion 4.) Value of Canadian Oil Sands Trust - investment \$0.0295 billion 5.) Present value of Royalty income \$0.0509 billion

Net Asset Value \$14.341 billion 97

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It will be observed that this approach does not make use of the lower and probably more correct method of gold reserve valuation based upon total production cost. Moreover, it must be observed that no allowance has been made for the payment of corporate income taxes. In any case, the net asset value calculated in this optimistic manner is:

\$14.341 billion

Newmont Mining diluted shares outstanding are currently \$447.060 million. Its most recent trading price is \$48.99 per share. Hence, the market value of Newmont Mining is \$21.901 billion or 52.7% above a very generous approach to the calculation of its asset value.

One might well wonder why the shares would trade at this valuation. The answer can only be given in terms of a theoretical abstraction. Of course, it was always possible that a mutual fund would purchase gold bullion as an alternative to the purchase of the shares of gold companies. The operational complexities of such an action could clearly be surmounted. Yet, gold was not a sufficiently important asset class and not a sufficiently large investment as a percentage of any fund such that such an action was advisable. Such an action would merely have raised the expense ratio of a given fund. In any case, gold company shares did not trade at an excessive valuation. Hence, the action was not necessary. The existence of the Street Tracks Gold Trust ETF changes the situation. A fund can now own bullion in preference to shares with no change to its fund charter, no board approval and no increase in expense ratio.

It will be argued that gold shares should trade at a substantial premium to net asset value to embrace the possibility that gold reserves will increase as gold prices rise. If one merely holds bullion, reserves are kept at a constant amount. This is mathematically true.

Nonetheless, it is logically equivalent to the argument that the trading price of a given security should reflect its most positive outcome. If the trading price of Newmont reflects the possibility that reserves will increase given a material increase in the price of gold, then it must follow that the anticipated appreciation in share price from such a possible scenario belongs to the current seller of the share and not to the current buyer of the share.

Moreover, although it is true that a positive scenario might develop, it is also true that negative scenarios might develop. Obviously, gold prices might decline and hence the reserve figure might decline. If Newmont trades at a substantial premium to its net asset value, none of this is reflected in the price of the Newmont share. In addition, there can be such events as environmental accidents and associated liabilities, mining accidents, an unanticipated increase in cash production costs or unfavorable political developments in some of the various nations in which the Newmont mining properties are located.

Examples of nations with a sufficient degree of political risk to disrupt mining operations might be Peru and Indonesia.

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© Horizon Research Group, 2003 Page 99 The valuation of the shares of gold related companies can markedly affect the correlation of those shares with the gold trading price. For example, according to the Gold Institute, the following table depicts the average gold trading price during the period 1970 – 1979.

Average Gold Trading Price per Ounce 1970 – 1979

Source: Gold Institute

1970	\$36.02	1971	\$40.62	1972	\$58.42	1973	\$97.31	1974	\$154.00	1975	\$160.86	1976	\$124.74
1977	\$147.84	1978	\$193.40	1979	\$306.00								

According to Moody's, the average trading price of the Newmont Mining share was in the range of $37\frac{1}{2}$ - $21\frac{1}{4}$ in 1970. By 1979, the average trading price of a Newmont Mining share was $39\frac{3}{4}$ - $21\frac{1}{2}$. In fairness, during this period, Newmont Mining had more exposure to copper, zinc and oil than is currently the case.

However, each of those commodities was also dramatically increasing in price during this time period.

In 1980, the gold price for a brief period exceeded \$800 per ounce. It was only in this year that the Newmont Mining share price traded in the range of $60\frac{7}{8}$ - $30\frac{1}{2}$. There were no share splits during the 1970 – 1980 time period.

Consequently, it is not correct to believe that the Newmont Mining share price is necessarily

correlated with the corresponding commodity trading price.

None of the above should be interpreted as a forecast that gold will not rise in price. It is quite possible that inflationary pressures will accelerate. It is quite possible that the U. S. dollar will collapse. It is quite possible that gold is an excellent hedge against the U.S. dollar. Opinions upon such matters are simply beyond the scope of this paper. However, one can quite correctly forecast the price of gold and quite incorrectly forecast the trading price of Newmont Mining. The valuation of the shares simply reflect a very optimistic outcome. Indeed, that 99

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© Horizon Research Group, 2003 Page 100 forecast is far more optimistic than is suggested by the existence of a gold price that exceeds \$450 per ounce.

A reasonable investor, if such a being actually exists, now has an interesting choice. If such an investor wishes to include a gold investment in a portfolio, such an investor can be very correct insofar as a gold forecast in concerned and very wrong insofar as a Newmont Mining forecast in concerned. Alternatively, such an investor can purchase the Street Tracks Gold Trust ETF as a gold investment. By definition, the ETF trading price will accurately reflect the gold price, thus making certain that a correct gold price forecast will be reflected in investment value.

The owner of conventional gold stocks as opposed to the newly created gold ETF will no doubt argue that investors will long retain the habit of owning gold mining shares as a proxy for gold investments. This may be true. In any case, this paper can in no sense offer a refutation of such a position.

However, it is worthy of note that the president of Newmont Mining, Mr. Pierre Lassonde, has been a substantial seller of shares in recent weeks. These are shares that result from stock option exercise. Mr. Lassonde can only be described as a brilliant gold investor. He founded Franco-Nevada Mining with almost no resources some two decades ago and in the worst gold investment environment in history built this firm into one of the great mining firms of the world until it was acquired by Newmont Mining. (See Contrarian Research Report on Franco-Nevada Mining dated June 18, 1999). The selling activity by Mr. Lassonde is very unusual. Readers are directed to a Form -4 SEC filing dated November 12, 2004.

Insider selling activity among other members of the Newmont Mining management team, including the chairman Mr. Wayne Murdy, has been uncharacteristically high.

It is also worthy of note that in recent months, Mr. Rob McEwen, the former chairman of Goldcorp, has asked the board of directors to find a successor. Goldcorp, itself, has been a rather unusual gold mining firm inasmuch as it has withheld gold production from the market. It recently sold all of its gold holdings. Goldcorp has been the subject of various Contrarian Research Recommendations. (See original report dated June 24, 2002 and five subsequent updates). Goldcorp chose to sell its previously accumulated 8.3 tons of gold in late 2003. It could be that the action of well informed and historically successful insiders, the creation of the Street Tracks Gold Trust ETF and the fact that gold related equities trade at very substantial premiums to net asset value are entirely coincidental events.

Nevertheless, even a coincidence is frequently worthy of notice.

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© Horizon Research Group, Page 102 Palladian Research LLC, 2004 High Yield Investing In the Absence of High Yield

Investment Thesis

The painfully obvious fact about the current state of the high yield bond market is that it is named quite inappropriately. However, the obvious issue of nomenclature obscures a far more important issue. The salient question is whether or not this phenomenon is to remain as it is for some prolonged period of time or whether yields will soon be restored to a more traditional level. It is difficult to imagine a circumstance in which the “high yield” market will provide even lower yields. This is because it is difficult, but not impossible to envisage lower treasury yields. If treasury yields do not decline, the only circumstance in which “high yield” instruments can provide a lower absolute yield would be if the “spread” between “high yield” and Treasury securities were to further contract. It is already remarkably low.

A 20-year Treasury bond yields roughly 5.12%. A BBB industrial credit with a 20-year maturity

yields 6.28%. A BB twenty-year industrial credit yields 7.03% A single B twenty-year industrial credit now yields only 7.64%. In order to find higher yields one must descend into the CCC range or even lower. It will be observed that there is not a large quantity of paper that is currently rated below CCC.

One can obtain a higher yield without undue credit risk in one of the closed-end high yield funds. An example would be the Blackrock High Yield Trust (BHY-NYSE listed). This is a fund that trades at a premium of 22.4% to its net asset value. It does have a trailing 12 month yield of 9.4%. However, it is 35.7% leveraged or, alternatively expressed in hedge fund jargon, 135.7% gross invested. Obviously, if the shape of the yield curve were to change with short term rates rising, this fund would at minimum experience a decline in income payout as the cost of carry for the leveraged portion of the portfolio continues to increase.

Thus, it seems evident that the high yield choices, such as these are, do not provide high yield and do certainly provide investment risk. This cannot be a situation that will permanently endure. Essentially, there are two possibilities. The first possibility is that the situation will be restored to normalcy. That is to say that the "spread" between "high yield" and Treasury issues will widen. The second possibility might appear to be somewhat absurd. Yet, it might be said to logically follow from the current yields of lower rated bonds. This is that the default rate on bonds of low quality will be permanently lower than was previously the case. Consequently, the yield has become properly lower and this merely reflects an appropriately lower risk premium for this variety of financial instrument.

It should be arithmetically evident that it is not possible to earn a double digit rate of return on high yield bonds as a class of investment. If the average yield of such an 102

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© Horizon Research Group, Page 103 Palladian Research LLC, 2004 instrument in 8.5% and the average instrument trades near or above par value, the best that can possibly happen is that none of the higher coupon issues are called for redemption, none of the issues default and that long term rates do not increase. In such a circumstance one might earn the average yield of 8.5%. In fact, in light of the average yields noted in the opening section of this paper, a yield of 8.5% is almost certainly an overestimate of what is actually obtainable. Since the default rate will not be zero, the actual return will be comprised of the average yield minus losses due to the average default rate. Hence, a portfolio of B-rated 20-year paper will yield only 7.64%. Losses due to default of over 67 basis points, in a portfolio context, will reduce the return to below 7%. If a 3% position within a given portfolio were to decline by 25% due to default or the threat of default, the portfolio will experience a 75 basis point loss. Thus, the margin for investment error in a portfolio context is very small.

Since it would appear that there are no significant return prospects from yield, one must necessarily seek return, if it can be found, in bond price changes. This leads once again to the remaining possibilities. If high yield spreads widen, high yield bond prices will decline. Alternatively, if the high yield default rate is to be permanently low, perhaps there exists appreciation potential in the related equity securities. The remainder of this paper will examine the two possible scenarios.

I. Investment Tactics in the Spread Widening

A possible trade that would benefit in the spread widening scenario is as follows: Sell short Nextel Communications 7.375% notes due 8/1/2015. Buy an equal dollar basis in the U. S. 20-year Treasury bond. The Nextel bonds currently trade at 109 1/2 bid 110 1/2 ask. The current redemption price is 107.375. These bonds are unconditionally callable on or after 8/1/2008 at 103.688%. The current yield is only 6.7%. Given the call prices as well as the maturity date, the Nextel bonds must eventually decline in price. The trade is a negative carry trade on a cash basis insofar as the Nextel bonds in question yield 158 basis points more than the 20-year U.S. Treasury. However in the event of a call at 103.688 in 2008, the trade would manifest a marginal profit.

Nextel is a highly leveraged company active in the very competitive sphere of mobile communications. This is a \$1 billion issue and considered to be one of the rather liquid issues in the world of so-called high yield. If the spread to Treasuries were to widen from the current 158 basis points to a more historical norm of perhaps 450 basis points the bonds would trade at roughly 77.63 for a current yield of 9.5% and a yield to maturity of 11.8%. This has the possibility of providing a 28% return if the spread widening occurs within the next 12 months. If the spread widening occurred within 24 months, the potential return would decline to roughly 12 1/2%. If the spread widening occurs within 3 years, the potential return would still be roughly 9.5%. This would be far superior to that which is to be expected if spreads were to remain at their current low level. Although time does work against such a trade inasmuch as there is a fixed possible return. As the time 103

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© Horizon Research Group, Page 104 Palladian Research LLC, 2004 required to earn such a return increases, the compound annual rate of return does decline.

Yet it is only a failed trade if spreads do not widen within the next three years. Accordingly, as the risk appears minimal, it does possess sufficiently asymmetrical return characteristics such that it is recommended as an action.

A similar type of "spread -widening" trade is possible in the case of L-3 Communications 7 5/8% notes due 8/1/2012. This is a general unsecured issue. The face amount of the bonds is \$750 million. The bonds trade at 109 bid 110 asked. The current yield is 6.9%. The yield to maturity is obviously much lower. This yield to maturity would be on the order of 5.2% or quite comparable to the current 20-year U.S. Treasury.

The recommended trade is to sell short the L-3 Communications bonds and purchase an equal dollar amount of U.S. Treasury 10 year bonds. This is a slightly negative carry trade on a current yield basis. It is approximately break even on a yield-to-maturity basis. The L-3 Communications bonds are callable on June 15, 2007 at 103.813. These, then, gradually decline in call price to the level of par on 6/15/2010. It is readily conceivable that if the current interest rate environment were to persist until June 2007, these bonds would trade at their call price. In this circumstance, the trade would manifest a very marginal profit.

If spreads were to widen to only 375 basis points on a current yield basis, the bonds in question would trade at 84.75. If spread widening occurs within a year, the return on such a trade could still exceed 20%. At a price of \$85.75 the L-3 Communications bonds would exhibit a yield to maturity of approximately 10.2%.

It should be noted that L-3 Communications is a very leveraged enterprise. Its long term debt does now exceed \$2.5 billion. Although shareholders' equity also exceeds \$2.5 billion, goodwill and other intangible assets exceed \$3.6 billion. L-3 Communications is a very acquisitive defense electronics firm. Acquisitions are notoriously difficult to orchestrate successfully. The growth rate of the company is dependent upon future acquisitions. A failed acquisition could easily restore the credit spread with respect to this issue to its customary level which would, at current levels of rates, result in an L-3 Communications yield of over 10%.

Time does work to the disadvantage of this trade as it does with all spread trades of this type. Nevertheless, given the current yield of the L-3 Communications bonds in question, there does not appear to be appreciable risk of appreciation of the short position in this trade. The asymmetrical risk reward character of this trade is sufficiently alluring such that this trade is recommended.

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II. Investment Tactics if Spreads Never Again Widen

Let us simply assume that spreads will never widen or, at least, not widen for years. This is, in investment community parlance, the functional equivalent of never. This is rather similar to the assertion that defaults will remain rather low. Let us use a recent Distressed Securities Report recommendation as an illustration of investment tactics.

It will be recalled that the Distressed Securities Report recently recommended purchase of Calpine 4 3/4% Convertible Notes due 11/15/2023. (See Distressed Securities Report dated October 7, 2004). At the time of recommendation only several weeks ago, these Calpine Convertible bonds traded at a yield-to-put off a pproximately 15% per annum.. (These bonds are puttable to the company on 11/15/2009 at par). In recent days these same bonds traded at 84 for a yield-to-put of 8.80%. The issue is convertible into 153.8462 shares of Calpine per \$1000 per value. At current Calpine share prices, this is equivalent to a conversion premium of 50.8% . Calpine common shares trade at a 35% discount to book value. This is probably due to the conventional view that Calpine has too much debt in its capital structure. It is believed that the associated debt service will be difficult to pay and hence losses will ensue that will erode the book value per share. A reduction in shareholder equity would imply that the company will experience a decline in credit worthiness. Yet, the low 8.8% yield-to-put of the bond in question logically is at variance with such a view. In fact, the former 16% yield-to-put would be far more consistent with the notion that Calpine is not credit worthy.

In order that the bond market and the equity market reflect a logically consistent view, then either the bond must return to its former level (roughly 68) or the shares should trade at a higher level. Insofar as the share price is concerned, a trading level such that the shares would trade at least at book value is much more consistent with the current yield-to-put of the bond.

Since the premise of this section is that current yield spreads will not widen let us assume that the shares will ultimately return to book value. The 4 3/4% coupon payment each year represents only a minimal rate of return. If one were to invest the coupon into Calpine call options, one could considerably enhance the return of the security in question.

In the precise scenario that Calpine common shares were to trade at book value by January 2007, the 4 3/4 Convertible bonds would trade at least at parity value. Parity would equal:

$\$9.90 \text{ (Book Value)} \times 158.8462 \text{ shares (conversion rate)} = 152.3 \text{ per bond.}$

This equals a principal-only return of 81.3%. The coupon return would add 11.3% to this return.

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Let us assume that in December 2004 and December 2005 the coupons of 4.75% were invested in Calpine January 2007 2 1/2 call options. In order to create an artificially simple calculation, it will be presumed, no doubt incorrectly, that Calpine shares will reach the level of \$9.90 in January 2007, not gradually but suddenly in that month. The shares, for calculation purposes, are therefore presumed to trade at the current price until such time.

A coupon of 4.75% can purchase \$4,750 worth of call options each year for two years. The January 2007 5 call options now trade at \$1.00 per contract. One year from now, assuming that Calpine were still to trade at \$3.49 per share, the price of the Calpine January 2007 5 Call Option can be simulated by the current trading price of the January 2006 5 Call Option since, in one year, the 2007 Call Option will be a one year option, as is the case with the January 2006 5 Call Option at present. Thus, at the current time \$4750 of coupon income will purchase 45 contracts. In one year, \$4,750 of coupon income should be sufficient to purchase 75 contracts.

Consequently, in January 2007 the Calpine convertible investor will be long the following quantities of securities, assuming an \$84,000 initial investment for 100,000 face amount of bonds. Once again, it is assumed that Calpine shares trade at book value of \$9.90 each.

1). 100,000 face Calpine 4 3/4 % Conv. 11/15/2023 at 152.3 = \$152,300

2). 45 Calpine January 2007 5 Call Options x Intrinsic
value per contract of \$4.90; Value = \$22,050

2). 75 Calpine January 2007 5 Call Options x Intrinsic
value per contract \$4.90; Value = \$36,750

Total Value = \$211,100

Compound annual rate of return = 58.5%

If coupons are collected in cash and the same scenario were to take place, the compound annual rate of return would only be 38.8%.

Calpine Convertible 4 3/4% Notes 11/15/ 2023 remain recommended for purchase.

In addition, it is recommended that coupon income be invested in Calpine call options.

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It should be self-evident that the current yield environment makes the traditional double digit return from high yield investing arithmetically unlikely. Historically, only some of the double digit rate of return was actually generated by coupon income. Much return was generated by price appreciation. At the current levels of trading prices, such appreciation is quite simply not obtainable. Since it does not exist, it must be artificially created.

It is possible to generate such return in both an environment of widening credit spreads as well as an environment in which credit spreads do not widen. This paper has been an exercise in tactics for both environments. It is beyond the scope of this paper to predict credit spreads. However, the current situation is certainly not immutable. The trades illustrated in this paper are a first attempt to assess the opportunities that exist as spread conditions evolve.

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www.kineticsfunds.com Horizon Kinetics Asset Management LLC Portfolio Managers Fund Characteristics Portfolio Allocation The Kinetics Paradigm Fund Paradigm Fund Overview a U.S. focused global all-cap fund comprised of investments in misunderstood or underappreciated companies whose businesses exhibit long product life cycles and have substantial barriers to entry. a Proprietary in-house research sources original investment opportunities in companies for which investors fail to distinguish between permanent and transitory problems. a The strategy seeks to capture long-term results commensurate with the returns of companies' underlying businesses, and has low turnover. The Paradigm Fund No Load Class inception date is 12/31/99. Figures include changes in principal value, reinvested dividends and capital gains distributions. Performance data quoted represents past performance and does not guarantee

future results. Investment return and principal value will vary, and shares may be worth more or less at redemption than at original purchase. Call 1-800-930-3828 or visit us at www.kineticsfunds.com for the most recent month-end performance data. Current performance may be lower or higher than the performance data quoted. Performance data does not reflect the deduction of the sales load or fee which, if reflected, would reduce the performance quoted. You will be charged a redemption fee equal to 2.00% of the net amount of the redemption if you redeem or exchange your shares 30 days or less after you purchase them. *Calculated such that all securities issued by one issuer are counted as one position. Top 10 Holdings (%) Statistics Fund*

Growth of \$10,000 Returns Since Inception Annualized Returns Class Information 3Q 2024 Murray Stahl Chief Investment Officer 46 years of management experience Co-Manager of Fund since inception (1999) Peter Doyle Senior Portfolio Manager 39 years of management experience Co-Manager of fund since inception (1999) as of 09/30/2024 Performance (No-Load Class) S&P 500 as of 09/30/2024 3 Month YTD 1YR 3YR 5YR 10YR 20YR Since Inception Fund (WWNPX) 19.81% 53.28% 47.72% 19.93% 20.61% 14.81% 12.05% 11.69% S&P 500 Index 5.89% 22.08% 36.35% 11.91% 15.98% 13.38% 10.71% 7.68% MSCI ACW Index 6.61% 18.66% 31.76% 8.09% 12.19% 9.39% 8.33% 5.71% Class Ticker Cusip 12b-1 fee Net Gross Adv. A KNPAX 494613854 0.50% 1.89% 1.93% Adv. C KNPCX 494613821 1.00% 2.39% 2.43% Inst. KNPYX 494613797 - 1.44% 1.63% No Load WWNPX 494613607 - 1.64% 1.68% Common Stocks 87.2% Unit Investment Trust 9.7% Cash and Cash Equivalents 3.1% Warrants 0.0% Preferred Stocks 0.0% Texas Pacific Land Corp. 65.5% Grayscale Bitcoin Trust BTC 8.8% Brookfield Corp. 3.3% Landbridge Co. LLC 3.1% Live Nation Entertainment, Inc. 2.4% CACI International, Inc. 2.1% Howard Hughes Holdings, Inc. 1.9% Franco-Nevada Corp. 1.8% Cboe Global Markets, Inc. 1.1% Grayscale Bitcoin Mini Trust BTC 1.0% Beta 1.00 1.00 Standard Deviation 21.30 15.34 Up Market Capture Ratio 1.11 - Down Market Capture Ratio 0.88 - Sharpe Ratio 0.46 0.38 Weighted Avg. Mkt. Cap. (\$mil) \$20,853 \$999,694 Median Market Cap. (\$mil) \$4,694 \$38,143 Price to Book 6.40 4.87 Price to Earnings 36.21 27.98 Return on Equity 37.10% 29.70% Active Share 99.56% - ■ Majority employee owned, SEC-registered investment adviser, initially established in 1994. ■ Manages approximately \$8.3 billion in assets. ■ Oversees investments in mutual funds, separate accounts and alternative products. ■ Investment decisions based on fundamental research, all of which is conducted in-house. ■ Horizon Kinetics Asset Management is the Fund's adviser. Previously, Kinetics Asset Management was the Fund's adviser. Total Net Assets \$1119.9 million Total Number of Positions 51 Turnover Ratio 2% Investment Style Global Equity Market Cap Focus All Cap Minimum Purchase \$2,500 \$154,321 '99 '01 '02 '03 '04 '05 '06 '08 '09 '10 '11 '12 '13 '15 '16 '17 '18 '19 '20 '22 '23 '24 \$0 \$50,000 \$100,000 \$150,000 \$200,000 108

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Definitions: We Do Our Research Active share: A measure of the percentage of holdings in a portfolio that differ from a benchmark index. It is calculated by taking the sum of the differences of the weight of each holding in the portfolio and the weight of each holding in the benchmark index and dividing by two. Active share is measured against the iShares S&P 500 ETF. Beta: A statistic that measures the volatility of the fund, as compared to the overall market. The market's beta is set at 1.00; therefore a beta higher than 1.00 is considered to be more volatile than the market, while a beta lower than 1.00 is considered to be less volatile. It is important to note that a low beta for a fund does not necessarily imply that the fund has a low level of volatility, a low beta signifies only that the fund's market-related risk is low. Down Market Capture Ratio: A statistical measure of an investment manager's overall performance in down-markets. The down-market capture ratio is used to evaluate how well or poorly an investment manager performed relative to an index during periods when that index has dropped. The down-market capture ratio has been calculated since inception. Median Market Capitalization: The median market capitalization (value of outstanding shares) of a basket of stocks. Unit Investment Trusts are excluded from the calculation. Price to Book Ratio: The harmonic weighted average of the price/book (P/B) ratios of the equity securities referenced. The ratio calculated by dividing the current price of the stock by the company's book value per share (assets minus liabilities). Unit Investment Trusts are excluded from the calculation. Price to Earnings Ratio: The harmonic weighted average of the price/earnings (P/E) ratios of the equity securities referenced. The P/E ratio is calculated by dividing the current price of the stock by the trailing one year earnings. Unit Investment Trusts are excluded from the calculation. Return on Equity: The amount of net income returned as a percentage of shareholders equity. Return on equity measures a corporation's profitability by revealing how much profit a company generates with the money shareholders have invested. Unit Investment Trusts are excluded from the calculation. Sharpe Ratio: A statistical measure that uses standard deviation and excess return to determine reward per unit of risk. A higher Sharpe ratio implies a better historical risk-adjusted performance. The Sharpe ratio has been calculated since inception using the 3-month treasury bill for the risk-free rate of return. Standard Deviation: A statistical measure of portfolio risk used to measure variability of total return around an average, over a specified period of time. The greater the standard deviation over the period, the wider the variability or range of returns and hence, the greater the fund's volatility. The standard deviation has been calculated since inception. Turnover Ratio: A measure of how frequently assets within a fund are bought and sold by the managers represented over a one year period. Up Market Capture Ratio: A statistical measure of an investment manager's overall performance in up-markets. The up-market capture ratio is used to evaluate how well an investment manager performed relative to an index during periods when that index has risen. The up-market capture

ratio has been calculated since inception. Weighted Avg. Mkt. Cap: The mean market capitalization (value of outstanding shares) of a basket of stocks, taking into account the relative weight of each investment. Unit Investment Trusts are excluded from the calculation. Historical Total Return (No-Load Class) 2000 2001 2002 2003 2004 2005 2006 2007 2008 2009 2010 2011 2012 2013 2014 2015 2016 2017 2018 2019 2020 2021 2022 2023 Fund (WVNPX) 4.0% 2.0% -4.6% 47.8% 20.9% 16.1% 27.8% 21.1% -53.2% 41.0% 17.4% -14.3% 21.8% 44.1% -0.8% -8.3% 20.5% 28.4% -5.6% 30.5% 3.3% 38.2% 29.2% -16.9% S&P 500 Index -9.1% -11.9% -22.1% 28.7% 10.9% 4.9% 15.8% 5.5% -37.0% 26.5% 15.1% 2.1% 16.0% 32.4% 13.7% 1.4% 12.0% 21.8% -4.4% 31.5% 18.4% 28.7% -18.1% 26.3% MSCI ACW Index -14.2% -16.2% -19.3% 34.0% 15.2% 10.8% 21.0% 11.7% -42.2% 34.6% 12.7% -7.4% 16.1% 22.8% 4.2% -2.4% 7.9% 24.0% -9.4% 26.6% 16.3% 18.5% -18.4% 22.2% The Kinetics Paradigm Fund In April 2019, Kinetics Asset Management LLC ("KAM") and Kinetics Advisers, LLC ("KA") reorganized into Horizon Asset Management LLC ("HAM"), following which HAM was renamed Horizon Kinetics Asset Management LLC ("HKAM"). KAM, HAM and KA were all wholly-owned subsidiaries of Horizon Kinetics LLC, and HKAM will remain a wholly-owned subsidiary. HKAM is the Fund's adviser, previously, KAM was the Fund's adviser.

Neither the portfolio managers of the Fund nor the Fund's investment objective and investment strategy have changed. The Growth of \$10,000 chart illustrates the performance of a hypothetical \$10,000 investment made in No Load Class shares commencement of operations (12/31/99). It assumes reinvestment of capital gains and dividends. This chart is not intended to imply any future performance. Holdings are expressed as a percentage of total net investments and may vary over time. Equity holdings are subject to change, and may not be indicative of actual market position due to the use of call and put options. The Top 10 Sectors and Top Countries displayed represent a percentage of the net assets and excludes cash equivalents and other assets in excess of liabilities. You should consider the investment objectives, risks, charges and expenses of the fund carefully before investing. For a free copy of the fund's prospectus, which contains this and other information, visit our website at www.kineticsfunds.com or call 1-800-930-3828. You should read the prospectus carefully before you invest. As a non-diversified fund, the value of its shares may fluctuate more than shares invested in a broader range of companies. Non-investment grade debt securities, ie., junk bonds, are subject to greater credit risk, price volatility and risk of loss than investment grade securities. Options contain special risks including the imperfect correlation between the value of the option and the value of the underlying asset. In addition, investing in foreign securities involves more risk than just U.S. investments, including the risk of currency fluctuations, political and economic instability and differences in financial reporting standards. There may also be heightened risks investing in non-investment grade debt securities and the use of options. There are also risks associated with investing in small and medium sized companies. Unlike other investment companies that directly acquire and manage their own portfolios of securities, the Fund pursues its investment objective by investing all of its investable assets in a corresponding portfolio series of Kinetics Portfolios Trust. The Fund holds investments that provide exposure to bitcoin. The value of bitcoins is determined by the supply of and demand for bitcoins in the global market for the trading of bitcoins, which consists of transactions on electronic bitcoin exchanges ("Bitcoin Exchanges"). Pricing on Bitcoin Exchanges and other venues can be volatile and can adversely affect the value of the bitcoin. Currently, there is relatively small use of bitcoins in the retail and commercial marketplace in comparison to the relatively large use of bitcoins by speculators, thus contributing to price volatility that could adversely affect a portfolio's direct or indirect investments in bitcoin. Bitcoin transactions are irrevocable, and stolen or incorrectly transferred bitcoins may be irretrievable. As a result, any incorrectly executed bitcoin transactions could adversely affect the value of a portfolio's direct or indirect investment in bitcoin. Only investors who can appreciate the risks associated with an investment should invest in cryptocurrencies or products that offer cryptocurrency exposure. As with all investments, investors should consult with their investment, legal and tax professionals before investing, as you may lose money. S&P 500 Index statistics chart data is based on the iShares S&P 500 Index Fund for market cap, price to book, price to earnings, return on equity, and active share. iShares is a product of BlackRock, Inc. The MSCI ACWI (All Country World Index) Index is a free float-adjusted market capitalization weighted index that is designed to measure the equity market performance of developed and emerging markets. The Standard & Poor's 500 Index represents an unmanaged, broad-based basket of stocks. It is typically used as a proxy for overall market performance. The S&P 500 Index returns assume that dividends are reinvested. An investor cannot invest directly in an index. Murray Stahl is a member of the Board of Directors of Texas Pacific Land Corporation ("TPL"), a large holding in certain client accounts and funds managed by Horizon Kinetics Asset Management LLC ("HKAM"). Officers, directors and employees may also hold substantial amounts of TPL, both directly and indirectly, in their personal accounts. HKAM seeks to address potential conflicts of interest through the adoption of various policies and procedures, which include both electronic and physical safeguards. All personal and proprietary trading is also subject to HKAM's Code of Ethics and is monitored by the firm's Legal and Compliance Department. Distributor: Kinetics Funds Distributor LLC is not an affiliate of Kinetics Mutual Funds, Inc. ©2024 Horizon Kinetics LLC ® All Rights Reserved 3Q 2024 as of 09/30/2024 Top 10 Sectors (%) The Gross expense ratios listed are as of 12/31/2023 as reported in the 4/30/2024 prospectus. The Fund's adviser has voluntarily agreed to waive management fees and reimburse fund expenses so that net annual fund operating expenses do not exceed certain levels, not including acquired fund fees and expenses (AFFE), through April 30, 2025 and may be discontinued at any time by the Fund's adviser after April 30, 2025. Top Countries (%) Mining, Quarrying, and Oil and Gas Extraction 70.7% Finance and Insurance 14.4% Real Estate and

Rental and Leasing 2.4% Arts, Entertainment, and Recreation 2.4% Professional, Scientific and Technical Services 2.1% Information 1.7% Management of Companies and Enterprises 0.9% Accommodation and Food Services 0.6% Food Services and Drinking Places 0.6% Transportation and Warehousing 0.5% 0 20 40 60 80 100 United States Canada United Kingdom Cayman Islands Japan 90.0% 6.7% 0.4% 0.1% 0.0% 109

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PCS Research Services 100 Wall Street, 20th Floor New York, NY 10005
research@pcsresearchservices.com (212) 233-0100 www.pcsresearchservices.com Research Team Murray Stahl Steven Bregman Rich Begun Thérèse Byars Ryan Casey James Davolos Peter Doyle Matthew Houk Utako Kojima Eric Sites Fredrik Tjernstrom Steven Tuen

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Penske Automotive Group, Inc. (BUY)

Price (close): \$47.63 Ticker: PAG 52-Week Range: \$41.26 - \$53.81 Dividend: \$1.58 Shares Outstanding: 81.1 million Yield: 3.3% Market Capitalization: \$3.86 billion

Data as of January 9, 2020

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Investment Thesis

Penske is the second largest retail automotive dealership company in the United States. It operates 149 dealership franchises in the U.S. and 184 internationally. Over the last decade, the company has also expanded into commercial and heavy-duty truck retail, as well as truck rental/leasing. The truck leasing business, of which Penske owns 28.9%, has performed well over the years, and now accounts for almost 30% of overall pre-tax income.

The earnings of Penske will naturally ebb and flow with the sale of new and used vehicles, which is a cyclical industry. At the moment, new vehicle sales have plateaued and most investors believe the industry will contract next year. This prospective weakness would almost certainly be accelerated by any noticeable deterioration in the overall economies of either the United States or Europe.

While this is clearly a possibility, such declines have historically been only temporary, as the natural wastage rate of vehicles compels consumers to purchase a new vehicle over time. Viewed in this light, the long-term record of Penske has been reasonably good. For instance, while the company’s earnings did decline dramatically during the 2008/2009 recession, the revenue growth rate since 2007 has been 5.4% annually, while net earnings have expanded by 13.2% per annum. The modest share repurchase program of the last decade has reduced shares outstanding by roughly 14%, creating an annualized earnings per share increase of 14.3% during this time.

Nonetheless, investors anticipate a flat, or possibly declining base of earnings over the next two years. It should be noted that the Penske shares were previously recommended for purchase in December 2015 by this publication. At the time, they were priced at slightly under \$44 per share. That price represented an earnings multiple of just under 11x the then forward earnings and judged to be undervalued in light of its demonstrated earnings growth and that it traded at a discount to the broader market.

While the shares have appreciated modestly over the past four years, it would appear that the tepid performance was less of a reflection on Penske than it was the result of weakness in auto-related shares generally. For example, the First Trust NASDAQ Global Auto Index ETF (CARZ), an exchange-traded fund that tracks the performance of global automakers, actually declined, albeit slightly (i.e., less than 10%), over the same time frame.

Yet, the company's revenues have risen steadily, if modestly, and operating and net income have increased rather robustly. For whatever attributable reasons, the valuation has been contracting while the company appears to be operating consistently in all respects, and now has a base of earnings that is considerably higher than four years ago.

Currently, the shares trade at 9.1x the 2019 consensus earnings forecast, and 8.3x the 2020 estimate. There is a reasonable possibility that the low valuation accurately reflects a

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temporary period of lower earnings. However, this raises an interesting theoretical exercise, since the company is 57.8% owned by Roger Penske (founder and current CEO/Chairman) and Mitsui – leaving a float-adjusted market capitalization of only \$1.6 billion.

If Penske could borrow at 6%, it could pay a 100% premium, or \$3.2 billion, for the remaining publicly-held shares outstanding, resulting in additional annual interest expense of \$192 million. Since the company generated \$604 million of pre-tax income in 2018, there would be adequate interest coverage to engage in such a transaction (although cyclically depressed earnings would clearly lower this coverage ratio). This is not to presume that a management-led buyout will necessarily occur, but simply to illustrate how a low valuation provides some value-creating pathways at the company's disposal.

In the absence of a broader transaction, it is not difficult to construct a scenario in which the Penske shares could return at least 10% per annum. The current \$1.58 per share dividend amounts to a 3.3% yield. In light of the company's longer-term 14.3% per share earnings growth record, if even half of this rate were generated, the implied annual return would exceed 10%. It is important to point out that Penske only has a 1% share of the U.S. automotive retail market. Its strategic policy of gradually consolidating both local and national franchises through acquisitions, cash flow permitting, could continue for decades, given the number of independently owned dealership franchises. Thus, this industry dynamic suggests that there is more than ample long-term capacity for Penske to expand at the historical rate.

Moreover, the company's discounted valuation is roughly 35% below the average P/E since 2007 of 12.9x. If this were viewed over the life of the auto retail cycle, there would then be an additional 55% of potential valuation-associated return if the shares ultimately were to trade at a normal earnings multiple.

Therefore, if one is willing to accept the possible decline in the Penske earnings over the near-term, which, in the paradoxical way of the market, appears to be already reflected in the valuation, a longer time horizon could well be rewarded with a reasonable low double-digit rate of return. Accordingly, the Penske Automotive Group shares are recommended for purchase.

Company Description

Penske operates through three segments, which are Retail Automotive Dealership, Retail Commercial Truck Dealership, and Commercial Vehicle Distribution. In addition, the company owns a 28.9% interest in Penske Truck Leasing.

Through the Retail Automotive segment, which generates slightly over 90% of total company revenue, Penske is the second largest automotive retailer in the United States, as

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measured by sales (AutoNation is the largest). It operates 149 franchises in the U.S. and 191 internationally. The domestic market is extremely fragmented, with an estimated 16,700 franchised new car dealerships, and twice as many independent used car dealerships¹. Based on this, Penske has only 0.9% market share. Even AutoNation, which operates 326 new vehicle franchises in the U.S., only captures 1.9% of the overall market. Given this, it is easy to understand why the primary growth strategy of these companies is to continually acquire dealership franchises and consolidate the industry.

At the end of 2018, about 70% of the Penske automotive retail sales were derived from luxury brands such as Audi, Mercedes-Benz, and Jaguar. Specifically, BMW/MINI accounted for 23%

of all automotive revenues, as shown below.

Exhibit 1: Penske Automotive Revenue, by Brand

The focus on international luxury automobile brands is logical, as the company has a strong presence in the U.K. market. At the end of the third quarter of 2019, 55% of its dealership franchises were located outside of the U.S., such that almost half of overall company revenues were generated internationally (at the end of 2018, geographical revenues were: U.S. 56%, U.K. 35%, Germany/Italy: 6%, Australia/New Zealand 2%, Canada: 1%).

Penske also operates 15 used vehicle supercenters in the U.S. and U.K. that adopt a one price, “no haggle” customer strategy. In the U.S., these locations, six in total, operate under the CarSense brand primarily in the Philadelphia and Pittsburgh markets. In the U.K., Penske owns nine supercenter locations under the CarShop brand.

The vehicle manufacturers effectively regulate the margin dealers may earn on new car sales. Typically, dealerships obtain financing from the captive finance units of the manufacturers to purchase the vehicles directly and are then allowed to sell the vehicles at the suggested price. At the moment, Penske generates about \$3,000 in gross profit per new vehicle sold. The margin produced on used vehicles depends on the price at which the dealership can

1 According to the AutoNation 2018 Form 10-K Brand % of Auto Revs BMW/MINI 23% Audi 13% Toyota 10% Mercedes-Benz 9% Land Rover/Jaguar 9% Source: Company reports 113

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acquire the vehicle from the owner (or, in the case of wholesale transactions, the spread between the acquired auction price and the end retail consumer price). The current gross margin Penske records on used vehicles is roughly \$1,300 per vehicle.

Thus, if Penske can only manage \$3,000 in gross profit on a \$40,000 new vehicle (an 8% gross margin), in order for auto retail to be a sustainable business, other sources of revenue are required. Along these lines, the service departments of most dealerships generate the bulk of earnings. For instance, in 2018 new vehicle sales represented 46% of the auto segment's total revenues, but only 24% of gross profit. In contrast, service and parts accounted for 10% of revenues and 42% of gross profit – clearly, a much higher margin business.

Additionally, Penske generates financing income when its customers obtain credit through its network of lenders. Importantly, Penske does not directly lend, and therefore does not have the associated balance sheet risk attached to a large receivables portfolio. The company simply collects an origination fee from the lender. This is very high margin – in 2018, finance produced 3% of segment revenues and 20% of gross profit. In the aggregate, service and finance represented 62% of total segment gross profit in 2018, highlighting the importance of these businesses to the dealership operating model.

Exhibit 2: Penske Automotive Segment Profit Mix (2018)

Penske's second operating segment is Commercial Truck Retail, branded in the marketplace as Premier Truck Group (“PTG”). This is still a relatively small segment, comprising 6% of total company revenues in 2018. Nevertheless, PTG operates 25 locations throughout Utah, Idaho, Texas, Oklahoma, Tennessee, Georgia – and also a small operation in Canada. At these dealerships, Penske sells medium and heavy-duty trucks, primarily offering Western Star and Freightliner brands.

Whereas the company's new auto sales amount to about \$40,000 per vehicle, the truck group generates almost \$112,000 per unit. Since the gross profit dynamic is not dramatically Total Gross Revenue Source Profit Contribution Service and Parts 42% New Vehicle 24% Finance and Insurance 20% Used Vehicle 13% Fleet and Wholesale 1% 100% Source: Company reports 114

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different than that of new auto sales—it generates roughly \$4,500 in gross profit per truck sold—Penske relies heavily on service revenue to generate additional income. For example, during 2018, new truck sales accounted for 63% of segment revenues, and only 19% of gross profit. On the other hand, service and parts comprised 67% of segment gross profit.

The third primary segment is Commercial Vehicle Distribution. Penske is an importer and distributor of heavy-duty trucks, primarily the Western Star (a Daimler brand), MAN (a VW Group brand), and Dennis Eagle, across Australia and New Zealand. In addition to the large

semis, it also distributes buses and refuse collection vehicles. The other component of this business is the distribution of engines and power systems from brands such as Allison, Detroit Diesel, and Rolls Royce. The company's network includes delivery to over 70 dealership locations as well as several other branches and field locations. At the end of 2018, this segment produced 2.5% of total company revenues.

Lastly, the company owns a 28.9% interest in Penske Truck Leasing Co. L.P. ("PTL"), which originally was a partnership with GE Capital. GE recently divested its interest in this business, allowing Penske to increase its ownership, as well as expand the interest held by the Japanese conglomerate, Mitsui. A separate entity (also controlled by Roger Penske) known as Penske Corporation owns 41.1% of PTL, while Mitsui now owns the remaining 30%.

PTL provides a wide range of transportation and supply chain services, including truck rental/leasing, distribution center management, and dry van truckload management. This has become a substantial business over the last decade, and given the company's increased ownership, now accounts for a rather significant amount of Penske's overall earnings. Under the equity method of accounting, PTL contributed \$130 million of income in 2018, which was 28% of total pre-tax earnings.

Penske: Since the Last Recession

As a cyclical business, sales of new vehicles in the U.S. have declined in almost every recession since 1970. While automobiles are a necessity for much, although not all, of the population, it is a large expenditure that generally can be delayed during weak economic cycles. During the 2008/2009 recession, the revenues of the two largest new car retailers declined between 25% and 34%. Interestingly, CarMax, which is one of the largest used car retailers, only experienced a 9% reduction in revenues during that time, as perhaps more consumers chose to acquire a less expensive used vehicle during the recession.

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Exhibit 3: Car Dealership Revenues During the 2008-2009 Recession

After a sharp recovery that began in 2010, light vehicle sales have remained at the 16 million to 18 million annual unit range since 2015, as shown below. As this sales plateau has persisted for almost five years, many observers expect revenues for the industry to either remain at this level, or perhaps decline should recessionary pressures or consumer credit weakness appear.

Exhibit 4: U.S. Light Vehicle Sales

Source: St. Louis Fed Year Penske AutoNation CarMax 2007 12,782 \$ 16,385 \$ 8,200 \$
2008 11,637 13,376 6,974
2009 9,523 10,758 7,470
Cum. Decline % -25.50% -34.30% -8.90% (\$ in millions) Source: Company reports 116

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Irrespective of one's opinion on the future outlook, Penske has performed rather well since the lows reached in 2009. From that base of \$9.5 billion in revenues, the company has steadily increased sales since, to the current \$23 billion. Prior to the recession, Penske recorded revenues of \$12.8 billion in 2007. The near doubling of revenues since that time has produced an annualized expansion rate of 5.4% - a reasonable growth rate for a cyclical, highly competitive, and quasi-margin constrained (meaning, the margin permitted by the auto manufacturers to the dealerships) industry. With some slight variations in certain years, the gross margin has remained consistent at roughly 15%.

Exhibit 5: Historical Revenues and Gross Margins (2007-2018)

Similarly, operating income has risen annually by 6.3% during this time, and net income by 13.2%. The higher rate of profit growth can largely be explained by the increasing contribution of Penske Truck Leasing, which is an equity method investment. While there has been a very modest degree of operating margin improvement (2.7% in 2007 versus 2.9% in 2018), this alone cannot account for the dramatically higher net profit growth rate.

The earnings from Penske's 29% share of Penske Truck Leasing directly add to pre-tax income (i.e., 29% of PTL's net income is captured below operating income on the Penske income statement). As this business has grown over the last decade, and in conjunction with the

additional interest purchased from GE Capital recently, the overall Penske earnings have benefited quite considerably. It should also be noted that the lowering of the U.S. corporate Year Revenues Gross Margin 2007 12,782 \$ 14.80% 2008 11,637 15.40% 2009 9,523 16.60% 2010 9,943 15.90% 2011 11,127 15.80% 2012 13,164 15.30% 2013 14,483 15.20% 2014 17,232 15.00% 2015 19,285 14.90% 2016 20,119 14.70% 2017 21,387 15.10% 2018 22,785 15.00% Annl'zd %: 5.40% (\$ in millions) Source: Company reports 117

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tax rate (resulting in a sizeable one-time tax-related gain for Penske in 2017) has also contributed to net profit growth since 2017.

Exhibit 6: Historical Earnings Growth (2007-2018)

Recently, though, the leveling of overall light vehicle sales both in the United States and in Europe has begun to apply some pressure to the company's revenues. Since Penske to some degree is continuously reshaping its dealership portfolio through acquisitions and disposals, it is important to evaluate year-over-year comparisons on a "same-store" basis to eliminate the distortion caused by frequent asset transactions.

During the first nine months of 2019, Penske suffered a 6.6% decline in new vehicle unit sales. While the average vehicle price did rise by 0.5% per unit, the increase was not sufficient, leading to a 6.1% erosion in overall new vehicle revenues.

Operating Operating Year Income Margin Net Income 2007 339 2.70% 120 \$
2008 242 2.10% n/a 209 2.20% 76
2010 253 2.50% 108
2011 296 2.70% 177
2012 365 2.80% 186
2013 430 3.00% 244
2014 500 2.90% 287
2015 567 2.90% 326
2016 575 2.90% 343
2017 611 2.90% 613
2018 665 2.90% 471
Annl'zd %: 6.30% 13.20% (\$ in millions) Source: Company reports 118

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Exhibit 7: Penske Retail Automotive Dealership Year-to-Date New Vehicle Sales

In somewhat of a contrast, the company's used vehicle business experienced unit sales growth of 0.8%, but the average selling price declined by 0.9%. Since the gross margin of this business is largely determined by the price at which the dealership can acquire the used vehicle, it appears that pricing dynamics in both the wholesale auction and consumer trade-in channels have tightened noticeably this year.

Exhibit 8: Penske Retail Automotive Dealership Year-to-Date Used Vehicle Sales

Valuation

As noted previously, new and used vehicle retail is a very low margin business. Limited operational and organizational efficiencies can be achieved through consolidation, but on balance the owners of dealerships are still bound by prices established by the auto manufacturers.

Despite being one of the largest and most effectively-managed companies in the industry, Penske still generates a net profit margin in the 1% to 2% range. It is really difficult to imagine how this can ever be a substantially more profitable business unless manufacturers cede more profit to the dealers – a notion that seems highly unlikely. 2019 2018 % Change Retail unit sales 163,418 175,001 -6.60% Retail revenue (in millions) 6,710 \$ 7,149 \$ -6.10% Revenue per unit 41,063 \$ 40,852 \$ 0.50% Gross profit per unit 3,040 \$ 3,090 \$ -1.60% Source: Company reports 9 Mos. Ended Sept. 30 2019 2018 % Change Retail unit sales 215,720 213,903 0.80% Retail revenue (in millions) 5,446 \$ 5,495 \$ -0.90% Revenue per unit 25,244 \$ 25,690 \$ -1.70% Gross profit per unit 1,307 \$ 1,509 \$ -13.40% Source: Company reports 9 Mos. Ended Sept. 30 119

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Even so, a low margin business can still be a successful investment over the long run. If profitability can be maintained, and revenues increase at a reasonably high rate, the earnings will expand accordingly. In the case of Penske, it has also managed to repurchase about 14% of its shares outstanding over the last 10 years, which has assisted in creating a double-digit rate of per share earnings growth.

Exhibit 9: Penske Margin and Earnings per Share History (2007-2018)

Penske's largest competitor, AutoNation, similarly produces net profitability that does not greatly exceed 2%. It is interesting to observe CarMax, which is one of the largest retailers of used vehicles. CarMax produces a net margin of 4% to 4.5%, or double the profitability of the new car dealers. While the margin characteristics of used vehicle retail are not dramatically different and, in some case, inferior, CarMax does have a sizeable lending portfolio. It has chosen to use its balance sheet to facilitate vehicle financing. This has the effect creating a higher margin overall business but, of course, also exposes the company to credit risk. Despite the two distinctly different operating models, during periods of generally increasing or even slightly stagnant industry sales, the margins remain very consistent.

Year	Net Margin	EPS
2007	0.90%	1.27
2008	n/a	n/a
2009	0.80%	0.83
2010	1.10%	1.18
2011	1.60%	1.94
2012	1.40%	2.05
2013	1.70%	2.70
2014	1.70%	3.17
2015	1.70%	3.63
2016	1.70%	3.99
2017	2.90%	7.14
2018	2.10%	5.53

Ann'd %: 14.30% Source: Company reports 120

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Exhibit 10: Peer Industry Historical Profitability (2009-2018)

The valuations of the few, large publicly-traded car dealership companies are expectedly low. Penske is forecasted to earn \$5.68 per share in profit in 2020, implying a forward P/E ratio of just 8.3x – although it is noteworthy that this estimate would suggest a prospective increase of 8% over the current year forecast of \$5.25 per share.

Since 2007, Penske has traded at an earnings multiple of as high as 16x to a low of roughly 8x. The average multiple during this 12-year period has been 12.9x, which is mostly reasonable for a cyclical, low margin company. Nevertheless, this does demonstrate that during a normal cycle, the shares could be valued, on average, at a level that is 55% higher than at present.

It might also be mentioned that over the past four years, the Penske share price has remained relatively unchanged at the \$44 to \$50 range. Yet, the company's revenues have risen steadily, if modestly, and operating and net income have increased rather robustly. For whatever attributable reasons, the valuation has been contracting while the company appears to be operating consistently in all respects, and now has a base of earnings that is considerably higher than four years ago.

Year	CarMax*	AutoNation
2009	3.70%	2.20%
2010	4.20%	1.90%
2011	4.10%	2.10%
2012	4.00%	2.00%
2013	3.90%	2.20%
2014	4.20%	2.20%
2015	4.10%	2.10%
2016	4.00%	2.00%
2017	3.90%	2.00%
2018	4.60%	1.90%

Source: Bloomberg Net Profit Margin data represents the FY2019-FY2010 reported results *CarMax operates on a Feb. 28 fiscal year, so the above 121

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Exhibit 11: Historical P/E Multiple (2007-2018)

In comparison, AutoNation trades at 10.8x the forecasted earnings of 2020. However, CarMax trades at nearly twice the valuation of Penske – 17.3x consensus 2020 profit estimates. CarMax has, at first glance, an advantage with respect to its financing business. The company does not

have to solely rely on third party financing providers that may or may not extend credit to prospect car buyers. If done properly, CarMax can use its balance sheet as a tool to manage revenue and earnings growth. On the other hand, the credit quality of the loan portfolio can be placed into question at any point in time, particularly during recessions or after extended periods of unrestrictive consumer credit access. At the moment, this appears to be a risk that CarMax investors are willing to accept without much question.

Possible Return Outcomes Penske is expected to generate \$464 million of earnings in 2020. Of this, the company currently pays a dividend of \$129 million, representing a 3.3% yield relative to the \$3.8 billion market capitalization.

Since 2007, the company's long-term revenue growth rate has been 5.4% annually. Some modest margin improvements allowed operating income to expand at a 6.3% annual rate during this time. If this can be sustained, in addition to the dividend yield, it is not difficult to arrive at a reasonable 10%-type long-term return from the shares.

However, net income and per share earnings have been increasing at a far higher rate of 13% due to the increasing amount of income collected from the Penske Truck Leasing business, Year P/E 2007 14.3x 2008 n/a 2009 16.9x 2010 14.3x 2011 11.5x 2012 16.1x 2013 15.9x 2014 15.3x 2015 8.6x 2016 13.6x 2017 7.3x 2018 8.5x Source: Bloomberg 122

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and from corporate share repurchases. Moreover, the currently depressed earnings multiple is 35% below the recent long-term average. When added to the base outcome, these factors could very well allow the investor to receive a return that noticeably exceeds 10% per annum over an extended period of time, irrespective of the perceived nearer-term earnings risk.

In a completely different scenario, it is worth examining the share ownership dynamic by the company's founder, Roger Penske Sr. Mostly through Penske Corp., he owns 42% of Penske Automotive Group. When adding the 15.8% ownership of long-time partner, Mitsui Corp. of Japan, the two collectively own 57.8% of the company, leaving public shareholders with 42.2% of the available float.

With a float-adjusted market capitalization of only \$1.6 billion, consider the following scenario. Penske could theoretically pay a 100% premium for the remaining shares held by the public, representing a total value of \$3.2 billion. If it could borrow at a 6% rate, the annual interest expense would be \$192 million. Relative to pre-tax earnings of \$604 million in 2018, the company could easily cover the interest payments in a normal industry cycle.

In this theoretical example, Penske would "go private" as a leveraged company. It currently has \$2.4 billion of debt, \$78 million in cash, and \$2.6 billion of shareholders' equity. In order to finance the purchase of its vehicles from the manufacturers, it has borrowed \$3.9 billion in floor plan notes payable. This debt is generally, but not always, non-recourse to the company and, as stated previously, often extended by the finance divisions of the large manufacturers. Nevertheless, adding another \$3.5 billion of recourse debt to the balance sheet to buy out public shareholders would certainly place the company in a leveraged position.

Therefore, this is not to imply that the company will engage in such a transaction. Rather, it is meant to demonstrate the possibilities that are available while the shares trade at this low valuation. Apart from the normal earnings cyclicalities, shareholders could experience a not unsubstantial return in any number of ways.

Summary and Recommendation

Penske is a well-managed operator of new and used vehicle dealerships. In the last decade or so, it has diversified into heavy-duty truck sales as well as truck leasing and distribution. Given that these are all cyclical businesses, the investor community currently believes that a decline in vehicle sales is imminent. Since this would undoubtedly damage the Penske earnings, the shares trade at less than 9x the 2020 earnings forecast. However, this short-term focus provides for a range of possibilities over a longer time horizon – all to the advantage of the investor. In the absence of a more substantial or strategic transaction, Penske appears quite capable of producing at least a 10% return per annum merely through 123

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the maintenance of the current operating strategy. Accordingly, the Penske Automotive Group shares are recommended for purchase.

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PENSKE AUTOMOTIVE GROUP, INC. CONSOLIDATED CONDENSED BALANCE SHEETS
September 30, December 31, 2019 2018 (Unaudited) (In millions, except share and per share amounts) ASSETS

Cash and cash equivalents \$ 77.5 \$ 39.4 Accounts receivable, net of allowance for doubtful accounts of \$5.6 and \$5.4 967.3 929.1 Inventories 4,043.9 4,040.1 Other current assets 98.8 86.6 Total current assets 5,187.5 5,095.2 Property and equipment, net 2,309.3 2,250.0 Operating lease right-of-use assets 2,384.4 — Goodwill 1,876.5 1,752.0 Other indefinite-lived intangible assets 547.4 486.2 Equity method investments 1,362.8 1,305.2 Other long-term assets 21.6 15.9 Total assets \$ 13,689.5 \$ 10,904.5 LIABILITIES AND EQUITY

Floor plan notes payable \$ 2,444.1 \$ 2,362.2 Floor plan notes payable — non-trade 1,410.6 1,428.6 Accounts payable 672.3 598.2 Accrued expenses and other current liabilities 722.2 566.6 Current portion of long-term debt 96.8 92.0 Liabilities held for sale 0.5 0.7 Total current liabilities 5,346.5 5,048.3 Long-term debt 2,282.5 2,124.7 Long-term operating lease liabilities 2,334.7 — Deferred tax liabilities 620.7 577.8 Other long-term liabilities 446.0 519.0 Total liabilities 11,030.4 8,269.8 Commitments and contingent liabilities (Note 11)

Equity

Penske Automotive Group stockholders' equity:

Preferred Stock, \$0.0001 par value; 100,000 shares authorized; none issued and outstanding —
— Common Stock, \$0.0001 par value, 240,000,000 shares authorized; 81,066,927 shares issued and outstanding at September 30, 2019; 84,546,970 shares issued and outstanding at December 31, 2018 — Non-voting Common Stock, \$0.0001 par value; 7,125,000 shares authorized; none issued and outstanding — Class C Common Stock, \$0.0001 par value; 20,000,000 shares authorized; none issued and outstanding — Additional paid-in capital 317.2 477.8 Retained earnings 2,607.7 2,365.8 Accumulated other comprehensive income (loss) (286.0) (234.5) Total Penske Automotive Group stockholders' equity 2,638.9 2,609.1 Non-controlling interest 20.2 25.6 Total equity 2,659.1 2,634.7 Total liabilities and equity \$ 13,689.5 \$ 10,904.5

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PENSKE AUTOMOTIVE GROUP, INC. CONSOLIDATED CONDENSED STATEMENTS OF INCOME

Three Months Ended Nine Months Ended September 30, September 30, 2019 2018 2019 2018
(Unaudited) (In millions, except per share amounts) Revenue:

Retail automotive dealership \$ 5,155.4 \$ 5,148.5 \$ 15,442.9 \$ 15,900.0 Retail commercial truck dealership 692.3 385.3 1,451.4 1,016.5 Commercial vehicle distribution and other 119.9 124.8 393.5 429.3 Total revenues 5,967.6 5,658.6 17,287.8 17,345.8 Cost of sales:

Retail automotive dealership 4,407.9 4,386.3 13,159.0 13,561.5 Retail commercial truck dealership 605.7 327.5 1,251.9 859.7 Commercial vehicle distribution and other 84.3 92.2 287.9 317.8 Total cost of sales 5,097.9 4,806.0 14,698.8 14,739.0 Gross profit 869.7 852.6 2,589.0 2,606.8 Selling, general and administrative expenses 672.8 662.8 2,008.1 2,001.3 Depreciation 27.5 25.9 81.0 77.2 Operating income 169.4 163.9 499.9 528.3 Floor plan interest expense (21.4) (20.2) (64.2) (59.0) Other interest expense (32.9) (28.3) (93.2) (86.7) Equity in earnings of affiliates 43.3 41.7 109.6 95.0 Income from continuing operations before income taxes 158.4 157.1 452.1 477.6 Income taxes (42.4) (27.1) (118.6) (104.7) Income from continuing operations 116.0 130.0 333.5 372.9 Income from discontinued operations, net of tax 0.1 0.1 0.3 0.2 Net income 116.1 130.1 333.8 373.1 Less: (Loss) income attributable to non-controlling interests (0.1) (0.1) (0.4) 0.2 Net income attributable to Penske Automotive Group common stockholders: \$ 116.2 \$ 130.2 \$ 334.2 \$ 372.9 Basic earnings per share attributable to Penske Automotive Group common stockholders:

Continuing operations \$ 1.42 \$ 1.53 \$ 4.02 \$ 4.37 Discontinued operations 0.00 0.00 0.00 0.00 Net income attributable to Penske Automotive Group common stockholders \$ 1.42 \$ 1.53 \$ 4.03 \$ 4.37 Shares used in determining basic earnings per share 81.6 84.9 83.0 85.2 Diluted earnings per share attributable to Penske Automotive Group common stockholders:

Continuing operations \$ 1.42 \$ 1.53 \$ 4.02 \$ 4.37 Discontinued operations 0.00 0.00 0.00 0.00 Net income attributable to Penske Automotive Group common stockholders \$ 1.42 \$ 1.53 \$ 4.03 \$ 4.37 Shares used in determining diluted earnings per share 81.7 84.9 83.0 85.3 Amounts attributable to Penske Automotive Group common stockholders:

Income from continuing operations \$ 116.0 \$ 130.0 \$ 333.5 \$ 372.9 Less: (Loss) income attributable to non-controlling interests (0.1) (0.1) (0.4) 0.2 Income from continuing operations, net of tax 116.1 130.1 333.9 372.7 Income from discontinued operations, net of tax

0.1 0.1 0.3 0.2 Net income attributable to Penske Automotive Group common stockholders \$
116.2 \$ 130.2 \$ 334.2 \$ 372.9 Cash dividends per share \$ 0.40 \$ 0.36 \$ 1.17 \$ 1.05 126

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PENSKE AUTOMOTIVE GROUP, INC. CONSOLIDATED CONDENSED STATEMENTS OF CASH FLOWS

Nine Months Ended September 30, 2019 2018 (Unaudited) (In millions) Operating Activities:
Net income \$333.8 \$ 373.1 Adjustments to reconcile net income to net cash from continuing operating activities:

Depreciation 81.0 77.2 Earnings of equity method investments (77.8) (69.5) Income from discontinued operations, net of tax (0.3) (0.2) Deferred income taxes 48.2 64.3 Changes in operating assets and liabilities:

Accounts receivable (38.0) 6.1 Inventories 119.9 205.2 Floor plan notes payable 108.7 (165.2)

Accounts payable and accrued expenses 128.6 26.2 Other (43.3) 18.7 Net cash provided by continuing operating activities 660.8 535.9 Investing Activities:

Purchase of equipment and improvements (188.8) (188.5) Proceeds from sale of dealerships 7.3

58.4 Proceeds from sale-leaseback transactions 7.3 10.7 Acquisitions net, including repayment of

sellers' floor plan notes payable of \$138.5 and \$25.8, respectively (326.9) (168.6) Other (2.3)

(3.5) Net cash used in continuing investing activities (503.4) (291.5) Financing Activities:

Proceeds from borrowings under U.S. credit agreement revolving credit line 1,458.0 1,163.0

Repayments under U.S. credit agreement revolving credit line (1,383.0) (1,335.0) Net

borrowings of other long-term debt 95.3 131.0 Net repayments of floor plan notes payable —

non-trade (18.0) (59.6) Repurchases of common stock (174.1) (55.8) Dividends (97.3) (89.7)

Other 0.1 (6.2) Net cash used in continuing financing activities (119.0) (252.3) Discontinued operations:

Net cash provided by discontinued operating activities 0.2 0.3 Net cash provided by discontinued

investing activities — — Net cash provided by discontinued financing activities — — Net cash

provided by discontinued operations 0.2 0.3 Effect of exchange rate changes on cash and cash

equivalents (0.5) (0.5) Net change in cash and cash equivalents 38.1 (8.1) Cash and cash

equivalents, beginning of period 39.4 45.7 Cash and cash equivalents, end of period \$ 77.5 \$ 37.6

Supplemental disclosures of cash flow information:

Cash paid for:

Interest \$139.6 \$ 127.4 Income taxes 69.8 36.5 Non cash activities:

Deferred consideration — \$ 6.8 Contingent consideration \$ 10.6 — 127

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PENSKE AUTOMOTIVE GROUP, INC. CONSOLIDATED BALANCE SHEETS

December 31, 2018 2017 (In millions, except share and per share amounts) ASSETS

Cash and cash equivalents \$ 39.4 \$ 45.7 Accounts receivable, net of allowance for doubtful

accounts of \$5.4 and \$5.5 929.1 954.9 Inventories 4,040.1 3,944.1 Other current assets 86.6

81.8 Total current assets 5,095.2 5,026.5 Property and equipment, net 2,250.0 2,108.6 Goodwill

1,752.0 1,660.5 Other indefinite-lived intangible assets 486.2 474.0 Equity method investments

1,305.2 1,256.6 Other long-term assets 15.9 14.4 Total assets \$ 10,904.5 \$ 10,540.6

LIABILITIES AND EQUITY

Floor plan notes payable \$ 2,362.2 \$ 2,343.2 Floor plan notes payable — non-trade 1,428.6

1,418.6 Accounts payable 598.2 641.6 Accrued expenses 566.6 523.5 Current portion of long-

term debt 92.0 72.8 Liabilities held for sale 0.7 0.7 Total current liabilities 5,048.3 5,000.4 Long-

term debt 2,124.7 2,090.4 Deferred tax liabilities 577.8 481.5 Other long-term liabilities 519.0

540.3 Total liabilities 8,269.8 8,112.6 Commitments and contingent liabilities (Note 12)

Equity

Penske Automotive Group stockholders' equity:

Preferred Stock, \$0.0001 par value; 100,000 shares authorized; none issued and outstanding —

— Common Stock, \$0.0001 par value, 240,000,000 shares authorized; 84,546,970 shares

issued and outstanding at December 31, 2018; 85,787,507 shares issued and outstanding at

December 31, 2017 — — Non-voting Common Stock, \$0.0001 par value; 7,125,000 shares

authorized; none issued and outstanding — — Class C Common Stock, \$0.0001 par value;

20,000,000 shares authorized; none issued and outstanding — — Additional paid-in capital

477.8 532.3 Retained earnings 2,365.8 2,009.4 Accumulated other comprehensive income (loss)

(234.5) (146.5) Total Penske Automotive Group stockholders' equity 2,609.1 2,395.2 Non-

controlling interest 25.6 32.8 Total equity 2,634.7 2,428.0 Total liabilities and equity \$ 10,904.5

\$ 10,540.6 128

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PENSKE AUTOMOTIVE GROUP, INC. CONSOLIDATED STATEMENTS OF INCOME

Year Ended December 31, 2018 2017 2016 (In millions, except share and per share amounts)

Revenue:

Retail automotive dealership \$ 20,849.2 \$ 19,824.3 \$ 18,673.2 Retail commercial truck dealership 1,374.5 1,048.0 1,000.7 Commercial vehicle distribution and other 561.4 514.6 444.6

Total revenues 22,785.1 21,386.9 20,118.5 Cost of sales:

Retail automotive dealership 17,790.6 16,899.5 15,968.4 Retail commercial truck dealership 1,163.0 882.2 857.8 Commercial vehicle distribution and other 416.6 382.7 325.7 Total cost of sales 19,370.2 18,164.4 17,151.9 Gross profit 3,414.9 3,222.5 2,966.6 Selling, general and administrative expenses 2,646.3 2,516.0 2,302.0 Depreciation 103.7 95.1 89.7 Operating income 664.9 611.4 574.9 Floor plan interest expense (80.9) (63.4) (50.9) Other interest expense (114.7) (107.4) (85.4) Equity in earnings of affiliates 134.8 107.6 69.5 Income from continuing operations before income taxes 604.1 548.2 508.1 Income taxes (134.3) 64.8 (160.7) Income from continuing operations 469.8 613.0 347.4 Income (loss) from discontinued operations, net of tax 0.5 (0.2) (1.0) Net income 470.3 612.8 346.4 Less: (Loss) income attributable to non-controlling interests (0.7) (0.5) 3.5 Net income attributable to Penske Automotive Group common stockholders \$ 471.0 \$ 613.3 \$ 342.9 Basic earnings per share attributable to Penske Automotive Group common stockholders:

Continuing operations \$ 5.52 \$ 7.14 \$ 4.00 Discontinued operations 0.01 (0.00) (0.01) Net income attributable to Penske Automotive Group common stockholders \$ 5.53 \$ 7.14 \$ 3.99 Shares used in determining basic earnings per share 85,165,367 85,877,227 86,000,754 Diluted earnings per share attributable to Penske Automotive Group common stockholders:

Continuing operations \$ 5.52 \$ 7.14 \$ 4.00 Discontinued operations 0.01 (0.00) (0.01) Net income attributable to Penske Automotive Group common stockholders \$ 5.53 \$ 7.14 \$ 3.99 Shares used in determining diluted earnings per share 85,165,367 85,877,227 86,000,754 Amounts attributable to Penske Automotive Group common stockholders:

Income from continuing operations \$ 469.8 \$ 613.0 \$ 347.4 Less: (Loss) income attributable to non-controlling interests (0.7) (0.5) 3.5 Income from continuing operations, net of tax 470.5 613.5 343.9 Income (loss) from discontinued operations, net of tax 0.5 (0.2) (1.0) Net income attributable to Penske Automotive Group common stockholders \$ 471.0 \$ 613.3 \$ 342.9 Cash dividends per share \$ 1.42 \$ 1.26 \$ 1.10 129

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PENSKE AUTOMOTIVE GROUP, INC. CONSOLIDATED STATEMENTS OF CASH FLOWS

Year Ended December 31, 2018 2017 2016 (In millions)

Operating Activities:

Net income \$ 470.3 \$ 612.8 \$ 346.4 Adjustments to reconcile net income to net cash from continuing operating activities:

Depreciation 103.7 95.1 89.7 Earnings of equity method investments (89.0) (68.9) (50.0) (Income) loss from discontinued operations, net of tax (0.5) 0.2 1.0 Deferred income taxes 105.9 (108.7) 162.2 Changes in operating assets and liabilities:

Accounts receivable 30.4 (73.1) (92.5) Inventories (12.6) (419.9) 130.4 Floor plan notes payable 27.4 276.3 (162.8) Accounts payable and accrued expenses (17.1) 272.0 (17.8) Other (4.3) 37.2 (35.3) Net cash provided by continuing operating activities 614.2 623.0 371.3 Investing Activities:

Purchase of equipment and improvements (305.6) (247.0) (203.1) Proceeds from sale of dealerships 84.5 25.1 4.2 Proceeds from sale-leaseback transactions 10.7 22.2 — Acquisition of additional ownership interest in Penske Truck Leasing — (239.1) (498.5) Acquisitions net, including repayment of sellers' floor plan notes payable of \$58.2, \$101.6 and \$62.2, respectively (309.1) (449.7) (140.8) Other (5.7) (40.2) 1.5 Net cash used in continuing investing activities (525.2) (928.7) (836.7) Financing Activities:

Proceeds from borrowings under U.S. credit agreement revolving credit line 1,642.0 2,040.0 1,476.5 Repayments under U.S. credit agreement revolving credit line (1,784.0) (2,108.0) (1,396.5) Issuance of 3.75% senior subordinated notes — 3 0 0 . 0 — Issuance of 5.50% senior subordinated notes — — 500.0 Net borrowings of other long-term debt 235.5 42.0 42.9 Net borrowings of floor plan notes payable — non-trade 10.0 185.3 101.0 Payment of debt issuance costs (1.9) (4.0) (6.7) Repurchases of common stock (68.9) (18.5) (173.6) Dividends (121.2) (108.4) (95.1) Other (5.8) (5.8) (15.5) Net cash (used in) provided by continuing financing activities (94.3) 322.6 433.0 Discontinued operations:

Net cash provided by discontinued operating activities 0.5 0.5 1.6 Net cash provided by discontinued investing activities — 2.4 1.7 Net cash used in discontinued financing activities — (0.2) (0.2) Net cash provided by discontinued operations 0.5 2.7 3.1 Effect of exchange rate

changes on cash and cash equivalents (1.5) 2.1 (9.1) Net change in cash and cash equivalents (6.3) 21.7 (38.4) Cash and cash equivalents, beginning of period 45.7 24.0 62.4 Cash and cash equivalents, end of period \$ 39.4 \$ 45.7 \$ 24.0 Supplemental disclosures of cash flow information:
Cash paid for:
Interest \$ 190.2 \$ 163.2 \$ 129.8 Income taxes 39.6 (29.7) 48.6 Seller financed/assumed debt — 0.8 — Non cash activities:
Deferred consideration \$ 6.8 \$ — \$ — Consideration transferred through common stock issuance — 32.4 — Contingent consideration — 20.0 — 130

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June 25, 2019

Exclusive Marketers of
The European Contrarian Report

PCS Research Services 100 Wall Street, 20th Floor New York, NY 10005
research@pcsresearchservices.com (212) 233-0100 www.pcsresearchservices.com Research
Team Murray Stahl Steven Bregman
Rich Begun Thérèse Byars Ryan Casey James Davolos Peter Doyle Matthew Houk Utako
Kojima Eric Sites Fredrik Tjernstrom Steven Tuen

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London Stock Exchange Group plc (BUY)

Price: £54.80 Ticker: LSE LN
52-Week Range: £38.42 – £55.00 Dividend: £0.86
Shares Outstanding: 57.0 million Yield: 1.6%
Market Capitalization: £19.1 billion (\$24.1 billion)¹

Data as of June 19, 2019

¹ One British pound is equivalent to \$1.2641 as of June 19, 2019.

²The common shares of LSE LN trade on the London Stock Exchange.

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Investment Thesis

London Stock Exchange Group (LSE LN), or “LSE,” is one of the largest financial infrastructure companies in the world, operating equity, bond and derivatives markets and offering a variety of services that include market data, securities classification, indexing, clearing and settlement. The ongoing introduction of new products and services, combined with steady increases in the number of listed companies and the total notional value of securities traded, are expected to fuel future earnings growth of 10%-15% a year. There is significant near-term optionality, however, related to the potential acquisition of Euroclear, which would be enormously accretive to earnings. This scenario is not reflected in the company’s current valuation, nor is it a necessary condition for shareholders to earn an adequate return, thus creating an attractive, low-cost option.

LSE trades at 29x and 25x the consensus earnings forecasts for 2019 and 2020, respectively, which is within the range of its global peer group. Free cash flow multiples are on par with earnings due to relatively small capital expenditure requirements, thus implying that LSE trades at a yield of 3.7% based on 2019 estimates. Although these multiples seem high on an absolute basis, exchanges are typically high margin, economically vital, quasi-monopolistic businesses with strong balance sheets, and LSE is no exception. The positive attributes of this business model, combined with the negligible risk profile is therefore fairly reflected in the valuations of these exchanges.

LSE is somewhat unique in the industry, as it is one of the few exchanges with significant upside optionality, as it appears to be in the running to acquire Euroclear. Euroclear settles nearly €800 trillion in securities transactions each year, including LSE’s stock and bond trades, and LSE, recently purchased nearly 5% of the business. The rest could be acquired for an estimated £4.7 billion and could easily be integrated with LSE’s existing platform. Many of its standalone operating costs could be made redundant. Considering that Euroclear had operating expenses of €814 million last year, cutting just 50% of these costs (which is reasonable based on other recent mergers), would improve the business’s operating profit to €900 million (£800 million). Such a scenario would lead to share price appreciation of an estimated 50% assuming current valuation multiples remained constant.

There is a genuine motivation for the other owners, a consortium of banks, to sell Euroclear. The bigger question seems to be whether LSE will be the one to ultimately acquire it. However, again, none of this potential is reflected in LSE’s valuation, while the upside, conservatively estimated to be 50% over the short term, is hard to ignore. Without a deal, the company should continue to grow earnings by 10%-15% per year while paying a dividend of 1.6%, which is attractive given the risk. In this manner, the optionality on Euroclear is free. Based on these factors, shares of LSE are recommended for purchase.

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Company Description

London Stock Exchange Group is an international markets infrastructure business, operating a number of equity, bond, derivatives markets. These include the London Stock Exchange, Borsa Italiana, the European fixed income platform MTS, and the multilateral trading facility (or self-regulated trading venue) Turquoise. Service offerings such as The Yield Book, Mergent, SEDOL, UnaVISTA, XTF and RNS provide data services, research and analytics, and it is a leader in financial indexing, benchmarking and analytical services through FTSE Russell. Post trade and risk management services are provided through its majority ownership of the multinational clearing house LCH, the Italian clearing house CC&G, and the European custody and settlement business Monte Titoli. Lastly, LSE has a technology division that develops trading, market surveillance and post trade systems, both for the group’s own platforms and outside organizations and exchanges.

The history of the exchange dates back to 1571, when The Royal Exchange was first founded. The London Stock Exchange was formed in 1698, while the Group, as it exists today, was created in 2007, when London Stock Exchange merged with Borsa Italiana. In the years that followed, LSE grew by acquisition and through strategic partnerships, joining forces with Oslo Bors in 2008 to provide trading services to its various platforms, purchasing a majority stake in Turquoise in 2009, acquiring the remaining 50% of FTSE International in 2011, and closing on a majority stake in LCH Clearnet in 2013.

The CEO of LSE is David Schwimmer, who was hired in August 2018. Before joining the company, Mr. Schwimmer was employed for twenty years with Goldman Sachs, serving most recently as its Global Head of Market Structure. The Chairman of the Board, Don Robert, was appointed on January 1, 2019. He currently serves as Chairman of Experian plc (EXPN LN), having been appointed in 2014 after eight years as CEO. Mr. Robert will step down as Chairman of Experian in July 2019.

The Group is divided into four business segments; Information Services, Post Trade, Capital Markets, and Technology. As one can see in Exhibit 1, the Capital Markets division, which comprises the traditional exchange/trading business of the group, represented only 19% of total revenues (also called "income" by the company when including net treasury income) in 2018. Information Services and Post Trade operations represent the vast majority of the business, each accounting for nearly 40% of revenues. The contribution from the Technology segment is relatively modest at just 3% last year.

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Exhibit 1 LSE: Total Income by Segment; 2018

Source: Company reports.

Information Services This division supplies subscribers with market information including real time pricing data, product identification, analytics solutions and reporting and reconciliation services.

The FTSE Russell business accounts for 75% of the division's revenues, approximately 65% of which is derived from index data subscriptions. These include licenses, data services and analytics. The remaining 35% is earned from asset-linked fees, with approximately \$16 trillion benchmarked against the indexes that it provides.

Exhibit 2 LSE: Revenue Breakdown for the Information Services Division; 2018

Source: Company reports.

The Real Time Data service provides tick-by-tick pricing information, as well as the full depth of the order book, to traders, brokers and fund managers. Simple pricing data is also licensed for display on websites. This business has been in slight decline in recent years, as the number of professional terminals in use has been gradually shrinking.

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The Other Information segment accounts for 14% of revenues, but has been growing rapidly. UnaVista, which is LSE's regulatory reporting platform, has seen business expand with the increased reporting requirements brought about by MIFID II. SEDOL, which provides identification codes for global equity, derivatives and fixed income securities, has also experienced an increase in recurring licenses.

As shown in Exhibit 3, the segment's total revenues expanded by 9% last year, while operating profit grew by nearly 15% due to the significant operating leverage inherent in the data provider/exchange business model.

Exhibit 3 LSE: Operating Results for the Information Services Division; 2017-2018 (£ in millions)

Source: Company reports.

Post Trade Services - LCH While Post Trade Services accounted for 38% of the business last year, LCH, alone, represented 31%, or £662 million, of LSE's £2,135 million in revenues. This business has clearing operations in the UK, Eurozone, US and the Asia-Pacific region. As shown in

Exhibit 4, 41% of LCH's 2018 revenues were generated through OTC Clearing, 33% through Non-OTC Clearing, and 26% was earned through Net Treasury Income, which is effectively the interest earned on collateral lodged with the clearing house. LSE owns 82.61% of LCH Group Holdings Ltd., having acquired a 14.64% stake in December 2018 for €425 million. This implies a valuation for the entire entity of €2.9 billion (£2.6 billion) before factoring in any control premium.

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Exhibit 4 LSE: Revenue Breakdown for the Post-Trade Services - LCH Division; 2018

Source: Company reports.

The OTC Clearing operation is comprised of SwapClear, which is the global leader in OTC Interest Rate Swap clearing, ForexClear, the foreign exchange derivative clearing service, and CDSClear, which handles US and European credit default swaps. The OTC Clearing business experienced significant growth last year, due in part to the continued rollout of Uncleared Margin Rules, which require counter-parties to post margin for non-cleared OTC derivative trades. The implementation of these rules has led to a number of new clients for LCH, as counterparties have chosen to begin having these trades cleared rather than post additional margin. LSE believes these tailwinds are now largely behind the company.

SwapClear has seen an increase in volumes due to the changing interest rate environment in the US and UK. It also launched Non-Deliverable Interest Rate Swaps in five currencies in 2018, with three more currencies expected to be added in 2019. Furthermore, SwapClear offers third-party compression services, which has experienced significant growth in notional compressed values. With OTC derivative securities, simply reversing a trade does not close the position. Instead, it leaves one with a long and a short position. Compression services net these positions out, thus freeing up a significant amount of capital for clients. Finally, ForexClear and CDSClear are launching new products and platforms in 2019, which should lead to continued growth.

Non-OTC Clearing includes RepoClear, which clears across 13 government bond markets, Equity Clear, which clears for over 20 trading venues, and the Listed Derivatives business, which clears for London Stock Exchange Derivatives Market ("LSEDM") and Euronext Derivatives Market. LSEDM is leaving the platform in June 2019 in favor of CurveGlobal, which is an interest rate derivatives joint venture between LSE, CBOE and a number of leading dealer banks. Clearing of CurveGlobal products will be handled by LCH.

As one can see in Exhibit 5, revenues at LCH increased by 13% last year, while total income, or revenues plus net treasury income, expanded by 18%. The growth was derived primarily from the OTC business and from higher short-term interest rates that led to an increase in net treasury income. 137

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Exhibit 5 LSE: Operating Results for Post-Trade Services - LCH; 2017-2018 (£ in millions)

Source: Company reports.

Post Trade Services – CC&G and Monte Titoli (Other) The remaining operations within Post Trade Services consist of the Italian-based clearing house CC&G and Monte Titoli, the Italian-based Central Securities Depository, which provides settlement, custody, asset servicing, collateral management and issuer services.

Exhibit 6 LSE: Revenue Breakdown for Post-Trade Services - Other; 2018

Source: Company reports.

These operations generate only about one-fifth the revenue of LCH and, as shown in Exhibit 7, the business contracted slightly last year. The operating margins for this business are fairly robust at 138

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57% relative to 37% for LCH, due largely to a higher cost of sales at LCH given its offering of services such as third-party compression.

Exhibit 7 LSE: Operating Results for Post-Trade Services – Other; 2017-2018 (£ in millions)

Source: Company reports.

Capital Markets The Capital Markets segment comprises what most might consider to be the traditional exchange business. It can be broken down into Primary Markets, which includes admission fees charged to issuers and annual fees based on the market value of the securities issued, and Secondary Markets, which are fees based on the value traded for the UK Equities, Turquoise and MTS fixed income markets. For derivatives, the Italian equities market, and retail fixed income market, a fee is charged on a per trade or per contract traded basis. The Secondary Market segment is subdivided into Equities, which accounted for 41% of revenues in 2018, and Fixed Income, Derivatives and Other, which accounted for 31%.

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Exhibit 8 LSE: Revenue Breakdown for Capital Markets; 2018

Source: Company reports.

Total revenues increased by 4% last year, supported by modest growth across all segments, and operating margins were robust at 45.2%.

Exhibit 9 LSE: Operating Results for Capital Markets; 2017-2018 (£ in millions)

Source: Company reports.

This review of LSE's business should highlight just how integral it is to the financial markets it serves. It is not surprising, therefore, that its operating results display a significant degree of stability and growth. If anything, one might be surprised that revenues and operating profits are as stable as they have been considering that the business can be impacted by any number of exogenous factors such as the economic cycle, which may dictate when and how many companies decide to raise capital, or the market's unpredictable periods of volatility or stability, which have

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positive (or negative, respectively) implications for trading volumes. Increases in trading volumes can add disproportionately to earnings given the high degree of operating leverage in this business. This leverage can work against the company when trading slows, however.

Despite these vagaries, one finds that LSE has experienced consistent growth in revenues and operating profits over the last six years. Although it appears that 2014 was an outlier, the period shown represents only nine months of operations, as the company switched its reporting calendar from a March 31 st year end to a December 31st year end. Adjusted for this, 2014 was another year of expansion for the company. Earnings have been slightly more volatile due primarily to fluctuations in tax expenses, often revolving around the sale or purchase of assets.

Exhibit 10 LSE: Historical Financial Results; 2012-2018 (£ in millions)

Source: Company reports.

Growth in book value per share, adjusted for dividends, is on par with the annualized increase in earnings. This is particularly notable considering the company has used equity to make sizeable acquisitions, such as the rights issue that was used to acquire the Frank Russell Company in 2014. The rights issue granted shareholders the opportunity to purchase three new LSE shares for every eleven shares held at a 30% discount to the prevailing price. As one can see from the development in book value per share, the transaction was clearly accretive.

Exhibit 11 LSE: Total Value Created per Share; 2012-2018 (£ in millions)

Source: Company reports.

The company made an important acquisition in January 2019, when it purchased a 4.92% equity interest in Euroclear Holding SA/NV for £242 million. Euroclear settles nearly €800

trillion worth of securities transactions per year. It is believed that this equity holding, which will allow LSE a seat on Euroclear's board, will strengthen the existing relationship between the two entities. It also might be viewed as the first step towards a potential acquisition at some point in the future.

2012* 2013* 2014** 2015 2016 2017 2018 CAGR **Total Income** 871 1,210 1,044 1,419 1,657 1,955 2,135 16.9% **Operating Profit** 348 353 233 404 427 626 751 14.3% **Profit Attributable to Shareholders** 217 170 123 328 152 505 480 14.8% LSE's fiscal year ended on March 31st for the years indicated. The year-end figures shown are based on the March 31st figures for the following year. Reflects only 9 months of operations as the company switched from a March 31st year end to a December 31st year end. **The compounded annual growth rate is calculated using a period of 5.75 years, from March 31, 2013 (the figures used for 2012) through December 31, 2018** 2012 2013* 2014 2015 2016 2017 2018 **Equity Attributable to Shareholders** 1,531 1,527 2,527 2,744 3,106 3,227 3,343 **Shares Outstanding (millions)** 271 271 347 348 350 347 348 **Book Value per Share** £5.65 £5.63 £7.28 £7.88 £8.87 £9.31 £9.60 **Dividends** 77 81 56 116 130 159 189 **Total Value Created per Share** £5.93 £7.68 £8.60 £9.96 £10.87 £11.70 **CAGR: 2012-2018** 13.5% *LSE's fiscal year ended on March 31st for the years indicated. The year-end figures shown are based on the March 31st figures for the following year. **The compounded annual growth rate is calculated using a period of 5.75 years, from March 31, 2013 (the figures used for 2012) through December 31, 2018 141

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Valuation Analysis

London Stock Exchange is reasonably priced relative to forward earnings estimates, which implies that returns going forward might approach the 10%-15% annualized earnings growth it has generated historically. This is on par with increases in book value per share adjusted for dividends, and is likely to continue based on the number of new products and initiatives in its pipeline.

In addition, there is a degree of optionality regarding any future periods of extreme market volatility, which should temporarily boost earnings. The greater optionality, however, comes from the potential of another large acquisition, such as Euroclear. The increase in earnings from such a transaction would be massively accretive, as the existing operations could be simply integrated into LSE's existing platform, thus generating significant cost synergies. In the case of Euroclear, which posted 2018 earnings of €322 million on revenues of €1.3 billion, and operating costs of €814 million, the resulting margin expansion and earnings growth to a company like LSE would be significant. Interestingly, the company already settles stock and bond trades for LSE.

There is real potential for Euroclear to be taken over, as a number of existing owners, primarily member investment banks, have been pushing for a sale of the company. There may be other interested parties, however, as Intercontinental Exchange (NYSE: ICE) recently purchased a 10% stake in the company in two separate transactions that took place in 2017 and 2018. Still, the upside should not be overlooked, as it offers meaningful upside to an investment that is already poised to generate steady returns.

To appreciate this, let us first analyze LSE's current valuation. As shown in Exhibit 12, the company trades at 29.2x and 25.3x the consensus earnings forecasts for 2019 and 2020, respectively. While these multiples may appear rich, they are on par with other top exchanges such as CME Group (NASDAQ: CME). Furthermore, the business requires little in terms of ongoing capital expenditures (purchases of property, plant and equipment was £50 million in 2018 relative to depreciation and amortization of £287 million), which means that free cash flow is often greater than earnings.

Exhibit 12 LSE: Current Valuation Metrics (£ in millions)

Source: Company reports, Bloomberg, The European Contrarian estimates.

Share Price (6/19/2019) £54.80 Share Outstanding (millions) 348.2 Market Capitalization £19,080 Net Debt £1,860 Enterprise Value £20,940 2019E Earnings 654.0 2020E Earnings 755.1 Price/2019E Earnings 29.2x Price/2020E Earnings 25.3x 2019E EBITDA 1,193.0 2020E EBITDA 1,320.0 EV/2019E EBITDA 17.6x EV/2020E EBITDA 15.9x 142

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The relationship between earnings and free cash flow is shown in Exhibit 13, which, for the sake

of being conservative, includes the purchase of intangible assets in the free cash flow calculation. Although some of these charges might rightly be classified as growth expenditures (such as acquisitions, which contributed to the higher 2017 figure), ongoing investments in IT are also included in these figures. However, even when accounting for these outlays, total free cash flow from 2014 through 2018 was £1.7 billion relative to total profits of £1.6 billion. Based on this, the £654 million consensus earnings estimates for 2019 and 2020 translates into free cash flow yields of approximately 3.7% and 4.25%, which equals free cash flow multiples and 27x and 23.5x.

Exhibit 13 LSE: Historical Free Cash Flow, Operating Profit and Earnings; 2012-2018 (£ in millions)

Source: Company reports, The European Contrarian estimates.

A potential Euroclear acquisition is feasible, as the following analysis shows. Based on LSE's recent 4.92% purchase for £242 million, the implied a total valuation for Euroclear is £4.9 billion. LSE would need £4.7 billion to buy the remaining 95% of the company. Considering it has £1.5 billion in cash on hand, LSE has ample ability to borrow, or issue equity for, the remaining £3.2 billion.

The deal would also offer significant upside to earnings and LSE's valuation. As stated earlier, Euroclear's revenues in 2018 were €1.3 billion and operating costs of €814 million, for operating profits of €514 million (£458 million at today's exchange rate). If one assumes that half of these operating costs are redundant, which is conservative, this would imply that Euroclear could contribute operating profits of €900 million (approximately £800 million) as part of LSE.

Other acquisitions in this industry show us that these estimated cost savings are reasonable. In essence, albeit perhaps overly simplified, a securities exchange, aside from the requisite regulatory approvals and licenses, and the base level of employees and management structure, is little more than a computer platform that matches trades. Therefore, when one exchange acquires another, it can migrate the existing volume of trading and clearing to one system and forego the maintenance expense of the redundant one. Redundancies can include not only pure computer operating expense, but employees, real estate, electric power, and so forth. These are hardly insignificant. For example, when Intercontinental Exchange acquired NYSE Euronext in 2012, NYSE Euronext reported \$1.7 billion in operating expenses, while Intercontinental Exchange had operating expenses of \$536 million. By 2014, total operating expenses for the combined entity were \$1.64 billion, meaning that more than 100% of the Intercontinental Exchange's operating expenses were made redundant after integrating with the larger NYSE platform. If anything, this shows that the 2014 2015 2016 2017 2018 Cash Flow from Operating Activities 288 498 221 659 722 Purchase of Property, Plant and Equipment (16) (30) (34) (47) (50) Purchase of Intangible Assets (43) (87) (112) (143) (144) Free Cash Flow 229 381 75 469 528 Operating Profit 233 404 427 626 751 Profit Attributable to Shareholders 123 328 152 505 480 143

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above analysis of the cost savings and upside related to a Euroclear acquisition may be conservative.

If one adds £4.7 billion to the company's net debt to finance the acquisition, then adds £800 million to the 2019 EBITDA estimate (ignoring, for the sake of simplicity, any depreciation or amortization at Euroclear), one sees that LSE's EV/EBITDA multiple falls from 17.6x at present to 12.8x the 2019 EBITDA estimate pro forma for the Euroclear acquisition. The company would also trade at an estimated 17x pro forma earnings, which is significantly below the current multiple of 29x.

Exhibit 14 LSE: EV/EBITDA Valuation Pro Forma for Euroclear Acquisition (£ in millions)

Source: Company reports, The European Contrarian estimates.

Should an LSE/Euroclear ultimately trade at 17.5x forward EBITDA forecasts, which is not only the current multiple for LSE, but the current average multiple for the US peer group of CME Group, Intercontinental Exchange, NASDAQ (NASDAQ: NDAQ) and CBOE Global Markets (BATS: CBOE), this would lead to nearly 50% upside to LSE's valuation.

Exhibit 15 LSE: Fair Value Target Pro Forma for Euroclear Acquisition (£ in millions)

Source: Company reports, The European Contrarian estimates.

This upside is not reflected in LSE's share price. Nor is the acquisition a necessary condition for shareholders to earn a favorable return, as the business should continue to grow at a double-digit rate in its current state while carrying relatively little risk of long-term impairment. In this sense, shareholders are buying the Euroclear optionality for free, which is an attractive

proposition.

Share Price (6/19/2019) £54.80 Share Outstanding (millions) 348.2 Market Capitalization £19,080 Net Debt £6,560 Enterprise Value £25,640 2019E Pro Forma EBITDA 2,000.0 EV/2019E EBITDA 12.8x 2019E Pro Forma EBITDA 2,000.0 Target Multiple 17.5x Implied Enterprise Value £35,000 Net Debt £6,560 Implied Market Capitalization £28,440 Current Market Capitalization £19,080 Potential Upside 49% 144

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Conclusion

The world's large financial infrastructure companies are firmly established, economically vital assets and, as such, carry negligible risk. Growth is typically modest, but consistent over time, being generated from new products and the gradual expansion of capital in the markets, both of which lead to increases in the number, and notional value, of securities traded. This risk profile, combined with ongoing growth potential, is reflected in the current valuations of these companies, with LSE trading at a forward free cash flow yield of approximately 3.7%. Its earnings are forecast to grow 15% next year.

There is significant upside to these forecasts, however, as LSE seems interested in acquiring Euroclear. This business could be added to the company's LCH business with little additional overhead, thus making redundant a large portion of Euroclear's operating expenses. LSE could experience a 67% increase in EBITDA from such a transaction, leading to share price appreciation of an estimated 50% in the short term, while allowing for additional future appreciation as the debt used to finance the acquisition is repaid.

This optionality is clearly sizeable, yet investors are paying little for it, as the current valuation is reasonable and allows for adequate risk-adjusted returns even if the company failed to eventually acquire Euroclear. Based on this, shares of LSE are recommended for purchase.

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Exhibit 16 LSE: Consolidated Income Statement (£ in millions)

Source: Company reports.

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Exhibit 17 LSE: Consolidated Balance Sheet (£ in millions)

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Source: Company reports.

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THE STAHL REPORT The Contrarian Series July 9, 2020

Exclusive Marketers of
The Stahl Report

PCS Research Services 100 Wall Street, 20th Floor New York, NY 10005

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Carnival Corp (BUY)

Price: \$14.91 Ticker: CCL
52-Week Range: \$7.80 - \$51.94 Dividend: None
Shares Outstanding: 721m Yield: n/a
Market Capitalization: \$10,750m

Valuations within this text are based on
a \$14.91 share price

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Investment Thesis

Carnival, the largest cruise ship operator in the world, is currently suffering one of the worst periods in its history. Since March 2020, the company's cruise ships have been docked and are not expected to resume normal operations until at least September or October of this year. This means that no meaningful revenue is being generated during this COVID-19-related shutdown, and Carnival is sustaining a monthly cash burn rate of \$650 million.

Despite this predicament, it raised roughly \$10 billion, mostly in debt capital, in very short order. At the end of May 2020, Carnival had \$7.6 billion in cash and available liquidity. If operations are able to resume by October 1st, using the current monthly cash burn rate, approximately \$5 billion in remaining excess liquidity would be available. Clearly, in a best case scenario, Carnival could survive and gradually begin to rebuild its prior base of cash flow.

In this type of outcome, it is salient to assess the future earnings power, which will initially be below historic levels due to the additional interest expense associated with the new debt. The \$9 billion in debt raised since March will require approximately \$726 million in annual interest expense. If one were to use the \$3.3 billion in operating income generated in 2019 as a forward-looking earnings baseline, Carnival could produce after-tax income of \$2.3 billion (after making adjustments for the recent debt and equity capital raises). Assuming operations are fully restored, the company could perhaps earn \$3/share, which is meaningfully lower than the \$4.32/share reported in 2019. However, even at a P/E ratio of 12x, which is well within historical parameters, the shares would be worth \$36.13 – a 142% potential return from the current share price.

Although this return could be achieved in a short period of time (meaning, 1-3 years), it does not come without the obvious risks. In fact, the entire industry really faces a binary outcome. If health authority-mandated restrictions are eliminated, future cruise ship booking levels suggest a fairly rapid recovery in passenger attendance rates. On the other hand, health and safety concerns could certainly continue to delay a resumption in cruise activity. Even though the largest operators such as Carnival have sufficient liquidity to withstand a shutdown lasting throughout this year, a prolonged period of inactivity would deplete this cash and perhaps force an existential crisis throughout the industry.

In light of this, one must balance the observable risks with the prospect of an industry recovery and the substantial share price expansion that would follow. At the moment, it appears that a chance of recovery is still well within reason and, consequently, the shares are recommended for purchase.

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Company Description

Operations Carnival is by far the largest cruise ship operator in the world. In terms of passengers carried, it accounted for 43% of the entire industry in 2019. The second largest, Royal Caribbean, accounted for 22%. Thus, these two companies alone control nearly two-thirds of the whole industry. This is somewhat logical, because enormous amounts of capital are required to build the increasingly sophisticated ships (i.e., high barrier to entry). Moreover, ports do not have unlimited docking space, so in theory a flood of new entrants to the market would certainly pose some structural capacity issues.

Carnival operates a variety of cruise brands, each providing a different level of service, itinerary, and entertainment options. The flagship brand, Carnival Cruise Lines, serves mostly the North American market, with 18 ports throughout the United States. The 27 Carnival-branded ships provide numerous food, entertainment, and offshore excursion options, and have recently added celebrity dining features that include restaurants branded by Emeril Lagasse and Guy Fieri. In addition, the company has on order a new ship, the Mardi Gras, that will be docked in Long Beach, California, and will feature the first ever roller coaster at sea. Passengers typically book packages ranging from 2 to 24 nights to destinations that include the Caribbean, Bermuda, Canada, Mexican Riviera, Alaska and Hawaii.

The Princess line, with 18 ships, travels to all seven continents, providing 380 ports of destination around the world. The company markets this as a premium cruise package, with travel options that include South America, the Panama Canal, Europe, Asia, Mexico, Australia, and the Bahamas.

Holland America is a European-themed cruise experience that travels to over 425 destination ports. Guests are provided with European-inspired accommodations, dining, and entertainment options on vacations that can be booked for up to 128 days.

The Seabourn line, comprised of five ships, is an ultra-luxury offering, characterized by smaller ships that provide suite-only room accommodations. Passengers receive customized individual service, upscale dining options, and personalized offshore excursion opportunities. In fact, on the Seabourn Venture, a ship that is scheduled for delivery in 2021, two submarines are carried on this ship that take passengers on underwater excursions at various destinations. Seabourn carries passengers to exotic locations, typically on cruises that last from 7 to 100 days.

Carnival also operates various international cruise lines that are specific to each country. For example, Costa is an Italian brand that carries passengers from Italy to destinations around the world. Other international brands include P&O Australia, AIDA (Germany), and P&O U.K.

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While the international operations certainly diversify the revenue stream, North America remains the company's largest market. In 2019, 55% of sales were generated in this region, which is shown in the table below.

Table 1: Carnival Geographic Revenues, 2019

Region	Revenues	% of Total
North America	\$11,502	55.2%
Europe	6,318	30.3%
Australia/Asia	2,632	12.6%
Other	373	1.8%
Total	\$20,825	

(\$ in millions)

Passenger Growth Characteristics Although the industry has existed for decades, cruise ships as a form of vacation actually have a very low penetration or population utilization rate. For instance, in the United States and Canada, only 3.9% of the total population purchased a cruise ticket in 2018. In other parts of the world, the rates are similarly low: Australia/New Zealand 5.2%, U.K. 3%, Germany/Italy: 2.1%.

Despite this, the passenger growth rate over the last several years has been remarkably steady. At the industry level, the number of passengers carried has increased by 5.4% annually since 2007. Importantly, there has been positive growth in each year during this time, not interrupted even by the Credit Crisis of 2008/2009. Carnival has managed to expand its passengers carried by 4.4% per annum over this time period, somewhat closely matching the industry trend, as

shown below.

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Table 2: Carnival Passenger Growth Rate

Year	Industry	% Change	Carnival	% Change
2019	30,100,000	5.6%	12,900,000	4.0%
2018	28,500,000	6.7%	12,400,000	2.5%
2017	26,700,000	10.1%	12,100,000	5.0%
2016	24,250,000	4.5%	11,522,000	6.3%
2015	23,200,000	5.3%	10,837,000	2.6%
2014	22,040,000	2.3%	10,566,000	5.0%
2013	21,540,000	3.5%	10,061,000	2.4%
2012	20,820,000	1.6%	9,829,000	2.8%
2011	20,490,000	7.4%	9,559,000	4.5%
2010	19,070,000	8.4%	9,147,000	7.4%
2009	17,590,000	8.0%	8,519,000	4.1%
2008	16,290,000	2.1%	8,183,000	6.7%
2007	15,950,000	7.672,000		

Ann'l'd %: 5.4% 4.4%

The consistency of passenger rates is indicative of the customer base which, while relatively small, is quite loyal. Traditionally, cruise vacations appealed to the aging or retired segment of the population. This is because there is a convenience factor not found in most land-based vacation options. A cruise experience allows the passenger to visit multiple destinations, without having to fly or drive to different locations, change hotels/rooms, unpack multiple times, all of which require additional time that most individuals would rather spend on leisure activities (and most cruise companies contract with a travel agency that can also book air travel to the origination port, providing another element of convenience). Most cruises currently available offer the convenience of an all-inclusive package that appeals not only to elements of the older population, but also to families with children.

In fact, the availability of entertainment on cruise ships is actually somewhat astonishing. In addition to the customary unlimited amounts of food and beverages, standard features such as maid service, pool access, group activities, gym access, water slides, and child care provide traditional vacation/hotel amenities. Moreover, some ships now offer more exotic offshore excursions, comedy clubs, jazz bars, galas and balls, casinos, luxury spas, and roller coasters. All of this appears to be not only solidifying the recurring loyalty of the customer base, but also very gradually expanding the passenger market to new demographics.

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Operating History

The historically stable passenger growth rate discussed in the previous section has provided a solid base for Carnival's revenues. Since 2007, sales have expanded by 4% annually, which is more or less the historical rate of increase in carried passengers. Of note, as well, was the company's performance during the Credit Crisis. As a provider of a truly discretionary product, one might expect that the severe consumer financial stress during that period would have contracted the cruise ship business substantially. However, revenues declined by only 9.9% in 2009 (relative to 2008), and Carnival actually achieved \$1.8 billion in net income. While one might view the current loss of revenues being mandated by government agencies as an aberration, under normal recessionary pressures, the cruise business has proven to be somewhat durable.

Table 3: Historical Carnival Revenues

Year	Revenues
2019	\$20,825
2018	18,881
2017	17,510
2016	16,389
2015	15,714
2014	15,884
2013	15,456
2012	15,382
2011	15,793
2010	14,469
2009	13,460
2008	14,947
2007	13,033

Ann'l'd %: 4.0%

In contrast, the company's profitability has been far more variable, due in no small part to the cost of fuel. The glaring weakness in the business is that cruise companies have not been able to raise ticket prices to a sufficient degree to mitigate the impact of periodic increases in fuel prices. As shown below, this expense can consume as much as 15% of revenues, which was the case during the 2010-2014 period of higher oil prices. Last year, fuel costs represented only 7.5% of total revenues, demonstrating that this is a volatile item on the income statement.

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Table 4: Annual Carnival Fuel Expense

Year	Revenues	Fuel Expense	% of Revenues	2019	\$20,825	\$1,562	7.5%	2018	18,881	1,619	8.6%
2017	17,510	1,244	7.1%	2016	16,389	915	5.6%	2015	15,714	1,249	7.9%
2014	15,884	2,033	12.8%	2013	15,456	2,208	14.3%	2012	15,382	2,381	15.5%
2011	15,793	2,193	13.9%	2010	14,469	1,622	11.2%	2009	13,460	1,156	8.6%
2008	14,947	1,774	11.9%	2007	13,033	1,096	8.4%				

Naturally, there is an inverse correlation between fuel costs and earnings. During the 2011- 2014 period, Carnival's operating margin was only 9%-11%. Following the swift decline in oil prices, operating profitability improved to a high of 18.7% in 2016, when fuel accounted for only 5. 6% of overall sales. While this is largely out of the company's control, it is important to recall that Carnival has successfully managed this variability over time, and remained solidly profitable during normal operating conditions.

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Table 5: Historical Carnival Profitability

Year	Operating Margin	2019	15.7%	2018	17.6%	2017	16.0%	2016	18.7%	2015	16.4%	2014	11.2%
2013	8.7%	2012	10.7%	2011	14.3%	2010	16.2%	2009	16.0%	2008	18.3%	2007	20.9%

Valuation

The Carnival shares have lost nearly 70% of their value since February 2020. Despite this, the company still has an \$11 billion market capitalization, so it is apparent that investors believe there is a more than reasonable possibility the company will survive the present circumstance. The market capitalization is a reflection of not only the anticipation of a resumption in cruise service later this year, but also the historic earnings power. In the two preceding years, Carnival earned over \$4/share, and nearly doubled its profit since 2015, as presented below.

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Table 6: Historical Carnival Earnings Per Share

Year	EPS	2019	\$4.32	2018	4.44	2017	3.59	2016	3.72	2015	2.26	2014	1.56	2013	1.39	2012	1.67
2011	2.42	2010	2.47	2009	2.24	2008	2.90	2007	2.95								

Simplistically, the shares trade at 3.5x last year's reported earnings. However, it is virtually certain the company will not achieve these earnings this year or even in 2021. Any prospective return or valuation must be based on a reasonable estimate of forward-looking profits.

To this end, let us review the measures recently taken to address the lack of revenues since March. Carnival issued 72 million shares at a price of \$8, raising \$575 million (which will increase the outstanding share count to 756 million). An additional \$4 billion was raised through the offering of 11.5% notes due in 2023. Then, \$2 billion worth of 5.75% convertible notes due in 2023 was issued. Finally, an existing \$3 billion credit facility was fully drawn, providing a cumulative total of \$9.6 billion in capital since March 1. Consequently, any future earnings will be reduced by the added interest expense and per share earnings will be diluted somewhat by the new shares.

As of May 31, 2020, Carnival had \$7.6 billion remaining in available cash and liquidity. Currently, the monthly cash burn rate (i.e., cash operating expenses, capital expenditures and other capital commitments) is \$650 million. If, as anticipated, some operations resume by October 1st, this four-month period will consume \$2.6 billion in cash. However, it is not likely that all operations will resume simultaneously, so one should expect that there will be additional cash burn, even after October. As \$5 billion in remaining capital should be

1 Carnival is considering another \$3 billion in capital through a new loan facility that could be priced at LIBOR plus 7.5%. This likely will be used to refinance the \$2.4 billion in debt that is maturing in 2020. 157

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available at that point, it appears that sufficient liquidity exists to withstand reduced operating capacity for the remainder of 2020.

The salient consideration is the degree to which operations may begin to normalize in 2021. If this indeed transpires, it is not unreasonable to believe that within two or perhaps even three years the company's capacity utilization might return to historical levels. As to future earnings, operating income in 2019 was \$3.276 billion. Excluding any successful efforts to either reduce or refinance the current debt load, total interest expense might be \$946 million (\$220 million of existing expense plus roughly \$726 million related to recent offerings)².

After subtracting interest, pre-tax income in this outcome would be \$2.33 billion. As a Panama-incorporated company, Carnival pays little in the way of corporate taxes, generally less than 5%. Using the effective 2019 tax rate of 2.3%, Carnival would produce \$2.276 billion in net profit. The pro-forma share count should be 756 million, so per share earnings would be \$3.01.

For the purpose of this report, a modest 12x P/E multiple is used to determine the future share price. However, this historic average is closer to 17x, so there could well be considerable valuation optionality not included in this analysis.

Table 7: Historical Carnival P/E Ratio Year P/E 2019 10.1x 2018 13.0x 2017 19.9x 2016 14.9x 2015 21.3x 2014 28.2x 2013 28.2x 2012 23.2x 2011 12.5x 2010 18.1x 2009 14.9x 2008 6.3x 2007 15.1x

² It is roughly assumed that the \$3 billion credit facility drawn in March 2020, as it is a LIBOR-plus loan, might incur an interest rate of 5%. ¹⁵⁸

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Even at a 12x earnings multiple on the reduced profit base, the shares would be worth \$36.13, some 142% higher than the current price. This type of return provides allowance for an analytical error pertaining to the timing of a presumed recovery. For instance, if this were to take five years, the annualized profit would be 24.8%. Moreover, this does not account for future interest expense reduction. If Carnival is viewed as a much more healthy and creditworthy company, it should be able to refinance the emergency debt issued this year at rates far below this level (for example, if the 11.5% notes could be refinanced at even 7%, the company would save \$180 million a year in expense, or \$0.24/share).

Opponents of this line of reasoning surely will highlight the prospect of a longer-than-anticipated operational shutdown, which would clearly place the balance sheet at risk at some point next year. Even though Carnival has adequate access to capital at present, an increasingly negative operating outlook could reverse this circumstance quickly. This is an observable risk the investor must weigh, because the dynamics of health agencies' responses to COVID-19 are unpredictable. Some investors also question the passenger demand level even when Carnival places ships back into service.

To this point, the future demand for 2021 ticket purchases has been encouraging. Through May 2020, the company reports that next year's bookings are within historical parameters, indicating a somewhat normal demand level into 2021. Since the cruise companies are using the downtime to install medical grade ventilation systems and other health safety features, the loyal customer base may actually emerge rather unphased by the events of this year and resume normal ticket buying patterns.

Conclusion

Carnival is the largest cruise ship operator in the world. Its business has been devastated this year by the mandated stoppage of worldwide cruises in response to COVID-19. Despite no meaningful revenues since March, Carnival was able to raise nearly \$10 billion in capital to absorb the substantial operating losses that will be incurred until cruise service resumes later this year. If the company is able to normalize operations over a reasonable period of time, forward-looking earnings power appears to be around \$3 per share. Relative to the current \$15 share price, one could double one's money over the next couple of years if Carnival can merely restore prior passenger capacity. The risks to this outcome are well-documented throughout the financial media. Yet, if it is believed that the cruise ship industry is not permanently impaired and will adjust over time to current health safety recommendations, the Carnival share price could rise substantially. Thus, the shares are recommended for purchase.

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CARNIVAL CORPORATION & PLC CONSOLIDATED STATEMENTS OF INCOME (LOSS)
(UNAUDITED) (in millions, except per share data)

Three Months Ended	
February 29/28,	
2020 2019	
Revenues	
Passenger ticket \$ 3,234 \$ 3,199	
Onboard and other 1,556 1,474	
4,789 4,673	
Operating Costs and Expenses	
Commissions, transportation and other 766 709	
Onboard and other 471 467	
Payroll and related 610 557	
Fuel 396 381	
Food 277 268	
Other operating 1,001 759	
3,523 3,142	
Selling and administrative 678 629	
Depreciation and amortization 570 516	
Goodwill impairment 731 —	
5,502 4,287	
Operating Income (Loss) (713) 386	
Nonoperating Income (Expense)	
Interest income 5 4	
Interest expense, net of capitalized interest (55) (51)	
Other income (expense), net (7) (2)	
(57) (49)	
Income (Loss) Before Income Taxes (770) 338	
Income Tax Expense, Net (11) (2)	
Net Income (Loss) \$ (781) \$ 336	
Earnings Per Share	
Basic \$ (1.14) \$ 0.48	
Diluted \$ (1.14) \$ 0.48	

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CARNIVAL CORPORATION & PLC CONSOLIDATED BALANCE SHEETS (UNAUDITED) (in
millions, except par values)

February 29, 2020	
November 30, 2019 ASSETS	
Current Assets	
Cash and cash equivalents \$ 1,354 \$ 518	
Trade and other receivables, net 405 444	
Inventories 440 427	
Prepaid expenses and other 687 671	
Total current assets 2,885 2,059	
Property and Equipment, Net 38,023 38,131	
Operating Lease Right-of-Use Assets (a) 1,469 —	
Goodwill 2,176 2,912	
Other Intangibles 1,173 1,174	
Other Assets 1,216 783	
\$ 46,943 \$ 45,058	
LIABILITIES AND SHAREHOLDERS' EQUITY	
Current Liabilities	
Short-term borrowings \$ 1,004 \$ 231	
Current portion of long-term debt 2,196 1,596	
Current portion of operating lease liabilities (a) 168 —	
Accounts payable 904 756	

Accrued liabilities and other 1,754 1,809
 Customer deposits 4,690 4,735
 Total current liabilities 10,716 9,127
 Long-Term Debt 9,738 9,675
 Long-Term Operating Lease Liabilities (a) 1,312 —
 Other Long-Term Liabilities 887 890
 Contingencies
 Shareholders' Equity
 Common stock of Carnival Corporation, \$0.01 par value; 1,960 shares authorized; 658 shares at 2020 and 657 shares at 2019 issued 7 7
 Ordinary shares of Carnival plc, \$1.66 par value; 217 shares at 2020 and 2019 issued 359 358
 Additional paid-in capital 8,829 8,807
 Retained earnings 25,527 26,653
 Accumulated other comprehensive income (loss) ("AOCI") (2,028) (2,066)
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Treasury stock, 130 shares at 2020 and 2019 of Carnival Corporation and 60 shares at 2020 and 2019 of Carnival plc, at cost (8,404) (8,394)
 Total shareholders' equity 24,290 25,365
 \$ 46,943 \$ 45,058

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THE DEVIL'S ADVOCATE REPORT

December 2, 2019

Exclusive Marketers of
 The Devil's Advocate Report

PCS Research Services 100 Wall Street, 20 th Floor New York, NY 10005
 research@pcsresearchservices.com (212) 233-0100 www.pcsresearchservices.com Research Team Murray Stahl Steven Bregman Rich Begun Thérèse Byars Ryan Casey James Davolos Peter Doyle Matthew Houk Utako Kojima Eric Sites Fredrik Tjernstrom Steven Tuen Horizon Kinetics LLC ("Horizon Kinetics") is the parent holding company to registered investment advisers Horizon Asset Management LLC, Kinetics Asset Management LLC and Kinetics Advisers LLC. PCS Research Services ("PCS") is the exclusive marketer and an authorized distributor of this and other research reports created by Horizon Kinetics. This report is based on

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Credit Acceptance Corp.
(SELL)

Price: \$433.00 Ticker: CACC 52-Week Range: \$356.12-\$437.89 Dividend: Zero Shares
Outstanding: 18.8 million Yield: Zero Market Capitalization: \$8.1 billion
Data As of December 2, 2019

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Investment Thesis

Credit Acceptance Corporation (CACC) has an \$8.1 billion market capitalization. It is a consumer finance company that specializes in automobile financing for consumers with FICO scores of less than 650 or no FICO score at all. Approximately 96% of its customer base falls into this category. The business has expanded substantially in the past five years and revenue has increased by 89%. Because there is an economy of scale aspect to its business model, CACC's net income has increased more rapidly than revenue, by 127%, or almost 18% per annum and has generated close to a 40% average annual return on equity (ROE) over the past decade.

Growth and economic returns of this order explain why the company trades at 3.6x book value. By the same token, they have attracted increasing competition. This, in turn, has led to significantly narrower interest spreads for CACC; this measure has declined steadily from a high of 36% in 2009 to just 20% in the most recent quarter.

There is also the issue of market saturation, with a long period of sustained loan growth that well exceeds the expansion rate of the population of borrowers. To make auto loans more affordable, CACC has stretched out the average initial term for a loan to 58 months, compared to just 38 months in 2009. Industry-wide, the most commonly used auto loan had a 72-month term and the 84-month loan was the second most popular loan in the third quarter. There is even a 96-month term available. Thus, it is possible that CACC's average term will continue to stretch out as these longer-term loans become more mainstream. However, most new vehicles only retain around 30% of their value after eight years, and perhaps just half as much if the vehicle is sold in a wholesale /auction process after being repossessed.

The benefit of longer payback periods in enabling CACC to continue expanding its loan book at a greater rate than the supply of sufficiently creditworthy subprime borrowers, though, must be weighed against the increased default risk that is associated with such actions. When consumers could pay off their vehicle in 38 months, they very quickly reached a point at which they had significant equity in their vehicles and, therefore, were more reluctant to default. With 58-month terms—not to mention 96 months—it could take many years to reach the point at which the vehicle is worth meaningfully more than the remaining debt and, therefore, the propensity to default is considerably greater during those first few years, particularly if consumers experience economic hardship. Consequently, should CACC further lengthen its payback terms, it is unlikely to experience the same benefit relative to the risk assumed (from the vehicle's deteriorating collateral value) as has been the case in the past decade. This is an instance in which historical recovery rate data will not be reliable in assessing the credit and balance sheet risk during CACC's next credit downcycle experience.

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When a customer falls behind with payments, general practice is to renegotiate a loan even to the degree of permitting a customer to purchase a newer, albeit still used, vehicle for a longer-

term loan and reduced monthly payments. The ratio of under-water (negative equity) trade-ins has expanded from approximately 19% to 33% over the past decade¹. The result of this practice is that consumers get new vehicles and loans that are considerably greater than the value of the vehicles. This practice has helped the industry expand by forestalling defaults and increasing the value of loans outstanding, but it introduces substantial risks, since loan-to-value ratios increase, and the more negative equity consumers have in their vehicles, the higher the chance that they will default in a deteriorating economy.

The average yield on CACC's \$7.1 billion lending portfolio was 21% in the third quarter—a figure that has declined steadily from almost 35% in 2010. That said, since the company pays just 4% for its \$4.2 billion of outstanding debt, substantial profits can be realized. The problem is that the periodic provision of credit losses appears low given the creditworthiness of the customer base (or lack thereof). For example, in the first three quarters of 2019, CACC recorded only \$23 million of provision for credit losses, which is just shy of a rounding error for a \$7 billion credit portfolio comprised of the lowest FICO scores in the nation. In fact, provision for credit losses in the first nine months of 2019 was 17% lower than in the same period in 2015, despite the fact that the loan portfolio is 122% larger. These provisions represented approximately 3% of the loan in 2016-2017, but just about 1% since. Thus, while this reduction in annual provisions has added to earnings, a similar boost over the next few years is unlikely, unless provisions turn negative, which actually happened briefly in 2009. Conversely, should provisions expand back to the 3.3% recorded in 2017, pre-tax earnings in 2019 would be approximately 20% lower.

The Financial Accounting Standards Board (FASB) has issued an Accounting Standards Update, which includes an impairment model known as the current expected credit loss (CECL) model. As a result of this implementation, CACC's earnings are expected to decline 17% next year, based on Wall Street's consensus, as the company will have to assume higher default rates and charge offs. While this will not have an impact on free cash flow, it will impact reported earnings growth, GAAP earnings and adjusted earnings. To what extent, if any, the P/E ratio will be impacted remains to be seen. As of this writing, CACC trades at 12.3x consensus 2019 earnings but 15.0x consensus 2020 earnings. Should the 2020 EPS be valued at the company's 10-year average P/E of 10.6x, the shares would be worth \$366.

The past decade has been characterized by economic expansion, interest rates and unemployment rates that have declined to all-time lows, and a complete lack of recessions. Thus, it has been a highly accommodating environment for subprime lending. In an economic downturn, CACC's earnings could deteriorate as a result of higher default rates and increasing cost of capital. Even at the current earnings level, as if such an increase in

¹ Source: <https://www.caranddriver.com/news/a29831248/americans-not-researching-auto-loans/> 166

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default rates and the attendant increase in the company's borrowing rates did not negatively impact its earnings, the mere reversion of its P/E ratio to the 2009 average of 5.9x would cause the shares to decline by more than half. On the other hand, at 15x next year's projected earnings—a 42% premium to its 10-year average and almost twice its peer group's valuation—the upside appears limited. Therefore, shares of Credit Acceptance Corp. are recommended for sale or short sale.

Company Overview

Credit Acceptance Corp. (CACC), headquartered in Southfield, Michigan, provides dealer financing programs that enable automobile dealers to sell vehicles to consumers, regardless of credit history. This means that it attracts consumers with the lowest, if any, credit scores. The company was founded by Donald Foss in 1972 and came public in 1992. Mr. Foss, currently 74 years old, is the son of a used car salesman and, together with his wife, holds approximately 29% of the shares outstanding.

Of the more than 300 million Americans with a credit history, 32% have Fair Isaac Corporation (FICO) scores below 650 and loans to such individuals are considered subprime. For many years, 96% of CACC's borrower have had FICO scores below 650, or completely lacked a FICO score. While the delinquency and default rates in the sub-prime auto market are high, that is not an insurmountable problem in and of itself since the interest rate on loans to consumers with low FICO scores is set high enough as to attempt to compensate for the default rate. For example, average industry interest rates (but not necessarily CACC's rates, which are not disclosed) for certain brackets of FICO scores are as follows:

FICO Score	Rate Charged
640-659	13.1%
580-619	17.4%
560-579	19.6%
Less than 560	21.4%

Source: LendingTree

To its credit, CACC has resisted the impulse to leverage the balance sheet, at least to the degree that is possible for an interest rate spread-based financing business. Equity equals 34% of total

assets, which must be considered conservative in its class.

CACC has two programs: the Portfolio Program and the Purchase Program. The former loans accounted for two-thirds of the loan volume in the third quarter, a slightly decrease from 70% in calendar 2018, while the purchased loans accounted for the remainder. 167

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Portfolio Program Under this program, the dealer generally receives as payment for the vehicle:

- a down payment from the consumer;
- a non-recourse cash payment (advance) from CACC; and
- after the advance balance (cash advance and related dealer loan fees and costs) has been recovered by CACC, the cash from payments made on the consumer loan, net of certain collection costs and servicing fee charged by CACC.

CACC then records the amount advanced to the dealer as a dealer loan. The company generally requires dealers to group advances into pools of at least 100 consumer loans. All advances within a dealer's pool are secured by the future collections on the related consumer loans assigned to the pool. For dealers with more than one pool, the pools are cross-collateralized so the performance of other pools is considered in determining eligibility for dealer holdback.

Since typically the combination of the advance and the consumer's down payment provides the dealer with a cash profit at the time of sale, the dealer's risk in the consumer loan is limited. CACC cannot demand repayment of the advance from the dealer except in the event the dealer is in default of the dealer servicing agreement. For accounting purposes, the transactions described under the Portfolio Program are not considered to be loans to consumers. Instead, CACC's accounting reflects that of a lender to the dealer. Perhaps for that reason, CACC does not disclose interesting consumer metrics, such as its borrower's FICO score breakdown, delinquency and default rates, nor the salvage values for repossessed vehicles.

Purchase Program The Purchase Program differs from the Portfolio Program in that the Dealer receives a one-time payment from CACC at the time of assignment to purchase the consumer loan instead of a cash advance at the time of assignment and future dealer holdback payments. For accounting purposes, the transactions described under the Purchase Program are considered to be originated by the dealer and then purchased by CACC.

There are close to 60,000 auto dealers in the United States, and approximately 12,500 of them have been active in CACC's loan programs in the current year.

Originations CACC's originations, in terms of loan unit volume, increased by just 0.3% year-to-date, as of the end of the third quarter, whereas based on dollar volume, it increased 6.0%. The stagnation in unit volume could be a sign of intense competition and CACC's inability to 168

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expand without extending larger loans to ever-riskier borrowers for ever-longer payback periods.

During the third quarter, the average monthly payment of newly originated loans increased to \$403.84. This could be compared to the third quarter of 2015, when the figure was \$321.13—20.5% lower. Given the increase in loan repayment terms, CACC needs the average borrower of loans originated in third quarter to remain current for 15.6% longer while making 25.8% larger monthly payments, compared to the same period in 2015, for a total payment increase of 45.3%, to simply break even and recover the amount paid to purchase the loan.

CECL- New Accounting Standard Current Expected Credit Losses (CECL) is a new credit loss accounting standard model that was issued by the Financial Accounting Standards Board (FASB) on June 16, 2016 but is taking effect on December 15, 2019. CECL replaces the current Allowance for Loan and Lease Losses (ALLL) accounting standard. Under the new standard, preparers will need to consider not only their method for estimating CECL, but also the evidence and documentation their governance and internal control framework should produce to support their estimates. An entity will recognize an allowance for credit losses based on the difference between contractual future net cash flows and its estimate of expected future net cash flows. The new guidance also changes the scope of the special accounting for loans acquired with significant credit deterioration. CACC has announced that the adoption of ASU 2016-13 will have a material impact on its consolidated financial statements and related disclosures as the

company will be forced to change its accounting policies for loans.

While CACC's earnings could be more than 20% lower in 2020 as a result of CECL, the adoption of CECL should also result in higher volatility of the company's future earnings. On January 1, 2020, the company expects to increase its loans receivable balance by 30% to 40%, offset by an equivalent increase in allowance for credit losses. Going forward, CACC will record a provision for credit losses at origination between 12% and 15% of the amount of the loan because that is the amount that is estimated not to be repaid from the contractually owed amount. Wall Street's consensus forecast for 2020 EPS indicate a 17% decline relative to 2019. Interestingly, the company's competitors are not expected to be impacted by this accounting rule and they are all expected to grow their earnings year-over-year in 2020. Yet, CACC's P/E ratio is approximately twice as high as the average for the peers.

The Auto Loan Market The consumer finance industry in the United States had approximately \$4.11 trillion of outstanding borrowings as of the end of the third quarter, including vehicle loans and leases, credit cards, home equity lines of credit, private student loans, and personal loans. CACC's

2 Based on certain estimates by Wall Street 169

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primary focus is the vehicle finance segment, which accounts for approximately \$1.32 trillion of auto loans outstanding, as of the end of September. This figure has increased by 75% in the past nine years:

(In trillions of dollars. Source: NY Federal Reserve)

Drilling down further, the subprime segment, which is CACC's main focus, represents just over 20% of the market value, or approximately \$270 billion:

Source: New York Federal Reserve

Of this amount, auto finance companies such as CACC and Santander Consumer (a prior subject of this report) have approximately a 50% market share. According to the New York Fed, auto finance companies have a much higher percentage of subprime loans than banks and credit union. Banks have to be accountable to federal regulators who are reviewing safety and soundness as well as risks, whereas auto finance companies, lacking that type of regulation, can assume more risk so as long as they are achieving adequate returns and can raise capital from investors.

In 2018, there were 17.2 million new cars sold in the U.S. and approximately 85% of total new auto sales were financed. Following the decline during the financial crisis and the subsequent recovery, new auto sales appear to have plateaued over the past five years at just over 17 million units per year:

Source: Ward's Automotive Reports; U.S. Department of Commerce: Bureau of Economic Analysis 170

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Year to date, auto light vehicle sales are 1.1% lower compared to the same period in 2018, although hidden in this relatively stable figure is the fact that sales of passenger cars declined 16.4% year-over-year in the most recent month (October), which could be a warning sign. Pick-up trucks and SUVs, though, increased 6.4% y-o-y in the month. In terms of price, both new and used auto prices have increased around 15-20% in the past 8 years:

The value of auto loans outstanding has increased 4.3% year-over-year in 2019, even though new-vehicle unit sales has declined and used-vehicle unit sales have been lackluster. The increase in outstanding loans is almost entirely the result of higher average transaction prices for both new and used vehicles, rising loan-to-value ratios, lengthening average duration of auto loans and the rising popularity of leases by people who could otherwise would have paid cash for their vehicles.

Competition and Increasing Systematic Stress

Following the systemic stress the auto loan industry experienced in 2008-2009, lenders have generally responded by extending the initial term of low-grade auto loans. Most recently, CACC's average had increased further to 58 months, compared to the 38 months term average in 2009.

The term expansion has no doubt reduced the monthly payment for consumers relative to what they otherwise would pay, but given that the average amount of the auto loan has increased even more, the average monthly auto payment for a CACC borrower has increased to \$404 now, from \$334, while the payments continue for a 53% longer period relative to 2009. While the economic improvements, in terms of

3 https://www.marklines.com/en/statistics/flash_sales/salesfig_usa_2019 171

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unemployment and wage increases, justifies some of this, had it been easy for consumers to make the monthly payments, there would not have been such demand for ever longer-duration loans. The average amount of a CACC loan has increased from \$12,689 in 2009 to \$23,423 in the most recently reported quarter—an 85% increase:

(Right axis represents number of months)

This could be compared to the increase in the Consumer Price Index (CPI) of 19% and population growth of 8% during the period. Thus, on an inflation-adjusted per-capita basis, the burden of these loans has increased. Also, after being essentially unchanged from 2011 to 2014, the amount of the average consumer loan has increased 50% since the end of 2014. This, no doubt, has boosted CACC's growth rates, but will be very difficult to repeat that over the next five-year period since that would mean that the average consumer loan would have to increase to just over \$35,000 by 2024. While the increase in the average consumer loan amounts have been supported by longer payback periods as well as declining interest rates and declining unemployment rate in a steadily expanding economic environment, this appears to unlikely to offer the same support over the next decade.

According to Edmunds.com:

"The most common term currently is for 72 months, with an 84-month loan not too far behind. It's been creeping up: 10 years ago, the most common new-car loan term was 60 months, followed closely by 72 months. Loans for used cars are about as long: The most common term for a used car in 2018 was 72 months. Even though people are financing about \$10,000 less for used cars than they do for new cars, it takes them roughly the same amount of time to pay off the loan."

While 84-month loans are the second most common term, 96-month loans are generally also available. According to credit reporting agency Experian, the terms of loans reached record

Term (months)	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
\$0 - \$5,000	10	10	10	10	10	10	10	10	10	10	10
\$5,000 - \$10,000	10	10	10	10	10	10	10	10	10	10	10
\$10,000 - \$15,000	10	10	10	10	10	10	10	10	10	10	10
\$15,000 - \$20,000	10	10	10	10	10	10	10	10	10	10	10
\$20,000 - \$25,000	10	10	10	10	10	10	10	10	10	10	10
\$25,000 - \$30,000	10	10	10	10	10	10	10	10	10	10	10
\$30,000 - \$35,000	10	10	10	10	10	10	10	10	10	10	10
\$35,000 - \$40,000	10	10	10	10	10	10	10	10	10	10	10
\$40,000 - \$45,000	10	10	10	10	10	10	10	10	10	10	10
\$45,000 - \$50,000	10	10	10	10	10	10	10	10	10	10	10
\$50,000 - \$55,000	10	10	10	10	10	10	10	10	10	10	10
\$55,000 - \$60,000	10	10	10	10	10	10	10	10	10	10	10
\$60,000 - \$65,000	10	10	10	10	10	10	10	10	10	10	10
\$65,000 - \$70,000	10	10	10	10	10	10	10	10	10	10	10
\$70,000 - \$75,000	10	10	10	10	10	10	10	10	10	10	10
\$75,000 - \$80,000	10	10	10	10	10	10	10	10	10	10	10
\$80,000 - \$85,000	10	10	10	10	10	10	10	10	10	10	10
\$85,000 - \$90,000	10	10	10	10	10	10	10	10	10	10	10
\$90,000 - \$95,000	10	10	10	10	10	10	10	10	10	10	10
\$95,000 - \$100,000	10	10	10	10	10	10	10	10	10	10	10

Average Consumer Loan Average Advance
Payback Terms (right axis) 172

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highs in the second quarter, averaging 72.9 months for subprime new vehicle loans. Thus, CACC is actually below the average with 'just' a 58-month (but rising) average.

Already, a side effect of the longer payment terms is that some used vehicles are being traded in despite the owner being 'under-water' on the loan, which is to say having negative equity. Lenders and auto dealers have long accepted under-water trade-ins, but these have become increasingly popular. Also known as upside-down trade-ins, these simply add the remaining balance on the trade-in to the new loan required for the new vehicle (which could also be a used vehicle, but new to the buyer) being purchased.

In fact, almost a third of cars traded in in 2017 were worth less than the loans that had been financing them, according to car-shopping website Edmunds. All that does is strap more debt onto the consumer and makes it even more difficult to pay off the new vehicle. The Wall Street Journal recently wrote about one particular consumer who took out a loan to buy a car in 2017. In two years, he signed up for four auto loans, each time trading in the previous car and rolling the unpaid balance into the next loan. He recently bought a \$27,000 Jeep Cherokee with a \$45,000 loan from Ally Financial Inc. 4 This appears to be an example of 'extend and pretend' (that the loan will ever be repaid)—a strategy that cannot last.

Thus, while it was obviously possible to extend the repayment period from 38 months to 58 months in the past decade, it is highly doubtful that a similar extension can be implemented during the next 10 years since that would result in a payment period of more than 6.5 years. The problem with that is that the borrower would essentially have no equity in the vehicle for the

first few years, which would provide a greater incentive to default during that period. In contrast, borrowers with a 38-month term loan in 2009 quickly reached a point at which their equity in the vehicle was positive and growing, at which point they would be more reluctant to default.

According to carsdirect.com, “On average, a new vehicle depreciates 19 percent in the first year, half of which occurs immediately after you take possession. Fortunately, depreciation does not continue at this rate. You can expect a 15 percent drop in the second and third years.”. Using this depreciation schedule, during the 38-month term of the loans extended in 2009, the downside to the lender was limited, since the value of the vehicle, even if no down-payment was made, was always positive and rapidly expanding:

4 <https://www.wsj.com/articles/a-45-000-loan-for-a-27-000-ride-more-borrowers-are-going-underwater-on-car-loans-11573295400173>

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For a 58-month loan, though, again assuming zero down payment, the consumer has just over \$100 of equity in the vehicle after one year, whereas in the case above, with a 38-month loan, the same consumer would have more than \$1,200 of equity in the vehicle after one year. Thus, the propensity to default will be greater the longer the term of the loan, all else equal, since the consumer has much more to lose by defaulting on a shorter loan, should the vehicle get repossessed.

Most new vehicles only retain around 30% of their value after eight years, and perhaps just half as much if the vehicle is sold in a wholesale/auction process after being repossessed. For example, according to kbb.com, the 2011 Chevy Suburban originally had a Manufacturer's Suggested Retail Price (MSRP) of \$42,000, which could reach as high as \$65,000 with options for additional features, but now, the residual value is \$5,800-\$8,700 if this vehicle is being sold to a dealer. Similarly, a 2011 Ford Escape could originally be purchased new for an MSRP of \$30,500, which could range up to \$40,000 fully equipped. Today, the residual value is \$2,700-\$4,500 if sold back to a dealer. Consequently, 90% depreciation is possible in an 8-year period, which would render the consumer essentially without any equity in the vehicle until the very end of the loan term – particularly if they were upside-down on their trade-ins.

While it is difficult to assess the extent to which longer pay-back terms have helped reduce the default rate (because of lower monthly payments), attract additional customers (which can afford the lower payments) and support higher initial loan amounts, it appears this is unlikely to continue. Since an automobile is a depreciating asset and its collateral value must fall over time, there is a limit to how far the loans can be extended. Even if the trend of \$- \$5,000 \$10,000 \$15,000 1 3 5 7 9 11 13 15 17 19 21 23 25 27 29 31 33 35 37 Value / Loan Balance Value of Vehicle Loan Balance \$- \$5,000 \$10,000 \$15,000 1 4 7 1 01 31 61 92 22 52 83 13 43 74 04 34 64 95 25 55 8 Value / Loan Balance Value of Vehicle Loan Balance 174

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longer loan terms does not reverse but, rather, remains at the present length, the result should be an increase in defaults, fewer incremental new customers and a stagnation in the initial loan amount. Should the trend actually reverse, toward shorter payment periods, all these factors will should become exponentially worse. Normalization should lead to much lower profitability on these loans, with less revenue growth and a considerably lower valuation for the company.

During the past three years, the CACC unit increase (number of loans) has averaged 5.0% per year:

Since the amount of each loan has increased 9.4% on average, the dollar volume has increased at a rate of 14.3%, although as the chart above indicates, most of this increase was derived in the hump between the fourth quarter of 2017 and the third quarter of 2018. More recently, there has been just 0.2% year-over-year unit volume growth in the two most recent quarters, while dollar volume has increased 6.4%. However, the increasing dollar amounts have been manageable by the borrowers to a large degree because repayment terms have lengthened, which has kept the monthly payment relatively steady.

Delinquencies and Defaults on the Rise

Although it may be unintentional, it appears that the interest rate and credit system is now designed to gravitate towards increasingly extreme forms of financial behavior, until the stresses placed upon it become unbearable. Until this point, though, the economic development

has been very favorable to CACC: -10.0% -5.0% 0.0% 5.0% 10.0% 15.0% 20.0% 25.0% 30.0% 35.0% 40.0% Q1-2017 Q2 -2017 Q3 -2017 Q4 -2017 Q1 -2018 Q2 -2018 Q3 -2018 Q4 -2018 Q1 -2019 Q2 -2019 Q3 -2019 Unit and Dollar Volume, Y-o-Y Change Unit Volume Dollar Volume

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1. New car: JD Power Index, monthly data as of June 30, 2019 2) Used car: Edmunds' data, one quarter lag, data as of March 31, 2019 3) University of Michigan, monthly 4) US Bureau of Economic Analysis, one quarter lag, monthly as of March 31, 2019 5) US Bureau of Labor Statistics, monthly
2. Manheim, Inc. Indexed to a basis at 1995 levels; JD Power Used-Vehicle Price Index (not seasonally adjusted). 2) Auction Only – includes all auto-related recoveries including inorganic/purchased receivables from auction lanes only 3) Recovery Rate – Per the financial statements includes insurance proceeds, bankruptcy/deficiency sales, and timing impacts 4) Standard & Poor's Rating Services (ABS Auto Trust Data – two-months lag on data, as of April 30, 2019)

Yet, in the past five years there has not been any improvement in industry net loss rates or delinquencies, as the two charts below indicate.

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In fact, auto loan delinquency rates in the US, at a recent 4.71%, are higher than in any period dating back to December 2011. In the aftermath of the financial crisis, this measure reached 5.27% in the fourth quarter of 2010, but the current rate is equivalent to the rate experienced in the third quarter of 2009, at which time it was in the middle of a spike from just 2.1% in early-2006. In dollar terms, auto loans that are 90 days or more past due reached a historic high of \$62 billion as of the third quarter of 2019, according to the New York Fed:

Source: New York Federal Reserve. In billions of dollars.

As the chart indicates, the high during the financial crisis occurred in 2010 at approximately \$38 billion of delinquent loans. Similarly, a record seven million consumers are 90 days or more behind on their auto loan payments, according to the Federal Reserve Bank of New York, which actually represents around one million additional borrowers compared to 2010:

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Although the market is considerably larger now, so the facts that the dollar amount is higher and more consumers are delinquent do not necessarily mean that there are greater risks compared to 2010, the charts indicate that the stress levels in the subprime industry are at least elevated.

Interestingly, fewer than 1% of auto loans issued by credit unions are 90 days or more overdue, compared with 6.5% of loans issued by auto finance companies such as CACC. This is partly a reflection of the fact that credit unions make fewer subprime loans as a percentage of the total, but also partly because they appear to have higher credit standards.

Narrowing Margins

CACC provides limited disclosures regarding the creditworthiness of its borrowers; it only indicates that in every year since 2013⁶ the percentage of total loan unit volume with FICO scores below 650, or no FICO scores at all, has been 96%. While there has been no discernable improvement in that metric, CACC's allowances for credit losses have been declining steadily:

2019 reflects the first nine months

5 CACC does not disclose delinquency rates or default rates for loans that it owns or services 6

There are no disclosures of FICO scores prior to 2013 5.0% 10.0% 15.0% 2007 2008 2009 2010 2011 2012 2013 2014 2015 2016 2017 2018 2019 Allowance for Credit Losses, as percentage of Loan Portfolio 178

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The percentages of provisions declined by approximately half since 2007. Thus, should this measure return to the 2007 level, another \$500 million of allowances would be required. In relation to the current size of the portfolio, a full year's worth of net income would be eradicated.

In 2018, CACC's average loan portfolio balance was \$5.19 billion, and the company collected \$1.176 billion in interest. Note, this is not the interest charged but the interest collected. That amounts to an effective earned yield of 22.7%. As elevated as that might appear, this measure has been in a steady decline from 34.4%, which peak was reached in 2010:

CACC presumes that it will collect 63.6% of the sums that are dispensed as loans. Which means that it anticipates that about 36.4% of those monies will never be repatriated. The initial collection forecast has been in decline since 2009. CACC provides the following consumer loan metrics in its 10-K. It illustrates how both the initial projections for the collection percentage of loans by their yearly vintages have declined—both in terms of the initial forecasts as well as the most recently updated forecasts, which is the end of 2018:

7 Repayment forecasts include both principal and interest 15.0% 20.0% 25.0% 30.0% 35.0% 40.0% 2009 2010 2011 2012 2013 2014 2015 2016 2017 2018 Average Yield on Loan Portfolio 179

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For example, in 2013, the company expected to recover 72.0% of the capital extended, and by the end of 2018, it expected to recover 63.6%. It is entirely reasonable that the forecasted collection rates should decline as the company makes credit available to increasingly less creditworthy borrowers. However, that is not necessarily a problem as long as the cost of funding is sufficiently low, and the rate charged on loans is sufficiently high.

From that perspective, it should be concerning to CACC's investors that the initial forecast of a 63.6% collection rate for the 2018 loan assignment years represents the lowest in the past decade, and considerably below the 2009 loan assignment year (which was a period of financial distress for many consumers) of 71.9%. While the initial forecast turned out to be too conservative from 2009-2013 (the actual collections being better than the initial forecasts), this has been more mixed in the past five years, three of those years having experienced deterioration relative to the initial forecasts, while the updated forecasts for other two years have just improved marginally.

CACC's overall collection rate on a loan is derived by adding the 'advance' (the amount paid to dealers on consumer loans assigned and purchased) and the 'spread' (interest and fees collected). While CACC's advance has been relatively steady over the past decade at approximately 43-47% of the loan amount, all the variability in the overall collection rate has come from the spread (interest and fees collected), which has been in steady decline:

60.0% 65.0% 70.0% 75.0% 80.0% 85.0% 2009 2010 2011 2012 2013 2014 2015 2016 2017 2018 Consumer Loan Metrics Forecasted Collection Percentage Initial Forecast 180

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The spread, which was 35.7% in 2009 when the competitive environment was unusually favorable, has narrowed by more than 40% now, simply because the competition has become more intense. This measure reached a new low of 20.2% for the loans originated in the third quarter of 2019. In history, only in 2001 and 2007 was the spread at similar levels (approximately 21.5% at those points). As a result of the narrowing spread, the yield on the company's loan portfolio has declined more or less linearly from 34.4% in 2010 to 21.6% in the third quarter of 2019, as mentioned above.

The sub-prime loan system is able to function because the 70-80% of the borrowers who do pay their loans essentially subsidize the ones who default in one way or another. In order to grow, CACC must find additional borrowers, and those borrowers will likely to have low FICO scores or even no FICO scores. Since their default rates will continue to be ever greater, which is to say the company's collection rates will be ever lower—as has been the experience throughout this portion of the credit cycle—this will require an ever-greater subsidy from the paying contingent of sub-prime borrowers, or from the Federal Reserve in the form of yet more severe interest rate suppression. If the under-water borrowers start to default with loans that are considerably greater than the value of their vehicles, charge-offs could accelerate rapidly.

Valuation

Historical Valuation Since May 2008, CACC's shares have appreciated 18.6-fold. They now trade at about 4.6x book value, which is a very rare valuation level for any type of company engaged in lending activity and even more extraordinary for a company involved in high-risk lending. In the third quarter of 2009, at the time the CACC stock traded at approximately \$35, the company's unit loan volume was 26,069, compared to 86,331 in the same period this year, indicating a growth rate of 12.7% per annum. The stock's annual rate of appreciation, however, is 28.5% over the same period, the valuation expanding considerably more rapidly than the underlying business.

	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	Spread
16.0%	21.0%	26.0%	31.0%	36.0%	41.0%	2009	2010	2011	2012	2013	2014	2015
2016	2017	2018	2019	Spread	181							

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CACC's focus on the least creditworthy segment of borrowers has historically resulted in a low P/E multiple—not infrequently in the single-digits. While it is currently trading above 12x the 2019 Wall Street consensus, this is, in effect, a doubling compared to the levels experienced in 2009, and above the 10.4x average P/E since 2008:

Furthermore, given the expected reduction in earnings for 2020, CACC trades at a record-high 15.0x that consensus earnings estimate. Because of CECL, the 2021 CACC EPS could be as low as \$24 based on some Wall Street forecasts. At just 10.4x the 2020 earnings, in line with the historical valuation, CACC would trade at just \$300. Not much, in terms of an economic slowdown or increases in delinquency rates, appears to be required to make earnings considerably lower. Thus, should any but the most optimistic scenario materialize, both earnings and the P/E multiple could deteriorate rapidly.

Relative Valuation CACC, as a pure-play subprime auto lender, competes mainly with banks and credit unions, so most such companies are not similar enough for a relative valuation to be useful. However, compared to the only other subprime pure-play, Santander Consumer USA, a company whose stock this publication recently recommended for short sale, CACC trades at almost twice the 2020 P/E. It also trades at a substantial premium compared to Ally Financial and OneMain Holdings, while being the only company in the group without a dividend:

Company	Symbol	Price	2020 P/E	Div. Yield	P/Tan	BV	Market Cap	Ally Financial	ALLY
\$32.22	7.7x	2.1%	0.89x	\$12.8B	Santander Consumer USA	SC	\$23.46	7.8x	3.8%
\$8.3B	OneMain Holdings	OMF	\$43.00	6.4x	2.3%	1.43x	\$5.8B	Average:	7.3x 2.7% 1.15x

Credit Acceptance Corp.	CACC	\$430.00	14.9x	0.0%	3.39x	\$8.1B	5.0	10.0	15.0	20.0	2007
2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
2020	Historical	P/E	182								

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A significant portion of CACC's premium valuation might be explained by the company's fair value accounting, which provided significant earnings stability compared to peers. However, with CACC's adoption of CECL, significant earnings volatility could become the norm—a development that should also reduce CACC's valuation premium. Interestingly, the three peer companies are not expected to be impacted by CECL as they are all expected to expand between 5-13%, with an 8% average, whereas CACC's earnings are expected to be more than 17% lower compared to 2019. At the peer group's average P/E of 7.3x, CACC's shares would be worth \$211.

Ally Financial is not focused on sub-prime, since its average auto loan customer had a FICO score of approximately 700 in 2018, compared to 593 for Santander (CACC does not disclose this measure), and it has considerably higher revenues from the auto lending business and less than half the loan loss provisions (which appears justified given the higher creditworthiness of its borrowers). OneMain provides consumer loans, and though it does not provide an average FICO score, it discloses that almost exactly half of its borrowers have scores above 619 with the other being lower. Hence, assuming 619 is the average as well as the median, that is still

considerably higher than Santander's 593 average. Given CACC's similar market focus, it is reasonable to assume that CACC's average borrower consumer FICO score is almost identical to Santander's. Therefore, since it appears CACC's credit risks are greater than Ally's and OneMain's a discounted valuation might be warranted. If the company were valued in line with Santander, at 7.8x 2020 earnings, or \$225, its shares would decline by almost 50%.

Investment Summary

CACC, like the rest of the industry, has attempted to keep growing by extending payback terms, which has allowed consumers to assume higher loan amounts. With six-year terms now being the industry norm, with seven-year loans being the second more common and even eight-year loans being available, it appears that the limit will soon be reached. Another factor that has allowed the industry to expand is the increasing acceptance of under-water trade-ins, which saddles consumers with ever-increasing loan balances, and considerably higher loan-to-value ratios. That means that borrowers increasingly have no equity in their vehicles for ever-longer periods of time and, as such, should be more likely to default.

Even if these metrics do not reverse, they will no longer boost the industry as they no doubt have in the past decade. Recently, loan demand is declining and CACC's unit volume growth was essentially zero in the first nine months of the year, while the increase in loan amount was 6%. Without ever-expanding payback terms, it will be difficult to keep the expansion going. To the extent that it is possible, the company might be taking on greater risk while accepting lower returns, which is difficult to analyze given CACC's lack of FICO score 183

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disclosures. Yet, if it can be accepted that it is extending larger and larger loans to ever-less-creditworthy consumers, the company's premium valuation is at risk.

It is questionable whether current investors will be compensated for the risks that CACC assumes. During the past decade, the major economic measures have been favorable for used auto lenders:

- ☐ Rising GDP; ☐ Declining unemployment;
- ☐ Rising consumer confidence;
- ☐ Rising used vehicle prices, and
- ☐ Rising used vehicle recovery rates.

Yet, the industry has experienced rising sub-prime net loss rates and rising sub-prime delinquency rates. Despite that, CACC has recently recorded declining provisions for credit losses even as it has expanded its loan book. In an economic downturn, and with the entire balance sheet, more or less, in the lower echelons of sub-prime loans, CACC will be among the hardest hit lenders. The least creditworthy borrowers, who presumably are too risky to qualify for the standard 5% that prime auto loans carry, are instead expected to pay, in some cases, over 20%. Should the default rate rise on subprime in a material manner, which will happen eventually, the subprime auto lenders are in serious trouble. On the other hand, if the default rate does not rise, the limited size of the subprime market prevents meaningful growth. With a valuation that is considerably higher than its historical reference as well as its peer group, even though CACC's earnings are widely expected to decline significantly in 2020, the risk-reward appears to favor the short seller. Consequently, shares of Credit Acceptance Corp. are recommended for sale and short sale.

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CREDIT ACCEPTANCE CORPORATION CONSOLIDATED BALANCE SHEETS (UNAUDITED) (Dollars in millions, except per share data) As of

September 30, 2019

December 31, 2018 ASSETS:

Cash and cash equivalents \$ 24.7 \$ 25.7 Restricted cash and cash equivalents 329.8 303.6

Restricted securities available for sale 59.8 58.6

Loans receivable 7,072.8 6,225.2 Allowance for credit losses (509.1) (461.9) Loans receivable, net 6,563.7 5,763.3

Property and equipment, net 58.5 40.2 Income taxes receivable 10.8 7.9 Other assets 23.7 38.1

Total Assets \$ 7,071.0 \$ 6,237.4

LIABILITIES AND SHAREHOLDERS' EQUITY:

Liabilities:

Accounts payable and accrued liabilities \$ 184.6 \$ 186.4 Revolving secured line of credit 52.2 171.9 Secured financing 3,228.3 3,092.7 Senior notes 940.2 544.4 Mortgage note 11.5 11.9 Deferred income taxes, net 271.2 236.7 Income taxes payable 0.2 2.5 Total Liabilities 4,688.2 4,246.5

Commitments and Contingencies

Shareholders' Equity:

Common stock, 18,796,770 shares issued and outstanding 0.2 0.2 Paid-in capital 155.7 154.9 Retained earnings 2,226.0 1,836.1 Accumulated other comprehensive income (loss) 0.9 (0.3) Total Shareholders' Equity 2,382.8 1,990.9 Total Liabilities and Shareholders' Equity \$ 7,071.0 \$ 6,237.4

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CREDIT ACCEPTANCE CORPORATION CONSOLIDATED STATEMENTS OF INCOME

(Dollars in millions, except per share data) For the Three Months Ended

September 30,

For the Nine Months Ended

September 30, 2019 2018 2019 2018 Revenue:

Finance charges \$ 349.9 \$ 303.0 \$ 1,013.3

\$ 863.0

Premiums earned 12.9 12.2 38.2

34.2

Other income 15.9 16.8 51.6

45.8

Total revenue 378.7 332.0 1,103.1

943.0

Costs and expenses:

Salaries and wages 47.9 41.1 143.9

123.3

General and administrative 17.2 14.1 47.9

41.3

Sales and marketing 16.6 16.3 53.1

51.3

Provision for credit losses 19.3 14.0 49.2

39.2

Interest 50.4 41.1 145.2

114.3

Provision for claims 8.2 7.0 23.1

19.5

Total costs and expenses 159.6 133.6 462.4

388.9

Income before provision for income taxes 219.1 198.4 640.7

554.1

Provision for income taxes 53.7 47.4 146.5

132.0

Net income \$ 165.4 \$ 151.0 \$ 494.2

\$ 422.1

Net income per share:

Basic \$ 8.73 \$ 7.76 \$ 26.08

\$ 21.69

Diluted \$ 8.73 \$ 7.75 \$ 26.06

\$ 21.68

Weighted average shares outstanding:

Basic 18,944,672 19,465,563 18,948,140

19,456,389

Diluted 18,950,866 19,473,978 18,967,552

19,472,197

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August 2020

Exclusive Marketers of

The Fixed Income Contrarian Report

PCS Research Services 100 Wall Street, 20th Floor New York, NY 10005

research@pcsrsearchservices.com (212) 233-0100 www.pcsrsearchservices.com Research

Team Murray Stahl Steven Bregman

Rich Begun Thérèse Byars Ryan Casey James Davolos Peter Doyle Matthew Houk Utako

Kojima Eric Sites Fredrik Tjernstrom Steven Tuen

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Featured Companies

Weyerhaeuser Company (WY) PotlatchDeltic Corporation (PCH) CatchMark Timber Trust, Inc. (CTT) Acadian Timber Corporation (AND CN)

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Murray's Musings

CRYPTOCURRENCY AND CREDIT CARDS

In order to make cryptocurrency generally available to the global population, a means is required that will enable transfer and exchange. This has not developed yet because even the largest user of cryptocurrency lacks the means to create such a universal exchange system. It is not merely a matter of money. A *de novo* system would require years to construct, and even if it were to be completed, universal acceptance would be highly questionable.

An ideal solution is the use of an existing universally accepted transfer and exchange system for fiat (government-issued) currency. Credit cards qualify as such an ideal solution since they are already universally accepted. Towards this end, Coinbase Inc. obtained permission from Visa Inc. (V) in February to issue a Visa debit card that can spend the Coinbase bitcoin balance. Merchants need not accept bitcoin; the Coinbase bitcoin balance is effectively a U.S. dollar balance, since Coinbase converts crypto to U.S. dollars at time of purchase. It is not exactly the same as having a merchant directly accept bitcoin, but in practice, the bitcoin can be spent.

The relationship of the Swiss company Eidoo with Visa is a nother step closer to the acceptance of cryptocurrency. In this case, the company introduced a Visa debit card that uses regulated stablecoins for the crypto- to-fiat conversion. Stablecoins are digital currencies tied to some benchmark or combination of other assets or currencies that is sufficiently non-volatile in price so that the stablecoin can function as a practical medium of exchange. Perhaps the best known one, at least for a time, was Facebook's Libra, which was to be indexed to a basket of short-term government securities and bank deposits denominated in relatively stable currencies like the euro and U.S. dollar. Eidoo is designed to facilitate conversion, at the point of sale with a merchant, of bitcoin and ether into European currencies like the euro, the British pound, or the Swiss franc, so the conversion of the crypto creates a spendable Visa balance that can be debited for each transaction.

Very slowly, the cryptocurrency system is being integrated with the existing fiat currency payment system. Examples of these *de facto* crypto debit cards are Plutus, Wirex, Monolith, Revolut, Cryptopay, Bitcard, Uquid, 2gether, Coinbase, Bit Pay, Paycent, Crypterium, and the MCO Visa (formerly the Monaco Card).

Many of these cards have expanded beyond bitcoin. For instance, Paycent, which is based in Singapore, uses Litecoin, ether, Dash, and XRP in addition to bitcoin. The BitPay Card is a Mastercard; crypto can be loaded on it and converted into a fiat currency debit balance. Naturally, if one can convert crypto into a spendable fiat debit balance, it is possible to 189

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reverse the process and use a debit balance to purchase crypto. For example, CoinFlip Solutions, Inc., one of the leading operators of cryptocurrency ATMs, allows customers to use credit or debit cards denominated in fiat to purchase cryptocurrency.

According to the firm Cryptowisser, there are now at least 30 Visa and Mastercard debit cards from which to choose if one wishes to maintain a crypto balance in the existing accepted fiat currency network. However, even this list does not completely cover all the ways that fiat transfer and exchange methods are incorporating crypto. Crypto is also entering the digital wallet space of Apple and Google. A company known as Swipe sells a crypto banking application that gives access to 30 cryptocurrencies via a digital wallet. Balances can be transferred between banks using a Swift wire transfer or even via the Single Euro Payments Area (SEPA).

SEPA contains all members of the European Union plus the United Kingdom after Brexit, Iceland, Lichtenstein, Norway, Switzerland, Monaco, and Vatican City. The accepted transfer medium is the euro, but in practice, crypto is simply converted to euro for transfer, and upon receipt the recipient can convert to a wide assortment of possible cryptocurrencies. SEPA is important because the European financial institutions stand behind it. These institutions are the European Commission, the European Central Bank, the European Banking Federation, and the European Payments Council. The SEPA system is operative 24 hours a day, 7 days a week, just like the cryptocurrency market.

As one can clearly see, the credit/debit card network applied to cryptocurrency is both a custody system and a payment transfer system. Custody of funds plus the ability to transfer funds is a rudimentary banking system, although it is not a bank, properly speaking. For most people, it is a network that possesses the faculties of a bank.

If one were to accept that these rudiments of a bank constitute a de facto banking system, then it logically follows that what is developing is a banking system that operates outside of the formal banking system. Since, the traditional banking system is presently conducted within the context of zero interest monetary policies by the world's major central banks, then what is developing is a banking system that is operating outside the context of the current monetary policy and influence of the central bank.

The zero-interest monetary policy has created a nearly flat yield curve throughout most of the world. Within this system, it is difficult to earn an adequate spread on an otherwise good lending client. One could surely earn a larger spread on less creditworthy borrowers, but that is the pathway to default and loan loss. It is not a viable pathway for a conventional bank. Moreover, given zero interest as a permanent feature of banking, there is no incentive for deposit clients to retain funds in a bank, and the danger is that it will be difficult to attract funding.

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The emerging cryptocurrency custody transfer system might be called an informal banking system that can use the traditional bank infrastructure but is not bound by the central bank monetary policy. Under these circumstances, it would be a logical development for a market-based interest rate structure to evolve.

Indeed, this has already begun to happen. For example, at BlockFi.com, it is possible to earn interest on cryptocurrency balances denominated in bitcoin, ether, and Litecoin. The only reason for paying interest on crypto is that the institution in question wishes to lend crypto. There are already a variety of possible cryptocurrency-based loans. The most simplistic is the case in which a holder of crypto would like to borrow some fiat money against the crypto collateral purchased at much lower prices than is currently the case. Fiat money can always be generated by simply selling some cryptocurrency, but that would be a taxable event if that crypto has appreciated. It is arguably more tax efficient to hold the crypto as collateral against a fiat currency loan. If one believes that the crypto will appreciate against the fiat borrowed, it is also reasonable to assert that the earnings from the crypto collateral might comfortably exceed the value of the fiat loan, and in this sense, the fiat loan ultimately extinguishes itself.

There are other reasons, too, for a crypto loan market to develop. Suppose a given investor were to believe that crypto will appreciate against fiat. One might simply convert \$1,000 to bitcoin in such a case, and the worst possible scenario is that the bitcoin would be worthless. In this instance, the loan loss would be \$1,000 plus any lost interest on the \$1,000 sum; as a practical matter, lately that interest amount would be close to zero.

Let us say, as an alternative, one were to borrow 0.1078 bitcoin, the bitcoin equivalent of \$1,000 at the time of this writing. Let us fix the interest rate at 8% at the time of loan inception, and the term of the loan is one year. At loan maturity, one would owe interest amounting to $8\% \times 0.1078$ bitcoin or 0.0086 bitcoin. And, of course, the borrower would return the 0.1078 bitcoin.

If bitcoin were to decline by 90%, the interest, being that this is a bitcoin-denominated loan, would be paid in bitcoin, and that would cost only about \$8 to purchase for a period of one year. That \$8 figure is derived by calculating 8% on a \$1,000 loan balance which, again, being denominated in bitcoin, declined by 90%. Really, it is not a \$1,000 balance; perhaps it should be referenced as a btc1,000 balance. Therefore, one would be paying 8% in bitcoin, and when the USD-based borrower pays that interest, it would only cost about \$8 upon conversion from USD to btc. If bitcoin were not to decline, the interest expense would be \$80, or simply 8% on the \$1,000 loan.

On the other hand, bitcoin could instead appreciate by 90%. Let us vary the example, somewhat. In this instance, the borrower does hold 0.1078 bitcoin during the term of the loan, but the original collateral for the loan would be fixed at \$1,000. If the borrower 191

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could, in turn, re-lend the bitcoin at 8% to another USD-denominated borrower, but on a mark-to-market basis, it might bring in \$80 plus 90% appreciation on the interest owed, for a total of \$152 of interest received in U.S. dollars. The interest to be received by the owner of the bitcoin at loan maturity would be the \$80 fixed sum, which would be owed by the 1st borrower of the bitcoin. At maturity, the 0.1078 bitcoin would be returned to the original lender. The profit to the re-lender would be the \$152 interest received from the 2nd borrower minus the \$80 on the fixed loan, or \$72. In other words, the re-lender would have held \$1,000 collateral for a bitcoin loan and that would be \$72 or 7.2% on the U.S. dollar balance.

Let us make matters more interesting and assume that a borrower is required to post only \$500 collateral in U.S. dollars for an original one-year term loan on \$1,000 worth of bitcoin. In success mode, the profit of \$72 would be based on a \$500 collateral sum, so that the return is far more alluring.

Why might a lender require only \$500 of collateral instead of \$1,000? A lender might do this if the institution or individual is confident that the original bitcoin is safely invested, earning a floating rate as opposed to a fixed rate paid by the borrower. The volatility of bitcoin makes various profit scenarios possible in such a transaction. It is, in effect, a fixed versus floating rate swap.

In any event, given that central banks have stabilized rates at low levels and appear to have no intention of increasing interest rates, the conventional banking or interest rate differentials have ceased to exist. The result is that conventional profit on the customary banking transactions are most difficult to achieve.

The system of evolving transactional possibilities is known as DeFi in cryptocurrency parlance. DeFi is vernacular for decentralized finance. Essentially, it entails the recreation of much of the existing set of financial transaction possibilities outside of the controls and constraints imposed by the central banking system.

DeFi is not merely cryptocurrency in the conventional sense. Stablecoins, for instance, which might be tied or pegged to the U.S. dollar, are basically digital fiat currencies that act like money market funds. The difference is that they are not within the reach of the central banking system, so that in some cases one can earn a reasonable rate of interest.

Naturally, stablecoins come without many of the safeguards of the traditional banking system. It is surely possible that, based on open source code, the stablecoin blockchain or custody system could be penetrated and money stolen. Another possibility is that even a stablecoin, despite the intent of its design, can be used for speculative purposes. For instance, the Dai stablecoin is frequently used by holders to purchase ether on leverage. Dai is controlled by another coin known as MKR. The holders of this latter coin vote to set

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the Dai interest rate. One might call this a form of decentralized central banking in which speculation is actually encouraged to some degree. This sort of activity could have negative consequences one day, but in the meantime it is a profitable banking transaction of a sort.

The concept of DeFi, decentralized finance, has entailed the evolution of the “decentralized application,” dApp in cryptocurrency parlance. The salient characteristic of dApp is that the software is not managed by an institution. This can be a “smart contract” that, in practice, will run itself with no human intervention once it is deployed to the blockchain of the cryptocurrency in question.

A dApp is based on open source code that is visible to anyone who cares to examine the code. It is also permissionless in the sense that anyone can create a dApp and anyone can use one. It is this development that created the equivalent of a banking system outside the fabric and structure of the conventional banking system.

An example is a dApp known as Compound. It is a blockchain-based borrowing and lending application. One can lend crypto and earn interest. One can deposit crypto and borrow against the balance used as collateral for a loan. The dApp has the equivalent of a matching engine that adjusts interest rates based upon the supply and demand for loans. This is not the only conventional banking dApp. Dharma, dYdX, and LoanScan are other examples.

The central point is that a significant change is commencing because of the crypto evolution. At

one time, the universal belief was that the cryptocurrency world would develop its own applications, and transfer and payments system. This process is continuing, as one can readily see by the development of a wide variety of so-called dApps. Yet, there is a separate development occurring as well. If the banking system were to completely isolate itself from crypto, the risk is the evolution of a parallel banking system that might eventually completely displace the conventional banking system based upon fiat currencies that are continually losing purchasing power due to excess money creation. Therefore, the evolution of cryptocurrency-oriented debit cards is actually the opening of the conventional existing payment and transfer system to crypto. It should greatly accelerate the universal adoption of cryptocurrency.

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Industry Thoughts

COMMODITY-BASED ROYALTY TRUSTS IN A FIXED INCOME PORTFOLIO

The contemporary fixed income investor must contend with the following two circumstances. First, the historical return for the past 10 years is not high in absolute terms, even with the historically unprecedented monetary stimulus by central banks. Using the iShares Core U.S. Aggregate Bond ETF (AGG) as a proxy for the fixed income market, the annualized return for the 10-year period ended June 30, 2020 is 3.73%. The second reality is that the current yield to maturity of the index is 1.12%. In other words, if interest rates do not increase, the rate of return for the next 10 years is the current yield to maturity. This is less than the benign-case inflation projection of 2% per annum.

None of the monetary stimulus would have been possible without a 10-year period of decline in commodity prices. Ten years ago, crude oil traded at almost \$80 a barrel and now it is approximately \$40 a barrel. Natural gas was \$4.50 per million BTU 10 years ago and now it is about \$1.80 per million BTU. Wheat traded at over \$7 per bushel and now it is \$5.35 per bushel. Lean hogs are down by about 50%, uranium is down by 20%, coffee is down by roughly 35%, sugar is down by 30%, copper is more or less unchanged, and cocoa is down about 30%—all in the past decade.

This is starting to change. Oddly enough, the first commodity that exhibited a significant price increase is lumber. The price per 1,000 board feet has increased vis-à-vis about 10 years ago from about \$2 to \$5.33. Almost all of that increase occurred in the past 60 days. This certainly gives the bond investor some understanding of the volatility as well as the rapidity of commodity price movements. It is particularly strange in the case of lumber, since one might have been inclined to think that COVID-19 would have a negative impact upon construction and, therefore, upon lumber.

No one seems to know why lumber prices are increasing. Some theorize that many workers are unemployed as a result of the current pandemic and these people are using the time to undertake home repair projects. This seems logical, because remodeling demand constitutes a significant proportion of lumber demand. Yet, there is no hard data upon which to base such a theory.

A more empirically-grounded theory is that the new restrictions by the Forest Stewardship Council (FSC) and the Programme for the Endorsement of Forest Certification (PEFC) are having the impact of permanently reducing supply. Both of these international, non-profit organizations are oriented towards the development of sustainable forest management practices. This amounts to reducing the annual cut, or tree harvest. Many corporations will not purchase—or even cannot purchase—lumber that is not FSC or PEFC certified. 194

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Consequently, each year there has been an increase in the world forest land that has FSC and PEFC double certification, which means certification from both organizations. The following table produced by both organizations shows the increase in hectares that are certified.

Table 1: FSC and PEFC Double Certification

Hectares (millions) End 2016 70.144 Mid 2017 71.067 Mid 2018 86.656 Mid 2019 92.518 Source: FSC & PEFC

Double certification now exists in 33 countries, and in terms of single certification, meaning one or the other, about 430 million hectares are now covered. There are other forestry certification

organizations, as well.

Because the lumber price increases are so recent, none of the timber real estate investment trusts (REITs) have reacted in price. One can only presume, perhaps very incorrectly, that much of the timber REIT ownership occurs within the context of generalized REIT indexes or at least REIT funds. These funds have exhibited a series of issues since the start of the COVID-19 pandemic, especially an exceedingly low lease renewal rate due to the uncertainty of local health authorities' restrictions on property use going forward.

The four primary timber REITs in the U.S. are:

Table 2: Four Major Timbers REITS

Market Capitalization Yield (billions)

WY Weyerhaeuser Company \$17.68 5.74% RYN Rayonier Inc. 3.50 4.20% PCH PotlatchDeltic Corporation 2.65 4.04% CTT CatchMark Timber Trust Inc. 0.448 5.87% Source: Bloomberg, week of July 13, 2020

These REITs yield anywhere from the low 4% to the high 5%.

Of these four stocks, only Weyerhaeuser is in the S&P 500, at a 6 basis-point weight. The S&P 500 has essentially no positive exposure to lumber prices. There is, on the other hand, a great deal of negative exposure, because many S&P 500 companies might use large 195

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amounts of lumber in their business es if for no other reason than construction and renovation of facilities.

It might be rather unorthodox, but a 1% position in Weyerhaeuser would raise the yield of a bond index portfolio by 5 basis points and introduces a different kind of risk than the convexity risk that is a property of every modern bond portfolio.

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Facts & Figures

SUPPLY AND DEMAND IN LUMBER AND TIMBER LANDS

According to Statista, the total annual value of the U.S. industry shipments of wood products is \$283.7 billion. This is the sum if one were to add up all the wood of whatever type anyone buys. Note that the probable annual revenue of Microsoft Corp. (MSFT) for the fiscal year ended June 2020, is about \$143 billion. Microsoft, of course, is a global corporation and, in this case, we are comparing it to a mere domestic U.S. industry. The importance of the comparison will become clear shortly.

The price of 1,000 board feet of lumber was about \$221 from late 2015 to early 2016. It increased dramatically to over \$600 per 1,000 board feet in early 2018, and the price rapidly collapsed to slightly in excess of \$300 per 1,000 board feet during that same year. Some modest recovery was realized afterwards.

The next serious decline occurred in March 2020 with the issues arising from the current pandemic. The price per 1,000-board feet understandably declined to about \$230 in early April.

Since that date, the price has increased to \$539 per 1,000 cubic board feet. If this becomes a permanent price increase, it will cost U.S. consumers over \$300 billion. That should be very significant as a vector for a dramatic enhancement of an S&P 500 industry sector's earnings and valuation. Yet, the weight of forest products in the S&P 500, in the incarnation of Weyerhaeuser, is only 6 basis points, Microsoft is a 622-basis-point weight. The S&P 500 is supposed to represent the U.S. economy, not the global economy.

Statista reports that the certified forest area in North America is 49% of total area, though this statistic can vary, depending upon the reporting organization and the definitions employed. This means that the industry has a reduced capacity to respond to the extraordinary demand for lumber, because roughly one-half of all forest land has adopted sustainable cutting practices, and more adopt it every month. The proportion of the industry that is certified—and certification can apply both to entities that own timberland and to companies that engage in logging—will almost certainly increase, since buyers are under great pressure to purchase only certified

lumber, and in some cases, companies are legally compelled to do so .

The country with the largest forest area in the world, according to Statista, is Russia, with 815 million hectares. Even a fair amount in Russia is now certified. Other large forest nations are Brazil, Canada, and the U.S.

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Table 3: Largest Forest Nations

Hectares (millions) Russia 815 Brazil 487 Canada 347 United States 310 Source: Statista

Statista reports that 31% of total world forest area is now certified. These statistics should be considered in the context of political and social changes that are now in process with respect to environmental sustainability. China and India alone have 2.8 billion people, and a higher standard of living on a sustainable basis that includes eliminating plastics will involve the increased use of paper and wood products. Certainly, paper as an alternative to plastic for packaging will be encouraged and quite possibly required by much of the world. One would think that even just a few nations in the world moving away from plastic towards wood fiber-based packaging at a time when the wood harvest is being reduced constitutes a possible source of inflation, at least in lumber.

Science.howstuffworks.com, an objective and generally reliable source of data on many aspects of applied technology, reports the following.

1. Manufacturing paper actually produces 80% more greenhouse gases than the manufacture of plastics. This is even before any allowance is made for the CO₂ that would be absorbed by a tree that actually survives.
2. The manufacturer of a given quantity of paper requires 4x as much energy as the manufacturer of plastic for a given unit of output.
3. The Environmental Protection Administration asserts that in a landfill, paper decomposes only modestly more rapidly than plastic.

Since paper obviously creates environmental problems, a substance known as paptic has been invented. It is paper product made from wood fiber originating from a sustainably managed forest with an FSC certification, but has foldability and strength characteristics that allow it to be reused many times in place of various types of plastic bags and other plastic packaging, and it is recyclable.

It would seem that the long-term wood products trend is towards lower production (that is to say supply) but not necessarily lower demand. 198

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Featured Companies

INTRODUCTION: BONDS AND TIMBER REITS

There is precedent for a sudden, enigmatic increase in lumber prices. The price increase as well as the subsequent price decrease is rarely explained. In all likelihood, the current sudden jump in lumber prices will give rise to a sudden price decrease. If it does not happen, that would be an inflationary problem, which is to say, a problem for bonds.

The typical timber REIT yields four times the Core Aggregate Bond Index. Paper is ubiquitous as a form of product packaging. An increase in lumber prices must give rise to an increase in the cost of packaging, which in turn, will impact a virtually infinite array of products.

Therefore, if a bond investor were to substitute a 1% position in timber REITs for an identical position in bonds, it would be a diversifying action. The timber REITs declined by about 20% to 25% at the onset of the COVID-19 crisis and never recovered much. The stocks reflect diminished demand—and therefore diminished price—at the moment when the reality seems to be different.

Of course, the typical bond investor will not exchange 1% of a bond portfolio for 1% in timber REITs because of the formalization of lines of responsibility in asset allocation. However, that would be a yield-enhancing, diversifying move.

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WEYERHAEUSER COMPANY (WY)

Weyerhaeuser Co. (WY) has a \$17.68 billion market capitalization, and the shares yield 5.74%.¹ It controls 11.5 million acres of timberland in the United States and manages an additional 14 million acres in Canada in accordance with long term contracts. In the U.S., the timberland ownership amounts to 10.8 million acres. The timber harvest, as well as the inventory, is weighted heavily towards Douglas fir cedar and southern yellow pine. This is because of the weather and growing characteristics of the areas in which the property is located: Oregon, Washington, Arkansas, Louisiana, and Mississippi.

This is important because demand for wood varies by species, with some species more suitable for some purposes than others. Similarly, age of wood is important. In the Weyerhaeuser Oregon and Washington holdings, the average age of harvested timber is 50 years, hence, 2% of the timber is harvested yearly. On the other hand, hardwood and conifer are harvested at a more advanced 63-year age. The Weyerhaeuser practice is to harvest about 1% of these species each year.

As should be apparent, timber has much of the characteristic of a zero-coupon bond. When a seedling is planted, it has minimal value. As a seedling grows into a mature tree, its market value increases in a regular and predictable manner. In 2019, for example, a Douglas fir log sold at an average of \$665 and a southern pine log in the same year sold at \$328.

The lumber futures price curve is in backwardation at the current time. This means lumber prices are expected to decline. However, even the July 2021 futures, a year from now, trade at \$420.90 for 1,000 board feet. It is worthy of note that as late as 2010, lumber traded in the same range as it did in 1980, when the average price per 1,000 board feet was \$225. They had not increased in 30 years. From 2010, onward—at least point-to-point—lumber prices have been rising very gradually, as sustainable forestry practice becomes more widespread.

The Weyerhaeuser REIT has \$6.3 billion of long-term debt and another \$1 billion of short-term debt, as against \$1.45 billion of balance sheet cash. Shareholders' equity is \$8 billion and it is devoid of any intangibles. Most of the corporate cash flow is paid out in the form of dividends, with the exception of expenditures related to reforestation projects.

The REIT is far less dangerous than a high-yield bond. Right now, a typical high-yield bond might have a yield to maturity of 5.8%. A significant portion of a high-yield portfolio will come from B-rated or even CCC-rated credit, and there is always the risk of default. Such a portfolio, as a whole, has both individual credit default risk as well as the portfolio

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spread widening risk. A timber REIT injects a new and different element into the bond portfolio: the possibility of income increase.

The current yield in Weyerhaeuser is based upon a much lower share price. The price of Weyerhaeuser has remained unchanged since lumber prices increased, because the market presumes that the price of lumber will quickly normalize. If it does not, Weyerhaeuser's dividend payment will eventually be much higher.

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POTLATCHDELTIC CORPORATION (PCH)

PotlatchDeltic Corporation (PCH) has a \$2.65 billion market capitalization and yields 4.04%.² It became an REIT in 2006, and it was expanded in 2018 with the merger of Deltic Timber into Potlatch. Its paper manufacturing operations, Clearwater Paper Corp. (CLW), were separated

from the company via a spin-off in late 2008.

Potlatch's business is almost entirely the management of 1.9 million acres of timberland. However, the company still owns six sawmills, so one might say an industrial component still exists. The company owns the following forested land.

Table 4: PotlatchDeltic Forests Acres Tree species State 628,000 Primarily Douglas fir, red cedar Idaho 106,000 Primary pine, aspen Minnesota 929,000 Primarily Southern pine Arkansas 96,000 Southern pine Mississippi 92,000 Southern pine Alabama 6,000 Southern pine Louisiana
Source: Company filings

About 6 million tons of timber should be harvested in 2020. If that occurs, it would be an increase of about 7.4% versus the prior year. That degree of growth is very unusual in the current environment, which is why PotlatchDeltic yields less than Weyerhaeuser. It is perceived to have the ability to achieve modest growth for a period of time. That makes no allowance for a potential increase in lumber prices, which could dramatically increase the dividend.

It should be observed that timber REITs do not generally insure against timber losses due to fire, weather, disease of trees, and even new regulations. All of these factors could reduce cash flow and therefore dividends, even before consideration of the other factors clearly affecting the industry, such as weakness in the economy.

The reason for undertaking such an investment despite the self-evident risks is that the bond index yields only 1.12% and bears with it the presumed risk of a 2% annual inflation rate as well as the risk of even greater inflation, such that the debasement of the principal value of a bond portfolio is a certainty. In contrast, PotlatchDeltic's 4% yield and a presumed 2% annual inflation rate should lead to timberland appreciating at the rate of inflation in the fullness of time, and the income produced should grow at least at the same rate.

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The presumptive rate of return, therefore, is 6%, meaning, 4% yield with a 2% inflation rate, versus the 1.12% return in the bond portfolio. However, lumber prices have already increased substantially, and even if those prices stay where they are now, the dividend—the cash flow that must be paid out—could be very substantial.

The PotlatchDeltic balance sheet holds \$710 million of long-term debt and \$46 million of short-term debt against \$1.149 billion of shareholders' equity and \$79 million of cash. The debt should gradually be debased over time by the same factors that will serve to debase the purchasing power of the typical bond index.

PotlatchDeltic is not only the beneficiary of inflation, in a direct operating business sense, but also through the diminution of the value of its debt via inflation, such that the REIT is a de facto short position in debt. That has value, in and of itself, if added to a bond portfolio. If inflation ever accelerates due to excessive money creation, or if timber prices increase because of the forestland certification movement, this security will probably significantly outperform bonds and provide a much higher yield.

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CATCHMARK TIMBER TRUST, INC. (CTT)

CatchMark Timber Trust, Inc. (CTT) has a \$448 million market capitalization yielding 5.87%. CatchMark Timber is somewhat different than the other timber REITs. Like the others, it owns timberland, in this case, 435,500 acres located primarily in the southern United States. However, the company controls over 1.1 million additional acres via a number of joint ventures, and it also buys and sells timberland for its own account, meaning, it trades. CatchMark also manages timber assets for institutional investors so that, in a sense, it is partially a money management firm. About 11% of revenue is derived from this segment and that figure is growing rapidly. About two years ago, revenue from this segment was nearly zero.

That provides an element of optionality in CatchMark beyond that of other REITs in the timber arena. Many institutional investors lack any exposure to timber as an alternative investment. In view of the sustainability features being introduced into most timber management in the world and the increasing regularity of the timber harvest or cut, the cash flow from timber as an

alternative investment is becoming more predictable.

Timber asset management could conceivably significantly expand the amount of capital available at institutions to deploy in search of cash flow at higher returns than the meager and insignificant cash flow to be earned by bonds. In success mode, it is readily conceivable that such a business could be spun off to the public at a fairly high valuation.

There are two somewhat undesirable aspects of CatchMark Timber. The first problem is that the continued trading of timberland reestablishes an ever higher tax basis for the property. This is very different from most timber trusts that maintain a low basis. As an accounting convention, the company must charge depletion allowances to newly purchased timberland. In practice, timber is a renewable resource that does not actually deplete, but the depletion charges make the company seem as if it is not profitable. In reality, the company has positive free cash flow. Dividends are paid from free cash flow. However, shareholders' equity declines as an accounting consequence.

CatchMark thus appears leveraged, with only \$156 million of shareholders' equity financed by \$432 million of notes payable and lines of credit. In the first quarter of 2020, the company did repay \$20 million of debt. If CatchMark were able to reduce its formal leverage, it would most likely receive a higher valuation. This might actually happen if the company could create a publicly traded vehicle for its timber asset management business.

On the other hand, if this business expands within the context of CatchMark, it could produce quite substantial cash flows with no cash reinvestment requirement, since the

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timber assets are owned by institutional investors and they are responsible for any necessary capital expenditures. 205

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ACADIAN TIMBER CORP. (ADN CN)

Acadian Timber has a USD176 million market capitalization and it yields 8.24%.⁴ It holds 761,000 freehold acres of timberland in New Brunswick, Canada as well as another 300,000 freehold timberland acres in Maine, U.S. It also provides services to about 1.3 million additional acres of provincial timberland in New Brunswick.

This is the only one of the timber firms that reinvests some cash flow in expanding the business, although the payout ratio is still 74%. As a consequence, book value per share is generally increasing, although not at a particularly alluring rate. The shares also trade at a 30% discount to book value. The company probably could be liquidated at a premium to book value. One reason for the discount to book value and the relatively high yield is that the company appears to be unprofitable at the moment, when this is not the case. Acadian Timber must continually adjust the exchange rate value of some debt based on changes in the U.S./Canadian dollar ratio. In actuality, the shares are trading right now for about 12x free cash flow.

For many years, this company was 45% owned by Brookfield Asset Management Inc. (BAM). The presumption was that Brookfield would eventually integrate timber into its large private equity business. Brookfield is probably the largest private equity manager in the world and it specializes in tangible assets such as real estate, power plants, and other infrastructure projects. Although considerable funds can be invested in timberlands in principle, these would be relatively small sums in comparison with the scale of investment that is possible in infrastructure. Consequently, in August 2019, Brookfield sold its 45% interest to Macer Forest Holdings Inc. Macer is a private company that is known as a long-term investor in the forest products industry.

Of the CAD113 million debt owed by the company, over CAD91 million is due in October 2020. It is presumed that this will be refinanced without difficulty and this amount was originally denominated in U.S. dollars. The only other piece of significant debt is a CAD20.7 million term loan facility due in March 2025. Thus, the company should have no significant refinancing needs after October 2025. Balance sheet cash amounts to CAD12 million.

The timberland has been owned for a very long time, so there are no depletion charges and the firm only has a very small annual depreciation expense representing equipment that exists. This is the simplest example of timber companies with the highest yield. The revenue increase that

would result if the price of timber were to go up would require no additional expenses. The revenue increase effectively would become earnings and dividend

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increases, and it would be a viable alternative to a bond with a very low yield. The Acadia share yields over 8%. 207

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Post-Musings

THE TRAJECTORY OF PROCESS

In the field of energy, well -intentioned and perhaps even noble, non- governmental organizations are combining forces with many institutional investors to compel the reduction of the use of plastic in packaging in favor of more readily biodegradable wood fiber products. This will result in a diminution of plastic production.

In the field of wood fiber, well -intentioned and perhaps even noble, non- governmental organizations are combining forces with large corporate buyers of wood fiber and many institutional investors to compel the reduction in the annual timber harvest to sustainable levels. This has already resulted in a diminution of wood fiber production and will continue to do so.

In the field of monetary policy, the various central banks of the world are promoting maximum stimulus to increase worldwide consumer demand that will strengthen economies and increase employment. This will result in an increase in demand for packaging materials. Perhaps the packaging might be based on wood fiber or perhaps it will be based on plastics, but in any case, increased economic activity demands increased packaging.

In the field of investing, the well -intentioned thrust towards indexation has never been more powerful. In the indexes, wood fiber companies as well as energy companies have minimal exposure . At the same time, policies are adopted to increase demand for commodities while reducing production of those very same commodities to sustainable levels. One might be tempted to say that such sustainability policies are quite possibly not actually sustainable.

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From the Readers

CRYPTOCURRENCY WITH, NOT AGAINST, FIAT BANKING AND CREDIT

Q: You talked about cryptocurrency side by side with banking, and about crypto and Visa. Jack Dorsey of Square Inc. (SQ) is a proponent of bitcoin through his network— I forget the name of it, and Square also enables buying and selling bitcoins in using the Square app. Does something like this fit in with the points you made?

A: It fits in exactly with this. It is just one more example; of course, I did not list them all. The existing fiat infrastructure is now basically accommodating crypto, which is a n important change from a year ago. It looks like everybody is going in that direction, and the example you cite is just another instance of that.

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Money Manager Index From Aug 1983 to July 2020 Annualized return Year Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec Yr. End Index Yearly return (since inception) 1983 1.00 0.81 0.76 0.87 0.75 1983 0.75 (60.5)% (50.2)% 1984 0.75 0.71 0.70 0.66 0.67 0.67 0.61 0.83 0.79 0.76 0.67 0.65 1984 0.65 (13.5)% (26.5)% 1985 0.92 0.93 0.99 0.95 1.20 1.30 1.32 1.38 1.28 1.50 1.86 2.02 1985 2.02 211.8% 33.7% 1986 2.46 2.78 2.47 2.31 2.36 2.33 2.03 2.23 1.98 2.37 2.34 2.34 1986 2.34 15.9% 28.2% 1987 3.21 3.27 3.16 2.55 2.37 2.30 2.39 2.47 2.22 1.56 1.44 1.52 1987 1.52 (35.0)% 9.9% 1988 1.80 1.87 1.78 1.79 1.69 1.94 1.92 1.96 2.01 1.97 1.95 2.07 1988 2.07 36.0% 14.3% 1989 2.42 2.37 2.54 2.63 2.64 2.64 2.93 3.12 3.07 3.05 3.23 3.26 1989 3.26 57.8% 20.2% 1990 3.12 3.15 3.53 3.06 3.47 3.45 3.30 2.70 2.68 2.40 2.52 3.02 1990 3.02 (7.3)% 16.1% 1991 3.08 3.49 3.70 3.68 3.71 3.61 3.86 4.05 4.07 4.69 4.47 5.72 1991 5.72 89.4% 23.0% 1992 5.76 5.61 5.30 5.12 4.98 4.99 5.93 6.06 6.19 6.56 7.25 7.36 1992 7.36 28.6% 23.6% 1993 8.06 8.04 8.20 7.94 8.15 8.57 9.05 10.00 9.99 9.31 8.97 8.90 1993 8.90 21.0% 23.4% 1994 9.52 8.73 8.05 7.85 7.81 7.53 7.66 8.31 8.15 8.52 7.88 7.95 1994 7.95 (10.6)% 19.9% 1995 7.74 8.38 8.72 8.77 9.20 9.35 9.93 10.78 11.22 10.53 10.89 10.40 1995 10.40 30.8% 20.8% 1996 11.12 11.50 11.33 11.62 11.86 12.53 11.91 12.36 13.32 14.03 14.42 15.02 1996 15.02 44.4% 22.4% 1997 16.04 16.81 15.32 17.27 18.42 20.29 22.28 21.39 25.31 24.95 24.95 25.50 1997 25.50 69.8% 25.2% 1998 25.67 29.00 29.89 30.60 28.90 30.44 27.67 21.33 21.74 25.16 27.27 25.41 1998 25.41 (0.4)% 23.3% 1999 26.00 23.71 23.92 26.77 28.94 29.74 28.78 26.74 25.89 27.73 28.54 30.55 1999 30.55 20.2% 23.2% 2000 31.07 31.19 36.01 35.60 35.20 40.32 43.58 45.75 45.62 48.69 44.05 49.84 2000 49.84 63.1% 25.2% 2001 50.23 46.41 44.27 46.96 48.90 49.98 50.67 49.70 46.47 44.81 48.04 51.91 2001 51.91 4.2% 23.9% 2002 53.62 53.74 55.11 52.52 52.83 50.48 42.58 44.92 41.54 42.66 45.78 43.17 2002 43.17 (16.8)% 21.4% 2003 42.72 41.18 42.36 45.98 49.02 50.71 53.47 53.97 53.46 56.12 55.83 58.49 2003 58.49 35.5% 22.1% 2004 64.38 65.08 64.63 61.68 60.86 62.30 58.71 64.08 65.73 68.86 73.53 78.16 2004 78.16 33.6% 22.6% 2005 76.46 77.94 74.06 72.83 77.02 80.25 83.59 83.07 86.03 89.19 96.58 97.35 2005 97.35 24.6% 22.7% 2006 107.62 111.44 110.75 111.88 101.89 100.61 100.62 104.98 114.61 116.64 113.78 118.05 2006 118.05 21.3% 22.6% 2007 125.73 123.77 122.62 127.58 133.57 134.68 126.61 124.07 133.57 148.09 135.13 135.56 2007 135.56 14.8% 22.3% 2008 127.53 115.76 115.94 121.58 130.51 115.68 119.94 120.55 109.69 72.70 62.95 67.91 2008 67.91 (49.9)% 18.1% 2009 57.51 51.76 65.63 79.49 85.67 90.79 99.97 101.69 107.32 107.36 110.94 115.01 2009 115.01 69.4% 19.7% 2010 106.84 110.32 118.13 114.91 100.18 88.17 97.65 89.64 103.59 108.29 108.64 119.58 2010 119.58 4.0% 19.1% 2011 122.80 128.28 127.94 127.97 126.06 121.03 115.49 104.25 91.32 102.44 103.79 103.98 2011 103.98 (13.1)% 17.8% 2012 109.46 120.12 125.37 121.64 108.44 114.12 113.56 118.33 123.18 127.91 131.76 135.00 2012 135.00 29.8% 18.1% 2013 151.20 155.13 165.52 166.55 174.89 164.20 179.01 168.47 176.12 192.14 197.16 208.44 2013 208.44 54.4% 19.2% 2014 194.17 196.87 203.88 196.24 195.40 206.41 194.00 207.06 201.07 205.28 212.28 215.25 2014 215.25 3.3% 18.6% 2015 203.96 217.70 215.97 218.17 217.01 211.12 203.85 184.77 175.53 195.50 198.54 181.68 2015 181.68 (15.6)% 17.4% 2016 165.64 164.85 183.47 190.06 194.22 177.37 187.78 190.19 185.87 173.66 194.88 199.52 2016 199.52 9.8% 17.2% 2017 196.14 209.63 205.70 207.52 210.37 221.66 230.87 225.39 239.74 245.52 261.47 264.79 2017 264.79 32.7% 17.6% 2018 278.34 266.70 266.44 253.48 256.42 243.56 250.69 238.98 234.72 209.04 213.62 194.43 2018 194.43 (26.6)% 16.0% 2019 203.57 219.59 213.65 234.81 206.43 229.48 229.27 210.59 223.22 225.90 239.32 240.01 2019 240.01 23.4% 16.2% 2020 248.58 219.88 192.81 222.04 227.81 240.17 250.44 2020 250.44 4.3% 16.1% S.No. Ticker 1 AMG US Equity 94,711\$

2 BLK US Equity 953,087\$

3 WDR US Equity 23,555\$

4 EV US Equity 146,001\$

5 TROW US Equity 278,112\$

6 BEN US Equity 80,790\$

7 LM US Equity 23,105\$

8 FHI US Equity 58,152\$

9 PZN US Equity 33,102\$

2,206 6,317 3,998 2,014 4/30/1986 4/30/1985 8/31/1983 5/31/1998 10/31/2007 9/30/1999 3/31/1998 1/31/1986 1,263 462 \$2,423 \$908 \$1,000 \$26,381 Eaton Vance T. Rowe Price \$2,641 \$122,426 Franklin resources Legg Mason Federated Hermes Inc Pzena Investment Management Index Constituent Changes: 1. Nuveen Investments Inc (JNC US) was delisted from the US Security Exchange effective 11/14/2007 and has been removed from the index. 2. Alliance Financial Corp (ALNC US) was delisted from US Security Exchange effective 03/11/2013 and has been removed from the index. The divisor has been adjusted accordingly for each of these changes. 3. Fortress Investment Group (FIG US) was delisted from US Security Exchange effective 12/27/2017 and has been removed from the index. Amount Invested \$22,947 \$23,205 \$27,513 Name Affiliated Manager BlackRock Waddell & Reed Shares Purchased 1,377 1,658 1,587 Current Index ValueDate of Investment 11/30/1997 211

International Money Manager Index From Nov 1986 to July 2020 Annualized return Year Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec Yr. End Index Yearly return (since inception)

1986 1.00 1.02 1986 1.02 10.0% 10.0% 1987 1.25 1.37 1.48 1.48 1.37 1.33 1.39 1.40 1.33 0.81 0.76
0.73 1987 0.73 (27.7)% (23.3)% 1988 0.75 0.92 1.02 0.95 0.80 0.89 0.88 0.82 0.86 0.88 0.89 0.93
1988 0.93 26.4% (3.4)% 1989 1.03 1.02 1.06 1.17 1.19 1.18 1.25 1.16 1.17 1.20 1.21 1.28 1989 1.28
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2 5,786\$
3 19,649\$
4 15,642\$
5 44,307\$
6 7,836\$
7 5,394\$
8 19,364\$
9 94,463\$
10 560,192\$
11 50,588\$

Index Constituent Changes: 1.New Star Asset Management (NSAM LN) was delisted from the London Security Exchange effective 03/10/2009 and has been removed from the index. 2. Australia Wealth Management (AUW AU) was delisted from Australian Security Exchange effective 05/18/2009 and has been removed from the index. 3. Bluebay Asset Management/UNI (BBAY LN) was delisted from the London Security Exchange effective 12/20/2010 and has been removed from the index. 4.Everest Financial Group Limited (EFG AU) was delisted from the Australian Security Exchange effective 7/19/2011 and has been removed from the index. 5. RAB Capital Plc (RAB LN) was delisted from the London Security Exchange effective 9/2/2011 and has been removed from the index. 6. Invista Real Estate (INRE LN) was delisted effective 8/13/2012 and has been removed from the index. 7. F&C Asset Management Plc (FCAM LN) was delisted effective 5/8/2014 and has been removed from the index. 8. Charlemagne Capital Ltd (CCAP LN) was delisted effective 12/14/2016 and has been removed from the index.The divisor has been adjusted accordingly for each of these changes. 9. Henderson Group Plc (HGG LN) was delisted from London Security Exchange effective 5/30/2017 and has been removed from the Index. 10. Aberdeen Asset Management Plc (ADN LN) was delisted from London Stock Exchange effective 8/14/2017 and has been removed from the Index. Shares Purchased 1,153 Date of Investment 1/31/1991\$1,357 IGM CN Equity IGM Financial Inc \$1,000 IVZ US Equity

Current Index Value 31/11/1986 Ticker Name Initial Amount Invested 73 7/31/2004 EMG LN Equity 1/31/1996 6/30/19943,224 6,344 10/31/1994 1,346 12/31/20018739 JP Equity Sparx Group Co Ltd \$11,762 108 4,977\$21,908Azimut Holding Spa 10/31/2006 578 3/31/2006PGHN SW Equity Partners Group-Reg \$36,848 ASHM LN Equity Ashmore Group Plc. \$36,688 9,873 AZM IM Equity

\$2,585 Man Group Plc \$2,862 AGF/B CN Equity AGF Management Ltd-Cl B \$3,343 CIX CN

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Michael Wolleben Joanna Makris Robert Dunn Murray Stahl Steven Bregman February 13, 2019 11:58 AM EST

The Madison Square Garden Company (Pre-Spin) (BUY)

Current Share Price (2/12/19): \$290.39 Ticker: MSG Fair Value Estimate: \$362 per share
Dividend: Nil Shares Outstanding: 23.8 million Yield: N/A Market Capitalization: \$6.9 billion

MSG Entertainment

Fair Value Estimate: \$221 per share Ticker: TBD Shares Outstanding: 23.8 million Dividend: Nil
Market Capitalization: \$5.2 billion Yield: N/A

MSG Sports

Fair Value Estimate: \$139 per share Ticker: TBD Shares Outstanding*: 36.0 million Dividend:
Nil Market Capitalization: \$5.0 billion Yield: N/A

TBD – To be determined. N/A – Not applicable.

*Based on one-for-one share distribution ratio to MSG shareholders of record, and includes MSG Entertainment retaining a 33% ownership stake in MSG Sports. Note: Market capitalization is

based on fair value estimate for post-spin entities and current market cap for pre-spin MSG.

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INVESTMENT THESIS

On June 27, 2018, after the market close, The Madison Square Garden Company (NYSE: MSG) announced that its Board of Directors had authorized the company's management to explore a possible spin-off that would create a separately traded public company comprised of its sports businesses, including the New York Knicks and New York Rangers professional sports franchises. On October 4, 2018, MSG issued a press release announcing that the company had confidentially filed an initial Form 10 registration statement with the SEC related to the sports business spin-off.

It should be noted that the proposed transaction follows the company's spin-off of its sports broadcasting businesses into MSG Networks (NYSE: MSGN) in September 2015. Preceding the announcement of the current sports businesses spin-off, activist investor Silver Lake filed a 13-D with the SEC indicating that the company had built a 6.3% ownership stake. Silver Lake has subsequently increased its position to 9.7% of Class A shares.

If the company proceeds with the spin-off of its sports businesses, it would be structured as a tax-free transaction to all MSG shareholders. Upon completion of the contemplated separation, shareholders of record of MSG common stock would receive a pro rata distribution, expected to be equivalent, in aggregate, to an approximate two-thirds economic interest in the pure-play sports company. The remaining common stock, expected to be equivalent to an approximate one-third economic interest in the sports company, would be retained by the entertainment company. These shares are expected to be used to raise capital and/or be exchanged for the common stock of the entertainment company. James L. Dolan is expected to be the Executive Chairman and Chief Executive Officer of both companies.

One company would be a leader in live entertainment, with a growing portfolio of assets, including state-of-the-art music and entertainment-focused venues, MSG Productions, other strategic entertainment joint ventures, approximately one-third interest in the post-spin sports company, and approximately \$1 billion in cash. The live entertainment company will also continue to move forward with the development of MSG Sphere. The first MSG Sphere is expected to open in Las Vegas around the end of calendar 2020, followed by a second MSG Sphere in London approximately one year later.

In terms of rationale, the separation of Sports from Entertainment makes sense as the Entertainment company looks to expand its portfolio of owned venues to drive earnings growth. The posited structure of the separation, with Entertainment retaining a 33% stake in Sports and approximately \$1 billion in cash, will allow the company to largely finance its ambitious Spheres project without having to tap the debt markets or dilute Entertainment shareholders. The Entertainment business will be a hybrid of an asset-based stock (derived from the ownership of venues) and operating assets consisting primarily of a growing events promotion business that leverages the prime assets of the company (e.g. Madison Square Garden and the planned Spheres) and majority interest in the TAO Group.

The Sports business will be driven by the value of the NY Knicks and NY Rangers NBA and NHL franchises. Historical increases in the value of professional sports teams, and comparable sales transactions should provide a stable, yet growing, asset-based valuation at MSG Sports. Further, if the Dolan family were to sell an ownership interest in either of the sports franchises (or potentially take one or both teams private), significant upside optionality would arise, given the "whisper numbers" that have been reported on team valuations.

On a pre-spin sum-of-the-parts basis, shares of MSG are fairly valued at \$362 per share. With our fair value estimate approximating 25% upside potential from the current share price, shares of MSG are recommended for purchase prior to the spin-off. The fair value consists of \$151 per share in value assigned to post-spin MSG Entertainment and \$211 per share derived from post-spin MSG Sports. The post-spin fair value estimates are subject to change. It should be noted that in more bullish scenarios additional value could be realized. This could include valuing the air rights above MSG, which could be worth \$25 per share alone, or monetizing the Knicks and/or Rangers. Although the latter is highly subjective, if the Dolan "whisper number" for the Knicks is used, this would add \$1 billion in fair value to the pre-spin sum-of-the-parts assuming the Rangers were valued at just the adjusted Forbes estimate, which excludes any value associated with the arena. That said, the typical rebuttal on the valuation of a sports franchise is that it is only worth what someone will pay, and that is dependent on the current owners' willingness to sell. In the absence of two willing parties, the sports franchises will likely be valued at a reasonable valuation approximating the Forbes value.

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COMPANY DESCRIPTION

The Madison Square Garden Co. (NYSE: MSG) was first publicly traded in 2010 following its spin-off from Cablevision Systems Corp. (previously traded on the NYSE under the symbol “CVC”; was acquired by Altice NV in 2016). MSG controls various sports and entertainment franchises, including the National Basketball Association’s (NBA) New York Knicks and the National Hockey League’s (NHL) New York Rangers franchises. In addition, the company produces and promotes various live events, including the Radio City Christmas Spectacular starring the Rockettes. MSG owns or operates a number of entertainment venues, including Madison Square Garden Arena (The Garden), the Hulu Theater at Madison Square Garden, Radio City Music Hall, The Beacon Theatre (New York City), The Chicago Theatre, the Wang Theatre (Boston), and the Forum (Inglewood, California).

CVC came into being in the 1960s, when founder Charles Dolan began building cable systems in Manhattan and launched the cable channel HBO, both of which were eventually sold to Time Life Inc. Dolan funneled the proceeds from the sale into new cable systems in Long Island under the Cablevision brand name. Through a series of transactions, CVC expanded its cable systems across the country and began investing in other assets.

The majority of MSG’s operating assets and franchises were initially acquired in 1994, when Viacom (NASDAQ: VIA, VIAB) sold its interest in the Knicks, Rangers, and related assets to a partnership between CVC and ITT Corp. for \$1 billion. VIA itself had acquired the properties earlier in 1994 via the acquisition of Paramount Communications, and decided to sell The Garden, Knicks, and Rangers in order to pay down debt associated with the purchase. In 1997 CVC bought out ITT’s 50% ownership in the partnership for \$650 million.

James Dolan, the current Chairman and CEO of MSG, is the son of Cablevision founder Charles Dolan. MSG has a dual class structure. Class B shares carry 10 votes per share and elect 75% of the Board of Directors. Class A shares have one vote per share and elect the remaining 25%. Class B shares do not trade; as of September 30, 2018, they are 100% owned by the Dolan Family Group, which also owns 2.9% of Class A shares.

In F2018 (June 30 year-end), MSG generated \$1.6 billion in revenue and \$193.8 million in earnings before income taxes, depreciation, and amortization (see Exhibit 1). The company operates under two reportable segments: MSG Sports, which controls the sports teams; and MSG Entertainment, which promotes live entertainment in MSG-owned/leased venues. Depreciation on owned venues (The Garden, the Hulu Theater at Madison Square Garden, and the Forum) is reported under “Corporate & Other.”

Exhibit 1 The Madison Square Garden Company: Historical Operating Results (\$ in millions; fiscal years ending June 30)

Source: Bloomberg and company reports.

MSG Entertainment

MSG Entertainment presents and hosts concerts, family shows, and special events at MSG-owned or long-term leased venues, including The Garden (owned), the Hulu Theater at Madison Square Garden (owned), Radio City Music Hall (lease expires in 2023), The Beacon Theatre (lease expires in 2026), The Chicago Theatre (owned), and the Forum (owned). The company also has a booking agreement with the Wang Theatre in Boston. In addition, the company produces a limited number of shows, such as the Radio City Christmas Spectacular. Of F2015 F2016 F2017 F2018 1H F2018 1H F2019 MSG Entertainment Revenue 414.2\$ 415.4\$ 506.5\$ 780.7\$ 435.5\$ 479.5\$
Adjusted Operating Income 41.2 (14.6) 22.0 117.0 100.5 110.0
Depreciation and Amortization 10.3 9.9 11.3 18.5 8.5 8.3
Capital Expenditures (5.7) (5.0) (11.5) (24.4) (11.1) (14.3)
MSG Sports Revenue 656.7\$ 699.1\$ 812.0\$ 778.7\$ 346.0\$ 371.2\$
Adjusted Operating Income 98.0 131.0 143.0 149.5 82.1 49.2
Depreciation and Amortization 19.1 11.0 9.3 7.5 3.8 3.9
Capital Expenditures (4.5) (4.6) (2.4) (4.6) (1.6) (2.1)
Corporate & Other Revenue 0.7\$ 0.9\$ - (0.3)\$ - -
Adjusted Operating Income (20.5) (48.1) (67.3) (72.8) (33.5) (38.6)
Depreciation and Amortization 79.3 81.6 83.6 78.4 39.9 38.2
Capital Expenditures (53.9) (62.2) (30.4) (163.0) (115.0) (64.6)

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note, while MSG owns The Garden arena, the platform that The Garden is built on, as well as the transferable development rights (air rights) above it, the company does not own the ground underneath the arena and in fact operates under a NYC permit. In 2013, the operating permit

for The Garden was extended for 10 years.

The company completed a \$1 billion renovation at The Garden in late 2013 and reopened the Forum in January 2014, after acquiring the latter property in 2012 and investing an estimated \$100 million (in addition to the \$24 million purchase price) to renovate the arena. The “Transformation” (as the company refers to it) at The Garden took three years to complete, with work done mostly during the summer months to avoid disruption to the NY Knicks and NY Rangers seasons, and it should allow The Garden to better compete for entertainment events with newer stadiums such as the Barclays Center in Brooklyn, NY. Enhancements include improved amenities and luxury suites. Similarly, renovations at the Forum have allowed the venue to take market share of concerts in the Los Angeles area away from newer venues such as the STAPLES Center (according to Pollstar data).

In July 2016, MSG Entertainment acquired a controlling interest in Boston Calling Events (BCE), which operates the Boston Calling Music Festival, a three-day indie and pop rock music event that began in 2013. The festival was most recently held on May 25-27, 2018, in Harvard University’s athletic complex; the line-up included acts such as Eminem, The Killers, Queens of The Stone Age, and Tyler, the Creator. While the financial terms of MSG’s purchase of BCE were not provided by management, other than to say that it was at an “attractive” valuation, MSG filings indicate that BCE contributed \$13.7 million in revenue in F2018.

In January 2017, the company acquired 62.5% of TAO Group, which generated pro forma revenue of \$235 million and adjusted operating income (AOI) of \$43 million in calendar 2016, for \$181 million (plus potential earn-outs of \$25.5 million over five years). Including debt and a small preferred equity component, the deal valued TAO at ~\$400 million, implying a ~9x AOI purchase multiple. TAO operates dining and nightlife venues in New York City, Las Vegas, and Sydney, Australia, under well-known brands including TAO, Marquee, Lavo, Avenue, The Stanton Social, Beauty & Essex, and VANDAL. As part of MSG, TAO Group’s operating results, which are primarily comprised of food and beverage sales, are recorded on a three-month lag basis. TAO’s founders retain a 37.5% stake and continue to manage the operations under five-year employment agreements (which include global non-compete clauses). MSG is eager to expand the TAO footprint, with contracts to open five new venues in New York City and eight additional locations in new markets, including Los Angeles, Chicago, and Singapore.

Revenue is generated from licensing venues to third-party promoters and, to a lesser degree, from promoting, co-promoting, and/or producing shows at MSG venues. The company does not promote shows at venues not owned, leased, or under long-term booking contracts. Ancillary revenues are generated from facility and ticketing fees, concessions, sponsorship, and a portion of merchandise. Expenses consist of performers’ salaries and costs to set up, promote, stage, and put on performances.

The Entertainment business generated \$781 million in revenue and adjusted operating income of \$117 million in F2018 (see Exhibit 2). Revenue increased 54% in F2018, which was a direct result of the inclusion of revenues from the TAO Group (\$209 million), partially offset by a decrease in revenue (\$11 million), as the New York Spectacular did not take place (the project was short lived and was cancelled). Through 1Q F2019, revenue in prior years includes the TAO impact; revenue in 1Q F2019 was virtually flat versus the prior year, with fewer live events held at MSG’s venues and slight accounting rule changes offsetting higher per-event revenue in the previous year.

Exhibit 2 The Madison Square Garden Company: MSG Entertainment Segment Historical Operating Results (\$ in thousands; fiscal years end June 30)

Source: Company reports.

	F2016	F2017	F2018 1Q	F2018 1Q	F2019
Revenue	415,390	506,468	780,726	164,281	162,953
Direct Operating Expenses	341,637	378,325	483,263	105,564	107,785
SG&A	96,204	120,496	192,929	44,391	48,969
Depreciation and Amortization	9,884	11,339	18,515	4,161	4,482
Operating Income (32,335)	(3,692)	86,019	10,165	1,717	
Reconciliation to Adjusted Operating Income					
Share-based Compensation	7,870	14,323	12,500		
	3,901	2,841			
Depreciation and Amortization	9,884	11,339	18,515	4,161	4,482
Adjusted Operating Income (14,581)	21,970	117,034	18,227	9,040	
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MSG Sports

The MSG Sports segment owns and operates various sports franchises, and promotes, produces and/or presents other live sporting-related events. Owned franchises include the New York Knicks (NBA), New York Rangers (NHL), Westchester Knicks (NBA Development League), and the Hartford Wolf Pack (American Hockey League). Other live sporting events include the presentation of professional boxing, college basketball, professional bull riding (PBR), and e-

sports, among others.

Franchise sports teams are subject to numerous limitations enforced by the respective leagues. The most notable are collective bargaining agreements (CBAs), particularly with respect to salary cap restrictions and organized labor of players. The NBA imposes a “soft salary cap,” whereby the cap on any given team’s total player salaries can be exceeded, but the team then has to pay a multiple of the excess over the cap (in a tiered rate structure) as a so-called luxury tax. The Knicks historically were a luxury tax payer, with expenses totaling \$8.3 million, \$38.1 million, and \$5.1 million in the 2012 -2013, 2013-2014, and 2014 -2015 seasons, respectively. However, the Knicks did not pay a luxury tax in the 2015-2016, 2016-2017, and 2017-2018 seasons; in those years, the Knicks recorded proceeds of \$2.5 million, \$500,000, and \$2.2 million, respectively, from luxury-tax-paying teams. The NHL imposes a “hard salary cap,” whereby a team’s total player salaries are not allowed to exceed a pre-defined amount based on league-wide revenue, and thus the NHL does not impose a luxury tax. Both the NHL and NBA have revenue -sharing programs whereby a pool of funds from teams with above -average net revenue is distributed to teams generating below-average net revenue.

The NBA’s current CBA expires after the 202 3-2024 season, with an option (for either the NBA or the NBA Players Association) to terminate following the 20 22-2023 season. The NHL’s CBA expires in September 2022, with an option for the league or Players’ Association to terminate the agreement following the 2019-2020 season.

Failure to negotiate CBAs has caused issues in the recent past. In 2012, following the expiration of the NHL’s CBA at the time, the NHL declared a player lockout, causing a delayed start to the 2012-2013 season. The 2012-2013 NHL season was shortened by 41% to just 48 games (versus 82 in a full regular season), while the 2004 -2005 season was completely cancelled due to labor difficulties. The NBA was disrupted in the 2011-2012 season, which was shortened to 66 games, and the 1998-1999 season had only 50 games due to a player lockout. The NBA regular season is typically 82 games.

Revenue is generated primarily from ticket sales and license fees for suites at The Garden, proportionate shares of NHL and NBA television contracts and revenue sources, venue signage, and sponsorship, concessions, and merchandise sales. In addition, MSG Sports sells the telecast rights for the owned team’s home and away games. The largest Sports expenses are player and team personnel salaries, league operating expenses, revenue sharing, and the luxury tax.

In F2018, MSG Sports generated \$779 million in revenue and adjusted operating income of \$150 million (see Exhibit 3). Through 1Q F2019, revenue decreased by nearly 32%, while adjusted operating income totaled \$588,000 (see Exhibit 3). F2018’s revenue decline of 4% versus the year prior was primarily attributed to the NY Rangers missing the playoffs, resulting in six fewer post-season home games, an NHL expansion fee, and a decrease in non -recurring league distributions from the NHL. Despite the F2018 revenue decline, adjusted operating income benefited from lower SG&A expense, which was due to lower employee compensation due to a severance cost that occurred in F2017. It should be noted that the change in the revenue recognition rule primarily relates to suite license fees and network TV deals, with revenue now being recorded based on events occurring, versus the previous straight-line recognition over the course of the fiscal year. This change will likely result in more lumpy revenue quarter to quarter based on sports and other event schedules. Management did comment that it expects revenue recognition to increase in 2Q F2019, likely based on the number of events scheduled versus 1Q F2019. Prior-year periods have not been adjusted to reflect the accounting change.

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Exhibit 3 The Madison Square Garden Company: MSG Sports Segment Historical Operating Results (\$ in thousands)

Source: Company reports.

While e-sports is not a current earnings contributor, a potential earnings stream arises from the rapid growth of the business. MSG acquired Counter Logic Gaming (“CLG”) in November 2017. CLG is an American e-sports team organization that fields premier video game players for CLG’s teams. Teams compete in a number of popular video games, including League of Legends, Fortnite, and Smash Brothers, among others. With e-sports rapidly changing to more closely resemble traditional sports, with leagues and teams, MSG’s acquisition looks to position the company to capitalize on the hugely popular and growing business of e-sports. While it is difficult to value CLG, a notable data point for game manufacturers has been the sale of franchises, which in turn have received sizeable valuations from bankers. According to Forbes, Riot Games (the developer of League of Legends) sold franchises for \$10 million each in the summer of 2017; Activision Blizzard sold Overwatch League franchises for similar amounts. Depending on city, country, and the likely success of the franchises, valuations have reached as high as \$80 million. It is estimated that global e-sports revenue could grow from a current

estimate of \$906 million to \$1.65 billion in 2021. Sponsorship, advertising, and media rights all present a burgeoning opportunity akin to the early days of monetizing traditional sports franchises.

MSG Sphere

In April 2018, MSG held an event at the Los Angeles Forum where James Dolan unveiled plans to create a new concept for multimedia entertainment venues, dubbed the MSG Sphere. The Sphere, which promises a “new way to tell stories,” will include innovative audio and visual capabilities to entertain guests within a 350-foot round dome that is 100% covered in wraparound LED screens with resolution 100 times today’s HD TV technology. Additionally, the outside of the Sphere will be able to display visual images ranging from artwork to the actual performances occurring within. Sound capabilities include what is being described as targetable audio, where each attendee receives customized sound, with features including being able to broadcast in different languages to people standing next to each other without the use of headphones and with no sound bleeding between the two guests. Technologies used in construction and operation of the Sphere are being internally developed, acquired, and licensed, with few details divulged at this point. In November 2017, MSG announced that it had purchased Obscura Digital, a creative studio that was responsible for digital content used in the company’s Christmas Spectacular show. It appears that Obscura’s technologies will be heavily used in the LED video panels that will cover both the inside and outside of the Sphere.

While the company has not put a price tag on Sphere construction, it has been estimated in the media that each Sphere would cost \$1 billion to fully construct. Currently, MSG is in the process (albeit in early stages) of constructing two Spheres: one in Las Vegas, where ground was broken in summer 2018 with an expected opening date of New Year’s Eve 2020; and one to be built in London, which is currently posited to be open a year after the Las Vegas Sphere.

Separation of Entertainment and Sports Businesses

On June 27, 2018, after the market close, The Madison Square Garden Company (NYSE: MSG) announced that its Board of Directors had authorized the company’s management to explore a possible spin-off that would create a separately traded public company comprised of its sports businesses, including the New York Knicks and New York Rangers professional sports franchises. On October 4, 2018, MSG issued a press release announcing that the company had confidentially filed an initial Form 10 registration statement with the SEC related to the sports business spin-off.

It should be noted that the proposed transaction follows the company’s spin-off of its sports broadcasting businesses into MSG Networks (NYSE: MSGN) in September 2015. Preceding the announcement of the current sports businesses spin-off, activist investor Silver Lake filed a 13-D with the SEC indicating that the company had built a 6.3% ownership stake. Silver Lake has subsequently increased its position to 9.7% of Class A shares.

F2016	F2017	F2018 1Q	F2018 1Q	F2019	Revenue	699,062	811,984	778,653	80,934	55,352
					Direct Operating Expenses	396,220	473,590	457,694	16,728	15,319
					SG&A	182,131	209,941	186,914	41,976	42,217
					Depreciation and Amortization	10,957	9,319	7,481	1,906	1,942
					Operating Income	109,754	119,134	126,564	20,324	(4,126)
					Reconciliation to Adjusted Operating Income	Share-based Compensation	10,316	14,548	15,498	
						4,236	2,772			
					Depreciation and Amortization	10,957	9,319	7,481	1,906	1,942
					Adjusted Operating Income	131,027	143,001	149,543	26,466	588
						219				

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If the company proceeds with the spin-off of its sports businesses, it would be structured as a tax-free transaction to all MSG shareholders. Upon completion of the contemplated separation, shareholders of record of MSG common stock would receive a pro rata distribution, expected to be equivalent, in aggregate, to an approximately two-thirds economic interest in the pure-play sports company. The remaining common stock, expected to be equivalent to an approximately one-third economic interest in the sports company, would be retained by the live entertainment company. These shares are expected to be used to raise capital and/or be exchanged for the common stock of the entertainment company. James L. Dolan is expected to be the Executive Chairman and Chief Executive Officer of both companies.

One company would be a leader in live entertainment, with a growing portfolio of assets, including state-of-the-art music and entertainment-focused venues, MSG Productions, other strategic entertainment joint ventures, approximately one-third interest in the post-spin sports company, and approximately \$1 billion in cash. The live entertainment company will also continue to move forward with the development of MSG Sphere. The first MSG Sphere is expected to open in Las Vegas by the end of calendar 2020, followed by a second MSG Sphere in London approximately one year later.

Upon separation, MSG Entertainment is slated to include:

- World-class venues: New York's Madison Square Garden, the Hulu Theater at Madison Square Garden, Radio City Music Hall, and Beacon Theatre; the Forum in Inglewood, CA; The Chicago Theatre; and the Wang Theatre in Boston;
- MSG's bookings business, which fills MSG's venues with a wide variety of the most popular entertainment events. This business would also include the company's bookings of live sporting events such as college basketball and professional boxing, the results of which are currently reported as part of the company's MSG Sports segment;
- Productions, which includes the Radio City Rockettes and the Christmas Spectacular, the nation's number-one live family holiday show;
- Majority interests in TAO Group, a world-class hospitality group, and Boston Calling Events, producer of New England's Boston Calling Music Festival, a preeminent live music event;
- Strategic entertainment joint ventures—e.g., Tribeca Enterprises;
- An approximately one-third economic interest in the pure-play sports company; and
- Approximately \$1 billion in cash on hand.

Notably, prior MSG Entertainment segment reporting (see Exhibit 2) does not include the "bookings" business related to sporting events, and little disclosure has been provided on the actual revenue and profit contribution of that business. Commentary surrounding the "bookings" business suggests that it had a highly successful F2018, which "delivered yet another year of increased venue utilization with a record-setting number of events ... [with] double-digit increases in both multi-market and multi-night engagements." Additionally, the depreciation charges that are associated with the owned arenas (The Garden, the Hulu Theater, and the Forum) are not included in the Entertainment segment's results, and are instead embedded within results for the corporate segment.

The pure-play sports company would include:

- The New York Knicks professional NBA franchise and its development team, the Westchester Knicks;
- The New York Rangers professional NHL franchise and its development team, the Hartford Wolf Pack;
- Knicks Gaming, the official NBA 2K e-sports franchise of the New York Knicks, and a majority interest in Counter Logic Gaming, a leading North American e-sports organization; and
- A professional sports team training center in Greenburgh, NY.

Outlook

ENTERTAINMENT

Following the separation, the Entertainment business's valuation will largely be derived from the underlying value of the physical assets (venues), and to a lesser (yet not insignificant) degree from the value of earnings generated from the bookings business, TAO Group, and BCE, among other promotional businesses. Earnings growth will be a function of the number of events held at MSG venues, revenue per event, and the number of venues. It is assumed that utilization of the venues is fairly high (1Q F2019 saw a decline in utilization), so that earnings impact becomes a function of cost control, increased fees, or venue expansion. It has been noted that multi-night and multi-city engagements (for example, Phish's "residency" at The Garden for 13 nights in the summer of 2017) generate higher margins, as costs are leveraged over the multiple nights. It is unreasonable to expect consistent similar long-run engagements, which suggests that increased fees

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are currently the key revenue growth component. Management has commented that the live touring business has never been stronger, mostly as a factor of acts (bands, comics, etc.) being financially motivated to do so, implying that there could be room for increased revenue per event. However, that will eventually receive pushback, necessitating new venues to promote. MSG Sphere appears to be management's solution to this, albeit at what in all likelihood will be a hefty, as yet unquantified price tag.

It should be noted that it is assumed that the air rights associated with the MSG arena will remain with the entertainment company. While there is no exact information on number of square feet and the ultimate value of those air rights, those questions have been a constant subject of discussion among investors. Interestingly, on MSG's May 2018 conference call, the possibility of combining those air rights with Vornado Realty Trust's (NYSE: VNO) development rights to redevelop 2 Penn Plaza in New York City was brought up. Anecdotally, an investment thesis on VNO has focused on Vornado's potential redevelopment of the area around Madison Square Garden, which would lend credence to an opportunity for MSG Entertainment to raise liquidity for the Sphere project.

Funding for the Spheres will be available from the expected \$1 billion cash position and the one-third of MSG Sports that will be retained by the company. It can be noted that there is minor debt (~\$105 million) associated with the TAO Group. Management has commented that the timing and method of disposition of the MSG Sports shares has yet to be determined; however, a sale or exchange for shares of MSG Entertainment are distinct possibilities. The final method of share disposal will depend on the respective valuations of MSG Sports and MSG Entertainment, which could present an opportunity to raise capital without tapping the debt markets. The

company currently has two untapped revolvers that are tied to the sports teams—\$150 million for the Rangers and \$215 million for the Knicks.

SPORTS

Clearly, annual operating results at the Sports segment can be lumpy, depending on a wide range of factors, including team performance and player compensation. Reflecting that, we think the estimated values of MSG's two core sports franchises, namely the New York Knicks basketball team and the New York Rangers hockey team, are the more relevant metrics for investors to consider with respect to the segment's potential valuation. On that front, Forbes magazine publishes an annual list of sports team valuations; the most recent list estimates the value of the Knicks and Rangers at \$4.0 billion and \$1.55 billion, respectively, based on four key factors, including the franchise's value attributable to revenue-sharing agreements, city/market size, the franchise's arena, and its brand (see Exhibit 4). For context, these estimates suggest the Knicks and the Rangers are the most valuable franchises in their respective sports (ahead of the Los Angeles Lakers, valued at \$3.7 billion, and the Toronto Maple Leafs at \$1.45 billion, also shown in Exhibit 4).

Exhibit 4 MSG Sports: Valuation Breakdown of NY Knicks and NY Rangers
(\$ in millions)

Source: Forbes.

Notably, the value of both franchises, as estimated by Forbes, has steadily increased in recent years; in fact, since 2006, the estimated value of the Knicks and Rangers has grown at a compound annual rate of ~17% and 14%, respectively (see Exhibit 5). In our view, this has been driven not only by high-profile deals at higher than expected valuations but also by more sophisticated pricing strategies (for tickets, luxury suites, and sponsorships) and increasingly lucrative national and local TV contracts. On the latter, the NBA's television deal with Walt Disney Co. (NYSE: DIS) and Turner, which extends through 2024, saw its annual receipts jump to \$2.7 billion (from \$930 million) in its most recent renewal. Similarly, when the NHL renegotiated its deal with NBC in 2011, annual payments to the league jumped to \$200 million (from \$77.5 million), and we see the opportunity for a similar increase when that agreement expires in 2020, tangentially increasing payments to the Rangers.

Knicks	Lakers	Rangers	Leafs	Sport	611	\$ 587	\$ 148	\$ 151	\$
Market	1,831	1,866	668	672					
Arena	995	573	518	411					
Brand	563	674	215	216					
Total	4,000	\$ 3,700	\$ 1,549	\$ 1,450					
Revenue	443	\$ 395	\$ 253	\$ 232					
Multiple	9.0x	9.4x	6.1x	6.3x	EBITDA	155	\$ 147	\$ 107	\$ 94
Multiple	25.8x	25.2x	14.5x	15.4x	221				

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Exhibit 5 The Madison Square Garden Company: Historical Estimated Value of NY Knicks and NY Rangers (\$ in millions)
NY Knicks NY Rangers

Source: Forbes.

Even as valuations have steadily risen in recent years, the vast majority of recent comparable transactions have been consummated at a significant premium to their most recent Forbes valuation. In the past seven years, those premiums have been 42% for professional basketball teams and 35% for professional hockey teams (see Exhibit 6). In our view, this phenomenon speaks to the scarcity value of what are undoubtedly considered trophy assets. Indeed, since 2004, the number of global billionaires has grown, according to Forbes, to 2,754 (from 587), and during that time the overall number of NBA and NHL teams has remained stagnant. Amid this ongoing global concentration of wealth (and increased focus on broadening the appeal of U.S. sports worldwide), sports assets have seen an increased influx of foreign investment. As an example, in 2016, Jiang Lizhang, a Chinese real estate investor, purchased 5% of the Minnesota Timberwolves from owner Glen Taylor at a reported valuation of about \$1 billion, which was ~30% higher than the team's most recent Forbes valuation. As well, on October 27, 2017, it was reported that Brooklyn Nets owner Mikhail Prokhorov reached a deal to sell a 49% stake in the team to Joseph Tsai, the co-founder of Chinese-based Alibaba (NYSE: BABA), at a reported valuation of \$2.35 billion (or almost 30% above Forbes' 2017 valuation of \$1.8 billion). (Notably, the deal is said not to include ownership of the Barclays Center, the arena in which the Nets play; however, it does include an option to buy a controlling interest in the team within three years.) Underlying the growing number of potential purchasers for a limited number of sports properties, we would also highlight that U.S. sports teams possess advantageous tax attributes that allow ~90% of the team's purchase price to be allocated toward intangible assets, which can then be amortized against the owner's total income.

All things considered, we would not be surprised to see incremental upward revisions to the

franchise values of both the Knicks and the Rangers when the next Forbes valuation lists are released in February 20 20 and November 201 9, respectively, which could serve as ancillary upside catalysts for the shares of MSG.

Exhibit 6 The Madison Square Garden Company: Recent Transactions Compared to Most Recent Appraisal Value
(\$ in millions)

Source: Company reports, Bloomberg, and Institutional Research Group estimates.

While less quantifiable to the value of a sports franchise, it would be remiss not to mention that the legalization of sports gambling in states outside of Nevada has garnered a great deal of interest. Seven states currently allow sports betting, two states have partial laws in place 0.0% 10.0% 20.0% 30.0% 40.0% 50.0% 60.0% \$- \$200 \$400 \$600 \$800 \$1,000 \$1,200 \$1,400 \$1,600 \$1,800 2006 2007 2008 2009 2010 2011 2012 2013 2014 2015 2016 2017 2018 Value YoY % Change -10.0% 0.0% 10.0% 20.0% 30.0% 40.0% 50.0% 60.0% 70.0% 80.0% 90.0% \$- \$500 \$1,000 \$1,500 \$2,000 \$2,500 \$3,000 \$3,500 \$4,000 \$4,500 2006 2007 2008 2009 2010 2011 2012 2013 2014 2015 2016 2017 2018 2019 Value YoY % Change Team Date Sale price (millions) Forbes' estimated value Premium/ Discount Team Date Sale price (millions) Forbes' estimated value Premium /Discount Brooklyn Nets (49% stake) Oct-17 \$2,300 \$1,800 27.8% Phoenix Coyotes Dec-14 \$305 \$200 52.5% Houston Rockets Sep-17 \$2,200 \$1,650 33.3% New Jersey Devils Aug-13 \$320 \$205 56.1% Minnesota Timberwolves (5% stake) Jun-16 \$1,000 \$770 29.9% St. Louis Blues May-12 \$180 \$157 14.6% Atlanta Hawks Apr-15 \$730 \$425 71.8% Toronto Maple Leafs Dec-11 \$1,000 \$521 91.9% Milwaukee Bucks Apr-14 \$550 \$405 35.8% Dallas Stars Nov-11 \$240 \$227 5.7% Los Angeles Clippers Aug-14 \$2,000 \$575 247.8% Atlanta Thrashers May-11 \$170 \$135 25.9% Sacramento Kings May-13 \$534 \$300 78.0% Buffalo Sabres Feb-11 \$165 \$169 -2.4% New Orleans Hornets Apr-12 \$338 \$285 18.6% Average premium 34.9% Memphis Grizzlies Jun-12 \$377 \$263 43.3% Detroit Pistons Apr-11 \$325 \$360 -9.7% Washington Wizards Apr-10 \$550 \$313 75.7% Golden State Warriors Jul-10 \$450 \$315 42.9% Average premium 57.9% Avg. premium, ex-Clippers 41.7% 222

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pending further action, and 15 states have filed bills for the current legislative session to legalize sports gambling. While the ultimate impact on the value of a franchise from legalized gambling is unclear, and is most certainly not a doubling in value as Dallas Mavericks owner Mark Cuban has suggested, increased revenue from gambling-related advertisers has already been commented on by MSG management. Additionally, the ability to further engage team fans either in the arena or offsite appears to be attractive from a monetization standpoint, while acknowledging that the ultimate form of monetization has yet to be fully fleshed out.

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VALUATION ANALYSIS

The following valuation analysis has been derived without a public Form-10 filing covering the financials of the MSG Sports standalone company. Given the lack of transaction details typically provided within the SEC filings, the analysis assumes: (i) no additional debt is to be raised prior to the separation, with the majority of the current cash balance remaining with MSG Entertainment; (ii) shares of MSG Sports are distributed on a one-for-one basis to MSG shareholders; and (iii) MSG Entertainment retains a 33% ownership stake in MSG Sports, resulting in MSG Sports shares outstanding of 36 million. Management's latest commentary suggests that the company is still working out details on a long-term venue license agreement between The Garden and the Knicks and Rangers. The fair value estimates derived below are subject to revision based on finalization of intercompany agreements, finalized capital structures, and capital allocation policies.

MSG Sports

The value of MSG Sports will be primarily based on the valuation assigned to the New York Knicks and New York Rangers. The most easily accessible and widely used benchmark for valuing sports franchises is the annual publication of values in Forbes magazine. In the 2019 publication, the Knicks were valued at \$4.0 billion, while the Rangers were valued at \$1.55 billion in the most recent NHL valuations. As noted earlier, a component of that valuation includes what Forbes refers to as "value attributable to its arena." While Forbes does not provide specifics on what exactly that arena value is derived from, it does not appear to be from the ownership of the arena, as other franchise teams that do not own their stadiums also appear to have a piece of their valuation tied to their arenas. In the spirit of conservatism, and given the view that if the MSG's sports franchises were to be located in a different arena, this value attribute would certainly change and most likely decline, we view it as appropriate to adjust the

valuations for the arena component. Forbes assigns \$995 million and \$518 million to the arena for the Knicks and the Rangers, respectively, resulting in an adjusted valuation for the Knicks and Rangers that total \$4.0 billion, or \$112 per share based on assumed 36.0 million shares outstanding (see Exhibit 7).

A valuation of MSG Sports would be incomplete if future growth in value were not considered. The Knicks have grown on average at 18.2% annually since 2016, representing a 16.6% CAGR. Included in the averages are the 78.6% and 41.0% increases in value in 2013 and 2015, with 2017 through 2019 increases of 10.0%, 9.1%, and 11.1% (excluding the two large outliers, the annual average increase in value drops to 10.5%). The Rangers' valuation increased, on average, approximately 15% over the same time period, with a 2018 increase of 3.3%. Assuming growth of 10% and 5% for the Knicks and Rangers, respectively, which approximates the most recent reported values growth, the sports teams together would be valued at \$4.4 billion, or \$122 per share based on assumed 36.0 million shares outstanding (see Exhibit 7).

While James Dolan, MSG's Chairman (whose family controls 71% of the voting shares through the ownership of all the Class B shares), has not expressly indicated a desire to monetize the company's sports assets, we think his public comments suggest a belief that sports team revenue, generally, will decline as MVPD (multichannel video programming distributor) bundles continue to unwind. In addition, Mr. Dolan has shown a willingness to monetize assets, as demonstrated by the sale of Cablevision to Altice in 2015 and prior news reports suggesting that MSG Networks (NYSE: MSGN) has been exploring a potential sale. (Notably, Mr. Dolan's concerns regarding the impact of bundling seemingly reflect a long-term horizon, as MSG's current national cable deals with the NBA and NHL do not expire until 2025 and 2021, respectively, and include annual escalators. As well, MSG's local rights deal does not expire until 2035 and also includes annual escalators.) Assigning historical acquisition premiums of 30% and 25% to the Knicks and Rangers, respectively, would result in a \$5.7 billion valuation, or \$157 per share based on assumed 36.0 million shares outstanding (see Exhibit 7).

While the path to a sale of one (or both) of MSG's sports teams is unclear in the near term, the potential for a minority investment from a foreign partner as a way for MSG Sports to unlock value as well as continue to broaden its appeal abroad could be an ancillary possibility. As recently as December 2018, following an ESPN profile on James Dolan, it was reported that parties interested in purchasing the Knicks had sent out "feelers" in the range of \$5 billion. In the same article, Dolan was quoted as saying that there were no bona fide offers and that the Dolan family ownership does not want to sell the Knicks and Rangers, yet he has a fiduciary duty to his shareholders. While reports are unclear as to whether the \$5 billion figure included The Garden (differing reports have unclear and divergent phrasing, we assume it did not), if the \$5 billion was for the Knicks alone, it would equate to a \$139 per share valuation for MSG Sports based on assumed 36.0 million shares outstanding, and imply zero valuation for the Rangers in our fair value estimate. Notably, the Rangers at the latest Forbes valuation (with no adjustments) would add \$43 per share in value to MSG Sports based on assumed 36.0 million shares outstanding. Anecdotally, James Dolan commented that feelers are "...like a stock price. It's only important if you're going to buy or sell."

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Exhibit 7 MSG Sports: Fair Value Estimate of Sports Franchises Based on Adjusted Forbes Valuation, Growth, and Acquisition Premium (\$ in millions, except per share amounts; shares in millions)

N/A: Not applicable. Source: Company reports, Forbes, and The Spin-Off Report estimates.

It should be noted that our fair value estimate of \$139 per share for MSG Sports (based on the average of "2019" valuation and "bull case" scenarios) does not include any value for the various other businesses included in the standalone company. These include the company's e-sports teams (MSG acquired Counter Logic Gaming Inc in 2017 for an undisclosed sum), a professional e-sports team, and a training facility in Greenburgh, NY. Optionality from these businesses appears minimal in the near future; however, over the longer term they could prove valuable, although how valuable is not quantifiable to any reasonable degree at this point in time.

MSG Entertainment

Following the separation of MSG Sports, MSG Entertainment will have three main assets driving its valuation: (i) the owned entertainment venues; (ii) the operating businesses, which include owned productions such as the Christmas Spectacular and Boston Calling, majority ownership in TAO Group; and (iii) an approximate \$1 billion in cash and \$106 million in debt associated with TAO.

Given that the sports bookings business that will be included in MSG Entertainment, which handles live sporting events such as college basketball and professional boxing in MSG venues, is currently included in MSG's Sports segment, the following analysis, based on earnings of the

operating businesses, is inherently conservative, in that it will underestimate the true earnings power of the bookings business overall. MSG has made a confidential Form-10 filing; when the filing is made available to the general public, we expect further details on the business, and at that point in time we will adjust our earnings estimates as needed.

The Entertainment business presents or hosts live events, such as concerts, shows, and other events, at its portfolio of owned and leased venues. MSG Entertainment primarily licenses its venues to third-party promoters for a fee (but in some cases takes on the economic risk of promotion). As well, the company collects facility and ticketing fees and generates revenue from concessions, sponsorships, and merchandise. Considering the lumpiness of MSG's concert/venue management business and its dependence on relationships with landlords, we think the company's owned real estate portfolio could be viewed as the Entertainment segment's most relevant source of value. Among its real estate assets, Madison Square Garden, located between 31st and 33rd Streets and 7th and 8th Avenues in Midtown Manhattan, has been dubbed "The World's Most Famous Arena." The complex, which includes both The Garden and the Hulu Theatre, was appraised at \$1.2 billion by the New York City Independent Budget Office in a May 2014 tax assessment.

(Note: This analysis does not include any value for the so-called air rights, which are also known as development rights and refer to the transferable allowances of potentially buildable space, most commonly applying to the space above a building that could be used for additional construction. While MSG management implicitly concedes that its air rights have substantial value, it is quite cagey about the potential timing and magnitude of any monetization; thus, we exclude air rights from our valuation but urge investors to be cognizant of their potential as a valuable source of upside optionality. For context, we have seen estimates suggesting MSG could have 2.4-5.4 million square feet of potential development rights, and we note that, according to Tenantwise, air rights averaged \$292 per square foot in 2016— with values in prime locations, such as the High Line, Hudson Yards, and Times Square, in the \$400-\$800 per square foot range. Notably, real estate website cityreality.com states that the Department of City Planning has said that Manhattan air rights average \$225 per square foot, but did not give a date for the data. This suggests to us that even at a conservative \$250 per square foot, MSG's air rights alone could be valued at more than \$600 million, or ~\$25 per share.)

In 2008, MSG acquired The Chicago Theatre, with the iconic six-story CHICAGO marquee, for an undisclosed sum. The venue was previously purchased in 2004 by TheatreDreams Chicago LLC for \$3 million, which for simplicity's sake is the value we assign, despite our contention that the replacement value is far higher. The Forum, which is in Inglewood, CA, and was the former home of the Los Angeles Lakers (NBA) and the Los Angeles Kings (NHL), was purchased by MSG in 2012 for \$23.5 million. Subsequently, the company spent ~\$100 million to renovate the venue into a world-class concert destination. We think MSG's investment could be used as a suitable baseline for valuation, which, again, is likely conservative relative to its replacement value, considering its proximity to the planned \$2.6 2018/ Arena Annual 2019/ Bull "Whisper 2019 Value Adj. Value Growth 2020 Premium Case Number" New York Knicks, ex-arena 4,000\$ 995\$ 3,005\$ 10.0% 3,305.5\$ 30.0% 4,297\$ 5,000\$ New York Rangers, ex arena 1,549 518 1,031 5.0% 1,083 25.0% 1,353 N/A Total Value 5,549\$ 1,513\$ 4,036\$ 8.7% 4,388\$ 28.8% 5,650\$ 5,000\$ Shares Outstanding 36.0 Value per Share 154.15\$ 112.12\$ 121.90\$ 156.96\$ 138.90\$ 225

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billion Los Angeles Rams sports complex and reports that the T-Mobile Arena, a 20,000-seat venue in Las Vegas, cost ~\$375 million to construct.

All told, our base-case valuation for MSG's owned real estate is \$1.3 billion, or \$56 per share. If one were to take a more bullish stance by applying some modicum of value to the potential air rights above Madison Square Garden as well as 25% premiums to the base-case values of the Forum and The Chicago Theatre, value of almost \$2 billion, or \$82 per share, could be derived (see Exhibit 8).

Exhibit 8 MSG Entertainment: Estimated Valuation of Owned Venues (\$ in millions, except per share amounts; shares in millions)

Source: Company reports, Bloomberg, and The Spin-Off Report estimates.

Regarding MSG's other Entertainment assets, with stable attendance, high profitability, and minimal re-investment requirements, we view the Christmas Spectacular as a highly valuable asset for MSG. We note, however, that there is no direct comparison for valuation. Similarly, the investment made in TAO Group appears to be profitable and growing, at least in terms of the number of venues that the company is opening and operating, although management has noted that recent results have not met their financial expectations. In F2018, the MSG Entertainment segment generated revenue and EBITDA of \$780.7 million and \$117.0 million, respectively, representing a 15.0% margin. Through 1H F2019, the segment has increased revenue by 10%

(primarily from higher event-related revenues) and operated with a 23.0% EBITDA margin. For context, the segment generated a 23.1% EBITDA margin in 1H F2018. Given this year's revenue trends, and an assumption of moderating top-line growth to 5% in F2020, it could be forecast that a standalone MSG Entertainment would generate \$902 million in revenue in F2020. Assuming margin contraction in the back half of the year, similar to F2018, and accounting for estimated standalone corporate costs, we forecast that MSG Entertainment could earn \$126 million in EBITDA. Applying a 9x multiple, in line with the price paid for the TAO Group ownership stake, MSG Entertainment's operating businesses would be valued at \$1.1 billion, or \$48 per share (see Exhibit 9). It should be noted that the margin contraction modeled for F2020 implies that approximately \$30 million in corporate costs is assigned to the standalone entertainment company.

Exhibit 9 MSG Entertainment: Estimated Valuation of Operating Entertainment Assets (\$ in millions, except per share amounts; shares in millions)

Source: Company reports, Bloomberg, and The Spin-Off Report estimates.

Based on the above valuation exercises for a standalone MSG Entertainment, and including roughly \$1.1 billion in net cash and a 33% ownership stake in MSG Sports, we estimate that MSG Entertainment would be fairly valued at \$5.2 billion, or \$221 per share (see Exhibit 10).

Venues Madison Square Garden 1,200.0\$
The Forum 123.5
The Chicago Theatre 3.0
Total 1,326.5
Shares Outstanding 23.8
FVE \$ / Share 55.83\$
F2018 Revenue 780.7\$
Growth 10.0% F2019E Revenue 858.8
Growth 5.0% F2020E Revenue 901.7
EBITDA Margin 14.0% EBITDA 126.2
Multiple 9.0x Total 1,136.2
Shares Outstanding 23.8
FVE \$ / Share 47.82\$
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Exhibit 10 MSG Entertainment: Post-Spin Sum-of-the-Parts Fair Value Estimate (\$ in millions, except per share amounts; shares in millions)

Source: Company reports, Bloomberg, and The Spin-Off Report estimates.

The Madison Square Garden Company: Pre-Spin Sum-of-the-Parts

On a pre-spin sum-of-the-parts basis, shares of MSG are fairly valued at \$362 per share (see Exhibit 11). With our fair value estimate approximating 25% upside potential from the current share price, shares of MSG are recommended for purchase prior to the spin-off. The fair value consists of \$151 in value assigned to post-spin MSG Entertainment and \$211 per share derived from post-spin MSG Sports. The post-spin fair value estimates are subject to change. It should be noted that in more bullish scenarios additional value could be realized. This could include valuing the air rights above MSG, which could be worth \$25 per share alone, or monetizing the Knicks and/or Rangers. Although the latter is highly subjective, if the Dolan "whisper number" for the Knicks is used, this would add \$1 billion in fair value to the pre-spin sum-of-the-parts assuming the Rangers were valued at just the adjusted Forbes estimate, which excludes any value associated with the arena. That said, the typical rebuttal on the valuation of a sports franchise is that it is only worth what someone will pay, and that is dependent on the current owners' willingness to sell. In the absence of two willing parties, the sports franchises will likely be valued at a reasonable valuation approximating the Forbes value.

Exhibit 11 The Madison Square Garden Company: Pre-Spin Sum-of-the-Parts Fair Value Estimate
(\$ in millions, except per share data; shares in millions)

Source: Company reports, Bloomberg, and The Spin-Off Report estimates.

Venues 1,326.5\$
Entertainment Operating Businesses 1,136.2
Net Cash (Debt) 1,125.0
Sub Total 3,587.6
33% Ownership of MSG Sports 1,656.3
Market Capitalization 5,244.0
Shares Outstanding 23.8
FVE \$ / Share 220.72\$
MSG Entertainment (ex 33% stake in Sports) 3,587.6\$

MSG Sports 5,019.2
Total 8,606.8
Shares Outstanding 23.8
FVE \$ / Share 362.26\$
Current MSG Share Price 290.39\$
Implied Upside 24.8% 227

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CONCLUSION

In terms of rationale, the separation of Sports from Entertainment makes sense as the Entertainment company looks to expand its portfolio of owned venues to drive earnings growth. The posited structure of the separation, with Entertainment retaining a 33% stake in Sports and approximately \$1 billion in cash, will allow the company to largely finance its ambitious Spheres project without having to tap the debt markets or dilute Entertainment shareholders. The Entertainment business will be a hybrid of an asset based stock (derived from the ownership of venues) and operating assets consisting primarily of a growing events promotion business that leverages the prime assets of the company (Madison Square Garden).

The Sports business will value will be driven by the value of the NY Knicks and NY Rangers NBA and NHL franchises. Historical increases in value of professional sports teams, and comparable sales transactions should provide for a stable yet growing asset based valuation at MSG Sports. Further, if the Dolan family were to sell an ownership interest in either of the sports franchises (or potentially take one or both teams private), significant upside optionality would arise given the “whisper numbers” that have been reported on team valuations.

On a pre-spin sum-of-the-parts basis, shares of MSG are fairly valued at \$362 per share. With our fair value estimate approximating 25% upside potential from the current share price, shares of MSG are recommended for purchase prior to the spin-off. The fair value consists of \$151 in value assigned to post -spin MSG Entertainment and \$211 per share derived from post -spin MSG Sports. The post -spin fair value estimates are subject to change. It should be noted that in more bullish scenarios additional value could be realized. This could include valuing the air rights above MSG, which could be worth \$25 per share alone, or monetizing the Knicks and/or Rangers. Although the latter is highly subjective, if the Dolan “whisper number” for the Knicks is used, this would add \$1 billion in fair value to the pre-spin sum-of-the-parts assuming the Rangers were valued at just the adjusted Forbes estimate, which excludes any value associated with the arena. That said, the typical rebuttal on the valuation of a sports franchise is that it is only worth what someone will pay, and that is dependent on the current owners’ willingness to sell. In the absence of two willing parties, the sports franchises will likely be valued at a reasonable valuation approximating the Forbes value. 228

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Exclusive Marketers of
The Spin-Off Report

PCS Research Services 100 Wall Street, 20th Floor New York, NY 10005 (212) 233-0100
www.pcsresearchservices.com

Michael Wolleben Joanna Makris Robert Dunn

Murray Stahl

Steven Bregman

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A.P. Moller-Maersk A/S (Pre-Spin)

Current Share Price (4/2/19): DKK 8,854 Fair Value Estimate: DKK 11,276 per share Shares Outstanding: 20.8 million

Ticker: MAERSKB DC Dividend: DKK 150 Yield: 1.7%

Market Capitalization: DKK 182.3 billion

A.P. Moller-Maersk A/S (Post-Spin)

Fair Value Estimate: DKK 9,254 per share Shares Outstanding: 20.8 million

Ticker: MAERSK DC Dividend: TBD

Market Capitalization: DKK 192.5 billion Yield: N/A

The Drilling Company of 1972 A/S

Fair Value Estimate: DKK 2,022 per share Shares Outstanding: 20.8 million

Ticker: TBD

Dividend: TBD

Market Capitalization: DKK 42.1 billion Yield: N/A

TBD – To be determined. N/A – Not applicable.

Note: Market capitalization is based on fair value estimate for post-spin entities and current market cap for pre-spin A.P. Moller- Maersk A/S. 1 USD = 6.66 DKK

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Investment Thesis

On August 17, 2018, A.P. Moller -Maersk A/S (MAERSKB: DC) announced that, following a strategic review, the company would seek a separate listing of Maersk Drilling Holding A/S, which is to be named The Drilling Company of 1972 A/S. Maersk Drilling shares will be distributed to A.P. Moller-Maersk shareholders on April 5, 2019, the demerger record date.

Every A or B share in MAERSKB of nominally DKK 1,000 will receive 2 new shares of nominally DKK 10 and every A or B share of MAERSKB of nominally DKK 500 will receive 1 new share of nominally DKK 10 as of the cut -off date of April 3, 2019 . All shares of The Drilling Company of 1972 will have the same rights and a single share structure, and will trade on Nasdaq Copenhagen A/S beginning on April 4, 2019.

Maersk, a Danish conglomerate, is primarily an integrated container marine shipping and logistics company. The company's exit from its drilling operations business following the sale of Maersk Oil to Total S.A. is the culmination of a strategic refocus on the shipping and logistics industry. Notably, the Maersk Oil sale generated proceeds in both cash and Total stock, and the company has indicated plans to spin off a large portion of its Total ownership stake. It has been reported that Maersk sought bids for the drilling business prior to the de-merger announcement; however, management was underwhelmed by the resultant bids. Management has noted that in conjunction with the de -merger, \$1.5 billion in debt financing has been secured to ensure that Maersk Drilling has an adequate capital structure.

In conjunction with the strategic refocus of its business lines, Maersk also re -designated its segment reporting, and now operates under four segments: Ocean (primarily the Maersk Line business as well as strategic trans -shipment hub ownership); Logistics & Services (including logistics and supply chain management under the Damco brand, operations of inland service facilities including container storage, warehousing , and local transportation); Terminals & Towage (operation of landside port activities with customers including carriers and towage services under the Svitser brand); and Manufacturing & Other (production and sale of reefer and dry containers based out of factories in China, as well as other ancillary businesses). Maersk Drilling, which is now reported under discontinued operations, is the owner and operator of offshore drilling rigs, with a focus on ultra-deepwater and harsh environment locales.

When management began its strategic review almost two years ago, it was posited that a possible break-up would likely be more value-accretive if it were implemented by means of a spin-off as opposed to a sale or carve -out. Given the state of the energy sector at the time (energy prices have rebounded, yet Maersk's share price has not benefited from its exposure), any sale would risk taking place at depressed values, without accounting for the long-term earnings potential of those businesses. In a market that favors pure-play companies, management considered that the spin-off of the drilling company might unlock the most value

while retaining upside potential for existing shareholders.

On a pre-spin, sum-of-the-parts basis, shares of Maersk A/S are estimated to have a fair value of DKK 11,276 per share, consisting of DKK 2,022 per share for The Drilling Company of 1972 230

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A/S and DKK 9, 254 per share for post -spin Maersk A/S . With the pre-spin sum-of-the-parts estimate representing approximately 27% upside to Maersk's current consolidated share price (DKK 8,854 per share as of this writing), the transaction appears to unlock significant upside. In addition, a recovery scenario could generate upside of 48% from current levels. It is important to note, however, that there are several overhangs on the shares which may limit near -term appreciation. Foremost, the International Maritime Organization 's (IMO) 2020 sulphur cap regulations will likely cause marine fuel prices to continue to fluctuate widely this year. New regulations, changing economic conditions, and trade tensions could also potentially trigger significant and lasting marine fuel price volatility. That said, at current share prices, the shares appear sufficiently discounted for these risks. Notably, Maersk shares have declined 34% over the past 5 years, versus a 46% return for the OMX Copenhagen 20 Index over the same period. Year-to-date, the shares have appreciated 4% (versus a 12% return for the OMX Copenhagen 20 Index). At current prices, we think the shares are sufficiently discounted for industry and macroeconomic uncertainties. Longer term, the quality of the company's industry -leading shipping assets—combined with the earnings potential of the remaining businesses, and potential for regulatory changes to shift the supply/demand balance to more normalized levels—could offer meaningful upside potential with minimum downside risk.

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Company Description

Based in Copenhagen, Denmark, A.P. Moller-Maersk A/S, also known as Maersk, is a conglomerate with activities in the transport, logistics, and energy sectors. The company was founded in Svendborg, Denmark in 1904 by captain Peter Maersk Moller (1836-1927) and his son Arnold Peter (A.P.) Moller (1876 -1965) as Dampskibsselskabet Svendborg (Danish for "Svendborg Steamship Company").

With a current market capitalization of DKK 182 billion (USD \$27 billion) and 2018 consolidated revenue of USD \$39 billion, Maersk is the largest container ship and supply vessel operator in the world (see Exhibit 1) . The company has subsidiaries and offices across 130 countries and has around 88,000 employees. Maersk consists of four business segments: Ocean (2018 revenue of \$29.4 billion, 10.6% EBITDA margin). This segment , which represented 75% of 2018 revenue, consists of traditional ocean vessels and the activities of the Maersk Liner Business (Maersk Line, Safmarine, and SeaLand), in addition to the Hamburg Sud brands (Hamburg Sud and Alianca). Maersk Line, the largest operating unit within A.P. Moller-Maersk, is the world's largest overseas cargo carrier and operates over 600 vessels with 3.8 million TEU (twenty-foot equivalent) container capacity. Maersk controls approximately 18% of transportation market share by sea, with a capacity of 4 million TEUs (twenty-feet equivalents); the second largest carrier, MSC (Mediterranean Shipping Company), has a 3.2 million TEU capacity.

- Logistics & Services (2018 revenue of \$6 billion, 1.6% EBITDA margin) (16% of revenue). The segment includes supply chain management services, container inland services, inland haulage, trade finance services, and freight forwarding
- Terminals & Towage (2018 revenue of \$3.8 billion, EBITDA margin of 20.6%) . Maersk's APM Terminals business operates out of a separate headquarters in The Hague, Netherlands. This business is one of the world 's leading port facility operators. The Terminals business operates a global port, terminal, and inland services network, with interests in 57 ports and container terminals in 36 countries on five continents, as well as 155 inland services operations in 48 countries. This segment also includes gateway terminals and towage services under the Svitser brand . Svitser, founded in 1833, has a fleet of over 400 tugs, line-handlers, and other vessels. The company provides harbor and terminal towage services in over 100 ports and 20 oil and gas terminals globally. This business has been growing in the 4% range annually as a result of significant investments and new projects.
- Manufacturing & Other (2018 revenue of \$2.5 billion, 2.3% EBITDA margin). This segment consists of Maersk Container Industry activities within reefer containers. This segment's sales have been declining , and in January 2019, Maersk announced its intention to exit the dry container business.

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Exhibit 1 A.P. Moller-Maersk A/S: Income Statement Highlights and Segment Details
(USD in millions)

Source: Company reports.

Maersk Drilling Group

Founded in 1972 in Lyngby, Denmark, Maersk Drilling Group was established as part of the Maersk Group and is part of the global offshore drilling industry, providing offshore drilling services to E&P companies in support of their upstream exploration and development activities. Maersk Drilling Group owns and operates a modern fleet of offshore rigs and generates revenue by leasing rigs and providing services to E&P companies. It operates one of the youngest and most advanced fleets of jack-up rigs and floaters (defined below) in the world, and operates worldwide, including in the North Sea, where it has a market-leading presence (see Exhibit 2).

There are three main types of drilling rigs: jack-up rigs, semi-submersibles, and drillships—the latter two of which are referred to as “floaters.” Jack-up rigs tend to operate in shallower waters. The most advanced jack-up rigs can operate in depths of up to 492 feet, while floaters can operate in depths of up to 12,000 feet.

Jack-up rigs are mobile drilling platforms standing on the seabed, and are typically equipped with three steel legs and a self-elevating system that adjusts the platform height based upon the water depth. When the rig relocates, it will jack its platform down on the water until it floats, and will then be towed by tugboats to its next location or lifted by a heavy -transportation vessel if being moved over a greater distance.

The deck of the jack-up rig provides space for drilling equipment, supplies, and living quarters. Modern jack-up rigs typically have a drilling package mounted on a cantilever to drill away from the hull.

As of February 15, 2019, Maersk Drilling Group's fleet comprised 23 rigs in two operating segments: (1) jack-up rigs and (2) floating rigs (“floaters”), as described in more detail below:

- Jack-up rigs (63% of 2018 revenue) . Of the company's 15 jack-up rigs, Maersk operates the following: (1) eight ultra-harsh-environment jack-up drilling rigs; of these, five are operating and one is preparing for contract in the Norwegian part of the North 2018 2017 2018 2017 Ocean 29,366 22,023 2,547 3,007 2,279 2,831
Logistics & Services 6,082 5,772 98 139 47 54
Terminals & Towage 3,772 3,481 778 639 556 704
Manufacturing & Others 2,547 1,689 59 173 16 23
Unallocated activities, eliminations, etc. (1,748) (2,020) (136) (196) (22) (13)
A.P. Moller - Maersk consolidated - continuing operations 39,019 30,945 3,806 3,532
2,876 3,599
Revenue E BITDA CAPE X 233

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Sea, while one is operating and one is warm-stacked¹ in the U.K. part of the North Sea; (2) four harsh-environment jack-up rigs, of which two are operating in the Dutch part of the North Sea, while one is operating and one is preparing for contract in the U.K. part of the North Sea; (3) two premium jack-up rigs, of which one is operating in Brunei and one is cold-stacked in Singapore; and (4) one accommodation jack-up rig operating in the Danish part of the North Sea.

- Floaters (37% of 2018 revenue). Of the company's eight floaters, Maersk operates (1) four drillships, of which three are operating offshore in Ghana and one is warm-stacked in the Gulf of Mexico, and (2) four semi-submersible rigs, of which one is operating in Egypt, one in the Caspian Sea, one is preparing for contract in Timor -Leste, and one is warm-stacked in Trinidad and Tobago.

In addition, one jack-up rig (Maersk Giant) is currently held -for-sale and cold -stacked in Denmark ahead of disposal, with the sale expected to be completed in 2019.

Exhibit 2 Maersk Drilling Group: Income Statement for the Years Ended December 31, 2016

Source: Company reports.

¹ Warm-stacked rigs are rigs expected to be idle for a short-term period and are not under contract but are available for prompt use with minor preparation. Normal maintenance

operations are conducted so that the rig remains work- ready. Key members of the crew are retained, and the rig is considered part of the marketable supply. In contrast, cold-stacked rigs are not expected to be utilized in the near term, the crew is not on board, and systems have been turned off.

2018 2017 2016 Revenue 1,429 1,439 2,297
 Costs (818) (756) (916)
 Profit before depreciation, amortisation, impairment losses/reversals, and special items 611 683 1,381
 Special items (16) 2 16
 Profit before depreciation, amortisation, impairment losses/reversals, and special items 595 685 1,397
 Depreciation and amortisation (403) (468) (589)
 Impairment losses/reversals 810 (1,769) (1,510)
 Share of results in joint ventures (1) - - Profit/loss before financial items 1,001 (1,552) (702)
 Financial expenses, net (12) (19) (89)
 Profit/loss before tax 989 (1,571) (791)
 Tax (48) 49 1
 Profit/loss for the year 941 (1,522) (790)
 Earnings per share, USD 1,882 (3,044) (1,580)
 Diluted earnings per share, USD 1,882 (3,044) (1,580)
 For the year ended 31 December 234

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Impact of Tariffs and Trade

The shipping sector was one of the most significantly affected during the economic crisis that started in 2007, suffering from drastically reduced demand, declining freight rates, and lower profit margins. Added to the economic consequences of the recession, the industry has faced growing concerns in other areas. The first is environmental. Shipping-related carbon emissions currently represent approximately 3% of the world's carbon emissions; however, industry forecasts show that within the next 35 years, shipping emissions may increase by up to 250%. By 2050, shipping-related emissions are expected to represent over 14% of global carbon emissions.

With increasing pressure on profits and environmental performance, the industry responded by reducing vessel speed (known as “slow steaming”). By doing so, companies could lower fuel costs while ensuring profitability. Slow steaming is expected to lower carbon emissions by 1,000 tons and lower fuel consumption by up to 54%. Maersk has indicated that by halving its top cruising speed, it has reduced fuel consumption and greenhouse gas emissions by 30%. This allowed Maersk to offer reduced prices and an “environmentally friendly” alternative to companies not requiring immediate delivery.

The International Maritime Organization (IMO) will introduce in 2020 a global limit on sulphur in fuel oil used on board ships of 0.50% m/m (mass by mass). A change from the previous limit of 3.5% m/m, the new sulphur cap will necessitate significant investment and implies that ships will be required to use fuel that is likely to be three times more expensive. To combat increasing fuel prices, Maersk is exploring several ideas to cut marine fuel use over the years, from solar-powered sails to kites that tow a vessel.

With the IMO's 2020 sulphur regulation deadline looming, widely fluctuating marine fuel prices are expected to continue this year, with new regulations, concerns about economic conditions, and other market risks potentially triggering significant and lasting marine fuel price volatility.

Beginning in 2014, a decline in oil prices prompted a severe downturn in the container industry as oil companies cancelled or postponed exploration plans, reducing demand for offshore drilling rigs. Maersk suffered from a combination of a slower than expected rise in freight rates and increasing bunker fuel costs.

Accordingly, beginning in 2016, A.P. Moller-Maersk embarked on a transformational strategy to shed its energy-related assets and focus on its container transportation and logistics businesses. To this end, in March 2018, the company sold its energy business Maersk Oil to French oil-refining company Total S.A. for 4.95 billion in Total shares and US\$2.5 billion in Maersk Oil's assumed debt. In March 2019, Maersk Oil Trading and Investments, a subsidiary of Maersk, announced the sale of its position in Total (17.3 million shares), which generated EUR 884 million in gross sale proceeds.

More recently, the effects of the current trade war between the world's two biggest economies, the U.S. and China, have created formidable headwinds. Whereas prior administrations of both parties in the U.S. have followed trade policies that have increased the volume of cross-border

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trade while largely ignoring U.S. trade deficits, the current Trump administration has taken the opposite approach. Maersk has indicated that the effect of trade restrictions introduced last year would cut global container trade growth by between 0.3 and 1 percentage points annually this year. As background, on May 8, 2018, the U.S. announced that it would no longer participate in the JCPOA (Joint Comprehensive Plan of Action), an international plan of sanctions relief for Iran, on the assumption that the country will not comply with commitments to refrain from nuclear activities. Among the actions triggered by the U.S. withdrawal from the JCPOA is that the shipping insurance industry is banned from providing insurance to ships involved in transport to and from Iran. The U.S. withdrawal went into effect in May 2018; the result has been an increase in idle tanker capacity. In addition, licenses issued to companies such as Boeing Company (NYSE: BA) and Airbus S.E. (AIR: PA) to sell aircraft and spare parts to Iran Air and other Iranian airlines were revoked. Under the current sanctions regime, any vessel carrying Iranian energy products will not be allowed entry into a U.S. port. Simply put, the sanctions regime is both comprehensive and severe.

Another factor potentially weakening demand for oil products is currency depreciation against the U.S. dollar in emerging markets. As the U.S. dollar strengthens—a result of economic turmoil in Brexit-era Europe and rising interest rates in the U.S.—currency in developing economies cannot keep pace. Since many commodities are traded in U.S. dollars, its value weighs heavily on buying power in emerging markets. A strengthening U.S. dollar and weakening of local currencies often stalls economic growth and development potential. The purchase of crude oil and refined products is directly affected by global interactions. Barrels of crude are traded in U.S. dollars—if the Mexican peso, South African rand, and other currencies buy fewer U.S. dollars, those dollars are also limited in the quantity of oil they can purchase. This results in weakening demand for global crude oil and subsequent price volatility on the global oil stage for both crude and other refined products like marine fuels.

Transaction Rationale and Structure

Beginning in 2016, Maersk began an extensive restructuring under chief executive Søren Skou, in an effort to transform itself from a conglomerate into a container transportation and logistics company, concentrating on its shipping, port terminals, and logistics businesses and deciding to sell off its oil-related companies such as production, drilling, and tankers.

In March 2018, Maersk sold Maersk Oil to French oil company Total S.A. (TOTF.PA) for US\$7.5 billion (4.95 billion in Total shares and US\$2.5 billion in Maersk Oil's assumed debt). In addition, the company set a deadline to sell or list Maersk Drilling by the end of 2019. However, given the size of the company's drill fleet, valued in excess of US\$4 billion, finding a buyer proved challenging. An outright sale was also likely made more difficult by an oversupplied drilling rig market, which has not yet recovered despite a recent gain in oil prices. On August 17, 2018, after attempting to sell the business, the company announced its plan for a tax-exempt de-merger of the company's drilling activities (Maersk Drilling).

The cut-off date (last day of trading including the company's drilling activities) is April 3, 2019. April 4 is the first day of trading for The Drilling Company of 1972 A/S. Upon completion of 236

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the de-merger, Maersk shareholders will receive shares in The Drilling Company of 1972 that will have a single share structure. The shares will be distributed evenly among Maersk shareholders, so that every A or B share in MAERSKB of nominally DKK 1,000 will receive 2 new shares of nominally DKK 10, and every A or B share of MAERSKB of nominally DKK 500 will receive 1 new share of nominally DKK 10. All shares of The Drilling Company of 1972 will have the same rights.

Outlook: Post-Spin A.P. Møller-Maersk A/S

The ocean container shipping industry has been in a depression since 2008, having suffered from oversupply. Tanker capacity, once created, has a lifespan of roughly 25 years. Worldwide tanker capacity has grown only modestly in recent years. According to Statista, the worldwide tanker fleet grew 6% in 2018. One of the primary reasons for the modest growth is the time required to build a modern tanker (which can take roughly 1,000 workers a year to build). In addition, well-capitalized companies order tankers financed largely through low-cost debt. Because of low interest rates, these companies purchase tankers in order to lease them on the international market, with acceptable returns. In addition, some nations view their tankers as security assets. For example, in June 2018, Japan filed a WTO complaint against South Korea for granting state subsidies to the Korean tanker fleet. The U.S. in turn filed the same complaint against South Korea. As a result of these factors, tanker rates for very large crude carriers (VLCC) are currently about 25% of the level recorded two years ago. Management has indicated that it expects global container growth of between 1-3% in 2019, down from approximately 4%

in 2018 and significantly weaker than the near 6% expansion achieved in 2017.

For post-spin Maersk, upcoming regulatory changes surrounding reduced sulphur emissions may impact the supply/demand balance for container ships and reverse the recent historical trend of pricing declines. As mentioned previously, the IMO will cut emissions of sulphur either by reducing its content in marine fuel from 3.5% to 0.5% from 2020 or by requiring ships to remove sulphur content from exhaust fumes. Increased demand for lower sulphur distillate fuel would increase prices. The requirement to remove sulphur from exhaust fumes would require shippers to deploy scrubbers, which strip out sulphur emissions. The cost of this equipment, which can range from \$1 million to \$6 million, would render many older ships uneconomical, and in turn, increase scrappage rates. It is estimated that as many as 70,000 ships will be affected. The potential for reduced shipping capacity in the near term, and an ensuing fleet rebuilding cycle, could accelerate a restoration of the supply/demand balance for container ships, which could in turn increase pricing. Notably, should these changes take place, it would be reasonable to assume that better capitalized companies, and those with a younger average fleet age would be disproportionately advantaged.

Given the potential for regulatory changes to necessitate increased investment on the part of container shipping companies, Maersk's capitalization structure and owner-operator mentality may be perceived as key competitive advantages in a capital-intensive, cyclical industry. Notably, Maersk is not leveraged, has a low cost of capital, and has effectively reallocated its capital for growth. This operational conservatism has enabled the company to maintain a competitive fleet, acquire financially stressed competitors, and, going forward, should enable it

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to take advantage of an environment in which many competitors might have to reduce fleet size.

Notably, Maersk has tended, over the course of full shipping cycles, to emerge from recoveries in a stronger market share position than when it entered.

Outlook: Drilling Company of 1972 A/S

Performance of the post-spin drilling company will be closely tied to capital expenditures associated with oceanic drilling and crude oil production. Global energy demand and offshore oil and gas' share of this demand represents the primary demand driver for offshore drilling rigs. The International Energy Agency (IEA) forecasts primary energy demand growth of over 25% from 2017-2040, representing 1% growth annually over the forecast period.

As previously discussed, the global economic outlook also creates the potential for fuel price volatility, as economies that lead crude oil production and consumption—such as the U.S., China, Russia, Europe, and the nations of OPEC—are projected to experience weaker economic performance throughout 2019. One factor triggering this economic weakness is the lingering trade tensions between the U.S. and China, as both nations grapple with the implications of imposed tariffs. Though Presidents Trump and Xi decided in December 2018 to forego tariff hikes and negotiate more mutually beneficial trade terms, they continue to have difficulty reaching longer-term trade solutions. That said, at the end of 2018, Maersk Drilling's operations had one of the strongest contract backlogs in the industry, at \$2.5 billion.

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Valuation Analysis

Valuation: The Drilling Company of 1972 A/S

Following the separation, The Drilling Company of 1972 can be most aptly compared to its primary competitors in the oceanic drilling space. These include Diamond Offshore Drilling Inc. (NYSE: DO), Transocean Ltd. (NYSE: RIG), and Pacific Drilling SA (NYSE: PACD). Detailed comparables are shown in Exhibit 14.

As a starting point for deriving a valuation for the post-spin company, one can formulate projected revenue and earnings based on historical financials coupled with management's commentary and formal guidance. In 2018, the business generated revenue and EBITDA of US\$1.4 billion and US\$611 million respectively, representing a 43% EBITDA margin. Assuming consistent revenue and EBITDA growth in 2019 and 2020, The Drilling Company could generate 2020 revenue and EBITDA of US\$1.6 billion and US\$671 million, respectively (see Exhibit 3). Applying a comparable multiple of 12x generates an implied enterprise value of US\$8.0 billion for the business. Factoring in net debt of US\$1.0 billion (balance sheet as of December 31, 2018), and approximately 20.8 million post-spin shares outstanding, The Drilling

Company of 1972 A/S can be fairly valued at DKK 2,235 per share, based on an exchange ratio of 1 USD:6.66 DKK.

Exhibit 3 The Drilling Company of 1972 A/S: Fair Value Estimate Based on 2020E EBITDA (US\$ in millions, except per share data; shares in millions)

Source: Company reports, Bloomberg, and The European Spin-Off & Restructuring Report estimates.

2018 Revenue 1,429\$
EBITDA 611\$
2018 EBITDA Margin 43% Growth 5.0% 2019E Revenue 1,500
Growth 4.0% 2020E Revenue 1,560
EBITDA Margin 43.0% 2020E EBITDA 671
Multiple 12.0x Enterprise Value 8,052\$
Net Debt 1,065 Implied Market Capitalization 6,987\$
Shares outstanding 20.8
FVE \$/Share 336\$
FVE DKK/Share 1 2,235DKK
1 based on US to DKK exchange ratio of 6.66 Implied Market Capitalization 46,534DKK
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An implied valuation based on estimated 2020 earnings per share should also be considered. Based on 2020 estimated revenue of US\$1.6 billion (derived in Exhibit 3 above), and assuming a 40% operating margin, The Drilling Company could generate 2020 operating income of US\$624 million and net income of US\$376 million, or EPS of US\$18 per share. Given operating losses at among ocean drilling comparables, we apply a 15x multiple, in line with the low end of container and logistics comparables, which results in a fair value estimate of DKK 1,805 per share (see Exhibit 4).

Exhibit 4 The Drilling Company of 1972 A/S: Fair Value Based on 2020E EPS (US\$ in millions, except per share data; shares in millions)

Source: Company reports, Bloomberg, and The European Spin-Off & Restructuring Report estimates.

The above exercises generate an average implied market capitalization of DKK 42.0 billion for The Drilling Company of 1972. Based on an estimated 20.8 million shares outstanding and an exchange rate of 1 USD:6.66 DKK, this analysis generates a fair value estimate of DKK 2,022 for The Drilling Company of 1972 A/S (see Exhibit 5).

Exhibit 5 The Drilling Company of 1972 A/S: Valuation Summary (DKK in millions, except per share data; shares in millions)

Source: Company reports, Bloomberg, and The European Spin-Off & Restructuring Report estimates.

2020E Revenues 1,560.5\$
Operating Margin 40.0% 2020E Operating Income 624.2
Interest Expense 200.0
Pre Tax Income 424.2
Tax Expense 48.0
Net Income 376.2
Shares Outstanding 20.8
2020E EPS 18.07\$
Multiple 15.0x FVE \$ / Share 271\$
FVE DKK/Share 1 1,805DKK
Implied Market Capitalization 37,581DKK
1 based on US to DKK exchange ratio of 6.66 Average Implied Market Capitalization 42,057DKK
Shares outstanding 20.8
FVE DKK/Share 2,022DKK
240

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Valuation: Post-Spin A.P. Moller-Maersk A/S Following the separation, Maersk can be most aptly compared to its primary competitors in the container shipping and logistics industry. These include Hapag-Lloyd (HLA DE), CMA CGM S.A. (parent company is Merit Corp.), Hanjin Transportation Company Ltd. (002320 KS), Nippon Yusen (9101 TYO), Mitsui OSK Lines

Ltd. (9104 TYO), China Cosco Shipping Holdings Co. Ltd. (1919 HK), and Yang Ming Marine Transport Corporation (2609 TT), among others. Note that many of the leading shipping and logistics companies are privately held. These include American President Lines Ltd., Mediterranean Shipping Company S.A., United Arab Shipping Company, ZIM Integrated Shipping Services, and Siem Shipping, Inc., among others. Detailed comparables are shown in Exhibit 14. In addition, there are several shipping joint ventures, including Antarctic Ocean Alliance, the 2M Alliance (Maersk and Mediterranean Shipping Co.), and THE Alliance (Ocean Network Express, Yang Ming, and Hapag-Lloyd).

As a starting point for valuation, we use 2018 revenue trends and EBITDA margins to forecast 2019 and 2020 revenue. Post-spin Maersk should experience faster revenue growth compared to The Drilling Company, based on current increases in production and commodity pricing. Assuming sales growth of 4% in both 2019E and 2020E and stable margins, the post-spin company could reasonably generate 2020E EBITDA of \$4.2 billion. We note that our current forecast for flat margins may prove conservative given pricing trends. Applying multiples of 8x to 11x based on segment peers, post-spin Maersk would be fairly valued at US\$40.9 billion on an enterprise value basis. As of the December 31, 2018, and adjusting for the sale of 17.3 million Total shares, which netted gross proceeds of €884 million, or USD\$990 million (based on current exchange ratio of 1: 1.12), Maersk carried net debt of US\$8.1 billion (see Exhibit 6).

Exhibit 6 Post-spin Maersk A/S: Balance Sheet

Factoring in for the adjusted net debt, a market capitalization of US\$32.8 billion can be derived, or DKK 10,486 per share (see Exhibit 7).

Cash 1 2,765\$

Sale of remaining position in Total Holdings S.A. (17.3M shares on March 22, 2019) 884€

Gross proceeds, in USD 2 990\$

Total Cash 3,755.08\$

Debt 11,837

Net Debt 8,082\$

1 balance sheet as of December 31, 2018 2 EUR to USD exchange ratio of 1.12 241

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Exhibit 7 Post-Spin Maersk A/S: Valuation Based on 2020E EV/EBITDA (US\$ in millions, except per share data; shares in millions)

Source: Company reports, Bloomberg, and The European Spin-Off & Restructuring Report estimates.

One can also derive projected EPS for the post-spin company. Based on projected 2020 revenue of US\$42.5 billion (derived previously), and assuming an operating margin of 8.5%, post-spin Maersk A/S could reasonably generate net income of US\$2.4 billion, or US\$115 in earnings per share (see Exhibit 8). Applying a comparable 10x multiple generates a post-spin fair value estimate of DKK 7,642 per share.

Exhibit 8 Post-Spin Maersk A/S: Valuation Based on 2020E EPS (US\$ in millions, except per share data; shares in millions)

Source: Company reports, Bloomberg, and The European Spin-Off & Restructuring Report estimates Ocean Logistics & Terminals & Manufacturing Maersk Services Towage & Others Post-Spin 2018 Revenue 26,937\$ 6,082\$ 3,772\$ 2,547\$ 39,338\$

Growth 4.0% 4.0% 4.0% 4.0% 4.0% 2019E Revenue 28,014 6,325 3,923 2,649 40,912

Growth 4.0% 4.0% 4.0% 4.0% 4.0% 2020E Revenue 29,135 6,578 4,080 2,755 42,548

EBITDA Margin 10.6% 1.9% 21.5% 3.2% 9.8% 2020E EBITDA 3,088 125 877 88 4,179

Multiple 10.0x 11.0x 9.0x 8.0x 9.8x Enterprise Value 30,883\$ 1,375\$ 7,894\$ 705\$ 40,858\$

Net Debt 1 8,082

Market Capitalization 32,776\$

Shares outstanding 20.8

FVE \$/Share 1,574.48\$

FVE DKK/Share 2 10,486DKK

1 Balance sheet as of December 31, 2018. Includes cash associated with sale of Total Holdings stake (March 22, 2019) 2 based on US to DKK exchange ratio of 6.66 Implied Market

Capitalization 218,286DKK

2020E Revenues 42,548.0\$

Operating Margin 8.5% 2020E Operating Income 3,616.6

Interest Expense 473.5

Pre Tax Income 3,143.1

Tax Expense 754.3

Net Income 2,388.8

Shares Outstanding 20.8

2020E EPS 114.75\$

Multiple 10.0x FVE \$ / Share 1,148\$

FVE DKK/Share 1 7,642DKK
Implied Market Capitalization 159,091DKK
1 based on US to DKK exchange ratio of 6.66 242

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Given the industry volatility in revenue and earnings growth, a book -value-based multiple can also be considered. The consolidated shares currently trade at below book value (0.8x), despite being cash flow positive and not debt leveraged, suggesting that the shares carry substantial cyclical-recovery optionality. Applying a more normalized price -to-book multiple of 2. 5x (which may prove conservative given that the shares traded at 1.4x in 2014), the shares could be valued at DKK 8,271 per share (see Exhibit 9).

Exhibit 9 Post-Spin Maersk A/S: Valuation Based on Historical Price-to-Book Value
(US\$ in millions, except per share data; shares in millions)

Source: Company reports, Bloomberg, and The European Spin-Off & Restructuring Report estimates

Given the sharp declines in container shipping prices, an analysis of more normalized growth rates and the ensuing impact on future earnings and valuation should be considered. Assuming 8.5% revenue growth in 2020, post -spin Maersk could be estimated to generate revenues of \$44.2 billion (see Exhibit 10). Note that this projection is in line with the 8.51% CAGR projected for the global shipping container industry during the period 2018-2022. Applying an expanded multiple of between 12x and 14x earnings for more normalized growth generates a fair value estimate of between 9,585 and 11,115 per share.

Book Value (December 31, 2018) 10,332\$
Applied multiple 1 2.5x Implied Market Capitalization 25,830\$
Implied Market Capitalization 2 172,028DKK
Shares Outstanding 20.8
FVE DKK/Share 8,271DKK
1 premium to current multiple of 0.8x and 2014 P/B multiple of 1.4x 2 based on US to DKK exchange ratio of 6.66 243

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Exhibit 10 Post-Spin Maersk A/S: Valuation Based on Price Recovery Scenario

Source: Company reports, Bloomberg, and The European Spin-Off & Restructuring Report estimates

Finally, the company's post-spin dividend should also be considered. Based on estimated net income of US\$2.4 billion (derived previously), and assuming a 55% payout, Maersk would have cash of US\$1.3 billion available for dividends. At a comparable yield of 5.0%, this analysis generates a fair value estimate of DKK 8,407 per share (see Exhibit 11).

Exhibit 11 Post-Spin Maersk A/S: Valuation Based on 2019E Dividend Yield (DKK in millions, except per share data; shares in millions)

Source: Company reports, Bloomberg, and The European Spin-Off & Restructuring Report estimates.

2019E Revenues 40,911.5\$
Growth 8% 2020E Revenues 44,184.4\$
Operating Margin 8.5% 2020E Operating Income 3,755.7
Interest Expense 473.5
Pre Tax Income 3,282.2
Tax Expense 787.7
Net Income 2,494.5
Shares Outstanding 20.8
2020E EPS 119.93\$
Multiple 12.0x 14.0x FVE \$ / Share 1,439\$ 1,679\$
FVE DKK/Share 1 9,585DKK 11,115DKK
Implied Market Capitalization 199,358DKK 231,187DKK
1 based on US to DKK exchange ratio of 6.66 2020E Net Income 2,388.75\$
Dividend Payout 55% Cash available for dividend \$1,313.8 Shares Outstanding 20.8
Dividend \$63.11 Yield 5.00% FVE per share 1,262\$
FVE DKK/Share 1 8,407DKK
Implied Market Capitalization 175,000DKK
1 based on US to DKK exchange ratio of 6.66 244

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The above exercises generate an average implied post-spin market capitalization of DKK 192 billion for post -spin Maersk, or DKK 9,254 per share for post -spin Maersk A/S based on approximately 20.8 million shares outstanding (see Exhibit 12).

Exhibit 12: Post-Spin Maersk A/S: Valuation Summary
(DKK in millions, except per share data; shares in millions)

Source: Company reports, Bloomberg, and The European Spin-Off & Restructuring Report estimates.

On a pre-spin, sum-of-the-parts basis, shares of Maersk A/S are estimated to have a fair value of DKK 11,276 per share, consisting of DKK 2,022 per share for The Drilling Company of 1972 A/S and DKK 9,254 per share for post-spin Maersk A/S (see Exhibit 13). With the pre-spin sum-of-the-parts estimate representing approximately 27% upside to Maersk's current consolidated share price (DKK 8,854 per share as of this writing), the transaction appears to unlock significant upside. In addition, a recovery scenario could generate upside of 48% from current levels. As such, the pre-spin shares are recommended for purchase.

Exhibit 13: Maersk A/S: Pre-Spin Sum-of-the-Parts Fair Value Estimate
(DKK in millions, except per share data; shares in millions)

Source: Company reports, Bloomberg, and The European Spin-Off & Restructuring Report estimates. Average Implied Market Capitalization 192,492DKK

Shares outstanding 20.8

FVE DKK/Share 9,254DKK

Base Recovery Implied Market Capitalization - Drilling 42,057DKK 42,057DKK

Implied Market Capitalization - Maersk 192,492DKK 231,187DKK

Total Implied Market Capitalization 234,549DKK 273,245DKK

Shares outstanding 20.8 20.8

FVE DKK/Share 11,276DKK 13,137

Current Share Price (4/2/2019) 8,854DKK

Implied Upside 27% 48% 245

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Exhibit 14 Maersk A/S: Peer Comparables
(currency in millions, except per share data)

Source: Bloomberg, AP Moller - Maersk A/S Hapag-Lloyd AG Hanjin Transportation Co Ltd COSCO SHIPPING Holdings Co Ltd Nippon Yusen KK Mitsui OSK Lines Ltd Yang Ming Marine Transport Corp Diamond Offshore Drilling Inc Transocean Ltd Pacific Drilling SA Ticker MAE RSKB DC HLAG GY 002320 KS 1919 HK 9101 JT 9104 JT 2609 TT DO RIG PACD Price 8,854.00DKK ¥29.60 GBP 36,150.00 ¥3.42 1,660.00 € ¥2,478.00 \$8.61 \$11.11 € 9.09 \$14.00 Market Capitalization 180,289.6 5,202.5 432,883.8 70,866.7 282,291.5 298,917.7 20,001.2 1,526.9 5,548.2 1,050.4 Net Debt (Cash) (145,923.6) 5,371.4 972,009.4 124,827.8 1,008,849.0 1,114,028.0 71,350.4 1,520.0 7,825.0 2,853.9 EV 34,366.0 10,573.9 1,404,893.2 195,694.5 1,291,140.5 1,412,945.7 91,351.7 3,046.9 13,373.2 3,904.4 Revenue 2017 31,965.8 10,087.3 1,850,814.0 92,335.6 1,954,146.5 1,474,422.6 131,562.2 1,472.7 2,787.3 562.0 Revenue 2018 38,820.9 11,452.9 1,941,682.5 107,937.3 2,139,808.3 1,650,636.4 140,791.8 1,101.7 2,997.2 261.8 Revenue 2019E 40,322.6 12,187.9 2,122,072.5 132,343.9 1,801,087.3 1,244,858.7 143,907.5 998.6 3,153.1 289.5 Revenue 2020E 13,110.1 2,252,397.0 143,566.5 1,804,380.8 1,258,934.4 149,164.0 1,082.4 3,502.7 383.0 EV/Sales 2018 0.9x 0.9x 0.7x 1.8x 0.6x 0.9x 0.6x 2.8x 4.5x 14.9x EV/Sales 2019E 0.9x 0.7x 1.5x 0.7x 1.1x 0.6x 3.1x 4.2x 13.5x EV/Sales 2020E 0.8x 0.6x 1.4x 0.7x 1.1x 0.6x 2.8x 3.8x 10.2x 2018 Net Income 291.1 47.9 57,502.0 959.0 13,836.3 12,621.4 (7,161.5) (141.9) (514.3) (916.8) 2019E Net Income 785.0 253.0 13,206.0 1,556.0 (2,913.9) 19,802.1 (2,260.3) (275.0) (470.1) (319.5) 2020E Net Income 1,331.8 411.9 25,331.0 2,376.7 20,080.1 36,474.3 837.0 (200.3) (289.9) (292.8) 2018E Net Income Margin 0.7% 0.4% 3.0% 0.9% 0.6% 0.8% -5.1% -12.9% -17.2% -350.2% 2019E Net Income Margin 1.9% 2.1% 0.6% 1.2% -0.2% 1.6% -1.6% -27.5% -14.9% -110.4% 2020E Net Income Margin 3.1% 1.1% 1.7% 1.1% 2.9% 0.6% -18.5% -8.3% -76.5% 2017 EBITDA 3,636.6 1,005.9 108,443.0 6,872.3 85,048.0 77,201.0 6,146.3 568.4 1,227.4 29.5

2018 EBITDA 3,829.8 1,129.9 113,242.5 5,745.6 117,014.8 104,060.2 (110.0) 271.3 1,059.7 (4.3)
2019 EBITDA 5,232.0 1,419.4 133,550.5 9,454.9 95,370.4 120,191.9 2,225.5 145.2 1,043.7 (0.3)
2020 EBITDA 5,780.1 1,575.8 148,972.5 11,356.6 124,857.6 136,082.4 9,558.0 225.0 1,231.2
16.9
2017 EBITDA Margin 11.4% 10.0% 5.9% 7.4% 4.4% 5.2% 4.7% 38.6% 44.0% 5.2% 2018
EBITDA Margin 9.9% 9.9% 5.8% 5.3% 5.5% 6.3% -0.1% 24.6% 35.4% -1.7% 2019 EBITDA
Margin 13.0% 11.6% 6.3% 7.1% 5.3% 9.7% 1.5% 14.5% 33.1% -0.1% 2020 EBITDA Margin
12.0% 6.6% 7.9% 6.9% 10.8% 6.4% 20.8% 35.2% 4.4% 2019 EV/EBITDA 6.6x 7.4x 10.5x 20.7x
13.5x 11.8x 41.0x 21.0x 12.8x -13.0 14.6x 2020 EV/EBITDA 5.9x 6.7x 9.4x 17.2x 10.3x 10.4x
9.6x 13.5x 10.9x 23.1x Average, ex. MAERSK 17.5x 12.2x 2019 EPS \$35.07 \$1.64 \$1,111.14
\$0.13 (\$11.85) \$164.81 (\$0.25) (\$2.00) (\$0.81) (\$4.22) 2020 EPS \$35.07 \$1.64 \$1,111.14
\$0.13 (\$11.85) \$164.81 (\$0.25) (\$2.00) (\$0.81) (\$4.22) P/E 2019E 252.5x 18.1x 32.5x 25.7x -
140.0x 15.0x -34.2x -5.6x -11.2x -3.3x P/E 2020E 252.5x 18.1x 32.5x 25.7x -140.0x 15.0x -34.2x
-5.6x -11.2x -3.3x Average, ex. MAERSK -13.8x NM 2018 FCF Yield 0.1% 6.9% -84.4% -7.9%
2.9% 6.5% -37.5% 2019 FCF Yield 1.3% 10.4% 27.5% -6.2% 3.0% -10.8% -6.6% -3.3% -44.6%
2020 FCF Yield 381.5% 5.5% 30.8% -0.2% 19.7% -7.7% -11.0% -5.0% -10.5% 2018 FCF Per
Share \$12.50 \$2.04 (\$2.89) (\$130.54) \$0.32 \$0.59 (\$5.25) 2018 P/FCF 708.5x 14.5x -1.2x -
12.7x 34.2x 15.4x -2.7x 2019 Free Cash Flow 2,320.7 539.1 117,277.2 (2,169.3) 8,415.5
(32,001.0) (101.3) (181.3) (133.3)
2019 FCF Per Share \$111.78 \$3.07 \$9,935.85 (\$0.21) \$49.88 (\$267.59) (\$0.74) (\$0.30)
(\$6.24) 2019 P/FCF 79.2x 9.6x 3.6x -16.1x 33.3x -9.3x -15.1x -30.6x -2.2x Containers and
Logistics Drilling 246

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Conclusion

On a pre-spin, sum-of-the-parts basis, shares of Maersk A/S are estimated to have a fair value of DKK 11,276 per share, consisting of DKK 2,022 per share for The Drilling Company of 1972 A/S and DKK 9,254 per share for post-spin Maersk A/S. With the pre-spin sum-of-the-parts estimate representing approximately 27% upside to Maersk's current consolidated share price (DKK 8,854 per share as of this writing), the transaction appears to unlock significant upside. In addition, a recovery scenario could generate upside of 48% from current levels. As such, the pre-spin shares are recommended for purchase. It is important to note, however, that there are several overhangs on the shares which may limit near-term appreciation in the shares. Foremost, the International Maritime Organization's (IMO) 2020 sulphur cap regulations will likely cause marine fuel prices to continue to fluctuate widely this year. New regulations, changing economic conditions, and trade tensions could also potentially trigger significant and lasting marine fuel price volatility. That said, at current share prices, the shares appear sufficiently discounted for these risks. Notably, Maersk shares have declined approximately 34% over the past 5 years, versus a 46% return for the OMX Copenhagen 20 Index over the same period. Year-to-date, the shares have appreciated 4% (versus 12% for the OMX Copenhagen 20 Index). At current prices, we think the shares are sufficiently discounted for industry and macroeconomic uncertainties. Longer term, the quality of the company's industry-leading shipping assets, combined with the earnings potential of the remaining businesses, could offer meaningful upside potential with minimum downside risk.

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Exhibit 15 The Drilling Company of 1972 A/S: Balance Sheet at December 31, 2018, 2017 and 2016
(US\$ in millions)

Source: Bloomberg.

2018 2017 2016 Intangible Assets 56 85 109
Property, plant and equipment 4,845 4,270 5,974
Financial non-current assets, etc. 3 2 31
Deferred tax 2 30 16
Total non-current assets 4,906 4,377 6,130
Trade receivables 339 297 288
Loans receivables 2 3,390 4,134
Other receivables 37 60 96
Prepayments 58 79 101
Receivables, etc. 436 3,826 4,619
Cash and bank balances 372 49 460
Total current assets 808 2,875 5,079

Total assets 5,714 8,252 11,209
 Share capital 87 87 87
 Reserves 3,723 6,122 8,170
 Dividend proposed for the year - - 500
 Total equity 3,810 6,209 8,757
 As of 31 December 248

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Exhibit 16 The Drilling Company of 1972 A/S: Historical Income Statement
 (\$ in millions, except per share data)

Source: Company filings.

2018	2017	2016	Revenue	1,429	1,439	2,297
			Costs (818)	(756)	(916)	
			Profit before depreciation, amortisation, impairment losses/reversals, and special items	611	683	1,381
			Special items (16)	2	16	
			Profit before depreciation, amortisation, impairment losses/reversals, and special items	595	685	1,397
			Depreciation and amortisation (403)	(468)	(589)	
			Impairment losses/reversals	810	(1,769)	(1,510)
			Share of results in joint ventures (1) - - Profit/loss before financial items	1,001	(1,552)	(702)
			Financial expenses, net (12)	(19)	(89)	
			Profit/loss before tax	989	(1,571)	(791)
			Tax (48)	49	1	
			Profit/loss for the year	941	(1,522)	(790)
			Earnings per share, USD	1,882	(3,044)	(1,580)
			Diluted earnings per share, USD	1,882	(3,044)	(1,580)
			For the year ended 31 December	249		

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FIXED INCOME & CRYPTOCURRENCY COMPENDIUM

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□ Horizon Kinetics LLC® 2020

Murray's Musings

THE TRANSFORMATION OF THE FIXED INCOME ASSET CLASS

Fixed income is an investment that is a type of loan. A borrower is obligated to make payments of a predetermined amount, and the principal must be paid on the maturity date. The legal predetermination of the payments is the origin of the term fixed. In other words, fixed means predetermined. This legal definition will not change.

An exceedingly low level of interest rates changes the character of this type of investment in extraordinary and not so obvious ways. It should be evident that a fixed income investment that pays a coupon of less than 2% is, both before and after taxes, a negative real (after inflation) return instrumentality. For a New York City resident of means, even a fixed income instrument that yields 4% or less, is a negative real return instrumentality. Although there is surprisingly little discussion of this reality, it should be self-evident to investors.

It is less obvious—although nonetheless true—that bonds, in this historically low interest rate period, if sold short, are actually incredibly inexpensive put options. An example will serve to illustrate this point. Consider that the Bank of America 4.44% bonds due January 20, 2048 are now priced, on a yield-to-maturity basis, at 3.31%. They trade well above par value and at maturity must be worth par value. Prior to maturity, which is almost 30 years from now, a variety of circumstances could occur in which these bonds could be worth considerably less than par.

If, for instance, U.S. government bonds for some reason yield 5% instead of the present level of 2.2%, the Bank of America bonds would probably yield 7.45%—even assuming that credit quality does not deteriorate.¹ Quite obviously, the bonds would trade well below par value.

Hence, two possibilities exist for these Bank of America bonds. The first is that interest rates will never increase. As a result, the Bank of America bonds will not be priced at par value until the maturity date of January 20, 2048. The short seller will simply lose the 3.31% annual yield to maturity. A short seller in the 50% tax bracket (Federal, State, and local) will effectively lose half that sum after the benefit of tax deductions. Consequently, one would pay 1.66%—half of 3.31%—for a long-term put option on interest rates. Which is incredibly cheap—just try to price a multi-year put option on long-term interest rates with a broker-dealer.

¹ At the time of this writing, a 30-year Treasury bond yields 2.22%. If the yield of that bond

increases to 5%, it would produce 2.25x more yield. Increasing the 3.31% yield of the Bank of America bonds by 2.25x produces a yield of 7.45%. A bond with a 4% coupon that yields 7.45% would trade below par. 250

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FIXED INCOME & CRYPTOCURRENCY COMPENDIUM

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□ Horizon Kinetics LLC® 2020

The other possibility is that interest rates do increase. In that event, the bonds might trade substantially below par value—depending upon how high interest rates rise. These bonds are effectively a put option for which the premium is the yield to maturity. The potential loss for a bond (absent default), or in this case for a near-30-year put option on the bond, is limited to the yield to maturity. In other words, the loss is limited to the amount of the current price premium for the bond, just as for the option premium for a conventional put option.

This similarity to a put option is problematic on a broader systematic basis, because it changes the incentives for investing in bonds. Consider the matter from the perspective of an investor who might purchase a U.S. Treasury bond due in 2048 that has a yield to maturity of 2.23%. What if, alternatively, that bond were used as collateral to establish a small short position in a Bank of America bond, say a short equal to 10% of the long position. The investor puts \$100 into the U.S. Treasury bond yielding 2.23%, then takes \$10, or 10%, shorting that amount of Bank of America debt for which they must pay a yield to maturity rate of 3.31%. In terms of that investor's new risk/reward profile, the first aspect of this hedged position is that the net yield on it is actually a positive 1.9% carry.

An investor considering such a trade must contemplate whether or not it is better to sacrifice 33 basis points of annual income in exchange for the possibility that the creditworthiness of Bank of America might deteriorate. If Bank of America creditworthiness deteriorates, as occurred in 2008, the bank's debt might trade at 50% of par value, as it did at that time. A 10% short position with a loss of possibly 60% in price would produce a portfolio-level trading gain of 6%.

One might argue that such a trade is outside of the ambit of a conventional bond investor and therefore unlikely to be made. But it is not outside of the ambit of an investor who would be inclined to sell short Bank of America common stock. The shares now yield 2.17%, so that is a cost of the short sale. Moreover, the shares can appreciate. Thus far, in 2019, the shares have increased in value by 32.67%. That would have been the loss to a short seller.

Why might a short-seller have been attracted to the Bank of America shares? The net income for the first nine months of 2019 was \$19.25 billion—a decline of 2.07% versus the comparable period in 2018, when it was \$19.657 billion.

What if a hypothetical short seller were sufficiently astute to predict that in 2019, the net income of Bank of America would decline. This hypothetical short seller would have predicted, as well, the bank's difficulty with overnight borrowing in September of that year. Although this investor might reasonably have anticipated that Bank of America shares would decline, what could not have been reasonably anticipated is that its stock would in fact appreciate by 32.67%.

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FIXED INCOME & CRYPTOCURRENCY COMPENDIUM

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□ Horizon Kinetics LLC® 2020

Establishing a hedge against a banking-related issue with systemic dimensions can be very expensive, carry costs aside, in terms trading risk, until the problem finally manifests itself. Selling short a bank bond substantially eliminates the trading or directional risk, since the bond will ultimately mature at par value. That is to say, if an investor is worried about possible banking risks, short sale of bank stock is too blunt an instrument to use as a hedge, there being in many circumstances in which bank shares could rise considerably in price, creating an extremely expensive hedge.

Surveying Both Idiosyncratic and Systemic Risks in Low-Yield Bonds There exist many long-term bonds that, in principle, have high price convexity paired with extraordinarily low yields to maturity. One example might be San Francisco City & County Water 6.95% due November 1, 2050, which trades at a price of 159.21 with a 3.22% yield to maturity. Apart from higher interest rates, variables that could cause the bonds to decline are fluctuations in the creditworthiness of the city or an earthquake that could severely damage the water system, requiring extensive repairs.

All U.S. bonds trade in relation to Treasury bond yields. Yet, with less than a 100-basis point spread over Treasuries, it is clear that municipal bonds, project finance bonds, or corporate bonds have unique risks that are not being priced into them. For example, consider the Comcast

4.00% Notes due November 1, 2049, trading at 110.82 with a 3.41% yield to maturity. Major changes are clearly evident in Comcast's delivery of content to subscribers and in the company's ability to sell advertising. Although no investor is in a position to predict the financial conditions of Comcast over the 30-year life of this bond, the valuation is predicated upon the belief that there is no realistic possibility of a significant deterioration in its financial position.

The same statement might be made about AT&T, which is actually a fairly leveraged enterprise. Its 5.7% bonds due March 1, 2057—a 38-year life—are priced at 125.38 with a yield to maturity of 4.32%.

Universities are now issuing debt, despite their huge endowments, because the cost of debt capital is so low, while the likelihood is high that the university will ultimately repay the debt in inflated dollars. As an example, Princeton University 5.70% bonds due March 1, 2039 trade at 139.32 with a yield to maturity of 3%. Similarly, Harvard has issued a 4.875% bond due October 15, 2040 now trading at 130.80 with a yield to maturity of 2.9%.

If one sells short the Princeton or Harvard bonds, all one needs for a substantial profit is a modest rise in interest rates. A rise in long-term Treasury yields of perhaps 150 basis points for any reason would cause the Harvard bonds to trade at par value. A 30-year U.S. Treasury now yields 2.2%; 150 basis points higher, and it would yield 3.72%. If the Harvard bond were to trade a yield spread of 115 basis points above the Treasury, it would be priced at par value almost 31 points lower. In fact, because any meaningful rise in interest rates would 252

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probably severely reduce the market value of the Harvard endowment, the assessment of Harvard's creditworthiness would probably be reduced as well, such that the spread to Treasuries would probably rise as well.

The State of Texas 5.52% bonds due April 1, 2039 are priced at 138 with a yield to maturity of 2.82%. Surely, one could envisage a diminution of creditworthiness if, for some reason, extraction of fossil fuels were to be prohibited in the United States. In a less extreme scenario, the same outcome might be achieved by a severe reduction in oil and gas prices. The bond market does not appear very concerned about this possibility, at least judging by the present yield of the State's debt.

If the production of coal were to be forbidden in the U.S., it would negatively impact the fortunes of Union Pacific Corporation (UNP). Union Pacific managed to issue a 4.375% bond due November 15, 2065 which now trades at 109.85 with a yield to maturity of 3.91%. In fact, even a severe recession could lower an investor's perception of the credit worthiness of Union Pacific. The company also issued 3.84% bonds due March 20, 2060 presently trading at 102.32 with a yield to maturity of 3.73%.

Altria Group, Inc. (MO) managed to sell 6.20% bonds due February 14, 2059, presently trading at 117.33 with a yield to maturity of 5.16%. The mere existence of such a bond is more than a statement about the long-term real rate of return on bonds. It is effectively a statement that large-scale smoking in the U.S. will continue past the year 2059.

Here is where it becomes yet more interesting.

The University of California issued 4.77% bonds due May 15, 2115, now priced at 126.36 with a yield to maturity of 3.75%. It is virtually a certainty that the person buying this bond will not live to the maturity date. Ohio State University managed to sell 4.80% bonds due June 1, 2111, now trading at 129.45 with a yield to maturity of 3.68%.

Guardian Life Insurance Company of America has issued 4.85% bonds due January 24, 2077. they trade at 124.87, a yield to maturity of 3.78%. A life insurance company is an interesting case relative to long-term bonds. If interest rates increase, it would surely have a negative impact on the market values of the debt. On the other hand, if rates decrease sufficiently from this point, the company might experience difficulty earning enough from its bond portfolio to pay its annuity liabilities. Viewed from the perspective of the bond buyer, the best -case scenario is that interest rates remain at present levels and neither materially increase nor decrease for many decades.

If an investor buys a 57-year bond from a life insurance company whose entire investment portfolio is comprised almost entirely of long-term bonds, and if one of the many long-term 253

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bonds in its investment portfolio loses value, then the collateral for your bond will lose value, as well.

These bonds are only a small sample of many that have a similar character. The common denominator is that the rate of compensation above that of Treasury bonds is less than 200 basis points per annum and sometimes even less than 100 basis points, despite the fact that many of the bonds have maturity dates far longer than the 30-year Treasury. The University of California bond will mature in 91-years or 3x the maturity of the longest -term U.S. Treasury bond. It does not appear that much, if any, marginal compensation is paid for the extra 60-year life of the bond. In the light of such pricing, there is no room for any rise in interest rates or any deterioration of credit quality. Yet, history demonstrates that such eventualities frequently occur.

The expense of selling such bonds short is small relative to the reward possible if history repeats itself. Ergo, such bonds and many others are now effectively low-cost, low-risk put options with heretofore unimaginably lengthy times to expiration.

If shorting, I prefer to sell short bonds—not stocks—because they are a much more reasonable hedge against uncertainty. There is a limit to how much money one can lose, and one knows what that limit is—as opposed conventional hedge instruments, in which the potential loss is infinite.

Q: I have a hypothetical question relative to the circumstance of a 75-year-old investor who expects to live ten additional years and who invests in long-term bonds that yield 4.8% with good credit, as with the University of California issue. Playing devil's advocate, what would be wrong with that, if rates stay roughly the same over that period?

A: What would be wrong with that? If that investor has to live off the principal—not just the income—and the bond declines in market value, there would be plenty wrong with it. That person would be forced to sell the bond for money to live on, and the sale might not produce adequate funds. That person might die without enough money. An alternative risk is that this investor is fortunate enough to be in very good health and live longer than the expected additional 10 years, but with each passing year, the cost of living rises above the purchasing power of that bond's coupon income. That investor would not be very happy with that outcome. That is what's wrong with it.

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Murray's Musings

THE ACHIEVEMENT OF DIVERSIFICATION

It would seem to be arithmetically self-evident that if various constituent members of an index have such low weightings as to be statistically irrelevant, then they cannot, arithmetically, play any role in diversification. Eighteen members of the S&P 500 index now have only a 1-basis point (0.01%) weight. There are 38 index constituents that have a 2-basis point weight, 41 that have a 3-basis point weight, 41 with a 4-basis point weight, and 46 at 5 basis points.

These 184 constituents of the 500 S&P Index are arguably statistically irrelevant. To go further, one could designate members with 6-basis point weights as also statistically irrelevant. If one draws the boundary of irrelevance in this manner, there are another 36 such positions, bringing the total number of statistical irrelevancies to 220, or 44% of the names in the index.

If one wishes to assert, as an issue of statistical irrelevance, that at least 50% of the S&P 500 names fall into this category, one has only to include the 7-basis point positions. There are 29 constituent members with that weight. Thus, the grand total of statistically -irrelevant positions is 249, or 49.8% of the names in the S&P 500 index.

A summary of the index weightings of approximately half of the constituent members of the S&P 500 is shown in the following table.

Table 1: Selected S&P 500 Index Weightings

Number of Companies	Weighting (basis points)	Total Weight (basis points)
18	1 bp	18 bp
38	2 bp	76 bp
41	3 bp	123 bp
41	4 bp	164 bp
46	5 bp	230 bp
36	6 bp	216 bp
29	7 bp	203 bp
Total:		249

1030 bp or 10.3%

Source: iShares

Roughly one-half of the number of corporations in the S&P 500 index have, in aggregate, 10.3% of the index's market capitalization.

Interestingly, there are only 17 positions with an 8-basis point weight, and another 28 9- basis point positions. If these positions, in addition to those listed above, were considered as statistically irrelevant, the total number of irrelevant positions would equal 295, or 59% of 255

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the index and would have a collective weight of 14.27%. One could continue in this manner, because there are 20 10-basis point positions.

As noted, there are 295 positions of 9-basis points weight or less that comprise 59% of the number of members of the S&P 500 index and 14.27% of its market value. To assess the importance of these firms to the index, assume that—for only the 295 companies in question— a recession occurs that is equivalent in severity to the recession of 2008. In this scenario, assume that these 295 companies as a group decline in share price by (37)%, which was the S&P 500 return for 2008. The result is a 37% loss on 14.27% of the index or (5.28)%—which is not a significant decline during a major recession.

This is hardly the same thing, though, as saying that these companies are unimportant businesses or are statistical irrelevancies within the U.S. economy. One way of understanding index weight versus economic impact is to select ten well-known firms with statistically immaterial S&P 500 weights and calculate their economic impact by a metric other than stock market capitalization. I chose these ten companies that are familiar to investors: Discovery, Inc., News Corp., Sealed Air, Newell Brands, Franklin Resources, Mohawk Industries ; American Airlines , Carnival Corporation, Textron, and Lumen Technologies, otherwise known as CenturyLink, Inc.

Table 2: 10 Firms with Immaterial Weights in the S&P 500 Ticker Name Weighting DISCA Discovery, Inc. - Series A 3 bp NWSA News Corporation Class A 3 bp SEE Sealed Air Corporation 2 bp NWL Newell Brands 2 bp BEN Franklin Resources, Inc. 2 bp MHK Mohawk Industries, Inc. 2 bp AAL American Airlines Group, Inc. 2 bp CCL Carnival Corporation 3 bp TXT Textron Inc. 3 bp LUMN Lumen Technologies, Inc. (CenturyLink, Inc.) 3 bp Source: iShares

Those ten firms collectively have a 25- basis point weight in the S&P 500. One might compare them to Apple Inc. (APPL), which has a 6.45% index weight. In other words, the weight of Apple in the index is 25.8x the weight of these ten well-known companies.

The metric I have chosen to assess these firms' economic impact is the number of their employees. This statistic is significant, because in a horrendous recession some of the employees presumably would lose their employment. Together, these firms have 485,800 employees. In contrast, Apple has 137,000 employees.

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Table 3: Number of Employees in 10 Firms with Immaterial Weights in the S&P 500 Ticker Name Number of Employees DISCA Discovery, Inc. – Series A 9,000 NWSA News Corporation Class A 28,000 SEE Sealed Air Corporation 16,500 NWL Newell Brands 49,000 BEN Franklin Resources, Inc. 9,700 MHK Mohawk Industries, Inc. 42,100 AAL American Airlines Group 133,700 CCL Carnival Corporation 120,000 TXT Textron Inc. 35,000 LUMN Lumen Technologies, Inc. (CenturyLink, Inc.) 42,800 Total: 485,800 AAPL Apple, Inc. 137,000 Source: Company Reports

Some argue that if the ten aforementioned firms were to have difficulties—and obviously some of these firms now face serious challenges—the economic impact would be heaviest from those firms with the largest number of employees. In industry terms, a 20% layoff rate from the diverse group with a 25-basis point collective weight would be far more damaging to the U.S. economy than if Apple were compelled to undertake a 20% layoff. If the object of diversification is to reflect economic impact, then surely one can argue that the weightings of an index should reflect the impact of a company on the economy. Market capitalization weights reflect valuation rather than economic impact.

Tesla, Inc. (TSLA) has a \$392 billion market capitalization and, viewed from that perspective, is by far the largest automobile company on the planet. General Motors (GM) has a \$52.7 billion market capitalization, and Ford Motor (F), \$32 billion. When Tesla enters the S&P 500 index, it will obviously overwhelm the weightings of GM and Ford. GM, in a reasonably good year, would build 7.7 million vehicles, Ford might produce 2.4 million vehicles, and Tesla, somewhat

more than 400,000.

Thus, the question of the meaning of the term diversification arises. Diversification of the S&P 500 index is meant to reflect the U.S. economy, but the fact that the S&P 500 does not do so is plain to see. If diversification is based upon market capitalization, then diversification is effectively based upon valuation. The conclusion would logically follow that the index would have a bias in favor of higher valuation.

Valuation is not the only bias within a national stock market. Most large firms have become global multinational companies, regardless of their nation of origin. If the country of incorporation is a small one, then that country is probably financially irrelevant to a global multinational firm. The obvious example of this phenomenon is Switzerland. The large index members of the iShares MSCI Switzerland ETF (EWL) are surely not Swiss with respect to the geographic diversification of their businesses. The following table lists the ten largest positions in that index.

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Table 4: iShares MSCI Switzerland ETF (EWL) Ten Largest Positions Ticker Name Weighting
NESN Nestlé SA 21.07% ROG Roche Holding PAR AG 12.90% NOVN Novartis AG 9.73%
ZURN Zurich Insurance Group AG 3.87% ABBN ABB Ltd. 3.70% LONN Lonza Group AG
3.56% UBSG UBS Group AG 3.47% GIVN Givaudan SA 3.04% SIKA SIKA AG 2.88% CFR
Compagnie Financiere Richemont SA 2.82% Total: 67.04% Source: iShares

In 2019, Switzerland accounted for 1.26% of Nestlé's sales. For Roche, 1.94% of sales were attributed to Switzerland. The fact that these firms cannot possibly achieve an acceptable level of growth for their investors by confining their business activities to Switzerland should be apparent. For many years, Nestlé, Roche, and Novartis expanded outside of Switzerland. These three firms collectively represent 43.7% of the MSCI Switzerland index, which is still considered to be a diversified index. MSCI Switzerland is an index that cannot possibly be a successful investment without the success of Nestlé, Roche and Novartis.

The historical return of MSCI Switzerland, since inception on March 12, 1996, is 7.18% annually. Is this an acceptable return, with that level of concentration? The difficulty with weightings based on market capitalization is that eventually some company within the index will either become the best performing investment or, alternatively, achieve enormous size by raising capital via share issuance. There exists no objective limiting principle to reestablish diversification other than placing an arbitrary limit on the size of position. Most index providers would not allow a single position to become 25% or more of an index. Another reasonably popular limiting principle is to restrict the number of 5% positions to about 50% of the portfolio. All indexes, nevertheless, seem to have accumulated statistically irrelevant smaller holdings.

MSCI Switzerland has only 39 positions, therefore, concentration might be inevitable. The smallest position in the index is Banque Cantonale Vaudoise, with a 44 -basis point weighting.

An alternative approach, if one wants to invest in Switzerland itself on a diversified basis, might be to purchase Swiss real estate companies, on the theory that the real estate is, after all, located in Switzerland. Valuations of those companies must, therefore, reflect conditions in that country.

There are four primary publicly traded real estate companies in Switzerland: Swiss Prime Site, PSP Swiss Property, Allreal Holding and Mobimo Holding.

Table 5: Four Primary Traded Real Estate Companies in Switzerland 258

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Ticker Names Market Capitalization (CHF in billions) SPSN Swiss Prime Site AG 6.05 PSPN PSP Swiss Property AG 5.15 ALLN Allreal Holding AG 3.12 MOBN Mobimo Holding AG 1.75 Source: Bloomberg

Of these four, only Swiss Prime Site, with a 74-basis point weighting, qualifies for inclusion in the MSCI Switzerland index. One could argue that these four real estate firms have collectively become a more representative and diversified Swiss market index than MSCI Switzerland, by virtue of the total number of properties they hold and of their diversification by geography and property type.

The following table lists the number of properties held by the four primary Swiss real estate

companies.

Table 6: Holdings of Four Swiss Public Real Estate Cos.

Ticker Names Number of Buildings SPSN Swiss Prime Site AG 187 PSPN PSP Swiss Property AG 160 ALLN Allreal Holding AG 64 MOBN Mobimo Holding AG 138 Total: 549 Source: Company filings

This table raises a profound question about the nature of indexation. If one wishes to invest in the nation of Switzerland, is this best achieved by building a concentrated portfolio of giant multinational firms that have proportionately little business activity in Switzerland? Or, alternatively, can one devise a weighting system among the four publicly traded Swiss real estate firms that have their entire investment located within Switzerland? The fact should be obvious that a portfolio of Swiss real estate firms is a superior means of investing in Switzerland.

The limiting factor is that if specific Swiss diversification is to be achieved in this manner, this diversification is not scalable. The aggregate market capitalization of these four Swiss real estate companies is only 16.07 billion CHF. Nestle, alone, has a market cap of 300 billion CHF. Mass-scale indexation must necessarily be global multinational investing, because only global-scale firms have sufficient equity liquidity to support indexation at the trillion-dollar level.

Concentrated—not diversified—investing can be rewarding during a period of declining interest rates and increasing globalization. If these economic variables are not dominant in the future, an entirely different form of indexation will probably take their place.

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Murray’s Musings INFLATION AND ECONOMIC THOUGHT Historians write about inflation

in a very different way than economists.

Historians view inflation as a destabilizing force, whatever the cause. For instance, historians agree that a salient cause of demise of the Song Dynasty of China (960-1279) was the fact that it issued paper money, the first government to do so. The successor Yuan Dynasty lasted until 1368, and the inflation caused by fiat currency issuance is generally cited by historians as a factor in its demise. Similarly, the inflation caused by the influx of gold and silver into Spain from Spanish colonies in the New World in the 16th and 17th centuries is known to historians as “the price revolution.” The repercussions of the inflation caused by this dramatic rise in the supply of gold and silver undermined Spain’s international economic standing. There were other historical debasements, such as King Henry VIII of England continually reducing the gold and silver content of the coinage to pay for his lavish lifestyle. In British history, this is known as “the Great Debasement.”

None of these events or any of the many other historical debasements have seemed to attract the interest of even the greatest economists. Many financial experts regard John Maynard Keynes as the greatest economist of all time, yet in his magnum opus, *The General Theory of Employment, Interest and Money* published in 1935, the index shows only seven mentions of inflation in the entire book. Of these mentions, one simply states the fact of inflation in the Central Europe of the postwar era. The other references are largely definitional and used as foundations upon which the theory of employment is placed. Inflation basically is defined as the point beyond which monetary expansion does not produce increases in output. The word debasement is never used in the book. Keynes might be forgiven for such a view, since *The General Theory of Employment, Interest, and Money* was written in a period of history during which the challenge was to increase output and thereby increase employment.

In the magisterial three-volume biography of John Maynard Keynes by Robert Skidelsky, there are no references to inflation in the index to Volume I, only six references to inflation in the index to the 635-page Volume II, and a single reference in the index to Volume III, which is about as long as Volume II.¹ Ironically, Keynes became very wealthy by speculating on the decline of the German currency in the aftermath of the First World War. In other words, Keynes became wealthy because of German inflation and his understanding of it. He was very interested in inflation as an investor. He appears not to have been vitally interested in the subject as a scholar.

¹Robert Skidelsky, *John Maynard Keynes, Vol. 1: Hopes Betrayed, 1883-1920* (New York: Viking, 1983); *Vol 2: The Economist as Savior, 1920-1937* (New York: Viking, 1983); *Vol 3: Fighting for Freedom, 1937-1946* (New York: Viking, 2001).

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A similar phenomenon is observed in *A Monetary History of the United States 1867-1960* by Milton Friedman and Anna Jacobson Schwartz.² Since the book is about monetary issuance, one might expect some discussion of the subject of debasement. The word is not mentioned once in the index to this book that is 700-pages long excluding the appendixes. There are five references to inflation in the index. This should not be surprising. The object of the book is to study the optimal rate of currency issuance with a view to economic stimulus. Some degree of currency issuance makes government and individual debt more manageable, so used properly and in moderation, inflation is just a policy tool to increase economic growth.

In neither the Keynes book nor the Friedman book is there any advice as to what an investor might do to protect wealth from inflation’s destruction of purchasing power. Of course, neither book is designed for an investor. These books were written for policymakers and other economists. It would never have occurred to either economist that the policymakers be deprived of the power or policy tool to create money and, thereby, inflation.

Friedrich Hayek is unique among the major economists of the 20th century in that he advocated that national governments be deprived of the money creation function, which he describes in *The Denationalization of Money* (1976).³ The basic idea of the Hayek book is that governments cannot be trusted not to abuse the money-creation function. Therefore, Hayek advocates the creation of “private money.” The problem is that one cannot necessarily trust those in control of private money either, and as a consequence of this obvious objection, the Hayek book remained in obscurity until rediscovered by the proponents of cryptocurrency.

In the world of cryptocurrency and blockchain, it is possible that users of the currency can continually verify its issuance status. As an example, if bitcoin will issue slightly less than 2.5 million coins between the time of this writing and the year 2140, that is objectively verifiable as a fact by means of the blockchain, which forms the permanent public record of every bitcoin transaction since its creation. Bitcoin ease of transfer has yet to rise to the level of being accessible to the average person, but that function is constantly improving. Private money might be the solution to the inflation problem. Unfortunately, it is not yet ready for mass usage. In any case, the community of academic scholars tends to avoid the subject of cryptocurrency.

In 1975, the economist John Kenneth Galbraith wrote a popular account of the history of

money entitled *Money, Whence It Came, Where It Went*.⁴ Given the era in which the book

2 Milton Friedman & Anna Jacobson Schwartz, *A Monetary History of the United States, 1867-1960* (Princeton, N.J.: Princeton University Press for the National Bureau of Economic Research, 1963). 3 Friedrich A. Hayek, *Denationalisation of Money: The Argument Refined – An Analysis of the Theory and Practice of Concurrent Currencies* (London: The Institute of Economic Affairs, 1976).

4 John Kenneth Galbraith, *Money: Whence It Came, Where It Went* (Boston: Houghton Mifflin, 1975). 262

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was written, the author could hardly avoid discussing the subject of inflation. Galbraith strongly advocates for government wage and price controls or, alternatively, wage and price guidelines that are informal requests with which government pressure might force compliance even if they are not formal controls. Even a book that is directed towards the general public is really a call for action by policymakers.

The common factor among economists of a variety of philosophical orientations is that inflation is something that must be controlled by government action of some sort. The underlying assumption is that the inflation problem has nothing to do with existing government policy. Moreover, inflation is not considered to be a problem that can be confronted by the individual on behalf of the individual.

The problem with a government policy is that its effects on individual circumstances differ from person to person. As an example, consider a pension adjusted annually to a national consumer price index. Even if it is calculated in a manner to properly measure and account for inflation, it might not properly measure the increase in expenses of a given pensioner if that individual's expenditure is weighted heavily towards items experiencing above average inflation, such as health insurance or education.

The economic consequences of inflation, the historical documentation of which spans well over 2,000 years, are such that it merits its own asset class in investment planning. It should be self-evident that one cannot simply assume that common stocks *sui generis* are efficient or are a sufficient hedge against inflation. Labor is the most significant operating expense for most members of the popular stock indexes. Many people assume that a business can maintain profit margins by increasing wages at the same rate as general inflation. However, a wage increase of X% in a progressive tax system does not necessarily become an after-tax wage increase of that same X%, because in a progressive system, higher wages are taxed at a higher rate. The U.S., of course, follows a progressive tax structure. As a generalization, one requires a pre-tax wage increase above inflation to obtain an after-tax increase equivalent to inflation. Most businesses, under a condition of persistent inflation, are not able to maintain their margins once wage rates begin to inflate.

A good argument can be made for gold as an inflation hedge in a portfolio, but it must be remembered that gold can be highly idiosyncratic for very long periods of time. For example, from 1980 to 2000, gold actually declined, although inflation clearly existed during this period; the price of gold was unchanged from 1934 to 1967, as a consequence of government action, although price levels increased by 2.5x.

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The Federal Reserve Bank of Minneapolis calculates inflation in the United States back to 1800 and the data illustrates the same point.⁵ The First World War years in the U.S. were highly inflationary, as is evident in the following figures

Table 1: U.S. Inflation Rate, 1916 - 1920 1916 7.7% 1917 17.8% 1918 17.3% 1919 15.2% 1920 15.6% Source: Federal Reserve Bank of Minneapolis

The gold prices for those same years are as follows; clearly, the gold price did not keep pace with inflation at that time:

Table 2: Gold Prices, 1916 - 1920 1916 \$18.99 1917 18.99 1918 18.99 1919 19.95 1920 20.68 Source: piketty.pse.ens.fr

As a matter of historical curiosity, in 1717, during Isaac Newton's tenure as Warden and Master of the Royal Mint of the United Kingdom, he set the gold price at 3 pounds 17 shillings. It was still at that price in 1914, on the eve of the First World War.

On the other hand, one could have purchased gold in the year 2000 for \$279.11 an ounce and it now trades at \$1,908.07 an ounce. This is about a 10% annualized increase, considerably above most estimates of inflation for this period. If one can tolerate the idiosyncratic behavior of gold, it can be a good protection against inflation.

Far better than gold are the royalty companies that finance gold production by the purchase of revenue interests, or some effective equivalent, in precious metals mines. The two most successful are Franco Nevada Corp. (FNV) and Wheaton Precious Metals Corp. (WPM). Once the investment is made, the return is greatly enhanced if the price of the commodity rises, since the royalty companies experience no increase in expenses; theirs is simply a contractual relationship, and they require none of the capital equipment or physical field operations of a mining company. Other examples of gold royalty companies, are:

Table 3: Additional Gold Royalty Companies

⁵See minneapolisfed.org

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RGLD Royal Gold, Inc. SAND Sandstorm Gold Ltd. OR Osisko Gold Royalties Ltd. MMX Maverix Metals Inc. MTA Metalla Royalty & Streaming Ltd.

This is an example of a “hard asset” business model. The revenue is a contractual commitment by the producer of the commodity to the company providing the financing. The inflation beneficiary aspect is that the royalty company’s operating expenses do not increase in the event of inflation; in the event of inflation, a revenue increase is essentially a pre-tax profit increase.

The challenge to asset allocators and other investors is that an insufficient number of such hard asset companies exist to form an index. Also, the aggregate market capitalization is very small in relation to the great amount of assets in the world that require some inflation protection.

Many investors in the practical world, therefore, are compelled to use leverage as an inflation hedge, on the premise that the cost of debt capital is low and, in any case, even modest inflation will eventually erode the purchasing power of the obligations that must one day be repaid. That is, for investors employing leverage, paying back ‘in cheaper money’: the amount of the bond or loan due on the maturity date in five years or 10 remains the same as on the first day of the loan, but the nominal income of the borrower has presumably inflated over time.

John Maynard Keynes did not write extensively about inflation, but his personal portfolio was structured in such a manner to benefit from consistent inflation inasmuch as he made use of copious leverage. The following table, taken from Keynes’ Way to Wealth by John F. Wasik, will illustrate the point. ⁶

⁶ John F. Wasik, Keynes’s Way to Wealth: Timeless Investment Lessons from the Great Economist (New York: McGraw Hill Education, 2014). 265

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Table 4: John Maynard Keynes Leverage and Net Worth, 1919-1945

Loans	Net Assets	Loans	Net Assets
1919 £15,498	£16,315	1933 £78,859	£55,222
1920 20,837	(1,837)	1934 165,343	146,007
1921 N/A	N/A	1935 188,271	220,619
1922 2,720	21,558	1936 299,347	506,522
1923 1,200	34,364	1937 190,035	215,244
1924 1,200	63,797	1938 106,470	181,547
1925 1,200	43,610	1939 109,136	199,495
1926 2,200	40,800	1940 60,655	171,090
1927 46,900	44,000	1941 28,753	205,281
1928 25,790	13,060	1942 19,720	254,073
1929 14,000	7,815	1943 31,643	312,271
1930 65,000	12,525	1944 46,167	355,310
1931 11,965	15,100	1945 38,886	411,238
1932 19,774	21,722		

Source: John F. Wasik, Keynes’ Way to Wealth (New York: McGraw Hill Education, 2014), 119

There were years where Keynes was enormously leveraged. For example, in 1920, Keynes had £20,837 of borrowings, and his net assets were negative £1,837. One could say technically he was insolvent. Similarly, in 1928 on the eve of the stock market crash, he had £25,790 of leverage and £13,060 of net assets. In the year 1929, it was £14,000 of leverage and £7,815 of net assets. One can see this involves a fairly large multiple. In 1930, he had £65,000 of loans, and £12,525 of net assets, he was over 5x leveraged. It was only near the end of his life, during the war years, that Keynes had fairly modest leverage in relation to his net assets. The last full year of his life, 1945, he had only £38,886 of borrowings against net assets of £411,238.

It should be noted that Keynes drew living expenses from his personal account, so that the years of decline in net assets are not entirely from investment depreciation. The net asset values are not meant to illustrate investment performance. In any case, it is obvious that Keynes frequently made use of enormous leverage relative to his net assets.

This process has continued among investors and business people to this day. The contemporary world might be the most leveraged society in history. That means that the constituency supporting debasement of debt and money is very large. Inflation is the potential savior of the

debt leveraged. Therefore, it is prudent to develop inflation hedges.

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Murray’s Musings

THE IMPORTANCE OF THE OIL SERVICE INDUSTRY

The energy sector weighting of the S&P 500 at the time of this writing is 5.04%. Some two years ago, it was in the vicinity of 2.5%.

Almost one-third of that 5.04% is accounted for by Exxon Mobil Corp. (XOM), with a 1.35% weight. Chevron Corporation (CVX) accounts for 92 basis points. The ConocoPhillips (COP) weight is 47 basis points.

The largest energy service company weighting in the S&P 500 index is Schlumberger Ltd. (SLB) at 19 basis points. The next oil service company to make an appearance is Haliburton Co. (HAL), which comprises 9 basis points. It is the 255th largest company in the S&P 500. Baker Hughes Co. (BKR) is an 8-basis-point position. This group is the extent of the S&P 500 exposure to oil service. In total it amounts to 36 basis points.

The oil service industry obviously can be ignored as a purely statistical factor in the S&P 500 Index. However, without the seismic fracking and well construction services offered by these companies, oil could not be extracted, and the oil extraction companies are important in the

index. The oil service industry, as a generalization, has earned very poor returns on capital since 1981. From 2013 until very recently, it was in crisis, with numerous bankruptcies and substantial losses by virtually all industry participants.

Schlumberger, by example, had \$39 billion plus of shareholders' equity in 2003, whereas by the end of 2020, it had dropped to modestly in excess of \$12 billion. By year-end 2021, book value was slightly in excess of \$15 billion. As to fixed assets less accumulated depreciation, generally understood to be plant, property and equipment, this amounted to \$15.096 billion at the close of 2013. By the end of 2021, this figure had declined to \$6.429 billion. It was even lower at the end of June 2022: \$6.386 billion.

In the first six months of 2022, Schlumberger produced a 17.9% annualized return on shareholders' equity. Nevertheless, the company's \$664 million of capital expenditures in the first six months of 2022 is only 62.4% of the \$1.065 billion of depreciation and amortization recorded in this period, which means the balance sheet value of PP&E is still declining at a rapid rate. Schlumberger is not investing in itself even though it is the leading oil service company, with services that are critical to the global extraction of oil and natural gas.

The figures for Haliburton are similarly profound. It had \$15.581 billion of shareholders' equity at the end of 2013, and PP&E net of accumulated depreciation was \$11.322 billion. At the end of 2021, shareholders' equity had declined to \$6.713 billion, and net PP&E had declined to \$4.326 billion. By June, this last measure had declined to \$4.165 billion. The annualized return on shareholders' equity thus far in 2022 is 10.69%. Nevertheless, despite 268

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its profitability, Haliburton is making capital expenditures of only about 85% of depreciation expense. This is exclusive of the fact that the company actually sold \$116 million of equipment, and it also excludes yet another \$366 million of impairments in the first half of 2022.

In other words, Haliburton is also disinclined to invest in its business. All of this is quite understandable in the context of Haliburton history. At the end of 2013, the company had debt of \$7.816 billion against shareholders' equity of \$13.581 billion. At the end of 2021, long term debt was a higher \$9.12 billion against half as much shareholders' equity of \$6.713 billion. At the end of June 2022, debt had declined to \$8.525 billion against shareholders' equity of \$7.12 billion. Haliburton is still a leveraged company, although it should be noted that net current assets amount to \$5.496 billion.

Schlumberger stock trades at more or less the price reached in the summer of 2005. The same can be said for Haliburton, even though its shares had a slightly better outcome for the time period in question.

It might be argued that modern oil rigs are more productive than the oil rigs of the past, so that fewer rigs are needed to produce a given quantity of hydrocarbons. The U.S. Energy Information Administration (EIA) calculates drilling rig productivity over time using "new well oil production per rig" as the key metric. The data is contained in a drilling productivity report per region. The data is not so clear on this question.

The EIA data shows that the Anadarko region of Western Oklahoma's new well oil production per rig has more or less quintupled since 2012. On the other hand, the productivity increase since 2012 in a region such as Appalachia has been minimal. This contrast might be compared to the roughly 8x productivity increase in the Bakken region of North Dakota, but with minimal increases in the Haynesville area of East Texas and Western Louisiana.

The Permian Basin is perhaps the most prolific energy producing region in North America, and new well oil production per rig has advanced approximately tenfold there since 2012. Productivity reached a peak in 2020, with significant shutdowns of producing wells in all regions. Only the most productive wells remained in operation. This phenomenon is known as high grading. Since 2020, productivity has been in decline. This is not necessarily indicative of a loss of productivity per rig; it is likely that less productive land is now being brought into production.

Both Schlumberger and Haliburton arguably have now achieved an acceptable level of profitability. This is not true for many oil service firms. Baker Hughes, for example, is profitable on an operating basis, with an operating margin of 6.68%. Unfortunately, the company still reports impairments in non-operating losses. The balance sheet has \$14.88 billion of shareholders' equity against \$6.625 billion of debt, and balance sheet cash amounts to \$2.928 billion. However, goodwill and other intangible assets account to \$5.741 billion and \$4.049 billion. This total of \$9.79 billion can be subtracted from shareholders' equity to 269

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arrive at a hard book value of \$5.09 billion. Viewed in this light, Baker Hughes might still be regarded as a leveraged company, so it still displays signs of the damage caused by the long energy depression from June 2014 to June 2021.

The contraction in the number of operating offshore oil rigs has been far more extreme, and the duration of the contraction now exceeds four decades. There were 250 U.S. offshore crude oil and natural gas rotary rigs in operation at the end of 1980.¹ By the end of 2007, just before the financial crisis of the following year, the number of offshore rigs in operation had dropped to 62. In the next 24 months, this figure shrank to 37. The number of rigs in operation offshore then began a gradual increase to 58 in June 2014, on the eve of what was to become an energy depression that lasted for eight years. Data for August 2022 is the most recent, and at that time 18 oil and natural gas rotary rigs were in operation in the offshore U.S. waters.

A shortage of oil may be developing right now. The U.S. Strategic Petroleum Reserve has been used to balance supply in relative to the embargo on Russian oil. The SPR has declined to 416.4 million barrels as of September 30th, 2022, after a drawdown of 201.4 million barrels in the past 12 months. This rate of diminution is not sustainable. One possible solution is a resolution of the Ukraine conflict. That does not seem likely in the foreseeable future.

The other obvious solution is a meaningful increase in U.S. oil production, but in order to accomplish that, the remaining oil service companies must increase investment in their facilities to facilitate that production increase. The government plans, though, are still to replace hydrocarbons with other forms of renewable energy once the current crisis is resolved. Effectively, the proposition to the oil service companies is that they would need to make long term investments to solve what the government hopes is a short-term problem. If the Ukraine conflict is resolved, energy commerce might return to normal, in which case any oil service company that made investments to handle temporary U.S. oil production increases would be left with enormous amounts of redundant equipment.

The oil service companies have experienced about four decades of excess oil service supply and endured enormous losses in order to establish low break-even operating levels. Significant capital expenditure would inevitably raise break-even points for oil production levels that world governments would rather not be sustained. In fact, it is the stated intention of those governments to reduce hydrocarbon production as soon as it can be practically achieved.

The surviving oil service companies, therefore, have absolutely no incentive to increase investment in their output capacity. In fact, the entire industry continues to disinvest in their plant and equipment. It will be difficult, if not impossible, for hydrocarbon production to increase while the oil service companies are reducing their capital investments.

¹ Energy Information Administration 270

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This makes oil service a very interesting industry from an investment perspective. It is the only industry that is currently aligned operationally to endure a depression, simply because it has just experienced a depression. It is therefore a potential non-correlated asset in an equity portfolio. The reluctance of the industry to invest will eventually serve to greatly increase typical oil service company return on equity to very alluring levels.

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Murray's Musings

BITCOIN V. BONDS

The bond market thus far in 2022 has produced results that generate questions about the nature of the asset class itself. These questions arise from the year-to-date returns for salient aspects of the asset class. The returns can be visualized using popular exchange traded bond index funds (ETFs), as follows:

Table 1: Returns for Selected Bond ETFs, Dec. 31, 2021- Oct. 21, 2022

Return TLT iShares 20+ Year Treasury Bond ETF (36.11)% IEF iShares 7-10 Year Treasury Bond ETF (18.08)% AGG iShares Core U.S. Aggregate Bond ETF (16.62)% HYG iShares iBoxx \$ High Yield Corporate Bond ETF (14.09)% TIP iShares TIPS Bond ETF (13.42)% Source: iShares.com

The obvious question to be posed when noting the negative returns experienced this year is whether the long- term return prospects of the bond asset class are sufficiently large to compensate the investor during the inevitable declines in those periods when the Federal Reserve must increase interest rates.

One might begin to answer this question by studying the returns since inception for the aforementioned funds. This data appears in the table below.

Table 2: Annualized Returns, Selected Popular Bond ETFs, Inception through 9/30/22 Inception Date Return TLT iShares 20+ Year Treasury Bond ETF July 22, 2002 4.50% IEF iShares 7-10 Year Treasury Bond ETF July 22, 2002 3.52% AGG iShares Core U.S. Aggregate Bond ETF Sept. 22, 2003 2.90% HYG iShares iBoxx \$ High Yield Corporate Bond ETF April 4, 2007 4.08% TIP iShares TIPS Bond ETF Dec. 4, 2003 3.42% Source: iShares.com

It should be pointed out that the inception dates of all these funds to the most recent date coincide with the greatest bond bull market of all time by far. It is an observation to keep in mind as one reflects on these results.

An obvious problem with this type of analysis is the question of subjectivity. One might choose to compare these funds to the prevailing rate of inflation during the relevant time period, in order to understand what the real, inflation- adjusted returns have been . Unfortunately, the calculation of the rate of inflation has itself become controversial and no

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agreement on a universal inflation measure appears to exist. Nevertheless, the returns can certainly be compared to one another. Even this approach, however, is not very promising because the returns since inception are for different time periods and the periods may not be sufficiently long to arrive at meaningful conclusions. Also, the returns for the different periods and the different classes of bonds are not radically different.

One can say that the highest return is for long term U.S. Treasury bonds, at least for the time period in question. If duration is a primary measure of risk, then it is reasonable that the highest return category is long term U.S. government bonds. But, we cannot know if the greatest overall risk is to be found in the high-yield bond sector, which in prior eras would have been the natural presumption due to the inherent credit risk. That can no longer be presumed, because the Federal Reserve has pursued an extremely aggressive policy of monetary accommodation for most of the time period in question. The alteration of this stance only commenced in mid-June 2022. Some time must pass until the withdrawal of monetary stimulus results in a more historically normal incidence of credit defaults or near defaults.

Credit spreads will likely widen and higher yields will be available in the sector if defaults once again become a primary risk factor for high-yield bonds. In that case, after a period of low returns or quite possibly negative returns, the best performing bond segment might ultimately become high-yield bonds.

On a year-to-date 2022 basis, high-yield bonds slightly underperformed inflation-protected government bonds. Credit problems have not yet made an appearance.

Another means of viewing these funds is through the prism of after-tax returns, since individual taxpaying investors frequently make use of ETFs.

Table 3: After-Tax Annualized Returns, Selected Bond ETFs, Inception Through 9/20/22
Inception Return TLT iShares 20+ Year Treasury Bond ETF July 22, 2002 3.06% IEF iShares 7-10 Year Treasury Bond ETF July 22, 2002 2.36% AGG iShares Core U.S. Aggregate Bond ETF Sept. 22, 2003 1.79% HYG iShares iBoxx \$ High Yield Corporate Bond ETF April 4, 2007 2.07% TIP iShares TIPS Bond ETF Dec. 4, 2003 2.20% Source: iShares.com

The calculations in Table 3 were made by iShares, not Horizon Kinetics. To make it universally applicable to all U.S. residents, only federal taxes are assessed. Residents of various states with state income taxes or even municipalities with local income taxes will have different rates of return. However, the returns are at least standard in that none include state and local taxes. State and local income taxes would not apply any of the three Treasury ETFs. For the iShares Core U.S. Aggregate Bond ETF and iShares iBoxx High Yield Corporate Bond, the yields in the table are shown without the impact of state and local taxes.

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If one presumes merely a 2% annual inflation rate, the Core U.S. Aggregate Bond ETF did not provide a positive real rate of return after taxes. This is significant because this particular ETF is designed to represent the entirety of the bond market. The fund itself has \$75.64 billion of assets under management. It is the second largest bond ETF and the tenth largest of all ETFs in the United States. The largest bond ETF is the Vanguard Total Bond Market ETF (BND), with \$79 billion of AUM and an inception date in 2007. In composition and returns, it is very similar in character to the iShares Aggregate U.S. Bond ETF.

If the after-tax rate of return of the entire bond market is lower than inflation over a long period of time during which inflation was mostly at a low rate and relatively benign – in this case covering periods of 15 to 20 years – one must logically rethink one's approach to bond investing.

The composition of the Aggregate Bond ETF is as follows:

Table 4: Composition, iShares Core U.S. Aggregate Bond ETF Bond Type Percentage of AGG
Treasury 41.72% MBS 27.05% Industrial 14.41% Finance 8.16% Utilities 2.06% CMBS 1.85%
Agency 1.22% Supranational org. 0.92% Local Authorities 0.84% Cash 0.48% ABS 0.35%
Source: iShares.com

Segments of the bond market clearly did provide a positive real after-tax rate of return when measured against the conventional CPI inflation statistic. If such is the case, yet the broad index underperformed inflation after taxes, this would constitute the first relatively long-run empirical argument against broad indexation in favor of an actively constructed portfolio.

The market value of all publicly traded fixed income securities outstanding as of year end 2021, is \$53.017 trillion. ¹ Bear in mind this is not all debt; it is the value of all publicly traded debt securities, which is lower than the value of all debt. This is the largest asset class in the U.S. and it is considerably larger than the market capitalization of the S&P 500. In fact, fixed income issuance has continued to be robust in 2022 even though, at the moment, the market for initial

public offerings of equity is almost nonexistent.

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In principle, the best decision that could have been made over the course of the past two decades within the broad bond asset class to maximize returns would have been a 100% allocation to long-term treasury bonds. The after-tax annualized return is 3.06%. Assuming roughly 2% annual inflation, which, again is subject to some degree of dispute, the real rate of return was 1.06% per annum. In order to earn this rate of return, one had to accept a standard deviation of 14.03, an effective duration of 17.17 years, and a weighted average maturity of 25.58 years.

By way of comparison, the first reliable price quote for bitcoin was \$0.0495 on July 17, 2010, a bit less than 5 cents. 2 Bitcoin now sells for \$19,189. The current interim bitcoin inflation rate is 1.73% per annum and at the time the notes for this Compendium were prepared, 19,186,625 bitcoin were in existence. By the time this is published, there will be even more. A total of 1,813,375 bitcoin will be issued from the time the notes were prepared for this Compendium until the year 2140. The number yet to be issued will continuously drop. In 556 days, the bitcoin interim inflation rate will decline to 87 basis points per year. With each passing day, of course, there will be one less day until the bitcoin interim inflation rate drops to 0.87%. Moreover, the bitcoin interim inflation rate will decline by 50% every four years in accordance with the bitcoin issuance protocol. The projected 0.87% annual inflation rate will decline to 0.44% after four years.

Profits on bitcoin are taxed at the capital gains rate, not the statutory federal income tax rate that applies to interest on debt. This is because the IRS considers bitcoin to be property. Also, unlike interest on debt, capital gains tax is paid after sale at a profit.

What if a holder of a bond ETF, such as iShares Core U.S. Aggregate Bond ETF (AGG), were to have made the most marginal change possible to the investment allocation plan by investing 10 basis points annually into bitcoin? Structured that way, in the worst instance, the after-tax rate of return would have declined from 1.79% annually to 1.69%. That basically assumes a complete failure of the Bitcoin project, that bitcoin would have gone to zero.

An investment in bitcoin made in July 2010 would have increased 387,797x in value by today. In other words, a \$100 investment would now be worth \$38,779,797. If the investment had been made with only \$100 of a \$1 million bond portfolio even once in 2010 – that would be a 1 basis point investment, one-hundredth of 1% – the remaining \$999,900 that was invested in bonds compounding at an after-tax 1.79% per annum rate for 12 years would now be worth \$1,237,137. If the entire \$1 million had been invested for in bonds for the 12 years under discussion, the total capital sum held at this point would be \$1,237,261, or \$124 greater. That quantifies the risk of investing \$100 in bitcoin in a bond portfolio as a loss of about \$124. That \$124 difference would be the amount of loss in principal value after 12 years if bitcoin had failed completely.

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In actuality, those bitcoin returns are somewhat understated since the calculation gives no credit for the distribution value of the various bitcoin forks such as Bitcoin Cash and Bitcoin Gold.

The market capitalization of Bitcoin is now \$368.3 billion. Bitcoin is worth less than 7/10ths of 1% of the U.S. publicly traded bond market value. The value of the bond asset class is constantly expanding due to new issuance. If the bond asset class did actually protect investors against debasement caused by inflation, there would be no need for this instrumentality.

The bitcoin return should be genuinely extraordinary if, at some point in the future, the investment community comes to the conclusion that bitcoin will better protect an investor against inflation than the bond asset class which appears to provide no such protection.

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Murray’s Musings

DIAMONDS AS AN INVESTMENT

Although diamonds were known and gathered 4,500 years ago, diamond mining on a systematic basis could only proceed with the development of the modern sciences of geology and crystallography. This was possible only at the end of the 19th century, so it was then that the 19th century diamond extraction in South Africa commenced. This ultimately gave rise to the De Beers monopoly. De Beers controlled the diamond market until the end of the 20th century.

The De Beers monopoly was an obvious barrier to price discovery and transparency. Thus, the evolution of diamonds as an asset class only became possible in the 21st century. Even still, diamonds present valuation problems not ordinarily present in commodities. Gold and silver of given levels of purity, for example, are homogenous commodities, while all diamonds are heterogeneous; no two are identical. Accordingly, there developed a detailed grading system. Diamonds are commonly divided by the four criteria of color, cut, clarity, and carats, each with its own measures.

That grading system has existed for over 50 years, but it is only lately that modern computing technology has enabled the ability to assemble a diverse set of diamonds into packets that each have equivalent value in much the same way that ETFs are designed to combine sets of individual companies be representative of an industry exposure. Such packets, then, are homogenous, like other commodities. That is a significant change in the investment characteristics of diamonds as an asset class.

This section of the Compendium is intended to serve as a reference guide for and answer certain questions about what might be regarded as basic diamond investing. One such question might

be: Where is historical data on diamond prices kept? Another might be: Is historical diamond price data accessible to the public?

Historical price data is available via the Rapaport Group, which is a large Belgian broker of diamonds. This is the only database that I have seen that makes diamond prices readily available. Rapaport publishes the price lists, and one can purchase them from the company. For instance, one can buy the price list for April 27, 2007 for \$150.

Rapaport also publishes and sells a Diamond Price Statistics Annual Report for each year, starting with 1994, which is approximately the time that the De Beers monopoly was effectively dismantled. It might be recalled that the U.S. government charged both General Electric Company (GE) and De Beers, in 1994, with industrial diamond price fixing.

It is fair to assert that diamond prices rarely decline, at least according to the data. A notable exception was during the 2008 financial crisis, but even then the decline was modest in 279

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comparison with other assets, such as stocks and bonds. A brief drop in diamond prices also occurred when the World Trade Center was destroyed in 2001.

Constructing a diamond price index would require making certain assumptions regarding index weightings, and such assumptions can be subjective. It will not be attempted in these pages. In order to comprehend the complexity of building an index of diamond prices, one might parse the problem by considering the impact on price just of the differences in diamond cut .

Petra Gems publishes a diamond buying guide that illustrates the complex nature of this challenge. Oval or round cut diamonds are the most desirable and therefore the most expensive diamonds if all other factors are held constant, which of course, is rarely the case. In order of desirability:

- Oval cut diamonds are priced 5-15% lower than round cut diamonds.
- Princess cut diamonds are priced 17-26% lower than round cut diamonds.
- Cushion cut diamonds are priced 22-30% lower than round cut diamonds.
- Radiant cut diamonds are priced 20-27% lower than round cut diamonds.
- Emerald cut diamonds are priced 19-27% lower than round cut diamonds.
- Pear cut diamonds are priced 15-22% lower than round cut diamonds.
- Marquise cut diamonds are priced 23-32% lower than round cut diamonds.
- Asscher cut diamonds are priced 21-31% lower than round cut diamonds.

Another complexity is that the Rapaport Price List is based on ask prices rather than actual realized prices. A dealer frequently will demand and receive a 15% or even 20% discount to the Rapaport ask price. The market still lacks a certain level of transparency , since it is difficult to know actual realized prices.

A further problem is that even realized prices are semilogarithmic with respect to certain attributes, such as carat size. To illustrate this relationship, let us assume the existence of a round cut diamond of 1 carat, with color D (highest color grade) and clarity FL (flawless). If the selling price is X, a 1.5 carat round cut diamond with color D and clarity FL would sell for about 2.14x the price of the comparable 1 carat diamond. To be clear: if one compares the 1.5 carat diamond to a 1 carat diamond with these characteristics in common, the price would not be 50% higher just because it has 50% more weight; it actually would be 2.14x as expensive as the 1 carat diamond.

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The only difference between the two diamonds in this example is that the latter has 1.5x the weight of the former. The 2.14x greater price is because a 1.5 carat diamond is rarer than a 1 carat diamond. In other words, the value density of a 1.5 carat diamond is more than 1.5x as great as a 1 carat diamond's value density.

Value density will ultimately become extremely important as the concept of physical

commodities increasingly dominates the world of commodity ETFs. A 3 carat round cut diamond with D color and FL clarity could sell for \$170,000.

To be clear about the value density, a 3 carat diamond weight is only 6 grams, and there are 31.1035 grams in a troy ounce. Thus, a diamond with the aforementioned characteristics and price is selling for $\$170,000 \div 6 \text{ grams}$, or \$28,333 per gram. One troy ounce of diamonds of this caliber would theoretically sell for $\$28,333 \times 31.1035$, or \$881,255. In contrast, one troy ounce of gold sells for about \$1,757. Therefore, one ounce of diamonds with the aforementioned quality features is 501.6x more value dense than gold ($\$881,255 \div \$1,757$).

The largest physically backed gold ETF is SPDR Gold Shares (GLD), with \$50.736 billion of assets under management. Using the current \$1,757/ounce gold price for the calculations, this fund holds roughly 28,876,494 ounces of gold. If confidence in the financial system were to be lost, the investors ostensibly could demand to collect their 28,876,494 ounces of gold. What would that entail?

A troy ounce of gold has the following dimensions: 0.95 inches long, 1.65 inches wide, and 0.08 inches thick, which equals 1.254 cubic inches. An ounce of gold is easily transported and easily hidden in a crisis. However, transporting and hiding 29 million ounces is a far more complex logistical challenge – that would be 1.999 million pounds. One Honda Accord weighs about 3,300 pounds.

Contrast that value density with diamonds. If an ounce of diamonds is worth \$881,255, achieving a valuation of \$50,736,000,000, the market capitalization of SPDR Gold Shares, would require only $\$50.736 \text{ billion} \div \$881,255$ or 57,572.43 ounces, or 3,985 pounds. A human being could easily transport or hide over \$8 million by holding a mere 10 -ounce diamond packet.

One of the reasons investors wish to hold physical gold is that it cannot be easily debased like the U.S. dollar or other fiat currencies. However, all of the large central banks have extensive gold reserves, and it might be necessary for central banks to sell those reserves if a time comes when the fiat currency is not be trusted and might not even be an acceptable medium of exchange. In that case, large gold sales might depress the price of gold. On the other hand, central banks do not have reserves of diamonds to sell.

Gold has been readily available in sufficient supply at the requisite level of purity to use as a medium of transaction for thousands of years. On the other hand, a market for diamonds could not exist that was fair in any meaningful sense of the word until the modern science of crystallography developed. 281

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A factor in the performance of diamonds is the evolution of India's diamond industry, which is where much of the cutting and polishing of rough diamonds is now done. The Indian diamond industry employs 850,000 people. In fact, the Indian city of Surat in the western state of Gujarat has become known as Diamond City. Diamond cutting is a highly skilled art, and people who are expert in this field are well compensated in Western nations. Indian skilled labor is far less expensive. The Indian diamond export trade is centered almost entirely in the Bharat Diamond Bourse in Mumbai, which handles 98% of the export trade. It has a constructed area of 2 million square feet, with two basement complexes containing an extra 1 million square feet.

The transition of a rough diamond to a polished cut diamond is a time-consuming process. Indian labor has greatly reduced friction in this area. That inexpensive labor has served to restrain diamond prices in recent years.

The global market for diamonds is expanding, as a greater proportion of the world population enters the middle class and becomes part of the diamond consumers' universe. Existing diamond exchanges include the London Diamond Bourse, the Diamond Federation of Hong Kong, the Israel Diamond Exchange, the Dubai Diamond Exchange, the Amsterdam Diamond Bourse, the Bharat Diamond Bourse, the Tokyo Diamond Exchange, the Shanghai Diamond Exchange, the Antwerp Diamond Exchange, the Diamond Dealers Club of Australia, Borsa Diamanti d'Italia, and the Guangzhou Diamond Exchange.

A number of new diamond bourses are in the process of being established. These include the New York Diamond Dealers Club, Borsa Istanbul, Diamant Club Wien, Bangkok Diamonds and Precious Stones Exchange, the Diamond Exchange of Singapore, and the Korea Diamond Exchange. This does not include the World Federation of Diamond Bourses itself.

It is important to recall that the De Beers had a global monopoly, for all practical purposes, and was involved in price fixing that was only effectively challenged in 1994. This was over 300 years after the Buttonwood Agreement of 1792 that became the foundation of the New York Stock Exchange. The first important revision of that agreement did not occur until 1817. The U.S. Securities and Exchange Commission did not commence operations until 1934. The

Depository Trust Company (DTC), which does clearing, was only founded in 1973, as the central securities depository of the U.S. market system, 181 years after the New York Stock Exchange. Bitcoin began its existence as a white paper published on October 31, 2008, and was first used in 2009. In 2022, it still has no regulatory structure.

In other words, the modern diamond market is evolving more rapidly than the securities market.

The first recorded use of a diamond engagement ring was in 1477, when Archduke Maximilian of Austria proposed marriage to Mary of Burgundy. Diamonds were a rarity even for engagement rings until 1947, when De Beers engaged the now-defunct advertising firm N.W. Ayer & Son to create the campaign entitled “A diamond is forever.” This was 282

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one of the most successful advertising campaigns of all time and created the modern diamond market. The universal diamond grading system was only created by the Gemological Institute of America (GIA) in the 1950s. The Gemological Institute of America was only founded in 1931.

The diamond asset class is developing very rapidly. However, it is very unlike ordinary asset classes like securities, in which creation can be infinite – which is to say, if a certain kind of ‘paper’ is appealing to the public, issuers can and will produce, with virtual immediacy, as great a quantity of it as the market will bear. The global production of rough diamonds is in decline, and probably in inexorable decline. The Facts & Figures section in this Compendium contains more data on that.

An investable asset class is in the process of being created and, moreover, with an asset manifesting declining production supply. According to Statista, rough diamond production was 177 million carats in 2005 and 116 million carats in 2021. The most alluring returns from an asset class occur before it actually becomes an investable asset class.

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Murray's Musings

WEALTH AND CRISIS: THE ROLE OF DIAMONDS IN ASSET ALLOCATION

The modern understanding of asset allocation is based on the premise of social stability. If the assumption of social stability is removed, the conceptual foundation of modern asset allocation vanishes along with it.

For example, a basic building block of asset allocation is the notion that short -term government securities represent a risk-free asset. This is obviously untrue in an inflationary environment, since the asset will lose purchasing power. It is also untrue in an environment of war or civil war.

Inflation has not usually been a problem in the developed world since World War II, with the great exception of the 1970s decade , although inflationary episodes did occur in the aftermath of World War II, as one might expect.

Globally, however, inflation is not uncommon. Trading Economics recently produced a list of 10 nations with very serious inflation:

Table 1: Nations with Highly Problematic Inflation Country Annual Inflation Rate Venezuela 1198.0% Sudan 340.0% Lebanon 201.0% Syria 139.0% Suriname 62.3% Zimbabwe 60.7% Argentina 51.2% Turkey 36.1% Iran 35.2% Ethiopia 33.0% Source: TradingEconomics.com

This list is only comprised of the nations with the highest recorded rates of inflation. If problematic inflation is defined as a recorded rate that exceeds 10% per annum, this list would double. Twenty countries, or about 10% of the world's nations, are experiencing serious inflation.

Inflation, though serious, is not the most grave threat to wealth preservation. The most serious threats are war, civil war, and social unrest.

Civil war can last surprisingly long. In the Western world, investors generally ignore such conflicts. It is astonishing to learn that the Myanmar civil war has been raging since 1948. 285

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The country, once known as the British colony of Burma, achieved independence that year. Since then, seven states of that nation—representing a variety of ethnicities—have engaged in a continuous armed struggle for either autonomy or independence.

This is not the only example . A left-wing armed revolt against the government of the Philippines has been continuing since 1969. An armed struggle between the government of Colombia and the drug cartels has been ongoing since 1964. The civil war in Syria began in 2011. In Sudan, the war in Darfur has been continuing since 2003. The civil war in Yemen has been happening since 2014. Conflict still roils Afghanistan, a nation that's been in a continuous state of armed conflict since 1978.

One might be tempted to dismiss these conflicts as irrelevant to industrialized nations, but one must note that the low -intensity Ukrainian civil war that commenced in 2014 has evolved into an actual Russo-Ukrainian war.

None of the long-term studies of wealth and investment returns really take into account the influence of war and social unrest. Few studies of the impact of war upon wealth appear to exist. Many books have been written on the French Revolution and the Napoleonic wars, but the focus is on military strategy and social change. Similar remarks might be made about the First and Second World Wars. One reason for this is that the period from the Battle of Waterloo in 1815 to the beginning of the First World War in August 1914 is usually considered to be a

peaceful era, but it is only peaceful from a Eurocentric or American perspective. In fact, even the American perspective ignores the wealth impact of its own Civil War.

At least 50 19th century civil wars have been almost entirely ignored by historians. One example is the Seinan Senso or Southwestern War, so called because it was largely fought on the island of Kyushu in Southwestern Japan. Essentially, the samurai class was not initially opposed to the Meiji Reform that implemented imperial rule in Japan in 1868, but it became increasingly concerned with the direction the government was taking and established an independent military academy. This was viewed as a serious challenge to governmental authority, and eventually a series of incidents quickly deteriorated into civil war. The government eventually prevailed, but the cost of the conflict nearly bankrupted the country. The war forced Japan off the gold standard and effectively tripled the country's national debt.

This very brief survey of conflict should be sufficient to establish that the probability of destabilizing conflict is not zero. Therefore, an investor's asset allocation profile should contain one or more assets that could maintain value and possibly even increase in value in the event of such a conflict.

Cash, bonds, stocks, private equity, and real estate—the current standards in asset allocation strategies—are vulnerable in the event of war. Cash and bonds are vulnerable since they are obligations of the government, and the creditworthiness of the government is in question during such a crisis. Equities represent business enterprises. Amid conflict, an enterprise 286

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might be subject to much higher taxation. Its business prospects could be greatly diminished during war, and might even be seriously damaged by the ensuing violence. The same points would apply to private equity. Real estate can obviously be damaged during an armed crisis, as well. It is also a highly visible asset that is not movable and could be expropriated.

Given the problematic features of conventional assets during a true crisis, the obvious and conventional alternatives are gold and silver.

One could readily dismiss silver as a crisis asset on the question of value density—that is, the physical quantity of an asset needed to afford protection. To demonstrate, let us assume that the basic unit of account for wealth is \$1 million. Using the current silver price of \$23.56/ounce, 42,444.82 ounces of silver would be required to store \$1 million in value. Each troy ounce is 0.18 cubic inches, so this equals 7,674.8 cubic inches, or 4.44 cubic feet, of volume. The challenge of transporting 4.44 cubic feet of silver is not trivial ; it would weigh 2,910 pounds, or about that of a Nissan Sentra 4-door sedan.

This is not to assert that silver would n't appreciate in a crisis , but the primary wealth management problem in a crisis is not the maximization of return, but the prevention of seizure.

In the case of gold, seizure can be prevented if it is held outside of the regulated custodial system. In comparison with silver, a gold price of \$1,788 per ounce requires only 559.2841 ounces of gold for \$1 million of value, thus mitigating the transportation problem to some extent. Although this is far simpler than transporting 22,444.82 ounces of silver, weighing 38 pounds it still requires a transportation conveyance and is not easily concealed.

Art, in the form of a painting, is potentially value dense. The transportation problem is not necessarily complex and it does not necessarily require specialized conveyance. However, a painting is very fragile. It could be concealed, but maintaining the proper storage conditions to preserve the painting's value in a crisis is not a trivial problem.

This leaves diamonds as an important crisis asset that has not yet become an accepted asset class. Diamonds are light. One carat weighs one-fifth of a gram. Stated in other terms, one carat is equivalent to 0.007 ounces, and there are approximately 2,265 carats to a pound. The average person should have no problem locating a one -carat diamond worth \$2,000 for purchase, although much more expensive stones can be found. Even with a \$2,000 average per-carat price, one pound of diamonds—or 2,265 carats —is worth \$4,530,000.

In contradistinction, gold's value density is much lower. A pound contains 14.5833 troy ounces. At \$1,788 per troy ounce, the value of one pound of gold is 14.5833 (troy ounces) x \$1,788 (price/troy ounce), or \$26,075. Assuming the purchase of relatively inexpensive \$2,000 diamonds as the basis for the calculations , one pound of diamonds is worth \$4,530,000, whereas one pound of gold is worth \$26,074. The ratio of the two assets, \$4,530,000 versus \$26,074, means that diamonds are 172.58 times as value dense as gold. It should be noted that this exercise greatly understates the potential value density of diamonds. 287

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Investors might not understand the importance of the concept of the value density of diamonds, but the idea has been explored thoroughly in popular literature and film. For instance, in the short story titled “The Adventure of the Blue Carbuncle” by Arthur Conan Doyle, Sherlock Holmes investigates a theft of a priceless gemstone from the hotel suite of the Countess of Morcar. The story would have been impossible to write, had the premise been that the equivalent value of gold were stolen: there would be no reason to have such a quantity of gold in a hotel suite. Moreover, Doyle would’ve had to explain the logistics of removing a large quantity of gold from a hotel suite without being observed. Similarly, in the Alfred Hitchcock movie *To Catch a Thief*, a cat burglar steals jewels, not gold. The jewelry theft is the MacGuffin, or plot device, upon which Hitchcock weaves the story based on David Dodge’s novel of the same name. A burglar’s escape by rooftop while carrying a large quantity of gold simply couldn’t happen.

Accepting the value density aspect of diamond investing leads to the logical conclusion that diamonds should be the asset of choice for the circumstance when social cohesion is in question. Social cohesion has been in question many times in history, and it is now unraveling in at least 10% of the world’s nations. Nevertheless, diamonds have yet to attract institutional interest.

This is probably because most investing in the contemporary era is institutional investing, and institutional investing can only exist with sufficient social cohesion for the institution to continue to function. If a financial institution’s investment committee were ever to turn its attention to the potential loss of social cohesion, it almost necessarily follows that the committee also must reflect on the possible inability of the committee and the institution to function. For such a committee considering that possibility, it would essentially be preparing for its own demise, in which case it might have an enhanced interest in a diamond portfolio to access in a genuine state of emergency. That would be much like planning for beneficiaries while simultaneously imagining the demise of the management.

As difficult as it might be to plan for the loss of social cohesion, the great number of breakdowns of the institutional order in many nations just in the first part of the 21st century makes an allocation necessary to an asset that can endure, is universally accepted, is transportable, can be hidden and protected from seizure, and appreciates over the course of time. Such an asset is diamonds.

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Murray's Musings

TRENDS IN HEALTHCARE

According to Statista.com, in the calendar year 2022, there were 4.76 billion retail prescriptions filled in the U.S. The current U.S. population is 334.299 million. This amounts to about 14.24 prescriptions for every human being in the nation. This does not include pharmaceuticals dispensed in hospitals. It does not include pharmaceuticals dispensed in ambulatory care clinics. It does not include so-called over-the-counter drugs sold without a prescription.

This is one of the reasons that healthcare is a growth industry. The proportion of GDP devoted to healthcare is generally increasing. A recent calculation by the World Bank tabulates that in 2019, healthcare spending in the U.S. amounted to 16.8% of GDP. The Statista calculation for 2020 indicates that healthcare spending accounted for 18.8% of U.S. GDP. Obviously, the 2020 figure was impacted by extraordinary measures necessary to deal with the pandemic.

COVID-19 is a worldwide phenomenon, yet the United States is unique even among industrialized nations as for its healthcare spending as a proportion of GDP. Apart from the U.S., the highest such expenditures as a percent of GDP are by the following ten nations, the highest of which is Canada at 12.9%.

Table 1: Healthcare Expenditure as a Percentage of Gross Domestic Product (2020) Country Healthcare Expenditure / GDP Canada 12.9% Germany 12.8% France 12.2% United Kingdom 12.0% Switzerland 11.8% Austria 11.5% Sweden 11.5% Norway 11.3% Netherlands 11.2% Japan 11.1% Source: Statista

According to Statista, Costa Rica spends 7.9% of its GDP on healthcare. Hungary spends 7.3% of its GDP on healthcare. Luxembourg spends 5.7% of its GDP on healthcare. Mexico spends 6.2% of its GDP on healthcare.

One measure, although certainly not the only measure, of the effectiveness of these levels of spending is life expectancy. The World Health Organization (WHO) calculates life expectancy figures for nations annually.

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Table 2: Life Expectancy by Nation (2022) Country Years United States 78.203 Canada 82.847 Germany 80.989 France 83.229 United Kingdom 82.156 Switzerland 84.255 Austria 82.412 Sweden 83.505 Norway 83.393 Netherlands 82.451 Japan 84.820 Costa Rica 77.320 Hungary 74.958 Luxembourg 82.591 Mexico 74.832 Source: World Health Organization

When comparing national expenditures as a proportion of GDP, it is important, for meaningful comparison, to take account of any wide differences in per-capita GDP. One can say that the U.S. spends 2 1/2 times more on health care than Costa Rica, based on spending 18.8% of GDP on healthcare versus Costa Rica spending 7.3%. However, the per-capita GDP of the U.S. is much higher than that of Costa Rica. The estimated per capita GDP of the U.S. in 2022 is \$76,027, while for Costa Rica it's \$13,089. Those figures allow one to calculate the per capita dollar amount of spending on healthcare:

U.S.: $18.8\% \times \$76,027 = \$14,293.08$ Costa Rica: $7.3\% \times \$13,089 = \955.50

That reveals that the per capita expenditure of the U.S. on healthcare is 15.0 times that of Costa Rica. Nevertheless, the U.S. life expectancy is 78.203 years and the life expectancy of Costa Rica is 77.320 years. Viewed on a per capita basis, the vast excess of U.S. expenditure does not appear to be productive.

A problem with this type of calculation is that the various data sources do not agree on the life expectancy figures. The aforementioned life expectancy figure for Costa Rica is compiled by the World Health Organization. The Population Division of the Department of Economic and Social Affairs at the United Nations calculates the life expectancy of Costa Rica at 79 years. Macrotrends.net lists it at 80.56 years. The peer reviewed journal known as Proceedings of the National Academy of Sciences is the source of the macrotrends.net life expectancy figure.

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In any case, in the U.S., national healthcare expenditure routinely expands more rapidly than GDP. According to CMS.gov (Centers for Medicaid and Medicare Services), the annual spending figures for the past 10 years are as follows.

Table 3: U.S. National Healthcare Expenditure Year Annual Expenditure (in trillions) 2011 \$2.676 2012 2.783 2013 2.856 2014 3.002 2015 3.165 2016 3.207 2017 3.448 2018 3.604 2019 3.757 2020 4.114 2021 4.256 Source: CMS.gov

The compound annual expansion rate of spending is 4.75%. It is, of course, needless to say that the U.S. population does not grow at 4.75% per annum, and the GDP does not grow at that rate, either. The logical consequence of the trend is that the healthcare expense burden on the economy will become increasingly heavy.

In common investment parlance, the healthcare industry is frequently used as a synonym for pharmaceuticals. However, the U.S. healthcare sector is far more diverse. This is evident in the sectoral diversification of the largest healthcare ETF, known as The Health Care Select Sector SPDR Fund (XLV). This ETF is divided into five sectors, as detailed in the accompanying table.

Of these, biotechnology is an interesting category. It is obvious that the sector commands significant market capitalization. However, it does not produce large quantities of revenue. In any analysis of healthcare expenditure, it is more meaningful to determine the respective shares of that expenditure exclusive of biotechnology due to its limited revenue relevance. If this were done, the proportionate share of each of the four sectors would be as follows. While the pharmaceuticals sector accounts for somewhat over a third of the total, the balance is made up of health care providers and services, and two separate categories of companies providing health care equipment and supplies, and analytical tools, instruments, clinical trial services and research services.

Table 4: XLV Industry Allocation Sector Weight Health Care Equipment & Supplies 18.77% Health Care Providers & Services 22.36% Life Sciences Tools & Services 12.95% Pharmaceuticals 30.28% Biotechnology 15.64% Source: ssga.com 292

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Table 5: XLV Industry Allocation Excluding Biotechnology Sector Weight Health Care Equipment & Supplies 22.25% Health Care Providers & Services 26.51% Life Sciences Tools & Services 15.36% Pharmaceuticals 35.89% Source: ssga.com

In order for this system to continue to expand at its historical rate, the typical human being would have to devote more time to medical examination and analysis in order to determine if medical intervention is required and, should such intervention be required, the degree of intervention and type of treatment. Ultimately, the growth proposition for health care spending can be reduced to the internet/technology growth proposition that has previously been propounded in Compendium discussions: that in order for the historical rate of industry growth to be sustained, the average person must spend more time on the internet. In the medical sphere, the average person would have to spend increasing amounts of time interacting with the medical system.

The OECD (Organisation for Economic Co- operation and Development) maintains data on the length of the average hospital stay for most nations. It will be recalled from a prior table that the U.S. spends more on healthcare, as a proportion of GDP, than other industrialized nations. The following table compares the length of the average hospital stay in the U.S. versus the other industrialized nations.

Using length of hospital stay as a metric indicates that the U.S. uses its healthcare system less intensively than all nations on the list, excepting Sweden and the Netherlands. Even in these two instances, the intensity of hospital use is only marginally less than that of the U.S. Nevertheless,

the U.S. spends a far greater proportion of its GDP on healthcare than does either Sweden or the Netherlands. The relevant figures are that Sweden spends 11.5% of its GDP on healthcare, the Netherlands spends 11.2% of its GDP on healthcare, and the U.S. spends 18.8%. If one were to use life expectancy as a measure of healthcare outcomes, the U.S. obtains a less satisfactory result than either Sweden or the Netherlands, or even any other industrial nation.

If U.S. healthcare expenditures were to remain at 18.8% of GDP and not rise beyond this point, it must logically follow that the healthcare industry would be growing at the rate of GDP. This would be the nominal figure and not the real figure. Thus, if real GDP were to Table 6: Length of Average Hospital Stay (2021) Country Days Canada 7.7 Germany 7.4 France 5.5 United Kingdom 6.2 Switzerland 6.9 Austria 6.4 Sweden 5.3 Norway 5.9 Netherlands 5.2 Japan 16.4 Costa Rica 6.8 Hungary 5.4 Luxembourg 7.5 Mexico N/A United States 5.4 Source: OECD 293

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average 2% growth per annum and inflation were held at the level of 2%, and the healthcare industry maintains its current share of GDP, it would be growing at a 4% rate.

At the time of this writing, The Health Care Select Sector SPDR Fund (XLV) trades at 16.32x earnings and 4.67x book value, and the 3- 5 year estimated annual growth rate, as stated on the most recent fund factsheet, is 7%.

Those valuations make it fairly obvious that the stock market participants in this sector presume that the current circumstance, which is to say the historical rate of expansion, will continue. Implicitly, that embodies the presumption that the healthcare industry will represent an ever-increasing share of U.S. GDP. It presumes that the general populace will not be troubled by the fact that other nations have a healthcare industry that consumes a much smaller proportion of GDP; that foreign healthcare systems appear to be used more intensively by the population and with a far lower financial burden than is the case in the U.S. Moreover, the general populace will not be troubled that the outcome, at least as measured by life expectancy, however imperfectly, is lower in the U.S. than in other industrialized nations.

The alternative view to that of the investment community as expressed in the valuation of the health care sector, is that the current circumstance is unsustainable and that the unsustainability will have negative consequences for equity indexes, the U.S. GDP, and the national government solvency. In any case, the consensus does appear to be untroubled, an additional measure of that sentiment being the \$40.987 billion in assets under management of XLV, as of the most recent reckoning.

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HORIZON RESEARCH GROUP

June 2011

Horizon Kinetics LLC 470 Park Avenue South, 4th Floor
New York, NY 10016

Murray Stahl Steven Bregman
Thérèse Byars Derek Devens Peter Doyle
Michael Gallant Matthew Houk David Leibowitz Eric Sites Fredrik Tjernstrom Steven Tuen

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The Equity Yield Curve

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Introduction

Horizon Kinetics LLC (“Horizon Kinetics” or “Horizon”) began writing about the theory of the “equity yield curve” in the early 1990s and has since successfully applied it as a cornerstone of our investment philosophy. It validates the virtues—and elucidates one of the underlying enabling principles—of long-term investing. Very occasionally, for reasons elaborated upon below, the equity yield curve can be plotted, much like the more familiar bond yield curve. In our experience, the equity yield curve generally has a positive slope, again like the bond yield curve does most of the time, but is decidedly steeper. This suggests that investors require a higher rate of return from a security that may be unlikely to gain in the short term, even if it may be almost certain to gain significantly over a longer time period.

This is specifically observed in the trading behavior of the institutional investment community, which manifests an overwhelming predilection for securities perceived as having the potential for positive relative returns within a discrete time period—generally one year, particularly prior to the end of a given year. The reason for such a short-term focus is quite logical. The compensation structure of the investment management industry is broken down into artificial time horizons. The typical hedge fund manager’s bonus, or incentive fee, is based on year-end results. Thus, securities not perceived as having the ability to perform within that time period are typically avoided, even if their long-term potential may be extraordinarily attractive. In other

words, money managers in general have an asymmetric aversion to underperformance versus outperformance, and economic incentives lead them to focus on short-term results. Hence, a future event that will have a decidedly positive impact on the price of a certain security, undisputable though it may be, has limited utility to the professional manager if that event is expected to take place beyond his or her artificial time horizon. Accordingly, the value of that event is heavily discounted by the market.

It has been observed that the equity yield curve flattens out over the last few months prior to the expected event, at which point the security provides returns commensurate with Treasury bills. Ergo, the market is very efficient when very little time remains before the triggering event, but increasingly inefficient when the event is 12 months or more away, or increasingly indeterminate. Fund managers' reluctance to purchase such securities is expressed in the discount rate, which in some cases exceeds 35%. To take advantage of this aberration one merely has to extend one's time horizon. From the professional manager's perspective, this requires the willingness to take on time risk.

In the overwhelming majority of cases, equity valuation is highly subjective. In other words, the elements of valuation (such as the rate of increase in earnings, the normalization of a disrupted business activity or the appropriate valuation multiple) are not certainties but rather are subject to differing estimations and assessments. It is all arguable. After all, reasonable minds may differ. As a consequence, it is understandable that the notion of an equity yield curve, with implicit future dates and values, does not find its way into common discourse, and certainly not as a working model. Occasionally, though rarely, these factors do converge, at least with as much certainty as can be had in the investment world. The following examples, presented in order from most assured and quantifiable to least, show how certain securities can be employed to actually plot the equity yield curve.

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Johns Manville (circa 1990)

Johns Manville, a major manufacturer of packaging and buildings products (including, unfortunately, fiberglass insulation), was the first major asbestos liability-related bankruptcy. It emerged from a 6-year bankruptcy proceeding in 1988 with a reorganization plan that created two trust funds to compensate claimants. The plan ceded ownership to the trusts of 80% of the shares and 20% of annual profits, so that the claimants were in the unusual position of owning the company, with a natural self-interest in its financial success.

More importantly, the court permanently enjoined any further asbestos-related cases against the firm, removing this legal liability from the valuation equation. The company did not historically exhibit unusual business or credit risk: it was profitable before the 1982 bankruptcy and remained so, with a 1989 interest coverage ratio of 3.7x, a 45% debt/capitalization ratio, operating cash flow of \$413 million vs. long-term debt of \$802 million and a current ratio of 1.5x. In 1990, the company reached an agreement with the trusts to pay dividends to common shareholders.

There did exist, however, an unusual Manville security, which is the subject of this example. This was the Johns Manville preferred, which had a cumulative preferred dividend of \$2.70 per share and a \$25 liquidation preference. An odd feature of this preferred was that the first dividend was not payable until March 1994, well over three years into the future. Furthermore, it was not callable until March 1994, although it appeared that the company would most likely wish to do so since the stated dividend yield, at over 10%, was well above market rates.

Yet, as of year-end 1990, with over three years to a known date and future value, the preferred traded at \$9, which would provide a 38% compound price return to the March 1994 dividend commencement and call date, and a contingent (if not called) 30% dividend yield. The offered returns would seem excessive indeed, considering, by that time: the substantial resolution of the legal risks, the unique circumstances that the interests of the claimants were fully aligned with those of the management and minority shareholders (maximization of profits, share price and liquidity), the financial health of the company and the capacity to redeem the preferred at \$25 per share and refinance the 10.8% stated yield at lower rates. The following prices for the preferred shares were observed:

Date Price Terminal Price Call date Years Until Call Date

12/31/1990	\$9.00	\$25.00	3/1/1994	3.17	12/31/1991	\$16.50	\$25.00	3/1/1994	2.17	12/31/1992	\$20.75	\$25.00	3/1/1994	1.16	12/31/1993	\$24.88	\$25.00	3/1/1994	0.16
------------	--------	---------	----------	------	------------	---------	---------	----------	------	------------	---------	---------	----------	------	------------	---------	---------	----------	------

Source: Bloomberg, Horizon analysis

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Therefore, in this example, the equity yield curve can be plotted as follows:

Hypothetical Equity Yield Curve Johns Manville Preferred

The chart above should be read like a bond yield chart, indicating that with three years left before the dividend date, the highest potential annualized return of 38.1% could be realized, whereas with just a few months before the first dividend payment (closer to "0" on the horizontal axis), the annualized return had declined to just 3.1%, almost identical to the yield on a three-month Treasury bill at the time.

0.0% 5.0% 10.0% 15.0% 20.0% 25.0% 30.0% 35.0% 40.0% 0123 Expected Annualized Return
(%) Years Until Call Date 300

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Cellular Communications Inc. (circa 1991)

Cellular Communications, Inc. announced in July 1990 that it would combine its cellular interests in Ohio and Michigan in a joint venture with Pacific Telesis, as the first step in a phased buy-out by Pacific Telesis. Cellular Communications owned and operated one of the more concentrated, efficient cellular networks in the nation and was among the first to achieve positive operating cash flow in this sector. Pacific Telesis was one of the seven Regional Bell Operating Companies created from the 1984 split-up of the AT&T holding company. The merger agreement between them was restated in December 1990 and the merger closed on August 1, 1991.

Under the terms of the agreement, Cellular Communications shareholders received redeemable preference shares in exchange for their common shares. Beginning January 28, 1992, any of these preference shares which were traded in the open market would automatically convert to Series A stock. Pacific Telesis had pledged to purchase 10 million of the 40 million then-outstanding preference and Series A shares on October 27, 1995, at \$60/share, with priority given to the participating preferred shares. Therefore, a holder of preferred shares on January 28, 1992 could anticipate a proration of 25% (no outstanding A shares, 40 million preference shares) if all preferred shareholders held their positions until October 27, 1995, meaning that, on average, 25% of their shares would be purchased by Pacific Telesis at \$60/share, while they would continue to hold 75% of their shares in preferred stock. At year-end 1993, 23 months following the announcement, 8 million of the original preference shares had traded and therefore converted to Series A stock. As a result, 32 million preference shares remained. Holders of the preference shares could now anticipate a proration of 31% if they were to hold their position until October 27, 1995. During the 25 months remaining until October 1995, if another 27.5% of the original shares traded hands each year, the proration for preference shares would be 100%. As a reference point, annual share turnover among domestic telephone companies at that time fell within a range of roughly 21% to 69%.¹ Turnover of the preference shares in the first 23 months was consistent with this range. Therefore, a 100% proration at the end of the 3.75 year period, which is to say a full \$60/share payoff, was an entirely reasonable expectation for investors who were willing to hold onto their preference shares. The greater the propensity of others to trade their preference shares, the greater the likelihood that those who held their preference shares until October 27, 1995 would achieve 100% proration.

Given the 3.75-year forward purchase contract by an investment grade company (Pacific Telesis), and conservative assumptions applied to a range of proration scenarios, the extended return horizon would appear to have induced a sufficiently high discount rate to offer equity yield curve return possibilities. As a base-case example, a 10% compound return from the January 28, 1992 market price of \$29.20 would have resulted if only 50% of the preference shares had traded over the almost 4-year period (a 50% proration), and the unredeemed stock did not appreciate at all. In reality, the stock appreciated more than 60%.

A somewhat more realistic scenario, with 75% cumulative turnover, which would require only 20% annual turnover, would achieve 100% proration for those who held their preference shares and a 21% compound return. In an alternative scenario, 5% annual share price appreciation and a 50% proration for preference shareholders would result in an expected annualized return of 13%.

¹ Based on AT&T, Bell Atlantic, Century Tel, Cincinnati Bell, Sprint and Telephone & Data Systems 301

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Scenario Analysis for Preference Shareholders

Beginning Market Price Price Appreciation (%) Ending Market Price Cumulative Turnover
Proration % * Payout Price Years Until Payout Compound Annualized Return (%) \$29.20 -
\$29.20 50% 50% \$60.00 3.75 10.6% \$29.20 - \$29.20 75% 100% \$60.00 3.75 21.2% \$29.20 5%
\$35.06 50% 50% \$60.00 3.75 13.1% * Proration rate applies to holders of preference shares.
Source: Horizon research, Bloomberg

As noted above, actual stock price appreciation was much greater than the conservative assumptions used in the above scenario analysis. For the case where 100% proration is achieved, we can plot the following yield curve using observed market prices for illustrative purposes:

Hypothetical Equity Yield Curve Cellular Communications Preferred

Date Price Terminal Price Payout date Years Until Payout

1/28/1992 \$29.20 \$60.00 10/27/1995 3.75 12/31/1992 \$34.75 \$60.00 10/27/1995 2.82
12/31/1993 \$46.75 \$60.00 10/27/1995 1.82 Source: Bloomberg. Horizon research

10.0% 12.0% 14.0% 16.0% 18.0% 20.0% 22.0% 24.0% 234 Expected Annualized Return (%)
Years Until Payout 302

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Pacific Gas & Electric Preferred Shares (circa 2001)

In 2001, Pacific Gas & Electric (“PG&E”), California’s largest electric utility, stopped paying preferred dividends. This was the result of a voluntary bankruptcy to force a resolution of a flawed plan of deregulation that led to the California energy crisis catalyzed by the Enron debacle, but with damaging consequences that the state government did not wish to acknowledge. While preferreds ordinarily have little value in bankruptcy, this situation was atypical.

At the time of Horizon’s purchase recommendation in May 2002, there were two competing plans of reorganization—one by the company itself and the other by the California Public Utility Commission—which were to be put to creditor vote in June. Both plans, despite major strategic differences, were in agreement on two issues: they were structured to 1) achieve an investment grade rating for PG&E, and 2) result in no impairment to either the company’s debtors or preferred shareholders. This meant that interest and dividends would continue to accrue. These goals were supported by the inherent economics of the PG&E operating business, which was not injured and which produced copious cash flow. Accordingly, the credit risk issue for the preferreds of the PG&E subsidiary was essentially resolved. The company was expected to emerge from bankruptcy in January 2003.

Nine preferred series in total were recommended by Horizon, and the basic return profile included one known variable and two variables subject to reasonable estimation. The known variable was the amount of dividend arrearages. These cumulated to about 17% of the trading prices of the preferreds as of May 2002—the trading prices themselves were about one-third below face value.

Were the company to exit bankruptcy in June 2003, the annualized return from the dividend payments would have been approximately 25%.

This high indicated yield was due in part to uncertainty about the second variable, the time frame for exiting bankruptcy. If an additional six months were required, say until January 2004, the annualized return from the dividend arrearages would have been approximately 16%.

However, even in the case of a January 2004 payoff date, the expected return would be higher than that implied merely by the dividend arrearages, since the trading prices would also be expected to recover to some degree. Using as a proxy the yields at which a basket of preferreds of investment-grade utilities then traded (since the PG&E preferreds would regain investment grade status), a \$22-\$25 price range would be expected, representing an annualized total return of approximately 58% from the \$14-\$18 beginning price range assuming a January 2003 bankruptcy emergence, or 21% assuming a January 2004 bankruptcy emergence.

Securities with readily quantifiable fixed-income characteristics such as the PG&E preferreds, coupled with an underlying investment grade credit, should not be priced at, roughly speaking, a 1- to 2-year base yield as high as 14% to 21%. In this instance, while the time horizon was not exceptionally long, it was somewhat indeterminate and was reasonably expected to extend beyond the conventional 12-month investment horizon of most investment managers. Below are the actual observations:

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Series Price on May 21, 2002 Price on April 2, 2004 Arrearages Annualized Return

6.0% PCGpfA \$ 18.00 \$ 29.73 \$ 4.875 30.8% 5.5% PCGpfB \$ 16.00 \$ 28.26 \$ 4.688 35.6% 5.0%
PCGpfC \$ 14.50 \$ 25.51 \$ 4.0625 35.3% Source: Bloomberg, Company reports

In the actual event, emergence from bankruptcy occurred in April 2004. Using a basket of these preferred stocks, the return profile from May 21, 2002 – April 2, 2004 can be plotted below. The annualized return for the basket was almost 35%, as the shares appreciated more than modeled (to approximately their face value). The yield curve in this instance is not entirely upward sloping because the payoff date was uncertain—from the perspective of an institutional buyer in May 2002, it could have ranged from well within the standard investment horizon (8 months) to well beyond (two years or more, given the complexities of combined bankruptcy and utility regulation proceedings). Accordingly, a time-risk-sensitive investor would have priced the payoff as if long-dated; as events unfolded, the actual payoff occurred earlier than the investor might have anticipated, resulting in a higher annualized return, albeit the same dollar amount.

Average Price Basket of Pacific Gas and Electric Preferred Stocks

Source: Bloomberg, Horizon research

10.00 12.00 14.00 16.00 18.00 20.00 22.00 24.00 26.00 28.00 Preferred Stock Price Years Until
Emergence From Bankruptcy Actual Forecasted 304

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Plotted as an equity yield curve, the results are as follows:

Equity Yield Curve

Pacific Gas and Electric Preferred Shares

This case represents somewhat of an anomaly in terms of the equity yield curve as comparatively little of the total return (3.4%) was realized in the first seven months. It should be noted that during this time period, the deterioration of the electric utility industry continued, and PG&E's common shares declined by more than 33% between May 21, 2002 and December 31, 2002. Despite that, the preferred shares appreciated during that same time period, particularly in the ensuing six-month period. Again, the uncertainty regarding the length of the bankruptcy proceedings may have contributed to this.

0.0% 10.0% 20.0% 30.0% 40.0% 50.0% 60.0% 0.0 0.3 0.8 1.3 1.9 Expected Annualized Return
(%) Years Until Emergence From Bankruptcy Actual Forecasted 305

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Utilities Companies in the 2003-2007 Time Frame

In late 2002, a meaningful swath of the electric utility industry was on the brink of bankruptcy. These holding companies had involved themselves with Enron in one way or another, created or vastly expanded energy trading subsidiaries and operated them not for the purpose of hedging the legitimate input/output costs and prices of their regulated business, but to pursue standalone trading profits. Many sold long-term power contracts during a rising-price environment and became highly exposed before these prices declined precipitously.

The virtue they shared was significant positive cash flow from their core regulated utility subsidiaries. Additionally, since at the end of the day electric utilities are essentially a public good with a regulatory structure designed to ensure their stability, it appeared that the banks did not want to foreclose on and operate these complex and asset-intensive businesses. Rather, the banks would continue to support them if such support would enable their recovery and assure the integrity of their debt. This thesis was expounded on in many Horizon research reports in late 2002 and early 2003, and various utility securities were acquired for most of our strategies.

Our analysis was that the profitability of the industry was temporarily masked by the debt and the uneconomic supply contracts assumed during the Enron debacle and the ensuing collapse in energy prices. Recovery merely required the temporary cessation of dividend payments to common shareholders, so that the companies' substantial cash flow could be deployed toward debt repayment. During the 1990s, as a group, these companies had achieved net profit margins of 8.6% with low variability, and they had traded, on average, at 1.57x book value, compared to substantial discounts to book value at the time of our recommendation. Within five years, we estimated, the balance sheet debt ratios would return to normal and these companies could recover their investment grade debt ratings. This analysis was not groundbreaking, and was no

doubt obvious to the many Wall Street utility analysts. However, it was not of great interest, since the time horizon for realization of this dynamic was at least three years—which on a practical basis for an institutional analyst, or for the analyst’s institutional customers, might as well be a decade. In fact, the net profit margin for the group reached 7.5% in 2007 and 8.7% in 2008, slightly surpassing the average for the 1990s. The price/book ratio value for these companies had advanced to 2.3x, surpassing the average of the 1990s, and indicating the return to “normalized” conditions was completed. The following chart represents the return achieved from January 1, 2003 – December 31, 2007 for a basket of electric utility companies:

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Hypothetical Equity Yield Curve
Selected Utilities Companies* in the 2003-2007 Time Frame

- Selected utilities companies refers to a basket of stocks consisting of Aquila Energy, Centerpoint Energy, CMS Energy, Sierra Pacific Energy, Reliant Energy, TXU, El Paso and Williams Companies. The latter two, while natural gas transmission companies, were also rate-of-return regulated companies, regulated by the Federal Energy Regulation Commission.

As the chart illustrates, the highest return was realized in the first two years of the recovery of the utilities companies (i.e. when the time horizon to the “event” was 4-5 years); eventually the return stabilized around 20% for the last three years of the period and finally declined to 18.7% in the last year.

15.0% 20.0% 25.0% 30.0% 35.0% 40.0% 12345 Expected Annualized Return (%) Years
Remaining Until Recovery 307

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Allegheny Energy (circa 2002)

Allegheny Energy was a company-specific case study of the aforementioned energy industry phenomenon. Allegheny Energy, an Atlantic coast electric utility, came to the verge of bankruptcy towards the end of 2002/early 2003. This was mostly attributable to the disastrous results of the company’s non-regulated energy trading division in the wake of the Enron scandal. Allegheny’s circumstances were immeasurably improved on February 25, 2003, when its bankers agreed to a completely restructured \$2.4 billion credit facility. The credit agreement essentially removed the pending insolvency risk, permitting the sort of predictive quantitative valuation exercises to which regulated utilities typically lend themselves.

Allegheny was then trading at about \$5.00 per share (versus a high of \$43.86 in April of 2002) against a fully diluted book value of \$12.49 per share and expected forward earnings per share of \$1.21, based on conservative assumptions (as a regulated electric utility, there are certain relatively reliable assumptions, such as the allowed return on regulatory capital, which would have exceeded 10%). In a report dated March 24, 2003, Horizon was of the opinion that the stock was deeply undervalued, given a trading price of less than 5x forward (2003) earnings and approximately 40% of book value. Allegheny had traded at an average 1.83x book value during the 1993-2001 period and generated an average net profit margin of 10.3%, both figures with a fairly low degree of variability. Also, as discussed in the prior segment, the utility industry traded at an average book value multiple of 1.57x during the 1990s, indicating that not only was this an extreme valuation relative to Allegheny’s own historical experience, but also compared to competitors in the sector.

Since bankruptcy was no longer a consideration in early 2003, the company’s core business was intact and the problematic energy trading operations had ceased, the question was only how long it would take to regain normalized net profit margins and price/book value ratio.

The time frame for achieving normalized net profit margins was modeled by Horizon at approximately three years. The major detraction from net income was interest expense on the debt incurred during the company’s financial crisis. Contrary to common misconception, electric utilities are highly profitable enterprises; they are, however, slow-growth companies, for there is little room for expansion. Accordingly, they ordinarily distribute the great majority of their income as dividends rather than reinvesting them. The earnings recovery path for Allegheny was simply one of forgoing the common dividend for a period of time, freeing the substantial cash flow for the purpose of debt reduction. This would decrease the interest expense and in turn, raise earnings. For Allegheny, normalized earnings would be around \$300 million per year (versus a March 2003 market capitalization of \$665 million and net debt of \$4.6 billion), as had been the experience in the late 1990s, and, indeed, its core business had continued to

generate strong cash flow throughout the crisis. It would require, prospectively, approximately 3-4 years to reduce the debt-to-equity ratio from 3.0x to around 1.5x, which would be consistent with the typical investment grade utility. For plotting an equity yield curve, a 1.5x book value target price ($1.5 \times \$12.49 = \18.73) and 3-year time frame could be assumed. The derived 3-year equity yield was 52.5%.

As for the actual results, by 2006 the net income level of \$300 million had been regained and the share price finished that year over \$43. By 2007, Allegheny's net profit margin was 12.5%, compared to an average of 10.4% in the 1990s, indicating that the 'normalization' of its business had been completed. Furthermore, its price/book ratio had expanded to 3.47x relative to its 1.52x average during the 1990s. Thus, as the company's income normalized over the course of less than 308

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four years, the shares increased 8.5-fold. The observed annualized returns over the four-year period are plotted below:

Annualized Return of Allegheny Shares,
January 2003-December 2006

Based On Allegheny's Actual Returns 2003-2006.

In this case, the compound annual growth rate was 71% over the 3.7-year period. While the steady upward progression of the stock price might make one believe that the annual returns were steady as well, it turns out that most of the returns were once again available in the first year of the turnaround when the stock price appreciation approached 120%.

5.0% 15.0% 25.0% 35.0% 45.0% 55.0% 65.0% 75.0% 85.0% 95.0% 105.0% 1234 Annualized
Return (%) Years to Business "Normalization" 309

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Groupe Bruxelles Lambert SA (circa 2001)

Groupe Bruxelles Lambert ("GBL"), as detailed in our February 14, 2001 Contrarian Research Report, held a 22% stake in privately-held German media giant Bertelsmann, in addition to a number of publicly-traded investments. In fact, at the time of the report, the company traded almost exactly at the value of its underlying public holdings. Therefore, in essence, no value was placed on the Bertelsmann holding, even though this position could be conservatively valued above \$10 billion, or more than GBL's market capitalization at the time. Bertelsmann was perhaps the largest publisher in the world (a partial list of properties included: in books, Random House, Ballantine, Fodor's, Knopf, Modern Library and Springer, a publisher of science and mathematics texts; in magazines: Der Stern, Family Circle, McCall's, Parents and YM), the second-largest music distributor (including record labels BMG, Arista and RCA Music) and the largest CD manufacturer in the world.² The company also had significant interests in global television programming, broadcasting (it was the largest broadcaster in Europe), multimedia production and printing. Comparable media companies of this scale included Viacom and AOL Time Warner, but unlike those companies, Bertelsmann did not have a levered balance sheet. Even so, the valuation of these other media companies provided a reasonably consistent and conservative set of benchmarks by which to value Bertelsmann. Based on comparables, Horizon estimated Bertelsmann's valuation at the time was estimated to be between \$57 and \$78 billion, which indicated that GBL's 22% stake was worth around \$12.5-\$17.2 billion, relative to its then market capitalization of \$6.6 billion (the company also had over \$300 million in net cash, for an enterprise value of \$6.3 billion).

Additionally, GBL held the right to list its Bertelsmann shares on a public exchange no later than year-end 2006, or within 5.8 years of the date of our report, which was the triggering event in this case. The modeled annualized return over the ensuing 5.9 years (in dollar terms) was approximately 19.5% (from the \$6.6 billion market capitalization to \$18.8 billion, inclusive of the comparable company-based public market value of the Bertelsmann stake). This assumed no growth in the underlying businesses, merely the realization of the value of the Bertelsmann stake. In the actual event, the realized annual return was only 17% because Bertelsmann, which did not wish to be forced into a public listing, ultimately made a private arrangement with GBL to repurchase its stake. Accordingly, a public valuation was never achieved. Nevertheless, the equity yield curve was manifest in this investment, and due to the global reach of the company should be compared to the S&P 500's performance, which was flat over the same time period.³

2 Source: Company reports. 3 Source: Bloomberg. 310

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Summary

While the case studies above were selected to demonstrate the concept of the equity yield curve, many other examples exist in the market. For most equity securities, the timing of potential future valuation changes or payouts is not predictable, as it was in some of the cases presented here, but the underlying theme is the same. Given the lackluster and variable returns of many major stock index components over the last decade, the merits of long-term investing have been called into question. However, as has been shown here, if one selects securities carefully, relying on thorough analysis, positive returns may be realized if one is willing to extend the time horizon and if such extension results in an attractive discount rate, as suggested by the theory of the equity yield curve.

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Part VIII — Video and interviews

Editorial rule for this edition: interviews include **Murray Stahl's answers only**, with publication context in the headnote. Video chapters use **official Horizon Kinetics pages** and Vimeo embeds where available.

Video — official pages and embeds

Source list: `_stahl_video_embed_urls.txt` in the vault root.

Event / title (short)	Page (Horizon Kinetics)	Vimeo embed (where captured)
Murray Stahl at the Project Punch Card Conference	https://horizonkinetics.com/videos/murray-stahl-at-the-project-punch-card-conference/	https://player.vimeo.com/video/826950692
Murray Stahl at Hedgeye Macrocosm — Big Opportunities in Global Market (May 5, 2023)	https://horizonkinetics.com/videos/murray-stahl-at-hedgeye-macrocosm-big-opportunities-in-global-market-may-5-2023/	https://player.vimeo.com/video/827365338
Murray Stahl at Hedgeye Fall 2024 Investing Summit — Inflation and Hard Asset Investing (Oct 9, 2024)	https://horizonkinetics.com/videos/murray-stahl-at-hedgeye-fall-2024-investing-summit-inflation-and-hard-asset-investing-october-9-2024/	https://player.vimeo.com/video/1020004863

Event / title (short)	Page (Horizon Kinetics)	Vimeo embed (where captured)
Murray Stahl on Bloomberg <i>Inside Active</i> — “Horizons: Stahl on Letting Winners Run”	https://horizonkinetics.com/videos/murray-stahl-on-bloomberg-inside-active-horizons-stahl-on-letting-winners-run/	(No static Vimeo URL in the vault list; use the page player or capture embed after load.)

Transcript placeholders

For each row above, add a subheading and transcript **after** you transcribe audio. Suggested stub:

```
### [Title] – transcript (Stahl only)

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Print interviews and profiles

Assign from your non-PDF sources (e.g., *GuruFocus*-style Q&A, magazine profiles). Each piece gets the same headnote pattern as numbered chapters: outlet, date, interviewer, URL, and note on cuts (answers only).

Done when every embed row has either a transcript file linked here or an explicit “deferred” note with reason.

Subject index (stub)

Add entries as you draft Parts II–VII. Suggested groupings:

- Artificial asset class / passive / index concentration / Russell / float / dividend ETFs / international overlap → Ch. 9–13
- Croupiers (exchanges, clearing, liquidity, volume) → Ch. 14–17
- Inflation, royalties, real assets
- Bitcoin, monetary architecture
- Exchanges, MIAX, ICE, private markets
- Japan, JAPN, geography
- Oil services, energy
- Diamonds, crisis wealth

Each line can later link to chapter anchors in the exported master.

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565063	thematic_essay	20230319230440_ac600d0_Bitcoin-and-Legal-Tender-December-2017.pdf
1048576	commentary	20230326191840_163baf82_Q4-2022-Commentary_FINAL.pdf
2302317	commentary	20230504220203_689b5ef1_Q1-2023-Commentary_FINAL.pdf
224465	thematic_essay	20230523111445_2def7d8d_COEXISTENCE-OF-FIAT-AND-CRYPTOCURRENCIES.pdf
1377194	commentary	20230531095812_163baf82_Q4-2022-Commentary_FINAL.pdf
161806	core_musings	20230531095935_91881140_Spin-Off-Compendium-October-2022_The-Importance-of-the-Oil-Service-Industry.pdf
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2536764	commentary	20230606053316_30e9e19f_Q4-2021-Quarterly-Review_FINAL.pdf
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1048576	commentary	20231201030746_a0b21484_Q3-Commentary-FINAL.pdf
178026	thematic_essay	20231201032448_f9d677a5_VII_Fertile-Ground_July-2023.pdf
186988	letters	20240114124053_9a65a3e7_Horizon-Kinetics-Letter-from-Founders_Jan-2024_APPROVED.pdf
1048576	commentary	20240303121118_dfd39310_Q4-2023-Commentary_FINAL.pdf
1048576	commentary	20240303125629_3e1613ec_Q3-Commentary-FINAL.pdf
1672041	commentary	20240308002456_dfd39310_Q4-2023-Commentary_FINAL.pdf
621204	thematic_essay	20240329003752_1cf4eac6_VII_Worth-The-Time_Stahl_Feb-2024.pdf
3098088	commentary	20240509222013_1402e3d3_Horizon-Kinetics-Q1-2024-Commentary.pdf
1048576	commentary	20240522092508_1402e3d3_Horizon-Kinetics-Q1-2024-Commentary.pdf
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3998156	commentary	20240808131536_e5f35473_Q2-2024-Horizon-Kinetics-Commentary_FINAL.pdf
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3339037	commentary	20250528193622_11f6c5ca_Horizon-Kinetics-Q4-2024-Commentary_Final.pdf
2420283	commentary	20250823223649_d0492631_Horizon-Kinetics-Q2-2025-Commentary_Final.pdf
704893	thematic_essay	20250914113336_351813fb_JAPN-Portfolio-Highlight-6544JP-Japan-Elevator-Service-Holding_Aug 2025_FINAL.pdf
3930765	commentary	20251031062933_a3f8a8eb_Horizon-Kinetics-Q3-2025-Commentary-Final.pdf
6805699	thematic_essay	830350067-Compilation-of-Murray-Stahl-s-Writings.pdf
319480	letters	BCDF-2023-Annual-Letter.pdf
288451	letters	BCDF-Annual-Letter-2022-Final-APPROVED.pdf
309731	letters	BCDF-Annual-Shareholder-Letter-2025-Final.pdf
265573	thematic_essay	BITCOIN-CO-EXISTING-WITH-FIAT-CURRENCY-AND-THE-BANKING-SYSTEM_Nov-2020_FINAL.pdf

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360902	thematic_essay	Bitcoin-newsletter_October-2017.pdf
297920	thematic_essay	CFTC-Testimony_Feb-2018_Final.pdf
370094	thematic_essay	CRYPTOCURRENCY-BY-THE-NUMBERS_November-2025_Final.pdf
192490	core_musings	Contrarian-Compendium-December-2022_Diamonds-as-an-Investment.pdf
1201549	research_report	Credit-Acceptance-Corp-Devils-Advocate-Report-December-2-2019.pdf
210279	thematic_essay	Cryptocurrency-Mainstream-Legitimacy_A-Brief-Survey-of-Institutional-Acceptance_April-2020
352227	thematic_essay	Cryptocurrency-and-Exchanges_July-2025_final.pdf
35230	thematic_essay	December-2015_Cryptocurrencies-An-Emerging-Asset-Class.pdf
189684	core_musings	Devils-Advocate-Compendium-February-2023_Trends-in-Healthcare.pdf
181945	core_musings	Devils-Advocate-Compendium-November-2022_Bitcoin-vs.-Bonds.pdf
170872	core_musings	European-Contrarian-Compendium-January-2023_Wealth-and-Crisis_The-Role-of-Diamonds-in-Asse Allocation.pdf
405929	letters	Founders-Letter-2026_Final.pdf
875014	thematic_essay	GIRO_4.24.26_His-stream-of-mind-was-perpetual.pdf
523135	thematic_essay	GIRO_5.28.21_Reveille-in-Japan.pdf
547114	thematic_essay	GIRO_9.13.24_Inflation-has-legs.pdf
216002	letters	HK-Letter-from-the-Board_April-2026_Final.pdf
334310	letters	Horizon-Kinetics-Founders-Letter-2025_Final.pdf
3308027	commentary	Horizon-Kinetics-Q1-2026-Commentary-Final.pdf
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3339037	commentary	Horizon-Kinetics-Q4-2024-Commentary_Final.pdf
4155467	commentary	Horizon-Kinetics-Q4-2025-Commentary-Final.pdf
441814	thematic_essay	Horizon-Kinetics_Royalty-As-an-Asset-Class_April-2023-4.pdf
384118	letters	INFBN-Semi-Annual-Letter-2023-1.pdf
311082	letters	INFL-2024-Annual-Letter_Final_Feb2025.pdf
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465983	thematic_essay	INFL-5-Year-Anniversary-Update_Jan-2026.pdf
200413	letters	INFL-Annual-Letter-FINAL_2021.pdf
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387482	letters	INFL-Semi-Annual-Letter-2023_APPROVED.pdf
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193657	core_musings	InflationandEconomicThought_Nov2020.pdf
420732	commentary	Interim-Q1-Q2-Commentary-2022-with-TPL-addendum_FINAL.pdf
337915	commentary	NVIR-Semi-Annual-2024-commentary_Final.pdf
3407745	commentary	Q1-2025-Horizon-Kinetics-Commentary_Final.pdf
187558	thematic_essay	Revisiting-an-Old-Friend---The-Thesis-for-a-Defensive-De-Minimis-Investment-in-Bitcoin_Feb
229192	commentary	SPAQ-Semi-Annual-2024-Commentary_Final.pdf
161806	core_musings	Spin-Off-Compendium-October-2022_The-Importance-of-the-Oil-Service-Industry.pdf
188596	core_musings	TheAchievementofDiversificationNovember2020.pdf
51173	core_musings	TheTransformationOfTheFixedIncomeAssetClass.pdf
986642	thematic_essay	Why-JAPN_November-2025_Final.pdf

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Editorial consistency pass (checklist)

Use this file as a **single pass** before treating the manuscript as “frozen.” Check off items as you complete them.

Names, tickers, and institutions

- ☐ Horizon Kinetics / affiliate names match originals in each headnote.
- ☐ Ticker symbols and company names: pick one spelling (U.S. vs U.K.) per entity across chapters; align with the **original publication date** (do not silently modernize tickers that changed).
- ☐ “Murray’s Musings” vs “Murray’s musing” / section headers: match original casing in headnotes.

Citations and cross-references

- ☐ **In-text cross-links:** bond convexity, index concentration, inflation, diamonds, Bitcoin, oil services — add “see Chapter ...” only where it aids the reader (optional for v1).
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- ☐ One dash style (em dash vs hyphen) per chapter after paste-from-PDF cleanup.
- ☐ Tables recreated from originals: spot-check numbers against PDF for any chapter you intend to publish commercially.

Part VIII

- ☐ Every video row: transcript or explicit deferral.
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Extraction exceptions

- ☐ book/sources/extraction_missing_yes.txt — should list only items needing **manual** remediation. Current known blockers: password-protected PDF (see file).

Reader PDF / stitch audit (automated + spot-check)

- ☒ **Front table of contents** — export_reader_pdf.py runs Pandoc with --toc --toc-depth=1 so the HTML/PDF opens with a **Contents** block listing all top-level # headings (chapters, parts, appendix). Re-run PDF after manuscript edits.
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- ☐ **Appendix B page markers** — ----- Page N ----- lines are intentional extract artifacts; leave or strip in a dedicated pass.

Redundancy (by design vs accidental)

- **Intentional:** appendix-b-book-vi-compilation-full-text.md repeats the **entire** compilation extract while **Chapters 14–17** (and the Book VI spine table) already cover overlapping *Studies in Absurdity* / Croupier / exchanges material from standalone PDFs. Keep appendix for completeness; prefer **chapter** text for reading.
- **Intentional:** compilation_coverage.csv maps the same vault PDF to **two** Book VI TOC rows where the anthology used one commentary for both “Exchanges Versus Indices” and “Part II.”
- **Numbered chapters:** each **Chapter 1–17** cites a **different** primary vault PDF (no duplicate PDF across those seventeen files).
- ☐ **Diamonds:** Chapters **4** and **8** are different essays (December 2022 vs January 2023); not redundant—keep both headnotes clear.

Last reviewed: edition metadata in book/EDITION.md.

Note on textual variants

Some compendium PDFs exist in **multiple vault filenames** (different crawl timestamps) for the same underlying hash suffix. For drafting, this edition prefers **UTF-8 extracts** under `book/build/text/`, using the **manifest filename stem**; where a vault copy was truncated or empty, text may have been **repaired from a duplicate PDF** with the same content hash (see `book/scripts/repair_extracts_from_duplicates.py`).

Web vs PDF: if you recovered a document from the Wayback Machine or a live site, say so in the chapter headnote.

OCR: none of the Part I Musings chapters in this batch were OCR-only; if you add scanned letters, label OCR lineage in the headnote.

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1. At the time of this writing, a 30-year Treasury bond yields 2.22%. If the yield of that bond increases to 5%, it would produce 2.25x more yield. Increasing the 3.31% yield of the Bank of America bonds by 2.25x produces a yield of 7.45%. A bond with a 4% coupon that yields 7.45% would trade below par.↵□
 2. The source compendium uses the ticker symbol “APPL”; Apple’s ordinary listing on Nasdaq is **AAPL**.↵□
 3. Robert Skidelsky, *John Maynard Keynes*, Vol. 1: *Hopes Betrayed, 1883–1920* (New York: Viking, 1983); Vol. 2: *The Economist as Savior, 1920–1937* (New York: Viking, 1983); Vol. 3: *Fighting for Freedom, 1937–1946* (New York: Viking, 2001).↵□
 4. Milton Friedman & Anna Jacobson Schwartz, *A Monetary History of the United States, 1867–1960* (Princeton, N.J.: Princeton University Press for the National Bureau of Economic Research, 1963).↵□
 5. Friedrich A. Hayek, *Denationalisation of Money: The Argument Refined – An Analysis of the Theory and Practice of Concurrent Currencies* (London: The Institute of Economic Affairs, 1976).↵□
 6. John Kenneth Galbraith, *Money: Whence It Came, Where It Went* (Boston: Houghton Mifflin, 1975).↵□
 7. See minneapolisfed.org.↵□
 8. John F. Wasik, *Keynes’s Way to Wealth: Timeless Investment Lessons from the Great Economist* (New York: McGraw Hill Education, 2014).↵□
 9. [Statista.com](https://www.statista.com)↵□
 10. Security Industry Financial Markets Association (SIFMA).↵□
 11. [Bitinfo.com](http://bitinfo.com)↵□
 12. Energy Information Administration.↵□
 13. Norman Angell, *The Story of Money* (New York: Frederick A. Stokes, 1929), 110–112.↵□
 14. As cited in the original: www.economist.com/content/global_debt↵□
 15. As in original: <http://www.ishares.com/us/products/239563/ishares-high-dividend-etf> (June 10, 2015).↵□